Analog Electronics

Detailed Solutions

GATE ELECTRONICS AND COMMUNICATION ENGINEERING

(EC)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book works, in full, every question GATE has actually asked in Analog Electronics (EC), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, and every solution is taken step by step rather than asserted.

Each solution carries the same identifier as the question it answers in the companion questions volume, so you can move between practice and explanation without a lookup table. Where the examining authority revised an answer or awarded marks to all (MTA), that is noted with the question it affects.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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- 5+ years of teaching experience

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> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:

○○○ ●○○ Easy ●●○ Medium ●●● Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Unit Snapshot

Every unit opens with a **Unit Snapshot** box summarizing that unit's real question data:

Questions	the real total for this unit.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks this unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Unit Snapshot*

Stream: EC	**Questions:** 45	**MCQ:** 30	**MSQ:** 3	**NAT:** 12
1M: 20	**2M:** 22	**Descriptive:** 3	**Avg Weightage** ~4M	**Range:** 2M(2015)–6M(2021)

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

Contributors

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SOLUTIONS

Diode Circuits and Applications

Topics

1. Diode Models with open and short circuit tests
2. Operating Point and Small Signal Analysis of a Diode
3. Zener Diode as a Voltage Regulator
4. Series and Shunt Clipper Circuits
5. Two-Level Clipper Circuits
6. Clamper Circuits and their examples
7. Peak Detector and Average Diode Current
8. Voltage Multiplier Circuits
9. Half Wave Rectifier and Centre Tapped Full Wave Rectifier
10. Full Wave Bridge Rectifier and Capacitive Filters

Unit Snapshot*

Stream: EC	**Questions:** 66	**MCQ:** 39	**MSQ:** 2	**NAT:** 21
1M: 30	**2M:** 34	**Descriptive:** 2	**Avg Weightage** ~3M	**Range:** 1M(2020)–5M(2011)

**See preface for how Avg Weightage and Range are calculated.*

Q1.26.1 NAT | 2026 | 2M | Ans: 4 | ● ○ ○

Given input signal

$$v_i(t) = 6\sin(\omega t)$$

So maximum input = +6 V and two diodes act as clippers.

Case 1: For $v_i < -4$ V, D1 is **OFF**, D2 is **ON**

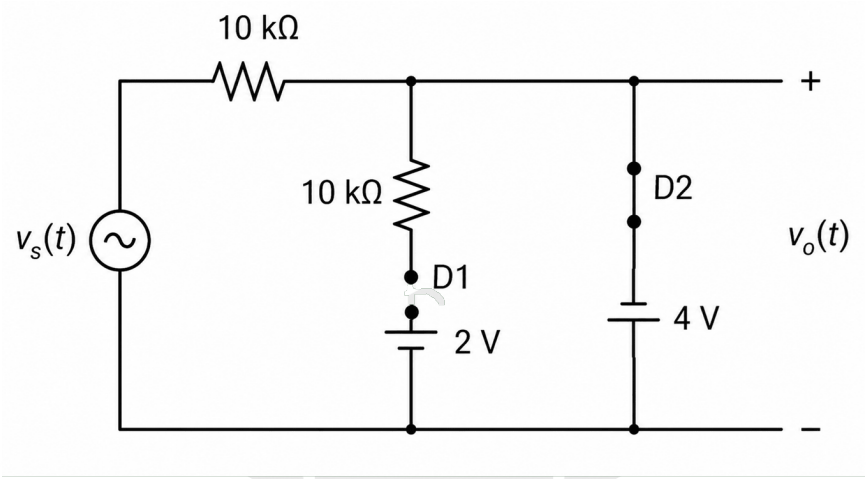

Fig. Solution Q1.26.1

Hence $v_o(t) = -4$ V only while $v_i < -4$ V. For $-4\text{ V} \le v_i \le 2\text{ V}$, both diodes are OFF and $v_o = v_i$.

Case 2: For $v_i > 2$ V, D1 is **ON**, D2 is **OFF**

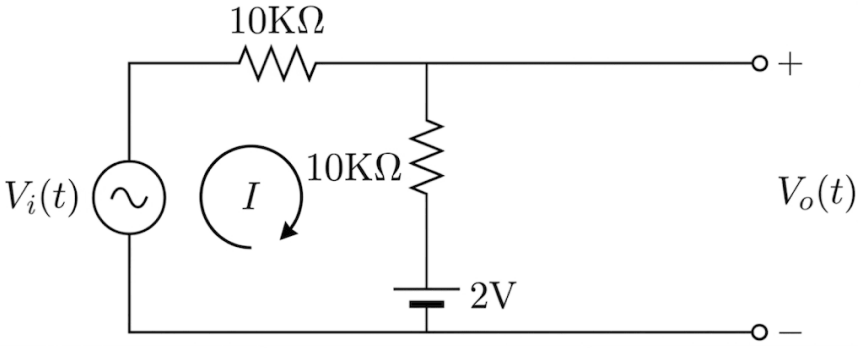

Fig. Solution Q1.26.1

$$I = \frac{6-2}{20\,\text{k}\Omega} = \frac{4}{20\,\text{k}\Omega} = 0.2\,\text{mA}$$

$$V_{0\,\max} = I \times 10\,\text{K} + 2\,\text{V}$$

$$= 0.2\,\text{mA} \times 10\,\text{K} + 2\,\text{V}$$

$$\boxed{V_{0\,\max} = 4\,\text{V}}$$

Key Points

- The above circuit is a 2-way clipper where the output voltage is clipped on both polarities due to diodes.
- Determine the voltages at both terminals of diode to find out whether it is ON or OFF.

Mistakes to be avoided

- Since $V_{i\,\max}$ is $6V$ don't assume $V_{0\,\max}$ is also $6V$ as it is clipped.

Q1.25.1 MSQ | 2025 | 2M | Ans: A,D | ● ○ ○

Given circuit is

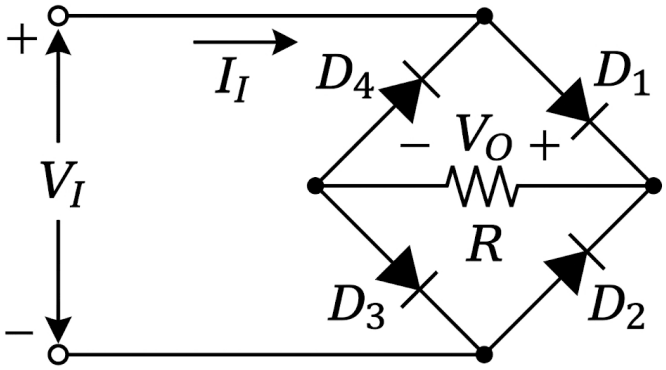

Fig. Solution Q1.25.1

Given, V_{in} is a sweep of $-M$ to $+M$.
For V_0 versus V_{in}:
Let $V_{in} = +M$.
In the bridge rectifier, D_1 and D_3 are forward bias, and D_2 and D_4 are reverse bias.

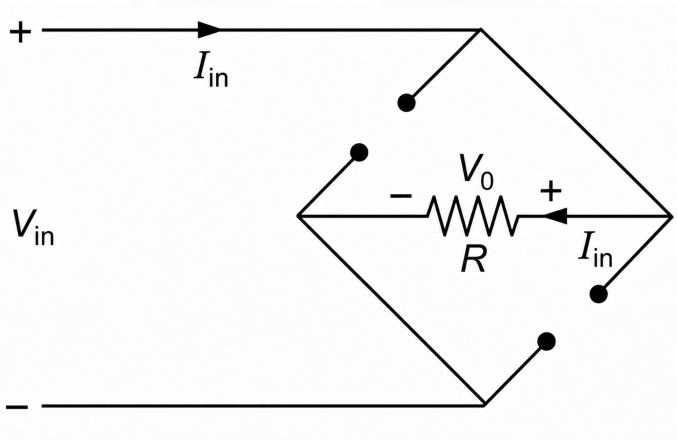

Fig. Solution Q1.25.1

Applying KVL,

$$V_{in} - V_0 = 0 \Rightarrow V_{in} = V_0$$

$$\therefore \text{For } V_{in} = +M \Rightarrow V_0 = M$$

and

$$-V_{in} + I_{in}R = 0$$

$$I_{in} = \frac{V_{in}}{R} = \frac{M}{R}$$

$$\boxed{V_0 = V_I, \quad I_{in} = \frac{M}{R} \times V_I} \quad \text{for } 0 < V_I < M$$

Let $V_{in} = -M$.
D_1 and D_3 are reverse bias, D_2 and D_4 are forward bias.

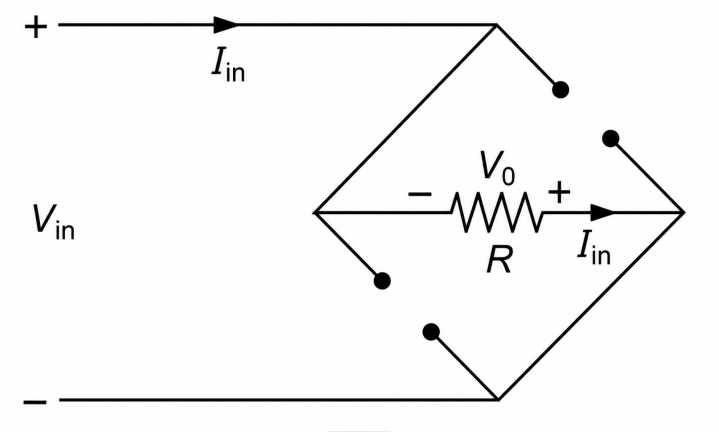

Fig. Solution Q1.25.1

$$\therefore V_{in} + V_0 = 0$$

$$V_0 = -V_{in} = -[-M] = M$$

$$-V_{in} + I_{in} \times R = 0$$

$$I_{in} = \frac{V_{in}}{R} = \frac{-M}{R}$$

$$\boxed{V_0 = -V_I, \quad I_{in} = \frac{M}{R} \times V_I} \quad \text{for } -M < V_I < 0$$

$$\boxed{A, D}$$

Q1.25.2 NAT 2025 2M Ans: 13.33 ●●○

The source frequency is 50 Hz, so one period is 20 ms. From the ideal diode/battery orientation the diode conducts when $v_i > -5$ V, and is OFF when $v_i < -5$ V. Let $\alpha = 100\pi t$. The two crossings of $10 \sin \alpha = -5$ occur at $7\pi/6$ and $11\pi/6$. Their times are 11.6667 and 18.3333 ms. Subtract the unrounded values:

$$t_{OFF} = \frac{20}{3} \text{ ms.}$$

$$t_{ON} = 20 - \frac{20}{3} = \frac{40}{3} \text{ ms.}$$

$$\boxed{t_{ON} = 13.33 \text{ ms}}$$

The requested two-decimal result is 13.33. Keep the crossing times unrounded until after subtraction.

Q1.24.1 MCQ 2024 1M Ans: A ●○○

Given,

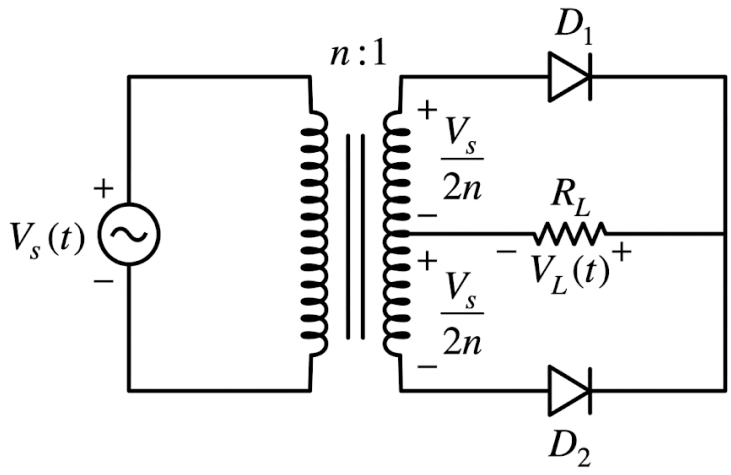

Fig. Solution Q1.24.1

Transformation ratio

$$k = \frac{V_2}{V_1} = \frac{N_2}{N_1} = \frac{1}{n}$$

$$V_2 = \frac{V_1}{n} = \frac{V_S}{n}$$

Case 1: During positive Half cycle. $D_1 \rightarrow$ ON and $D_2 \rightarrow$ OFF

$$V_L = \frac{V_S}{2n} \quad \ldots (i)$$

Case 2: During Negative Half cycle. $D_1 \rightarrow$ OFF and $D_2 \rightarrow$ ON

$$V_L = \frac{-V_S}{2n} \quad \ldots (ii)$$

The DC Output Voltage or Average Output Voltage can be calculated as follows:

$$V_0 = V_{dc} = \frac{1}{\pi}\int_0^{\pi}\left(\frac{V_S}{2n}\right)d(\omega t)$$

$$V_0 = \frac{1}{\pi}\int_0^{\pi}\frac{V_m}{2n}\sin(\omega t)d(\omega t)$$

$$V_0 = \frac{V_m}{2\pi n}\int_0^{\pi}\sin(\omega t)d(\omega t)$$

$$V_{0(\text{avg})} = \frac{V_m}{2\pi n}[-\cos(\omega t)]_0^{\pi}$$

$$V_{0(\text{avg})} = \frac{V_m}{2\pi n}[-\cos(\pi) + \cos(0)]$$

$$V_{0(\text{avg})} = \frac{V_m}{2\pi n}[1+1] = \frac{2V_m}{2\pi n} = \frac{V_m}{\pi n}$$

$$V_{0(\text{avg})} = \frac{2.5}{\pi} = \frac{V_m}{\pi n}$$

$$n = \frac{V_m}{2.5} = \frac{10}{2.5} = 4$$

$$\boxed{(A) \quad 4}$$

Key Points

- The transformation ratio relates the primary and secondary voltages of a transformer: $k = \frac{N_2}{N_1}$.
- For a full-wave rectified signal, the average (DC) voltage is calculated by integrating the waveform over one half-cycle: $V_{dc} = \frac{1}{\pi}\int_0^{\pi} V(\omega t)d(\omega t)$.
- In case of a centre-tapped transformer the secondary voltage is divided equally across the 2 halves of secondary winding.

Q1.24.2 NAT 2024 1M Ans: 0.25 ●●○

In the given circuit,
For $t > 0$, the capacitor starts discharging through 10 kΩ resistor and zener diode.
The voltage across capacitor is given by,

$$V_C(t) = 10e^{-t/RC}, \quad R \times C = 0.1 \text{ sec}$$

Current in the circuit is $i(t) = I_0 e^{-t/RC}$ where

$$I_0 = I(t=0) = \frac{V_C(0) - V_Z}{R} = \frac{10-5}{10\,\text{k}} = 0.5\,\text{mA}$$

Total energy dissipated in zener diode is,

$$\boxed{W = \int_0^{\infty} V_Z \times i(t)\,dt}$$

$$W = \int_0^{\infty} 5 \times 0.5e^{-t/RC}\,\text{mJ}$$

$$= 2.5[-0.1]\left[e^{-t/0.1}\right]_0^{\infty}$$

The total energy dissipated $W = 0.25$ mJ

Key Points

- The voltage across a discharging capacitor follows the exponential decay function $V_C(t) = V_0 e^{-t/\tau}$, where the time constant is $\tau = RC$.
- A Zener diode in the breakdown region acts as a constant voltage drop (V_Z). It will cut off and stop conducting when the driving voltage (V_C in this case) drops below V_Z.

Q1.23.1 MCQ 2023 2M Ans: A ●○○

Case I:

$$V_\gamma = 0.7\,\text{V}$$

For the +ve half cycle if input V_{in},

$$D_1 \rightarrow \text{ON} \quad \text{and} \quad D_2 \rightarrow \text{OFF}$$

For diode D_1:

$$V_{in} - 1\,\text{V} > 0.7$$

$$V_{in} > 1.7\,\text{V}$$

$$V_0 = V_{in} - 0.7$$

Case II: For the +ve half cycle if input V_{in},

$$D_1 \rightarrow \text{OFF} \quad \text{and} \quad D_2 \rightarrow \text{ON}$$

For diode D_2:

$$1 - V_{in} > 0.7$$

$$V_{in} < 0.3\,\text{V}$$

$$V_0 = V_{in} + 0.7$$

Case III:

$$0.3\,\text{V} < V_{in} < 1.7\,\text{V}$$

$$D_1 \rightarrow \text{OFF} \quad \text{and} \quad D_2 \rightarrow \text{OFF}$$

$$V_0 = 1\,\text{V}$$

Hence by combining all 3 cases option A matches the answer.

Key Points

- When both diodes are reverse-biased (OFF), the output node follows the potential of the constant reference source, yielding $V_0 = 1\,\text{V}$.
- The transfer characteristic is constructed by mapping V_0 as a function of V_{in} piecewise across the determined threshold regions.
- For this question we can also verify the options and find out correct answer instead of solving.

Q1.22.1 NAT | 2022 | 2M | Ans: 0.75 | ● ○ ○

When diode is OFF:

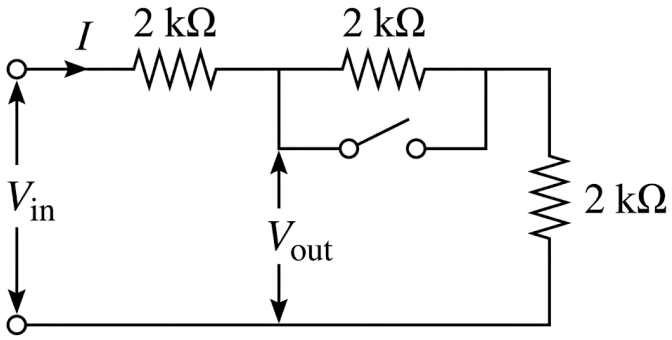

Fig. Solution Q1.22.1

$$V_{out} = 4I$$

$$V_{in} = 6I$$

$$A_v = \frac{V_{out}}{V_{in}} = \frac{4}{6} = \frac{2}{3}$$

When diode is ON, voltage drop across diode is 0.7 V.

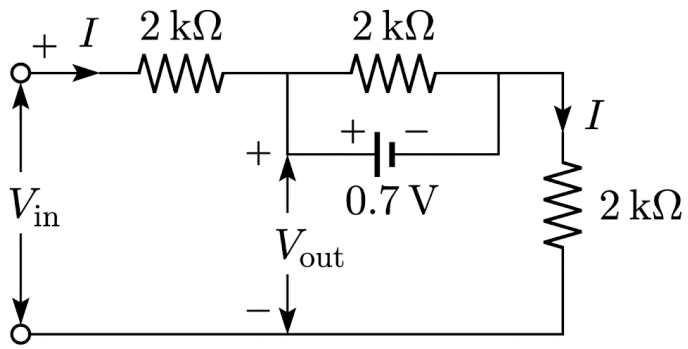

Fig. Solution Q1.22.1

$$V_o = 0.7 + (2\,\text{k}\Omega)I$$

$$V_{in} = 0.7 + (4\,\text{k}\Omega)I$$

$$I = \frac{V_o - 0.7}{2\,\text{k}\Omega}$$

$$V_{in} = 0.7 + (4\,\text{k}\Omega) \cdot \frac{(V_o - 0.7)}{(2\,\text{k}\Omega)} = 0.7 + 2V_o - 1.4$$

$$V_o = \frac{V_{in} + 0.7}{2}$$

$$A_v' = \frac{\partial V_o}{\partial V_{in}} = \frac{1}{2}$$

Maximum voltage gain, $A_v = \frac{2}{3}$ and Minimum voltage gain, $A_v' = \frac{1}{2}$

$$\frac{(\partial V_{out}/\partial V_{in})_{\min}}{(\partial V_{out}/\partial V_{in})_{\max}} = \frac{A_v'}{A_v} = \frac{1}{2} \times \frac{3}{2} = \frac{3}{4} = 0.75$$

Key Points

- From the graph the cut-in voltage of the diode is 0.7 V.
- When the diode is OFF, the circuit behaves as a simple linear resistive divider.
- During AC analysis DC voltage sources must be short circuited.

Mistakes to be avoided

- Not short circuiting the 0.7 V source while finding gain.

Q1.21.1 NAT | 2021 | 2M | Ans: 10 | ● ○ ○

$V_{in} = 10$ V: Diode is ON

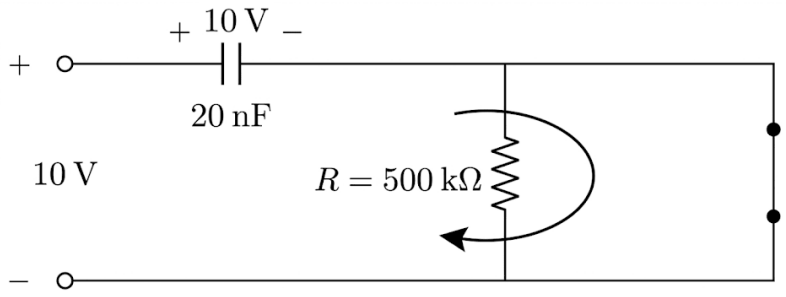

Fig. Solution Q1.21.1

$\therefore$ Capacitor charges upto 10 V, i.e, $V_C = 10\,\text{V}$
when $V_{\text{in}} = 0$; Diode is OFF

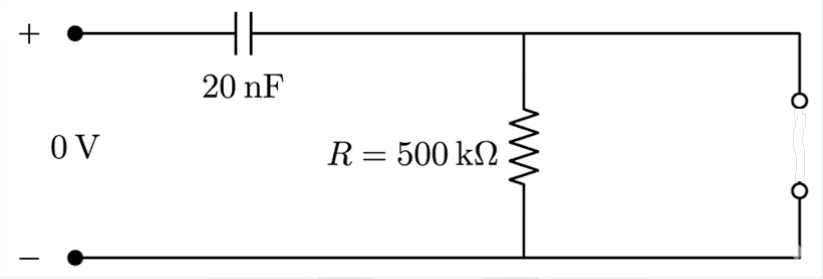

Fig. Solution Q1.21.1

$$\text{Discharging time constant} = R \times C = 10\,\text{m sec}$$

$$\tau_{\text{discharging}} \gg \tau_{\text{OFF}}$$

Hence capcacitor discharge is negligible. So, In steady state, $V_C = 10\,\text{V}$

$$V_{\text{out}} = V_{\text{in}} - V_C = V_{\text{in}} - 10\,\text{V}$$

When $V_{\text{in}} = 10\,\text{V}$

$$\Rightarrow V_{\text{out}} = 10 - 10 = 0\,\text{V}$$

$V_{\text{in}} = 0\,\text{V}$

$$\Rightarrow V_{\text{out}} = 0 - 10 = -10\,\text{V}$$

$$\boxed{V_{\text{out(max)}} - V_{\text{out(min)}} = 0 - (-10) = 10\,\text{V}}$$

Key Points

- The circuit operates as a clamper, shifting the input waveform by a DC level determined by the capacitor voltage.
- When the diode is forward-biased, the capacitor quickly charges to the maximum input voltage ($V_C = 10\,\text{V}$).
- Since the discharging time constant ($\tau = RC$) is much greater than the OFF time ($\tau_{\text{discharging}} \gg \tau_{\text{OFF}}$), the capacitor retains its charge and acts as a constant 10 V voltage source.

Q1.20.1 NAT 2020 1M Ans: 650.54 ● ○ ○

The ideal circuit is a voltage doubler: a clamper followed by a peak detector. The sinusoidal input is 230 V RMS, giving peak $V_m = 230\sqrt{2}$ V. In unloaded steady state the output magnitude is twice that peak:

$$|V_o| = 2V_m = 460\sqrt{2}\,\text{V},$$

$$|V_o| = 650.5382387\ldots\ \text{V}.$$

$$\boxed{650.54\,\text{V (two decimals)}}$$

The result rounded to two decimal places is 650.54.

Q1.19.1 NAT 2019 1M Ans: -3 ● ○ ○

Case 1: For $0 < t < T/2$, $V_s = 8$ V. So, diode will be off and can be replaced with open circuit

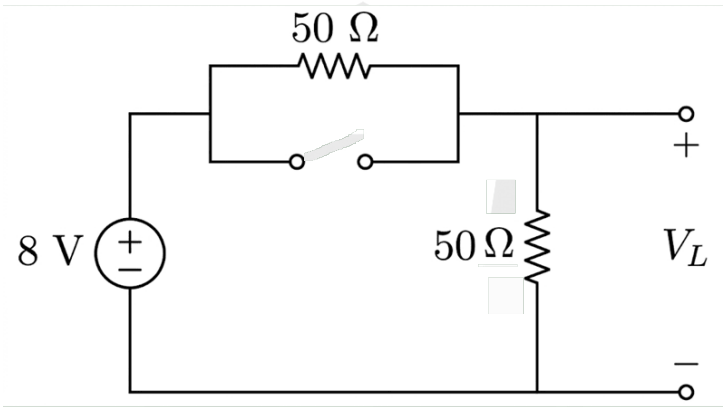

Fig. Solution Q1.19.1

$$\therefore \quad V_L = \left[\frac{50}{50+50}\right] \times 8 = 4\,\text{V} \quad \text{[by Voltage Division Rule]}$$

Case 2: For $T/2 < t < T$, $V_s = -10$ V. So, diode will be ON and hence it will be short circuited and R_1 can be neglected as it comes across short circuit.

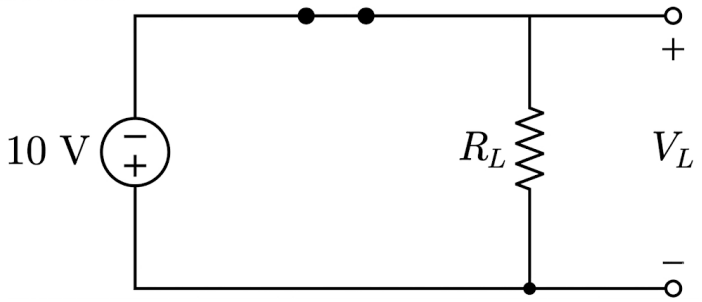

Fig. Solution Q1.19.1

$$\therefore V_L = -10\,\text{V}$$

Therefore, equivalent output waveform will be

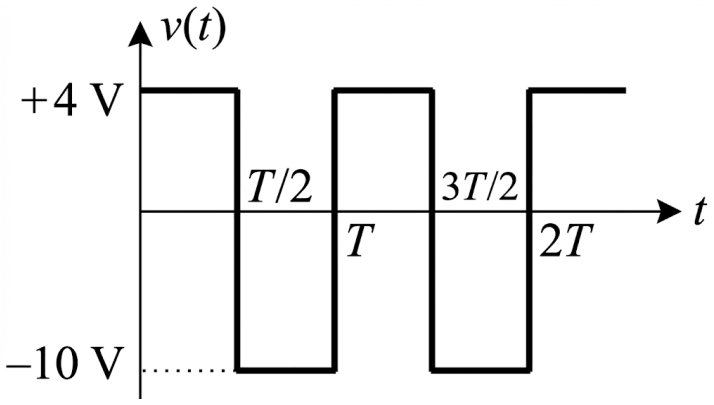

Fig. Solution Q1.19.1

Average value of $V_L(t)$ is given as

$$V_{av} = \frac{1}{T}\int_0^T v_L(t)dt$$

$$V_{L_{av}} = \frac{1}{T}\left[\int_0^{T/2} 4 \cdot dt + \int_{T/2}^{T} (-10)dt\right]$$

$$V_{L_{av}} = \frac{1}{T}\left[4 \times \frac{T}{2} - 10 \times \frac{T}{2}\right]$$

$$V_{L_{av}} = \frac{1}{T} \times (-6) \times \frac{T}{2} = -3\,\text{V}$$

(OR)

$$V_{av} = \frac{\text{Area of one period of wave from}}{\text{Time period}}$$

$$V_{av} = \frac{\left(4 \times \frac{T}{2} - 10 \times \frac{T}{2}\right)}{T}$$

$$V_{av} = \frac{\frac{T}{2}(4-10)}{T} = \frac{-6}{2} = -3\,\text{V}$$

The average value of V_L is -3 V.

Key Points

- Analyze the diode state by finding out voltages at both terminals.
- When the diode is forward-biased (ON) and ideal, it acts as a short circuit, bypassing any parallel resistor.

Q1.19.2 MCQ 2019 2M Ans: B ●●○

Given,

$$V_z = 20\,\text{V}$$

$$I_{z\,\max} = 60\,\text{mA}$$

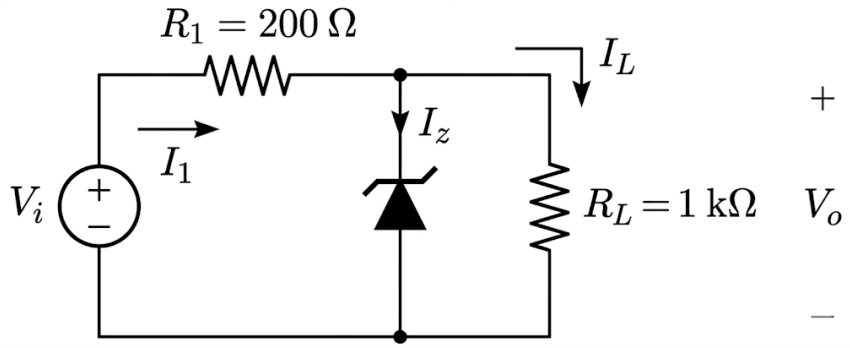

Fig. Solution Q1.19.2

Case 1: Finding Lower limit of V_i

Initially for lower values of V_i zener diode is OFF, so by applying Open Circuit Test,

$$V_o = \frac{V_i \times 1\text{k}}{0.2\text{k} + 1\text{k}} = \frac{V_i}{1.2}$$

Zener diode will be ON i.e. it goes into breakdown, when $V_L > V_z$ or $V_0 > V_z$

$$\frac{V_i}{1.2} > 20\,\text{V}$$

$$\boxed{V_i > 24\,\text{V}}$$

So V_i should be atleast 24V so that zener is maintained in ON state.

Case 2: Finding upper limit of V_i

When Zener diode is in breakdown region,

$$I_1 = \frac{V_i - 20}{0.2\,\text{k}\Omega} = \frac{V_i - 20}{0.2}\,\text{mA}$$

$$I_L = \frac{V_o}{R_L} = \frac{20}{1\,\text{k}\Omega} = 20\,\text{mA}$$

$$I_z = I_1 - I_L = \frac{V_i - 20}{0.2} - 20$$

For safe operation, $I_z \le I_{z\,\max}$, as if $I_z > I_{z\,\max}$ then zener will be damaged and no longer be considered ON

$$\frac{V_i - 20}{0.2} - 20 \le 60$$

$$\boxed{V_i \le 36\,\text{V}}$$

When V_i is at maximum value and load current is minimum($I_{L\,\min}$) then zener current is maximum($I_{Z\,\max}$)

$$I_{L\,\min} = \frac{20}{1\text{k}} = 20\text{mA}$$

$$\frac{V_i - 20}{200} \le I_{Z\,\max} + I_{L\,\min}$$

$$\frac{V_i - 20}{200} \le 80\,\text{mA}$$

$$\frac{V_i - 20}{200} \le \frac{80}{1000}$$

$$V_i - 20 \le 16$$

$$\boxed{V_i \le 36\,\text{V}}$$

Hence,

$$\boxed{\text{(B)} \quad 24 < V_i < 36\,\text{V}}$$

Key Points

- It is a voltage regulator because if there is a fluctuation in the load or input voltage , those are absorbed by zener as current.
- The minimum input voltage ($V_{i(\min)}$) required to turn the Zener diode ON is found by assuming the diode is open and calculating the voltage across it using the voltage divider rule.
- The maximum input voltage ($V_{i(\max)}$) is determined by the maximum current rating of the Zener diode ($I_{z\,\max}$).

Q1.18.1 MCQ | 2018 | 2M | Ans: B ●●○

Given $V_Z = 5\,\text{V}$, $I_{Z\,\min} = 2\,\text{mA}$, $R_L = 1\,\text{k}\Omega$.
Nominal input voltage $V_I = 6\,\text{V}$.
Variation in $V_I = \pm 5\% \Rightarrow V_I = 6 \pm 0.05(6) = 6 \pm 0.3\,\text{V}$.
Thus, $V_{I(\min)} = 5.7\,\text{V}$ and $V_{I(\max)} = 6.3\,\text{V}$.

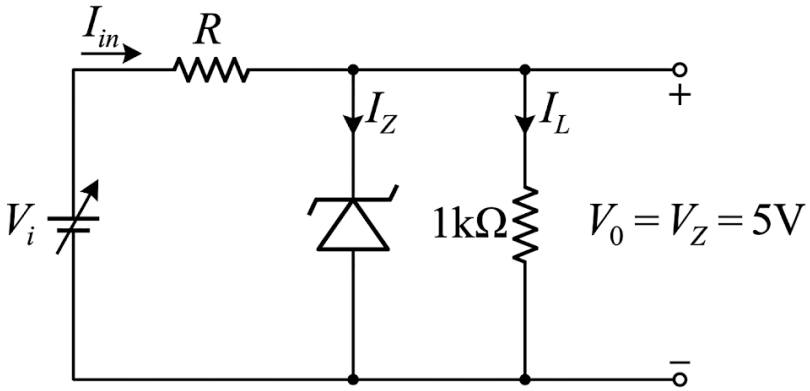

Fig. Solution Q1.18.1

Load current I_L is constant:

$$I_L = \frac{V_Z}{R_L} = \frac{5}{1\,\text{k}\Omega} = 5\,\text{mA}$$

The Zener diode requires a minimum current $I_{Z\,\min}$ to maintain regulation. The worst-case condition for maintaining regulation occurs when the input voltage is at its minimum. At this point, the total input current is minimum:

$$I_{in(\min)} = I_{Z\,\min} + I_L$$

$$I_{in(\min)} = 2\,\text{mA} + 5\,\text{mA} = 7\,\text{mA}$$

The value of the series resistance R is calculated using this minimum input current:

$$R = \frac{V_{I(\min)} - V_Z}{I_{in(\min)}}$$

$$R = \frac{5.7 - 5}{7 \times 10^{-3}} = \frac{0.7}{7 \times 10^{-3}} = 100\,\Omega$$

The maximum power dissipation in the Zener diode occurs when the Zener current is maximum. This happens when the input voltage is at its maximum. The maximum input current is:

$$I_{in(\max)} = \frac{V_{I(\max)} - V_Z}{R}$$

$$I_{in(\max)} = \frac{6.3 - 5}{100} = 13\,\text{mA}$$

The maximum Zener current is:

$$I_{Z\,\max} = I_{in(\max)} - I_L$$

$$I_{Z\,\max} = 13\,\text{mA} - 5\,\text{mA} = 8\,\text{mA}$$

The minimum required power dissipation rating of the diode is:

$$P_{Z\,\max} = V_Z \times I_{Z\,\max} = 5\,\text{V} \times 8\,\text{mA} = 40\,\text{mW}$$

The Zener must be rated to atleast withstand the $P_{Z\,\max}$

(B) 100Ω and 40 mW

Key Points

- In a Zener voltage regulator, the series resistor R must be sized such that the Zener diode receives at least its minimum required current ($I_{Z\,\min}$) even when the input voltage is at its minimum and the load current is at its maximum.
- The maximum power dissipation of the Zener diode must be evaluated under the condition of maximum input voltage and minimum load current (or maximum Zener current).

Mistakes to be avoided

- Considering minimum zener current for minimum power rating.
- In the question it is asked "Minimum Power **Rating** of Device", one must confuse it with minimum power dissipated across zener. So read the question carefully.

Q1.18.2 NAT 2018 2M Ans: 6.4 ● ● ●

Given, $I_{dc} = 26\,\mu$A, Ideality factor $\eta = 1$, $C_d = 0.5$ nF, $\omega = 2 \times 10^6$ rad/sec, $V_T = 26$ mV.
To find small signal component of diode current deactivate dc source and replace diode by its equivalent small signal model.

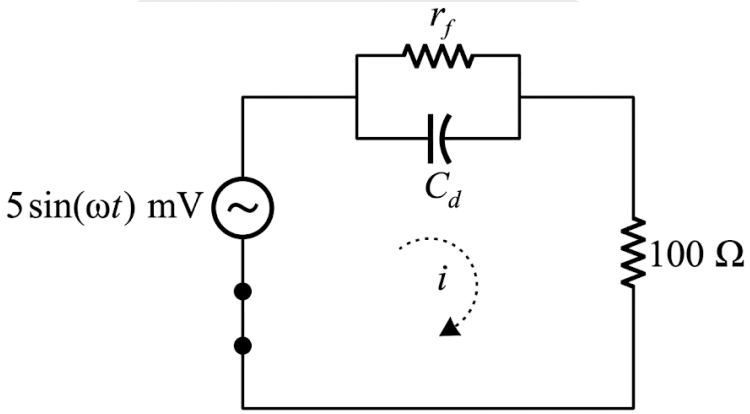

Fig. Solution Q1.18.2

Here,

$$r_f = \frac{\eta V_T}{I} = \frac{1 \times 26 \times 10^{-3}}{26 \times 10^{-6}} = 10^3\,\Omega$$

$$i = \frac{5\sin(\omega t)\text{ mV}}{Z_{eq}}$$

where,

$$Z_{eq} = \left(r_f \parallel \frac{1}{sC_d}\right) + 100$$

$$Z_{eq} = \left(\frac{r_f/sC_d}{(1 + sr_fC_d)/sC_d}\right) + 100$$

$$Z_{eq} = \frac{r_f}{1 + j\omega r_f C_d} + 100$$

$$Z_{eq} = \left[\frac{1000}{1 + j \times 2 \times 10^6 \times 10^3 \times 0.5 \times 10^{-9}}\right] + 100$$

$$Z_{eq} = \frac{1000}{1 + j} + 100 = \frac{1100 + j100}{1 + j}$$

$$Z_{eq} = \frac{\sqrt{(1100)^2 + (100)^2}}{\sqrt{(1)^2 + (1)^2}} \angle \left[\tan^{-1}\left(\frac{100}{1100}\right) - \tan^{-1}\left(\frac{1}{1}\right)\right]$$

$$Z_{eq} = 781.02 \angle -39.805^\circ$$

$$i = \frac{5\sin(\omega t)\ \text{mV}}{781.02 \angle -39.805^\circ\ \Omega}$$

$$i = 6.401\sin(\omega t + 39.805^\circ)\ \mu\text{A}$$

Hence amplitude of small signal current is 6.401 μA. Rounding correct to one decimal place, we get 6.4 μA.

The amplitude of small signal component of diode current is 6.4μA

Key Points

- The small-signal dynamic resistance of a diode is calculated as $r_f = \frac{\eta V_T}{I_{dc}}$.
- During AC small-signal analysis, independent DC voltage sources are replaced by short circuits.
- The small-signal equivalent model of a forward-biased diode consists of its dynamic resistance (r_f) in parallel with its junction capacitance (C_d).
- The equivalent AC impedance of the circuit (Z_{eq}) must be evaluated using complex arithmetic to accurately find the amplitude and phase shift of the resulting small-signal current.

Mistakes to be avoided

- Connecting diode capacitance in series.
- Calculation mistakes in complex analysis (rectangular to polar and vice versa).

Q1.17.1 NAT | 2017 | 1M | Ans: 3.18 | ● ○ ○

Case 1 : During positive half cycle, Diode is ON, hence it can be replaced by short circuit.

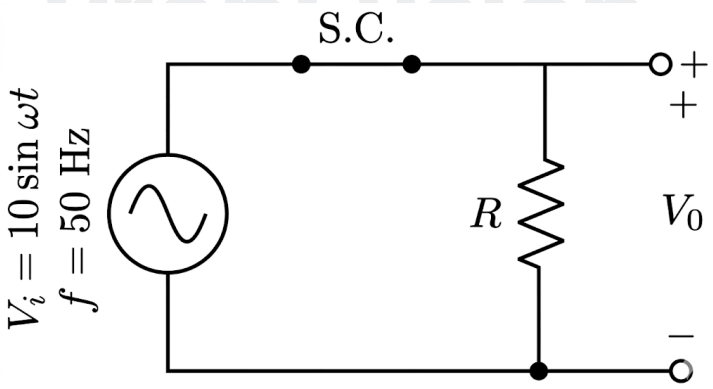

Fig. Solution Q1.17.1

Hence,

$$V_0 = V_i$$

Case 2 : During negative half cycle, Diode is reverse biased, hence it is OFF and therefore it can be replaced by open circuit.

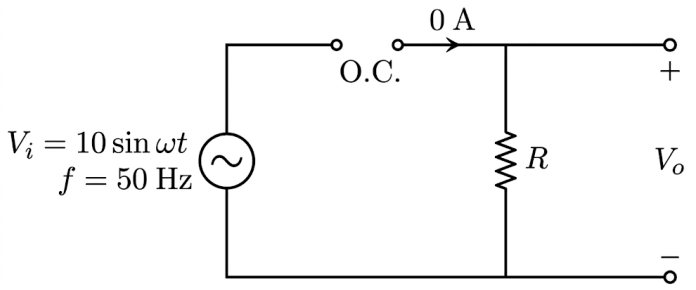

Fig. Solution Q1.17.1

Hence,

$$V_0 = 0\,\text{V}$$

Hence, from case 1 and case 2, V_0 can be expressed as, given below.

$$V_0 = \begin{cases} V_i; & V_i > 0 \\ 0; & V_i < 0 \end{cases}$$

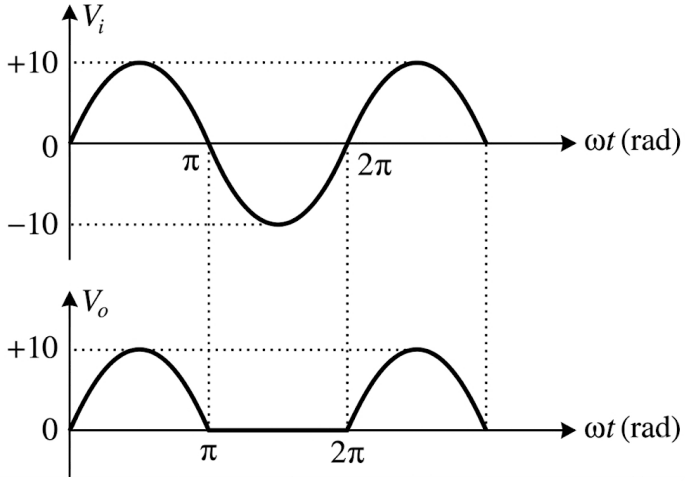

Fig. Solution Q1.17.1

Average value of V_0 is given by,

$$V_{0(\text{avg})} = \frac{1}{2\pi}\int_0^{2\pi} V_0(t)\,d\omega t$$

$$V_{0(\text{avg})} = \frac{1}{2\pi}\int_0^{\pi} 10 \sin \omega t\, d\omega t$$

$$V_{0(\text{avg})} = \frac{10}{2\pi}[-\cos \omega t]_0^{\pi}$$

$$V_{0(\text{avg})} = \frac{10}{2\pi}[-\cos \pi + \cos 0]$$

$$V_{0(\text{avg})} = \frac{10[1+1]}{2\pi} = \frac{10}{\pi} = 3.18\,\text{volt}$$

Hence, the reading on the DC voltmeter is **3.18 V**.

Key Points

- A DC voltmeter always reads the average (or DC) value of a time-varying periodic signal.
- The average value of a half-wave rectified sinusoidal waveform with a peak voltage V_m is derived as $V_{avg} = \frac{V_m}{\pi}$.

Q1.17.2 MCQ 2017 1M Ans: A

Case 1 : When $0 < t < \frac{T}{2}$, $V_{in}(t) = 14\,V$

Diode D_1 is in forward bias and hence it can be replaced by 0.7 V.

Zener diode D_2 is in reverse bias and because voltage across D_2 is more than breakdown voltage, hence it is operating in breakdown region. Therefore, it can be replaced by its breakdown voltage.

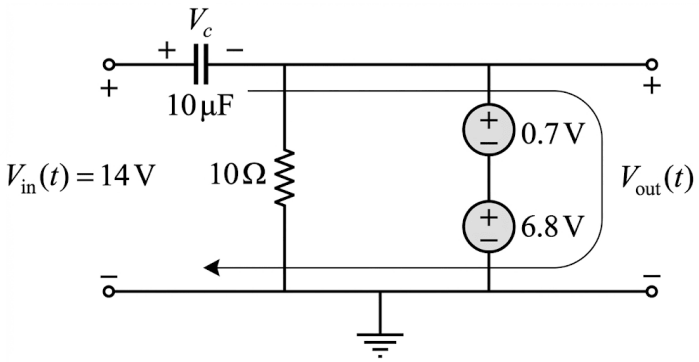

Fig. Solution Q1.17.2

Hence,

$$V_{out\,(max)} = 0.7 + 6.8 = 7.5\,V$$

Applying KVL in the loop shown in figure,

$$-14 + V_c + 7.5 = 0$$

$$V_c = 6.5\,V$$

At $t = \frac{T}{2}$, capacitor will charge upto 6.5 V due to small time constant ($10\tau \ll T$).

Case 2 : When $\frac{T}{2} < t < T$,

$$V_{in}(t) = -14\,V$$

Diode D_1 is in reverse bias and hence it can be replaced by open circuit. Zener diode D_2 is forward biased and therefore, it will act as normal diode. Hence it can be replaced by short circuit. Equivalent circuit is shown in below figure.

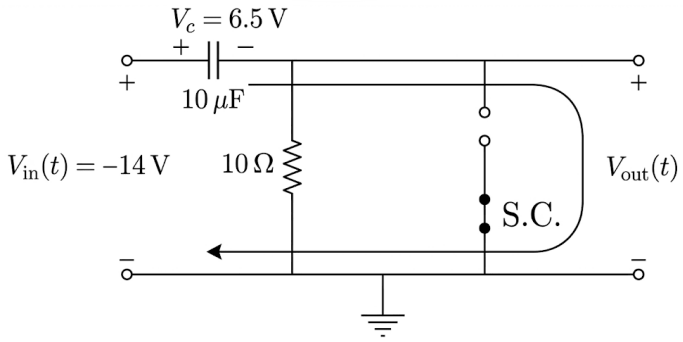

Fig. Solution Q1.17.2

Applying KVL in the loop shown in figure,

$$-V_{in} + V_c + V_{out} = 0$$

$$14 + 6.5 + V_{out} = 0$$

$$V_{out\,(\min)} = -20.5\ \text{V}$$

(A) 7.5 V and − 20.5 V

Key Points

- In the above clamper circuit with a sufficiently small time constant ($\tau \ll T$), the capacitor rapidly charges to its steady-state voltage during the phase when the diode is forward-biased.
- The voltage stored across the capacitor acts as a constant DC voltage source that shifts the entire input waveform up or down.
- When analyzing Zener diodes in series with regular PN junction diodes, carefully evaluate the polarity to determine if the branch operates at the forward voltage drop (e.g., 0.7 V) or the Zener breakdown voltage (V_Z).

Mistakes to be avoided

- Reversing the polarity of voltage of zener diode. Note that zener is more useful in reverse bias and PN diode is more useful in forward bias.

Q1.16.1 NAT | 2016 | 2M | Ans: 1 | ● ○ ○

Method 1

Given circuit is,

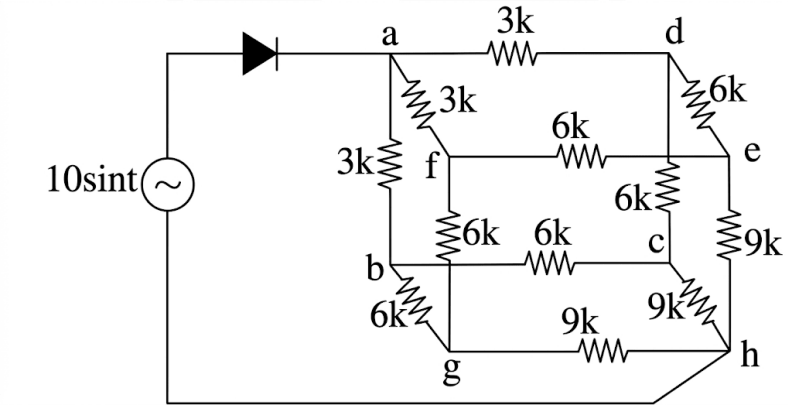

Fig. Solution Q1.16.1

Here if we draw a line between a and h nodes, the circuits on both sides of the line have exactly same number and values of resistors.

Here all the paths to reach from a to h encounter same total resistance. For example the paths $a \rightarrow d \rightarrow e \rightarrow h$ has total resistance of 3k + 6k + 9k = 18k and the path $a \rightarrow f \rightarrow g \rightarrow h$ also has total resistance of 3k + 6k + 9k = 18k.

So we can say that equal amount of current flows through all the 3kΩ resistors, hence nodes b,d,f have same potentials. Similarly nodes c,e,g have same potentials. So based on symmetry of potential, we can join the nodes that have same potential and the circuit becomes,

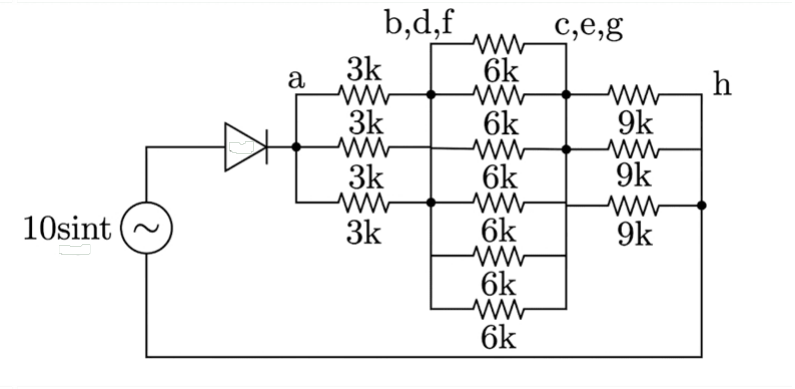

Fig. Solution Q1.16.1

By reducing the parallel combinations, the equivalent resistance R_{eq} of the network is:

$$R_{eq} = \frac{3\,\text{k}\Omega}{3} + \frac{6\,\text{k}\Omega}{6} + \frac{9\,\text{k}\Omega}{3}$$

$$R_{eq} = 1\,\text{k}\Omega + 1\,\text{k}\Omega + 3\,\text{k}\Omega = 5\,\text{k}\Omega$$

The equivalent circuit and its current waveform are shown below:

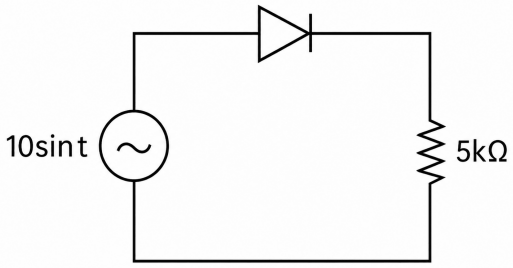

Fig. Solution Q1.16.1

The peak current (I_m) is calculated using Ohm's law:

$$I_m = \frac{10\,\text{V}}{5\,\text{k}\Omega} = 2\,\text{mA}$$

For the half-wave rectified current, the RMS value is:

$$I_{D(RMS)} = \frac{I_m}{2}$$

$$\boxed{I_{D(RMS)} = \frac{2\,\text{mA}}{2} = 1\,\text{mA}}$$

Method 2

The circuit is

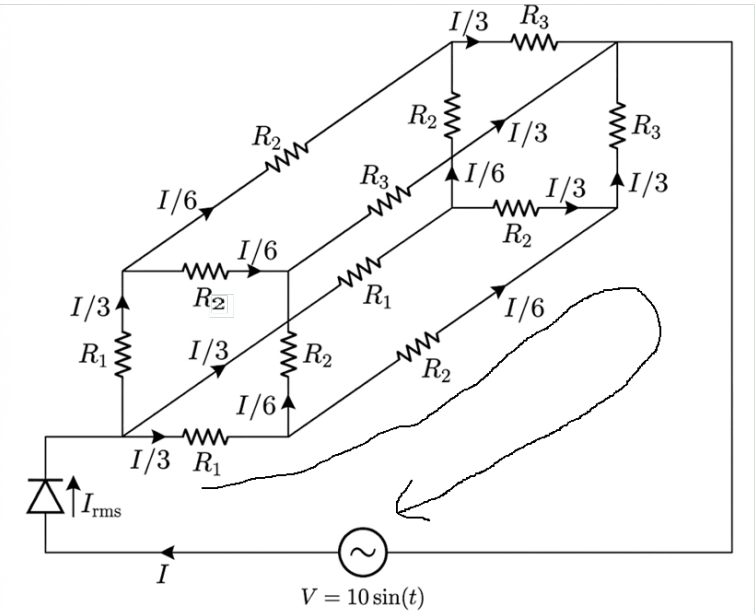

Fig. Solution Q1.16.1

Case 1 : During positive half cycle, Diode will be ON.
Applying KVL in the above loop,

$$V = \frac{I}{3} \times R_1 + \frac{I}{6} \times R_2 + \frac{I}{3} \times R_3$$

$$V = \frac{I}{3} \times 3 + \frac{I}{6} \times 6 + \frac{I}{3} \times 9 = 5I$$

$$\boxed{I = \frac{V}{5} = \frac{10 \sin t}{5} = 2 \sin t \text{ mA}}$$

Case 2 : During negative half cycle, Diode will be OFF.
Hence,

$$\boxed{I = 0 \text{ A}}$$

For the half-wave rectified current, the RMS value is:

$$I_{D(RMS)} = \frac{I_m}{2}$$

$$\boxed{I_{D(RMS)} = \frac{2 \text{ mA}}{2} = 1 \text{ mA}}$$

Key Points

- **Symmetry in Networks:** Nodes that are topologically equivalent and equidistant from the source/sink in a symmetrical resistor network are at the same equipotential. They can be shorted together to simplify the circuit into series-parallel combinations.

Q1.16.2 NAT 2016 1M Ans: 0.25 ● ○ ○

Assume diode D is ON, hence it can be replaced by cut-in voltage i.e. V_{on}.

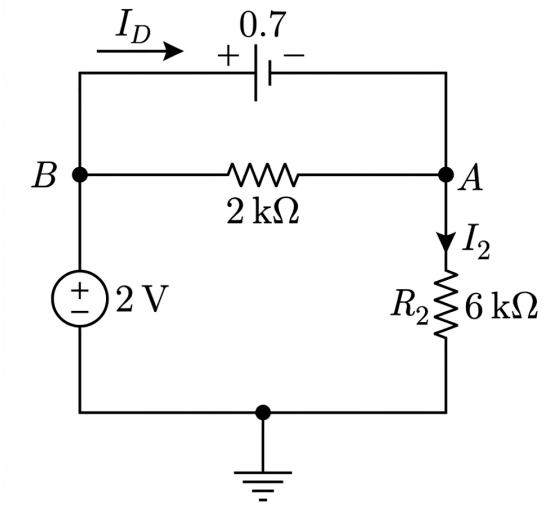

Fig. Solution Q1.16.2

From the above figure,

$$V_B = 2\,\text{V}, V_A = V_B - 0.7 = 1.3\,\text{V}$$

Applying KCL at node A,

$$\frac{V_A - V_B}{2} + \frac{V_A}{6} = I_D$$

$$\frac{1.3 - 2}{2} + \frac{1.3}{6} = I_D$$

$$\boxed{I_D = -0.13\,\text{mA}}$$

For diode to be ON, I_D must be positive from p to n.
From the above analysis, I_D is negative. So, diode will be OFF.
Hence, our assumption is wrong.
Since, diode is OFF, hence it can be replaced by open circuit.
Hence, the modified figure is shown below,

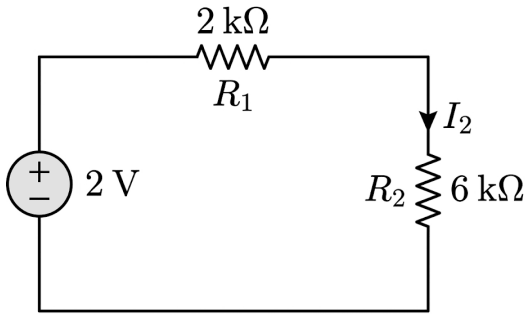

Fig. Solution Q1.16.2

Applying KVL in above figure,

$$I_2 = \frac{2}{8} = \frac{1}{4} = 0.25\,\text{mA}$$

Hence, the magnitude of the current I_2 is equal to **0.25 mA**

Key Points

- When analyzing ideal or constant-voltage drop diode circuits, always assume a state (ON/OFF), solve the circuit, and then verify the assumption.
- For a diode assumed to be ON, the calculated forward current (I_D) must be positive. If $I_D < 0$, the assumption is invalid and the diode is OFF.

Mistakes to be avoided

Proceeding with the final calculation without verifying the boundary conditions of the assumed diode state (current direction for ON state, or voltage polarity for OFF state).

Q1.16.3 NAT | 2016 | 2M | Ans: 2 | ●●○ | CLICK HERE FOR VIDEO SOLUTION

Given : $I_{DC} = I_0 = 1\text{ A}, C = 475\,\mu\text{F}, T = 1\text{ msec}$

Case 1: $0 < t < T/4$

when input is rising to its peak, the diode is ON, and the input is directly connected across the capacitor, so the capacitor instantly charges upto input voltage i.e, capacitor voltage follows input voltage.

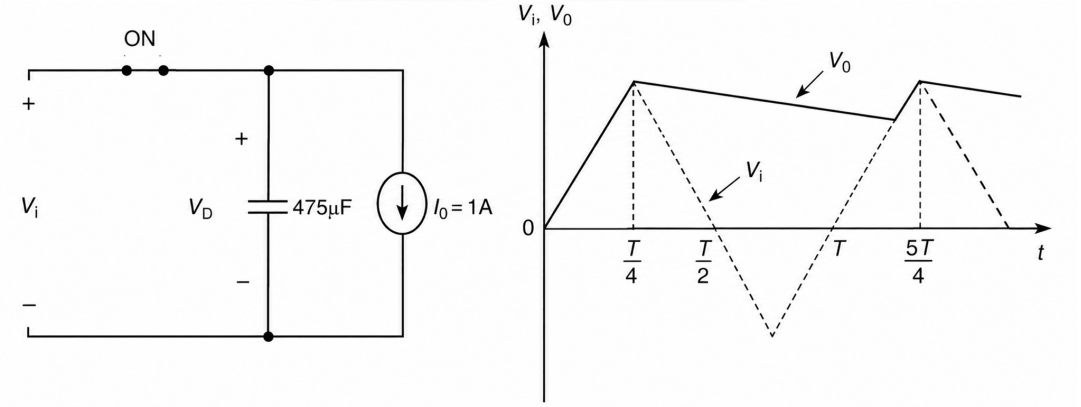

Fig. Solution Q1.16.3

$$\text{So, at } t = \frac{T}{4}, V_C(t) = 10\text{ V}$$

Case 2: $t > T/4$

When $t = (T/4)^+$ input voltage goes below 10V but capacitor is holding 10V so diode goes OFF and the circuit becomes,

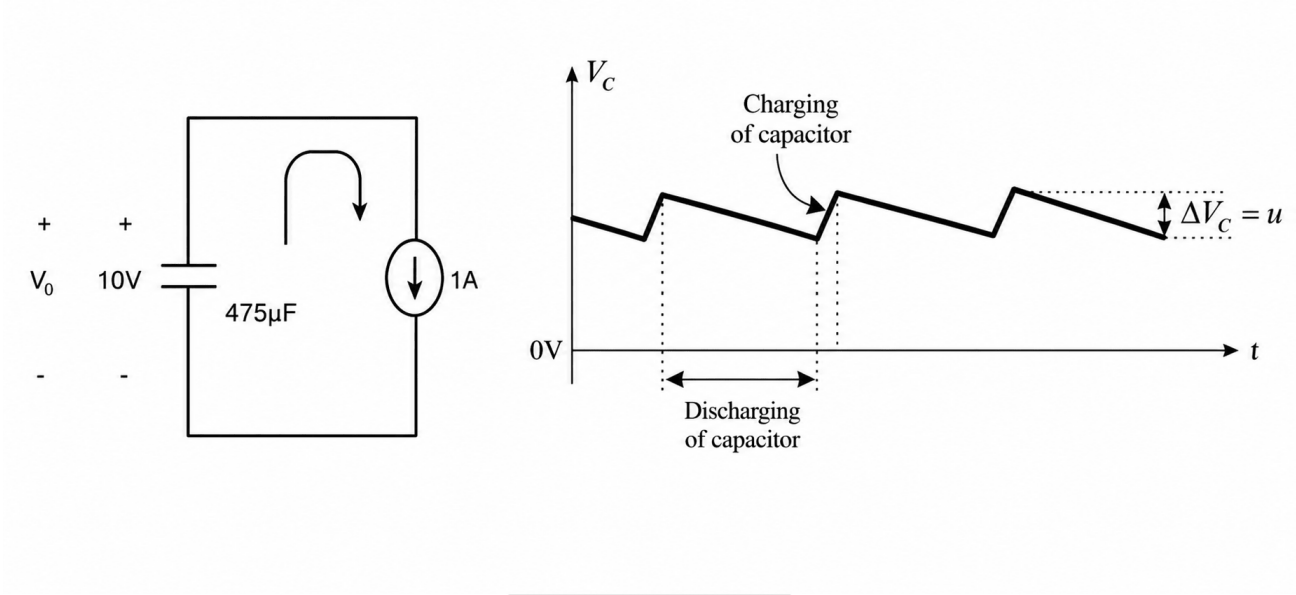

Fig. Solution Q1.16.3

After $t = T/4$ the capacitor starts discharging slowly through the current source(or we can say charging in reverse direction) and the voltage drops from 10V until it encounters a greater input voltage where it starts charging again. But from the figure the charging time is very small and negligible and we consider that capacitor is discharging throughout its time period. During discharging time, diode will be OFF and hence,

$$I_0 = C\frac{dV_C}{dt} \quad \text{[Ohm's law]}$$

$$dV_C = \frac{I_0}{C}dt$$

$$V_C = \frac{1}{C}\int_{-\infty}^{t} I_0(t)\,dt$$

$$= \frac{I_0}{C}\int_{-\infty}^{t} dt = \frac{I_0(t - T/4)}{C} \text{ (Since discharging starts from T/4)}$$

$$\therefore V_0(t) = 10 - V_C(t) = 10 - \frac{(t - T/4)}{475\mu}$$

$$V_0(t = \frac{5T^-}{4}) = 10 - \frac{T}{475\mu}$$

Now, ripple voltage u is given by,

$$\Delta V_0 = u = V_0(T/4) - V_0(5T/4)$$

$$= 10 - (10 - \frac{T}{475\mu}) = \frac{1m}{475\mu} = 2.105 \text{ Volts}$$

Hence, the value of the ripple u is **2.105 V**.

Key Points

- For a filter capacitor under a constant current load, the discharge profile is linear, allowing the use of the fundamental capacitor equation $I = C\frac{dV}{dt}$.
- When the charging time is negligibly small compared to the discharging time, the peak-to-peak ripple voltage u can be approximated as $u = \frac{I_0 T}{C}$, where T is the total wave period.

Q1.16.4 NAT | 2016 | 1M | Ans: 0 | ● ● ○ | CLICK HERE FOR VIDEO SOLUTION

The given circuit can be redrawn as shown below,

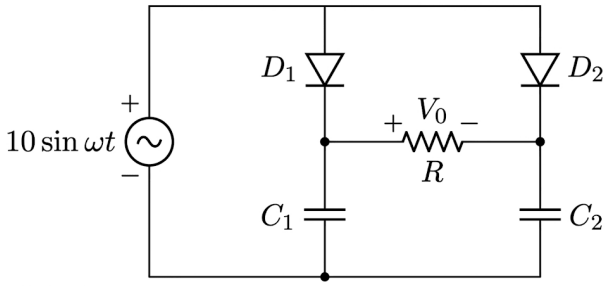

Fig. Solution Q1.16.4

Case 1 : $0 < \omega t < \frac{\pi}{2}$
Since, both the diodes will be forward biased, hence both diodes can be replaced by short circuits.

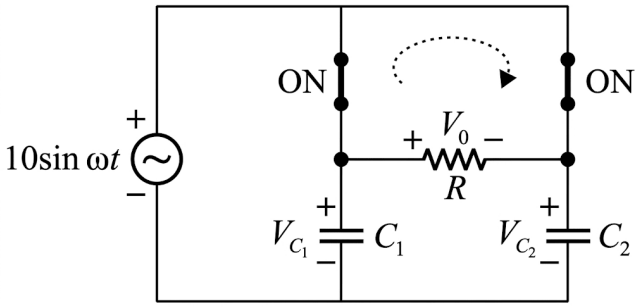

Fig. Solution Q1.16.4

From figure,

$$\boxed{V_0 = 0\,\text{V}}$$

At $\omega t = \frac{\pi}{2}$, $V_i = 10\,\text{V}$
Charging time constant for C_1, $\tau_C = 0\,\text{sec}$
Charging time constant for C_2, $\tau_C = 0\,\text{sec}$
From figure, $V_{C_1} = V_{C_2} = V_i = 10\,\text{V}$
Case 2 : $\frac{\pi}{2} < \omega t < \pi$

$$V_i < 10\,\text{V}, V_i < V_{C_1} \text{ and } V_i < V_{C_2}$$

Since, both diode will be reverse biased, hence both diode can be replaced by open circuit.

$$V_i < 10\,\text{V}, V_C = 10\,\text{V}$$

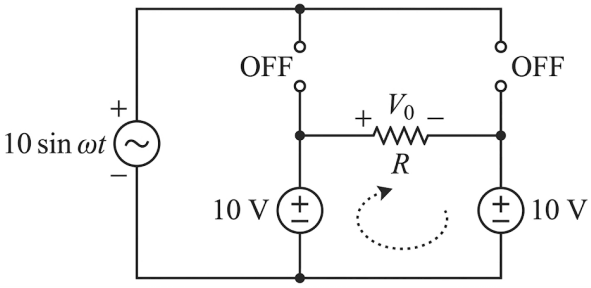

Fig. Solution Q1.16.4

Discharging time constant for C_1,

$$\tau_D = \infty$$

Discharging time constant for C_2,

$$\tau_D = \infty$$

Hence, both capacitor voltage fixed at 10 V.
Applying KVL in loop shown in figure,

$$-10 + V_0 + 10 = 0$$

$$\boxed{V_0 = 0 \text{ V}}$$

Key Points

- When ideal diodes are forward-biased by an AC source charging a capacitor with zero series resistance, the charging time constant is zero ($\tau_C = 0$), allowing the capacitor to instantaneously track the input voltage to its peak.
- Once the AC source drops below the peak voltage, the diodes become reverse-biased. If the discharge path is an open circuit ($\tau_D = \infty$), the capacitors hold their peak voltage indefinitely.

Q1.16.5 MCQ 2016 2M Ans: A ●●○

In the given circuit both zener diodes are in RB and they are connected in series.

$$V_{BZ_1} = 80 \text{ V and } V_{BZ_2} = 70 \text{ V}$$

D_1 has the least saturation current.
Initially when we vary the voltage, both diodes are off so least current flows through the circuit. But when the current crosses the saturation current of D_1 , it goes into breakdown region and allows any amount of current through it.
But because of D_2 we will take minimum current i.e. net current equals to reverse saturation current of D_2 after D_1 goes to breakdown.

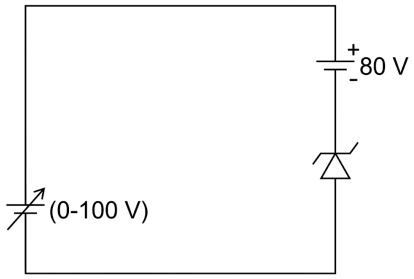

Fig. Solution Q1.16.5

when $V_{inmax} = 100\,\text{V}$ then $V_{D2} = 100 - 80 = 20 < V_{BZ_2}$

So D2 never gets a chance to enter into breakdown.

$$\boxed{\text{(A)} \quad D_1 \text{ only}}$$

Key Points

- In a series circuit with two reverse-biased Zener diodes, the diode with the smaller reverse saturation current determines the maximum current before breakdown.
- The diode with the lower reverse saturation current will experience a higher voltage drop, causing it to reach its Zener breakdown voltage first, while the other diode restricts the current to its own reverse saturation limit.

Mistakes to be avoided

- Considering the zener with least breakdown voltage first.

Q1.16.6 NAT 2016 2M Ans: 100 ●○○

Before opening the switch, the inductor is a DC short and the capacitor an open circuit, so $i_L(0) = 10$ A and $v_C(0) = 0$. After opening, the inductor transfers its stored energy to the capacitor through the ideal diode. With $L = 1$ mH and $C = 10$ microfarad,

$$\omega_0 = \frac{1}{\sqrt{LC}} = 10^4\,\text{rad/s}.$$

While the diode conducts,

$$i_L(t) = 10\cos(10^4 t)\,\text{A},$$
$$v_C(t) = 100\sin(10^4 t)\,\text{V}.$$

At $t_* = \pi/(2\times 10^4)$ s the current first reaches zero and the capacitor reaches 100 V. The diode then turns OFF: its current cannot follow the negative cosine lobe. The actual current remains zero and the isolated capacitor keeps its charge. A negative lobe would describe a hypothetical permanently closed LC loop; the ideal diode blocks that current. An independent energy check gives

$$\tfrac{1}{2}L(10)^2 = \tfrac{1}{2}CV_C^2,$$
$$\boxed{|V_C| = 100\,\text{V}}.$$

Q1.15.1 MCQ 2015 1M Ans: C ●○○

Diode circuit can be simplified as shown below,

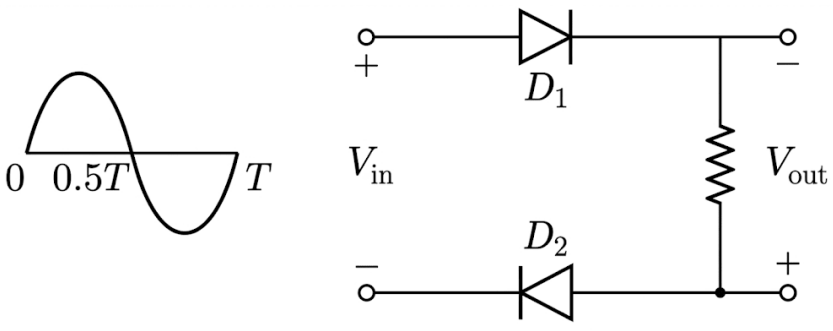

Fig. Solution Q1.15.1

Case 1 : During positive half cycle,

Both diodes D_1 and D_2 are ON. Hence,

$$V_{out} = -V_{in}$$

Case 2 : During negative half cycle,
Both diodes D_1 and D_2 are OFF. Hence,

$$V_{out} = 0\,\text{V}$$

Hence, from case 1 and case 2, the correct option is (C).

Option (C)

Q1.15.2 NAT 2015 1M Ans: 5

For the given regulator circuit, check whether Zener diode is in breakdown or not, by finding open circuit voltage across its terminals.
Let us assume Zener diode is not in reverse breakdown region, hence it can be replaced by open circuit.

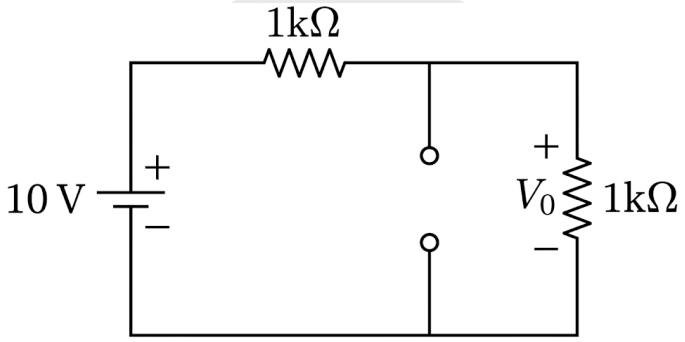

Fig. Solution Q1.15.2

$$V_0 = \frac{1\,\text{k}\Omega}{(1+1)\,\text{k}\Omega} \times 10 = 5\,\text{V}$$

For breakdown of Zener diode, $V_0 \geq V_Z$
Given, $V_Z = 6\,\text{V}$ and terminal voltage of Zener diode is less than V_Z, hence our assumption is correct i.e., Zener diode is not in breakdown region and can be replaced by open circuit.

Hence, the output voltage V_0 is **5 V**.

Key Points

- To determine the state of a Zener diode in a circuit, calculate the open-circuit Thevenin voltage (V_{th}) across its terminals by assuming the diode is OFF.
- If $V_{th} < V_Z$, the Zener diode does not enter the breakdown region and behaves as an open circuit.
- We can also solve the above problem assuming initially zener is in breakdown region and replacing it with 6V voltage source and finding the current through that branch. If current is entering the voltage source then zener is in breakdown region and if current is leaving then it is not in breakdown.

Q1.15.3 MCQ 2015 1M Ans: D ● ○ ○

For an ideal clamping circuit, once the capacitor is charged, it should not discharge.
Hence, discharging time constant (RC) must be much larger than the time period of input signal. i.e.

$$RC \gg T$$

(D) $RC \gg T$

Key Points

- In a clamping circuit, the capacitor charges quickly to the peak input voltage when the diode is forward-biased.
- To maintain this DC shift (clamp), the capacitor must hold its charge during the reverse-biased period. Therefore, the discharge time constant $\tau = RC$ must be significantly greater than the signal period T to prevent the capacitor voltage from drooping.

Q1.15.4 NAT 2015 2M Ans: 0.6 ● ○ ○

Assume the diode is OFF and check that assumption. The two outer paths are 2 + 4 = 6 and 3 + 6 = 9 kΩ, so their parallel combination is 3.6 kΩ. The 1 mA source then produces a top-to-bottom voltage of 3.6 V. Both bridge midpoints are at 2.4 V relative to the lower node. With zero central current the series 1 kΩ resistor also has zero drop, hence the diode anode-to-cathode voltage is zero. Zero volts is below the specified 0.7 V turn-on voltage: the diode is OFF. It is not strictly reverse biased; it is unbiased. This model check, rather than an unsupported rule about any zero-voltage branch, justifies opening it.

$$I_{4k} = \frac{3.6}{2+4} = 0.6\,\text{mA}.$$

0.6 mA

Q1.15.5 NAT 2015 2M Ans: 5 ● ● ○

Case 1 : During positive half cycle,
Diode D_1 is ON and D_2 is OFF. Then total input voltage appears across $10K\Omega$ as shown in figure.

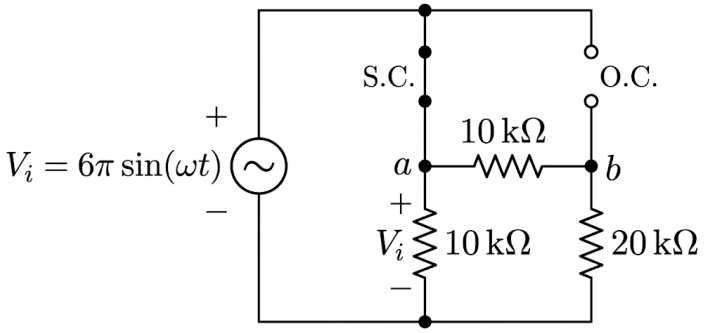

Fig. Solution Q1.15.5

$$V_{ab} = \frac{10}{10+20} \times 6\pi \sin \omega t \quad \text{[By VDR]}$$

$$V_{ab} = 2\pi \sin \omega t \text{ V}$$

Case 2 : During negative half cycle,
Diode D_1 is OFF and D_2 is ON. Then entire input voltage appears across $20K\Omega$ as shown below.

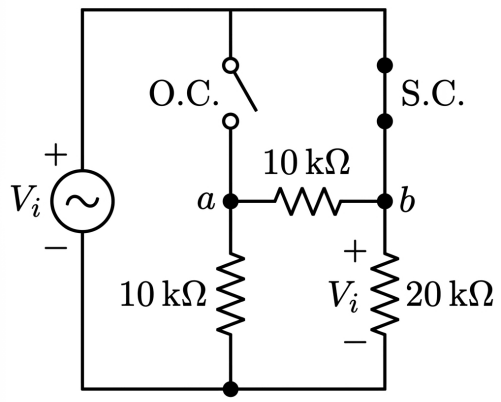

Fig. Solution Q1.15.5

$$V_{ab} = \frac{-V_i \times 10}{10 + 10} = \frac{-6\pi \sin \omega t}{2} = -3\pi \sin \omega t$$

[By VDR]

Hence, from case 1 and case 2 expression of V_{ab} is given by,

$$V_{ab} = \begin{cases} 2\pi \sin \omega t & ; \quad 0 < \omega t < \pi \\ -3\pi \sin \omega t & ; \quad \pi < \omega t < 2\pi \end{cases}$$

Average value of V_{ab} is given by,

$$(V_{ab})_{avg} = \frac{1}{T} \int_0^T V_{ab} d(\omega t)$$

$$(V_{ab})_{avg} = \frac{1}{2\pi} \int_0^{2\pi} V_{ab} d(\omega t)$$

$$(V_{ab})_{avg} = \left[\frac{1}{2\pi} \int_0^{\pi} 2\pi \sin \omega t \, d(\omega t) + \int_{\pi}^{2\pi} (-3\pi \sin \omega t) \, d(\omega t) \right]$$

$$(V_{ab})_{avg} = \frac{1}{2\pi} 2\pi [-\cos \omega t]_0^{\pi} + \frac{3\pi}{2\pi} [\cos \omega t]_{\pi}^{2\pi}$$

$$(V_{ab})_{avg} = [-\cos \pi + \cos 0^\circ] + \frac{3\pi}{2\pi} [\cos 2\pi - \cos \pi]$$

$$(V_{ab})_{avg} = (1 + 1) + \frac{3}{2}(1 + 1)$$

$$(V_{ab})_{avg} = 5 \text{ Volt}$$

Hence, the average value of voltage V_{ab} is **5 V**.

Mistakes to be avoided

- In both diagrams, don't consider resistances which are connected directly across voltage source.

Q1.15.6 NAT | 2015 | 1M | Ans: 100 | ● ○ ○

The given circuit can be redrawn as given below,

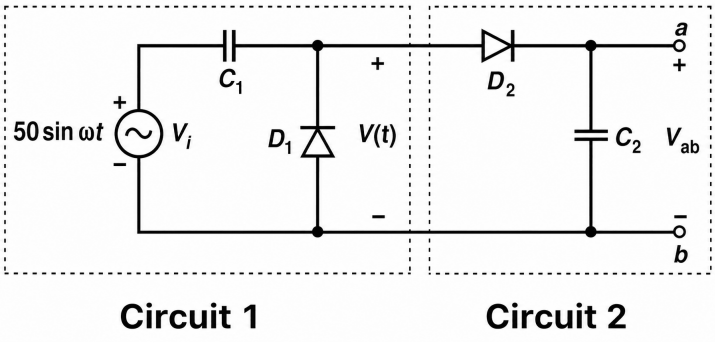

Fig. Solution Q1.15.6

In above figure, circuit (1) represents ideal positive clamper circuit and circuit (2) represents ideal positive peak detector circuit.

Hence, $V(t)$ is the output of positive clamper circuit and so, it will be shifted form of input in upward direction.

$$V(t) = V_i + V_m = 50\sin(\omega t) + 50$$

Output of positive peak detector circuit will be maximum positive value of $V(t)$.

$$V_{ab} = 100\,\text{V}$$

The average voltage V_{ab} is 100

Q1.14.1 MCQ | 2014 | 2M | Ans: D | ● ● ○

Given : (i) Voltage drop of PN diode = 0.7 V. (ii) Voltage drop of Schottky diode = 0.3 V.

Method 1

(i) Assume both diodes D_1 and D_2 are ON.

(ii) For both diodes to be ON, I_{D_1} and I_{D_2} should be greater than zero ($I_D > 0$ from p to n).

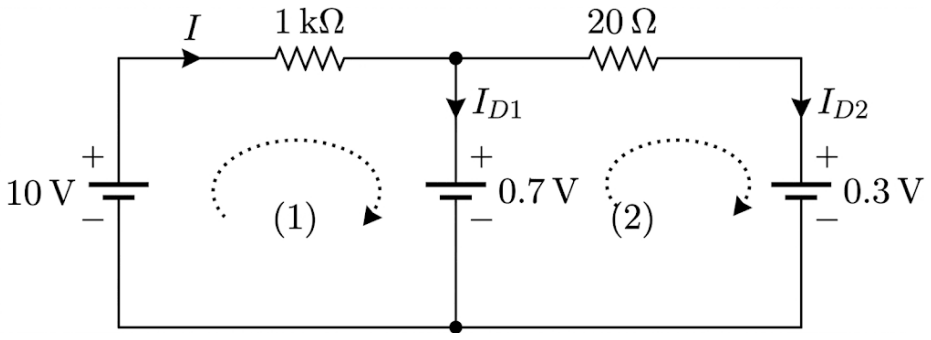

Fig. Solution Q1.14.1

Applying KVL in the loop (1),

$$-10 + 1 \times I + 0.7 = 0$$

$$I = 9.3\,\text{mA}$$

Applying KVL in the loop (2),

$$-0.7 + 0.02 \times I_{D_2} + 0.3 = 0$$

$$I_{D_2} = \frac{0.4}{0.02} = 20\,\text{mA}$$

$$I_{D_1} = I - I_{D_2} = 9.3 - 20 = -10.7\,\text{mA}$$

Since, $I_{D_1} < 0$ and $I_{D_2} > 0$ therefore, this assumption is not correct and D_1 is OFF and D_2 is ON.

(D) D_1 is OFF and D_2 is ON

Method 2

(i) Assume both diodes D_1 and D_2 are OFF.
(ii) For both diodes to be OFF, V_{D_1} should be less than 0.7 V and V_{D_2} should be less than 0.3 V (from p to n).

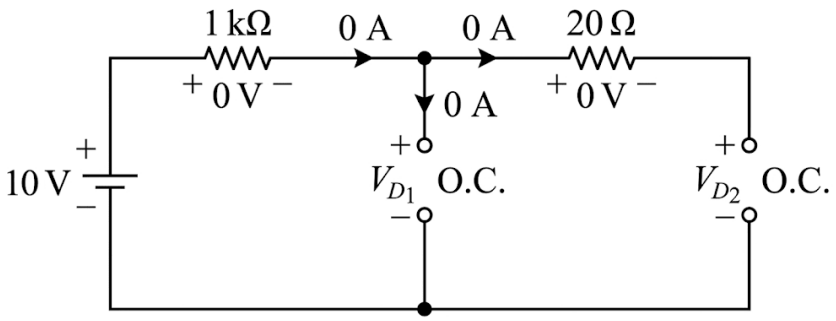

Fig. Solution Q1.14.1

From figure,

$$V_{D_2} = 10\,\text{V}$$

So D_2 is ON and the moment D_2 becomes ON, it is replaced by 0.3V source and the circuit becomes,

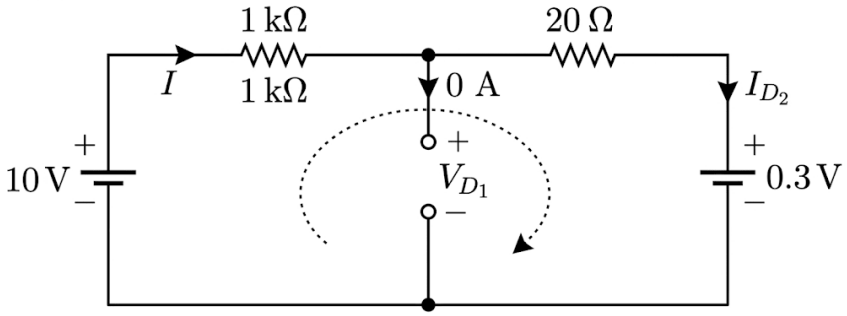

Fig. Solution Q1.14.1

Applying KVL in the above loop,

$$-10 + I + 0.02I + 0.3 = 0$$

$$I = \frac{10 - 0.3}{1.02\text{k}} = 9.50\,\text{mA}$$

$$I_{D_2} = I = 9.50\,\text{mA}$$

$$\therefore V_{D_1} = I \times 0.02 + 0.3$$

$$V_{D_1} = 9.50\text{m} \times 0.02\text{k} + 0.3 = 0.49\,\text{V}$$

Hence, $V_{D_1} < 0.7\,\text{V}$
Therefore, this assumption is correct and diode D_1 is OFF.

(D) D_1 is OFF and D_2 is ON

Key Points

- For a diode assumed to be ON, replace it with its respective forward voltage drop (V_{on}) and verify that the calculated current flows from the p-type to n-type material ($I_D > 0$).
- For a diode assumed to be OFF, replace it with an open circuit and verify that the voltage across its terminals ($V_p - V_n$) is less than its cut-in voltage ($V_D < V_{on}$).
- If an assumed state leads to a contradiction in verification, the assumption is incorrect, and the alternative state must be analyzed.

Mistakes to be avoided

- Check the resistance units carefully.

Q1.14.2 MCQ 2014 2M Ans: C ● ○ ○

Given : (i) $V_{on} = 0.7\text{ V}, V_i = 5\sin(\omega t)\text{ V}$
(ii) $V_{i(\max)} = 5\sin\omega t|_{\max} = +5\text{ V}$
(iii) $V_{i(\min)} = 5\sin\omega t|_{\min} = -5\text{ V}$
Case 1 : $V_{i(\max)} = +5$ V ,
Diode will be forward biased, hence it can be replaced by its cut-in voltage i.e. V_{on}.

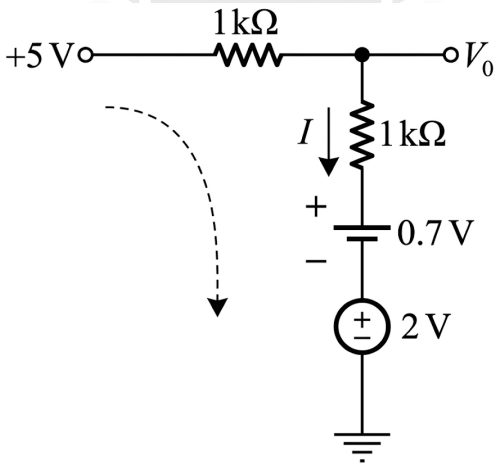

Fig. Solution Q1.14.2

Applying KVL in above figure,

$$I = \frac{5 - 0.7 - 2}{2} = \frac{2.3}{2}$$

$$V_o = \frac{2.3}{2} \times 1 + 0.7 + 2 = 3.85\text{ V}$$

Case 2 : $V_i = -5$ V ,
Diode will be reverse biased, hence it can be replaced by open circuit.

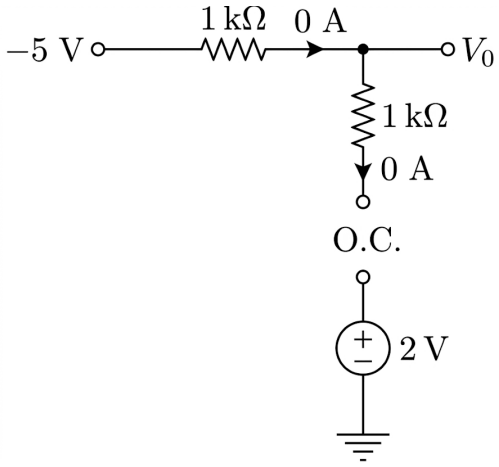

Fig. Solution Q1.14.2

From the above circuit, $V_o = V_i = -5$ V
From Case 1 and Case 2,

$$V_o = \begin{cases} V_{o(\max)} = 3.85\text{ V} & ; \quad V_i = +5\text{ V} \\ V_{o(\min)} = -5\text{ V} & ; \quad V_i = -5\text{ V} \end{cases}$$

(C) -5 and 3.85

Key Points

- To find the maximum and minimum output voltages in a clipping circuit, evaluate the circuit at the positive and negative peaks of the input signal.

Q1.14.3 NAT | 2014 | 1M | Ans: 0.1 | ● ● ○ | CLICK HERE FOR VIDEO SOLUTION

In periodic steady state the capacitor returns to the same voltage each cycle, so its average current is zero. KCL therefore gives

$$\bar{I}_D = \bar{I}_R = \frac{\bar{V}_C}{R}.$$

Here $RC = 100(4 \times 10^{-3}) = 0.4$ s, whereas the source period is only 0.02 s. The capacitor consequently keeps the load near the 10 V peak with a small ripple.

$$\bar{I}_D \approx \frac{10}{100} = 0.1\text{ A}.$$

A first-order ripple estimate gives

$$I_{dc} \approx \frac{V_m}{R + 1/(2fC)},$$

$$I_{dc} \approx \frac{10}{102.5} = 0.09756\text{ A},$$

which is still approximately 0.1 A. Exact switching is not instantaneous at the voltage peak: while the diode conducts its current is $v/R + C\,dv/dt$, so it ceases when this expression reaches zero.

Q1.14.4 MCQ | 2014 | 1M | Ans: D

In the given circuit make the diodes ideal and add their cutin voltages to the voltage sources directly and the circuit becomes

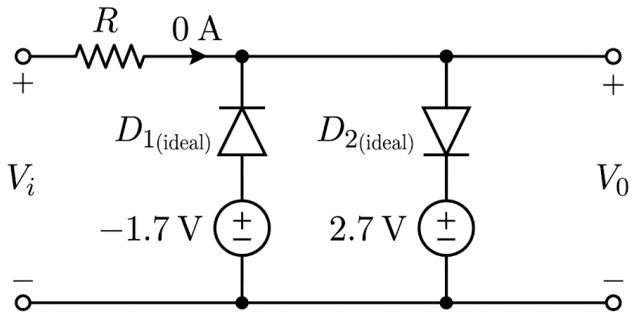

Fig. Solution Q1.14.4

Case 1 : During positive half cycle,

Diode D_1 will be reverse biased and hence, D_1 will act as open circuit.

(a) $\mathbf{0 < V_i < 2.7\,V}$ **:** Diode D_2 will be reversed biased and hence, it acts as open circuit.

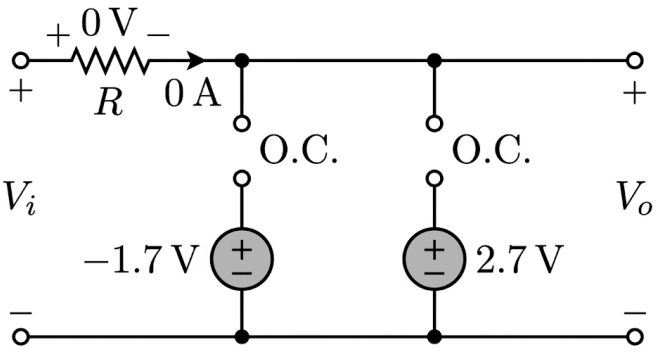

Fig. Solution Q1.14.4

Applying KVL, $-V_i + 0\,V + V_0 = 0$

Hence, $V_0 = V_i$

(b) $\mathbf{V_i > 2.7\,V}$ **:** Diode D_2 will be forward biased and hence, it can be replaced by cut-in voltage V_γ.

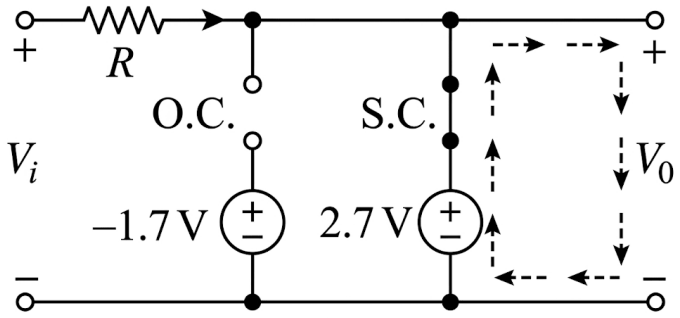

Fig. Solution Q1.14.4

Applying KVL, $-2.7\,V + V_0 = 0$

Hence, $V_0 = 2.7\,V$

Case 2 : During negative half cycle,

Diode D_2 will be reverse biased and hence, D_2 will act as open circuit.

(a) $\mathbf{-1.7\,V < V_i < 0\,V}$ **:** Diode D_1 will be reverse biased and hence it acts as open circuit.

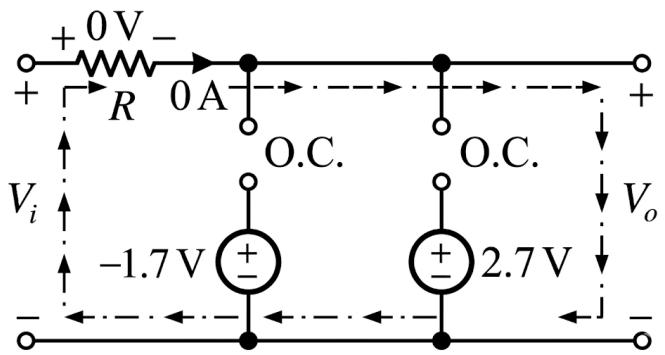

Fig. Solution Q1.14.4

Applying KVL, $-V_i + 0\,V + V_0 = 0 \implies V_0 = V_i$

(b) $\mathbf{V_i < -1.7\,V}$ **:** Diode D_1 will be forward biased and hence it can be replaced by cut-in voltage V_γ.

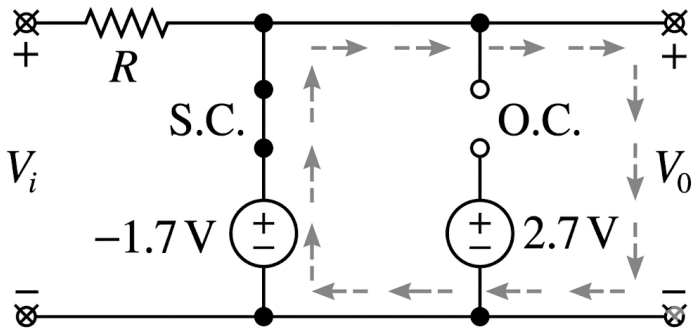

Fig. Solution Q1.14.4

Applying KVL in the above circuit,

$-1.7\,V + V_0 = 0$ Hence, $V_0 = -1.7\,V$

Hence, from Case 1 and Case 2, expression of output voltage can be written as given below,

$$V_0 = \begin{cases} V_i & ; \quad 0\,V \le V_i < 2.7\,V \\ 2.7\,V & ; \quad V_i \ge 2.7\,V \\ V_i & ; \quad -1.7\,V < V_i \le 0\,V \\ -1.7\,V & ; \quad V_i \le -1.7\,V \end{cases}$$

Hence, the output waveform is shown below,

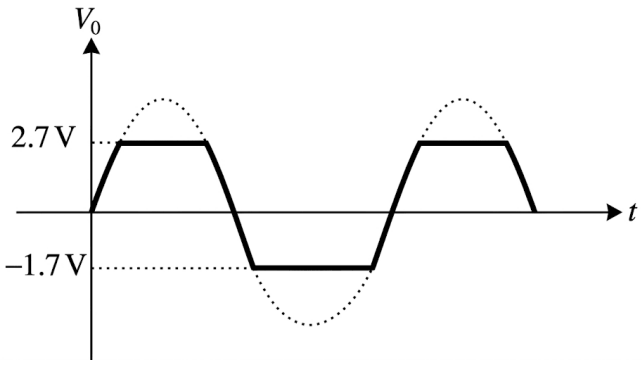

Fig. Solution Q1.14.4

Therefore when $-1.7 < V_i < 2.7$, then both D_1 and D_2 are OFF and hence $V_0 = V_i$.

(D) -1.7 V < V_i < 2.7 V

Q1.13.1 MCQ 2013 2M Ans: B ●●○

Given :
(i) Knee current of Zener diode $I_{Z(\min)} = 10\,\text{mA}$
(ii) $V_Z = 5$ Volt
Regulator circuit is shown below,

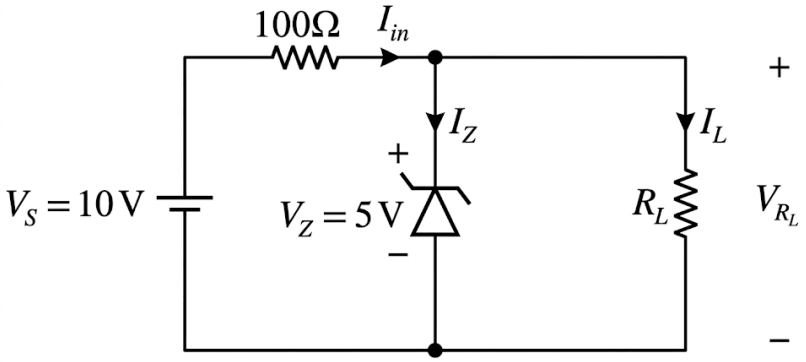

Fig. Solution Q1.13.1

For variable load, input current I_{in} is given by,

$$I_{in} = I_{Z(\max)} + I_{L(\min)} \quad \ldots (i)$$

$$I_{in} = I_{Z(\min)} + I_{L(\max)} \quad \ldots (ii)$$

(i) Calculation for minimum load resistance :
Input current I_{in} is given by,

$$I_{in} = \frac{V_S - V_Z}{R}$$

$$I_{in} = \frac{10 - 5}{0.1} = 50\,\text{mA}$$

From equation (ii),

$$I_{L(\max)} = I_{in} - I_{Z(\min)} = 50 - 10 = 40\,\text{mA}$$

From figure,

$$V_{R_L} = I_L R_L = V_Z = 5\,\text{V}$$

$$I_{L(\max)} R_{L(\min)} = 5\,\text{V}$$

$$R_{L(\min)} = \frac{5}{I_{L(\max)}} = \frac{5}{40} = 125\,\Omega$$

(ii) Calculation for minimum power rating :

Power dissipation across Zener diode is given by,

$$P_Z = V_Z I_Z$$

Minimum power rating of Zener diode represents maximum allowed power dissipation across Zener diode

$$P_{Z(\max)} = V_Z I_{Z(\max)} \quad \ldots (\text{iii})$$

For no load condition $R_L = \infty$, $I_{L(\min)} = 0\,\text{A}$

From equation (i),

$$I_{Z(\max)} = I_{in} = 50\,\text{mA}$$

From equation (iii),

$$P_{Z(\max)} = 5 \times 50 = 250\,\text{mW}$$

(B) 125 and 250

Key Points

- **Minimum Load Resistance ($R_{L(\min)}$):** Occurs when the load draws maximum current ($I_{L(\max)}$), which happens when the Zener diode operates at its minimum knee current ($I_{Z(\min)}$).
- **Maximum Power Rating ($P_{Z(\max)}$):** Dictated by the maximum power dissipation, which occurs when the load is disconnected ($R_L = \infty, I_L = 0$). Here, the entire input current flows through the Zener diode ($I_{Z(\max)} = I_{in}$).

Q1.13.2 MCQ | 2013 | 2M | Ans: D | ● ○ ○ | CLICK HERE FOR VIDEO SOLUTION

Given, $V_{YZ} = 1000 \sin \omega t$ and all diodes are ideal. Since there is no ground in the circuit, assume that the node Z is ground. Then the circuit becomes,

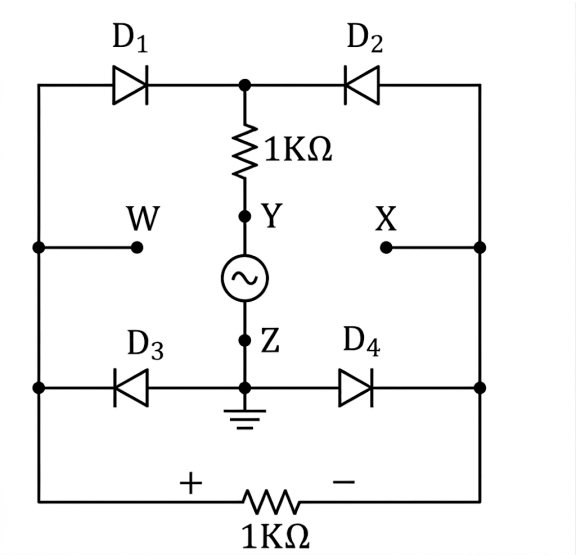

Fig. Solution Q1.13.2

Case 1 : During positive half cycle

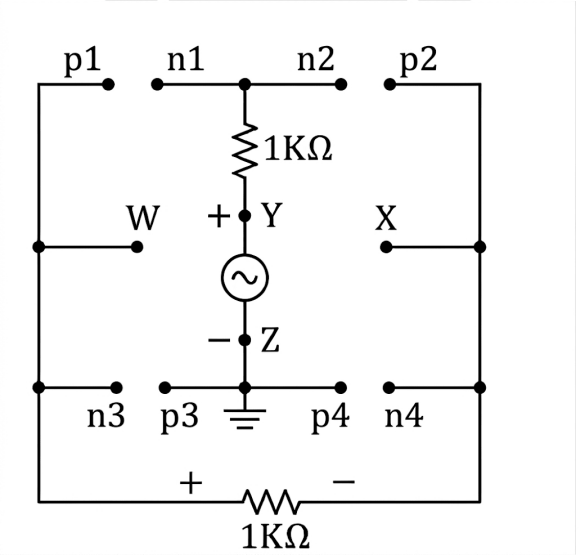

Fig. Solution Q1.13.2

By Open circuit test, open circuit all the diodes initially. Here node Y has positive voltage and Z is grounded. So n_1 , n_2 nodes have positive voltage and p_3 and p_4 have zero voltage. But we can't directly say that all diodes are OFF as we don't know the voltages at p_1, p_2, n_3, n_4 but we can certainly say that $V_{p_1} = V_{n_3} = V_W$ and $V_{p_2} = V_{n_4} = V_X$.

Now assume that $V_W > V_{n_1}$ and $V_X > V_{n_2}$, so D_1 and D_2 are ON and D_3,D_4 are OFF. But in this case the current through the resistor at the bottom flows in 2 different directions to keep the diodes ON which is not possible. Which means one of D_1 and D_2 are OFF to keep the current flowing in only one direction, and in such case there is no return path for current so other diode is also OFF.
So our initial assumption is correct all the four diodes are eventually OFF. so

$$V_{n_3} = V_{p_1} = V_{n_4} = V_{p_2} \implies V_W = V_X$$

Hence, $V_{wx} = 0\,\text{V}$ ... (i)

Case 2 : During negative half cycle

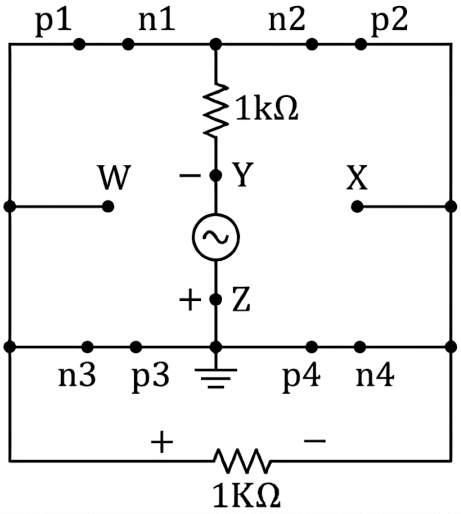

Fig. Solution Q1.13.2

During negative half cycle nodes n_1 and n_2 have more negative voltages but we can't directly say D_1 and D_2 are ON as we don't know the voltage at V_W and V_X. Similarly we can't directly say D_3 and D_4 are OFF.

Assume V_W and V_X are negative but grater than V_Y. In this case all four diodes are ON since their $V_p > V_n$ and the circuit is shown above. Here in this case, the voltage across bottom resistor is 0V and hence zero current flows through it. Hence all the four diodes are eventually ON and we can say that $V_W = V_X$

$$V_{wx} = 0\,\text{V} \quad \ldots (ii)$$

Note : We can't say $V_W = V_X = 0$ (since they are connected to ground), as we have assumed the ground there. Based on our location of assumption of ground, the absolute potentials at V_W and V_X can change.
From equation (i) and (ii),

$$V_{wx} = 0 \text{ for all } t$$

(D) 0 for all t

Key Points

- If ground is not given in the circuit, we can assume anywhere but note that absolute potentials at nodes will change based on our assumption.
- Here the potential difference between V_W and V_X is zero but not their absolute potentials.

Q1.12.1 MCQ | 2012 | 1M | Ans: D | ● ● ○

Given circuit is shown below,

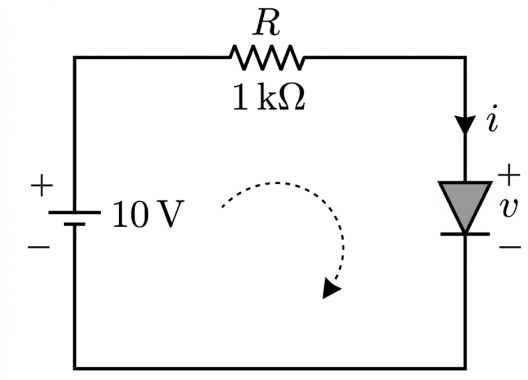

Fig. Solution Q1.12.1

The i-v characteristics is given as,

$$i = \begin{cases} \frac{v-0.7}{500}\ \text{A}; & v > 0.7\ \text{V} \\ 0; & v < 0.7\ \text{V} \end{cases}$$

Method 1

Assuming diode is ON, (i.e. $v > 0.7$ V)
Applying KVL in the loop shown,

$$-10 + iR + v = 0 \quad \ldots (i)$$

Put, $i = \frac{v-0.7}{500}, R = 1\ \text{k}\Omega$ [Given]
So, equation is becomes as,

$$10 = \frac{(v - 0.7)}{500} \times 1000 + v$$

$$10 = 2v - 1.4 + v$$

$$3v = 11.4$$

$$v = 3.8\ \text{V}$$

Here, we get $v = 3.8$ V which is greater than 0.7 V. It means our assumption is correct. So, current i,

$$i = \frac{v - 0.7}{500}\ \text{A}$$

$$i = \frac{3.8 - 0.7}{500}\ \text{A} \quad [\text{Put } v = 3.8\ \text{V}]$$

$$i = 0.0062\ \text{A} = 6.2\ \text{mA}$$

(D) 6.2 mA

Method 2

From given i-v characteristics,

$$i = \frac{v - 0.7}{500} \quad \text{for } v > 0.7\ \text{V} \quad \ldots (i)$$

i-v characteristics of forward bias practical diode is given by,

$$i = \frac{v - V_\gamma}{R_f} \quad \ldots (ii)$$

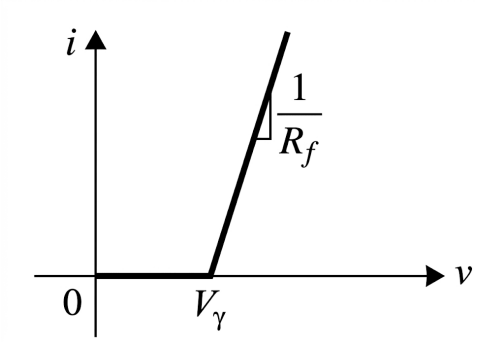

Fig. Solution Q1.12.1

R_f = Forward resistance of diode and V_γ = Cut-in voltage (offset voltage)
From equation (i) and (ii), we get

$$R_f = 500\,\Omega = 0.5\,\text{k}\Omega, V_\gamma = 0.7\,\text{V}$$

The given circuit is replaced by its equivalent as shown below,

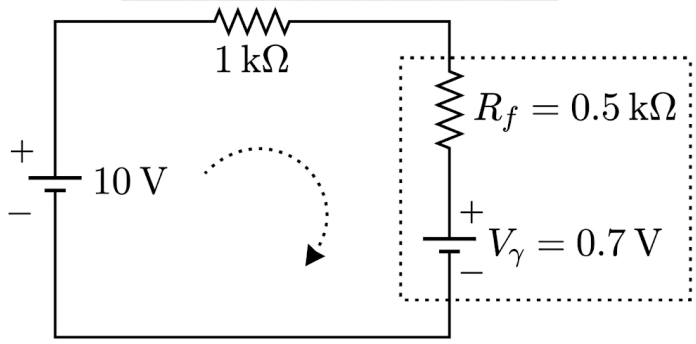

Fig. Solution Q1.12.1

Appling KVL in the above loop,

$$-10 + 1 \times i + 0.5 \times i + 0.7 = 0$$

$$1.5 \times i = 9.3 \implies i = \frac{9.3}{1.5} = 6.2\,\text{mA}$$

(D) 6.2 mA

Key Points

- A practical diode in forward bias can be modeled using a piecewise-linear equivalent circuit, consisting of an ideal voltage source equal to its cut-in voltage (V_γ) in series with a forward resistance (R_f).

Q1.12.2 MCQ 2012 1M Ans: A ●○○

The given circuit can be redrawn as given below,

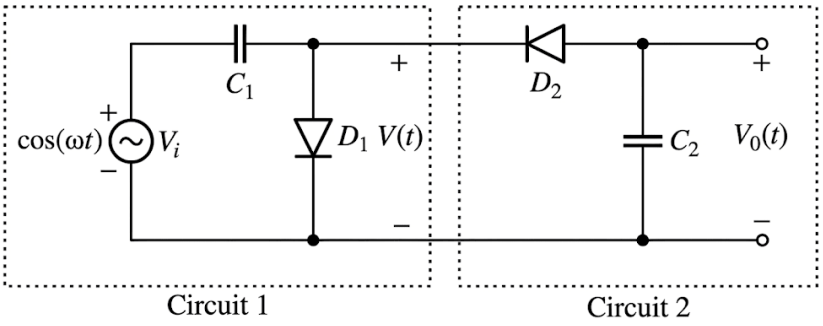

Fig. Solution Q1.12.2

In above figure, circuit (1) represents ideal negative clamper circuit and circuit (2) represents ideal negative peak detector circuit. Hence, $V(t)$ is the output of negative clamper circuit and hence, it will be shifted form of input in downward direction.

$$V(t) = V_i - V_m = \cos \omega t - 1$$

Hence, the correct option is (A).
Output of negative peak detector circuit will be maximum negative value of $V(t)$.

$$V_0(t) = -2\,\text{V}$$

(A) $\cos(\omega t) - 1$

Q1.11.1 MCQ 2011 1M Ans: B ● ○ ○

A Zener diode is specifically designed to operate in the reverse breakdown region for voltage regulation. In this region, the diode maintains a nearly constant voltage despite variations in current.

(B) reverse breakdown region

Q1.11.2 MCQ 2011 2M Ans: A ● ○ ○

Given :
(i) Forward voltage drop of diode = 0.7 V
(ii) $V_T = \frac{kT}{q} = 25\,\text{mV}$
(iii)Input signal $V_i = V_p \cos(\omega t)$ where, $V_p = 100\,\text{mV}$
For DC analysis, short circuit the AC voltage source.
By replacing all diodes by their cut-in voltage, we get the modified circuit as given below,

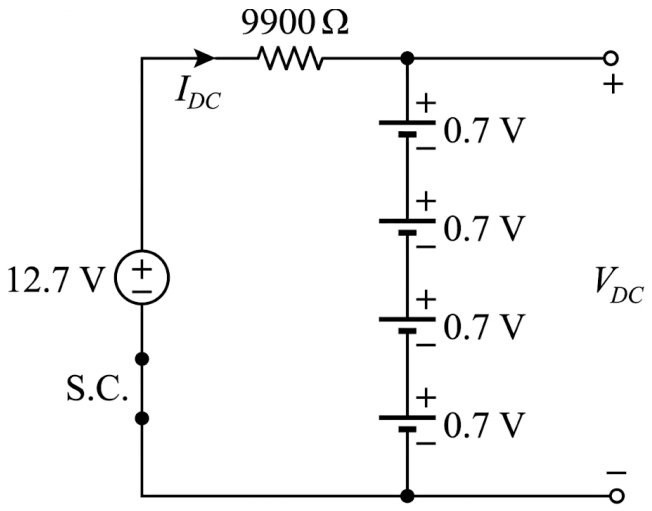

Fig. Solution Q1.11.2

$$V_{DC} = 4 \times 0.7 = 2.8\,\text{V}$$

$$\text{Current, } I_{DC} = \frac{12.7 - 2.8}{9.9} = 1\,\text{mA}$$

(A) 1 mA

Key Points

- When a circuit contains both DC and small-signal AC sources, the analysis is separated using the superposition principle.
- During DC analysis, AC voltage sources are turned off (replaced by short circuits).

Q1.11.3 MCQ 2011 2M Ans: B ● ○ ○

For AC analysis, short circuit the DC voltage source.

Dynamic resistance or ac resistance of diode is given by,

$$r_{ac} = \frac{\eta V_T}{I}$$

$$r_{ac} = \frac{V_T}{I} \qquad (\text{Assume } \eta = 1)$$

$$r_{ac} = \frac{25\,\text{mV}}{1\,\text{mA}} = 25\,\Omega$$

By replacing all diodes by their ac resistance, we get the modified circuit as given below,

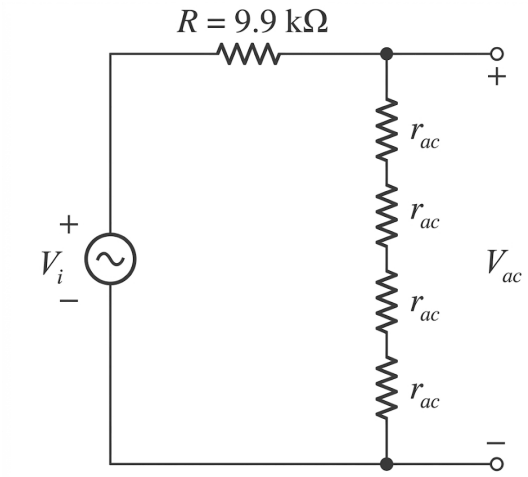

Fig. Solution Q1.11.3

$$V_{ac} = \frac{4r_{ac}}{R + 4r_{ac}} \times V_i \qquad \text{[By VDR]}$$

$$V_{ac} = \frac{4 \times 25}{9900 + 4 \times 25} \times V_i = \frac{100}{10,000} V_i$$

$$V_{ac} = \frac{1}{100} V_i = \frac{1}{100} \times 100 \cos \omega t$$

$$V_{ac} = \cos \omega t \text{ mV}$$

$$\boxed{\text{(B)} \quad 1 \cos(\omega t) \text{ mV}}$$

Key Points

- In AC small-signal analysis, DC voltage sources are replaced by short circuits, leaving only the AC sources active.
- A forward-biased diode is replaced by its small-signal dynamic resistance r_{ac}, calculated as $r_{ac} = \frac{\eta V_T}{I_{DC}}$, where I_{DC} is the quiescent current found during DC analysis.

Q1.09.1 MCQ | 2009 | 2M | Ans: A | ● ○ ○

Case 1 : Assume diode is ON,
Hence, diode can be replaced by short circuit.

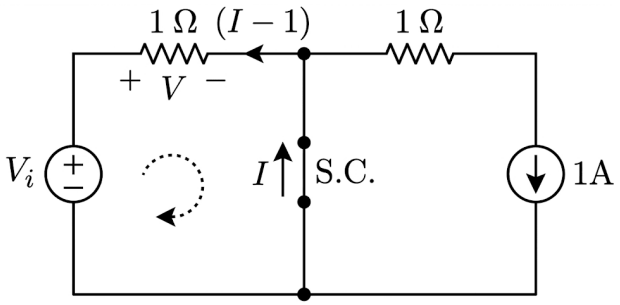

Fig. Solution Q1.09.1

Applying KVL in the loop shown,

$$-V_i + V + 0 = 0 \implies V = V_i \quad \ldots (i)$$

where, $V = -(I - 1) \times 1 = 1 - I$ or $I = 1 - V$
From equation (i),

$$I = 1 - V_i \quad \ldots \text{(ii)}$$

For diode to be ON, current I must be positive i.e. $I > 0$.

$$1 - V_i > 0 \implies V_i < 1\,\text{V}$$

Hence, for diode to be ON, the voltage V_i must be less than 1 V.
If $V_i = 0.5$ V, then $V = 0.5$ V

Option	V	Status
(A) $\min(V_i, 1)$	0.5 V	Correct
(B) $\max(V_i, 1)$	1 V	Wrong
(C) $\min(-V_i, 1)$	−0.5 V	Wrong
(D) $\max(-V_i, 1)$	1 V	Wrong

Case 2 : Assume diode is OFF,
Hence, diode can be replaced by open circuit.

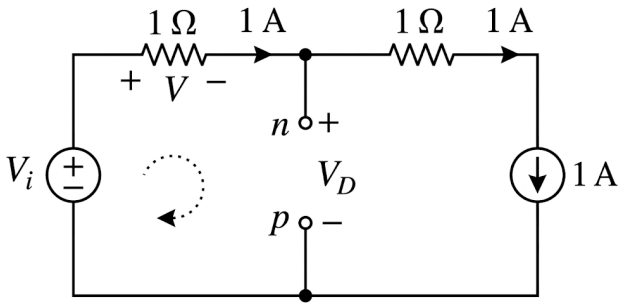

Fig. Solution Q1.09.1

From figure, $V = 1 \times 1 = 1$ V
Applying KVL in the loop shown,

$$-V_i + V + V_D = 0$$
$$V_D = V_i - V$$
$$V_D = V_i - 1$$

For diode to be OFF, V_D must be positive i.e. $V_D > 0$ V.

$$V_i - 1 > 0 \implies V_i > 1\,\text{V}$$

Hence, for diode to be OFF, V_i must be greater than 1 V and V must be equal to 1 V.
Let, $V_i = 1.5$ V and $V = 1$ V

Option	V	Status
(A) $\min(V_i, 1)$	1 V	Correct
(B) $\max(V_i, 1)$	1.5 V	Wrong
(C) $\min(-V_i, 1)$	−1.5 V	Wrong
(D) $\max(-V_i, 1)$	1 V	Correct

Hence, from the Case 1 and Case 2 it is clear that option (A) is common for both cases.

(A) $\min(V_i, 1)$

Q1.08.1 MCQ | 2008 | 2M | Ans: C

Case 1 : $\mathbf{V_{i(max)} = 10\ V}$

Since, voltage across Zener diode Z is higher than breakdown voltage, hence Zener diode Z will work in breakdown region and therefore, Zener diode Z can be replaced by its breakdown voltage.

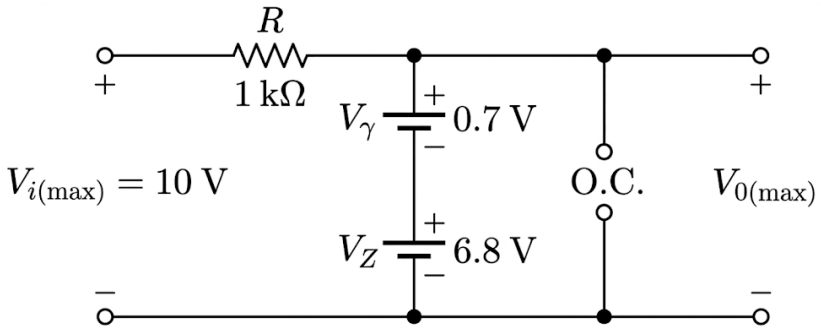

Fig. Solution Q1.08.1

Hence, $V_{0(max)} = V_\gamma + V_Z = 7.5\ V$

Case 2 : $\mathbf{V_{i(min)} = -10\ V}$

When voltage across diode D_2 will be higher than V_γ, D_2 will be forward biased and therefore, D_2 can be replaced by its cut-in voltage V_γ. Zener diode will be in forward bias and acts as a normal diode.

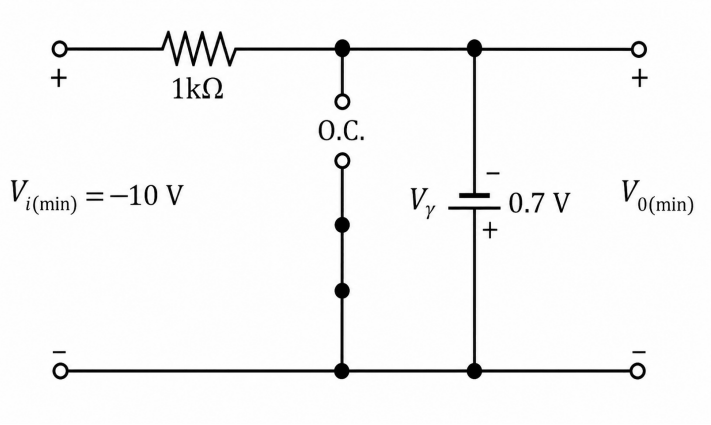

Fig. Solution Q1.08.1

Hence, $V_{0(min)} = -0.7\ V$

(C) 7.5, -0.7

Key Points

- During the positive half-cycle, D_1 is forward-biased and Zener diode Z is in reverse breakdown. The output voltage is clipped at the sum of their voltages: $V_0 = V_{D1} + V_Z$.
- During the negative half-cycle, D_2 becomes forward-biased, clamping the output voltage to its forward voltage drop: $V_0 = -V_\gamma$.

Q1.07.1 MCQ 2007 1M Ans: C ● ○ ○

From option (C),

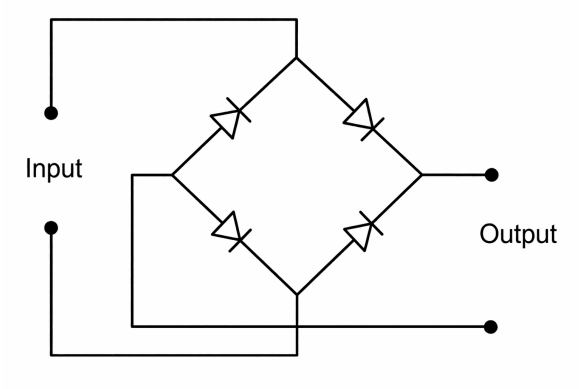

Fig. Solution Q1.07.1

Case 1 : During positive half cycle,

Diode D_2 and D_3 will be forward biased, hence these diodes can be replaced by short circuits. Diode D_1 and D_4 will be reverse biased, hence these diodes can be replaced by open circuits.

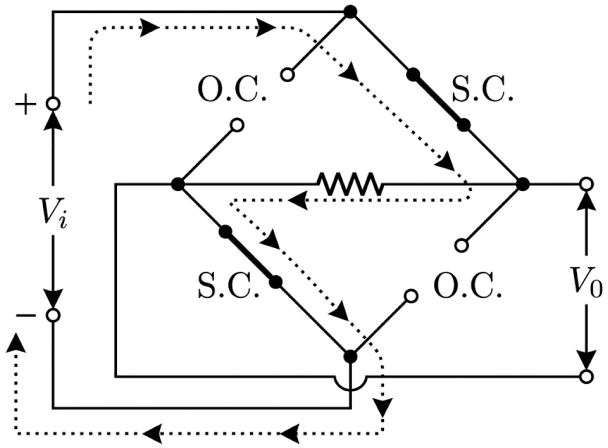

Fig. Solution Q1.07.1

Hence, $V_0 = V_i$ [V_0 = Voltage across resistor]

Case 2 : During negative half cycle,

Diode D_2 and D_3 will be reverse biased, hence these diodes can be replaced by open circuit. Diode D_1 and D_4 will be forward biased, hence these diodes can be replaced by short circuit.

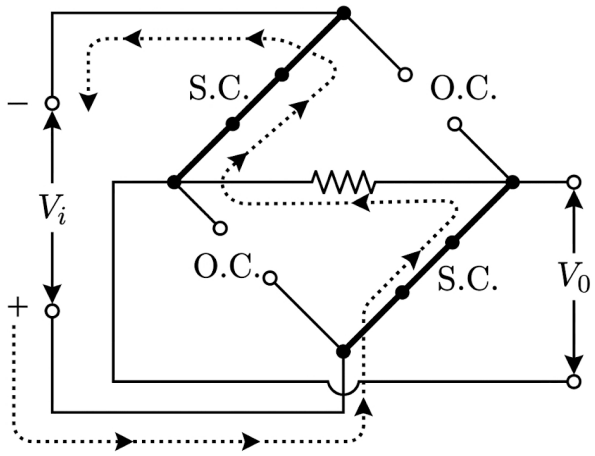

Fig. Solution Q1.07.1

Hence,

$$-V_i - V_0 = 0$$

$$V_0 = -V_i$$

Hence, from case 1 and case 2 the expression of output can be written as given below,

$$V_0 = \begin{cases} V_i & ; \quad V_i > 0\,\text{V} \\ -V_i & ; \quad V_i < 0\,\text{V} \end{cases}$$

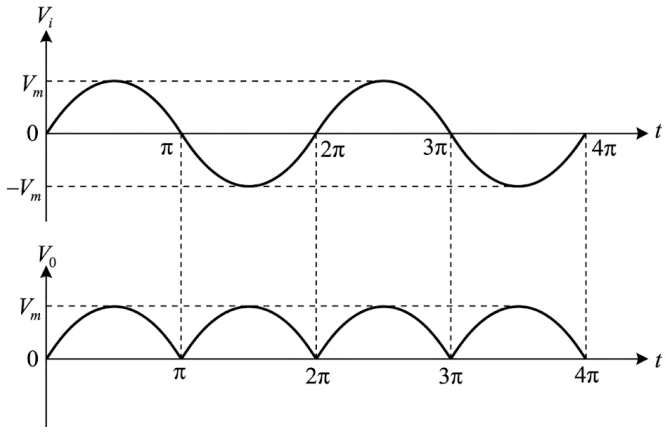

Fig. Solution Q1.07.1

Above waveform is output of full wave bridge rectifier.

Option (C)

Q1.07.2 MCQ 2007 2M Ans: C ●●○

Given,
Zener knee voltage = 7 V
Zener dynamic resistance = 10 Ω
$V_{in(\min)} = 10\,\text{V}$, $V_{in(\max)} = 16\,\text{V}$

Replacing Zener diode with dynamic resistance and Zener voltage in the given circuit as shown below,

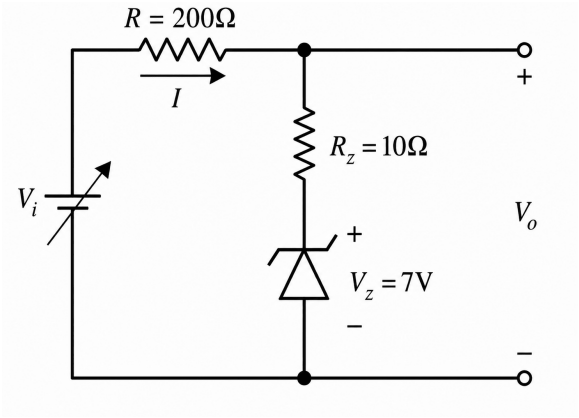

Fig. Solution Q1.07.2

Output voltage V_0 is given by,

$$V_0 = V_Z + IR_Z$$

where,

$$I = \frac{V_i - V_Z}{R + R_Z}$$

$$I_{max} = \frac{V_{i(max)} - V_Z}{R + R_Z} = \frac{16 - 7}{200 + 10} = \frac{9}{210}\text{ A}$$

$$I_{min} = \frac{V_{i(min)} - V_Z}{R + R_Z} = \frac{10 - 7}{200 + 10} = \frac{3}{210}\text{ A}$$

Output voltage V_0 will be maximum when I is maximum.

$$(V_0)_{max} = V_Z + I_{max}R_Z$$

$$(V_0)_{max} = 7 + \frac{9}{210} \times 10 = 7.43\text{ V}$$

Output voltage V_0 will be minimum when I is minimum.

$$(V_0)_{min} = V_Z + I_{min}R_Z$$

$$(V_0)_{min} = 7 + \frac{3}{210} \times 10 = 7.14\text{ V}$$

Thus, V_0 lies between 7.14 V and 7.43 V.

(C) 7.14 to 7.43 V

Key Points

- The total output voltage across a practical Zener diode increases slightly as the Zener current increases, given by the relation $V_0 = V_Z + I_Z R_Z$.

Q1.06.1 MCQ | 2006 | 2M | Ans: B | ● ○ ○

Case 1 : During positive half cycle,

(a) $0 < V_i < 6$ V :

When voltage across Zener diode will be less than its break down voltage, Zener diode will act as a normal diode in reverse bias. Therefore, Zener diode can be replaced by open circuit.

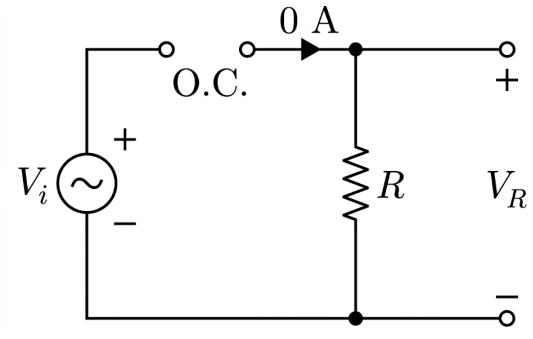

Fig. Solution Q1.06.1

From figure, $V_R = 0 \times R = 0\,\text{V}$

(b) $V_i > 6\,V$:

When voltage across Zener diode will be higher than its break down voltage, Zener diode will start to work in reverse bias break down region. Therefore, Zener diode can be replaced by its break down voltage.

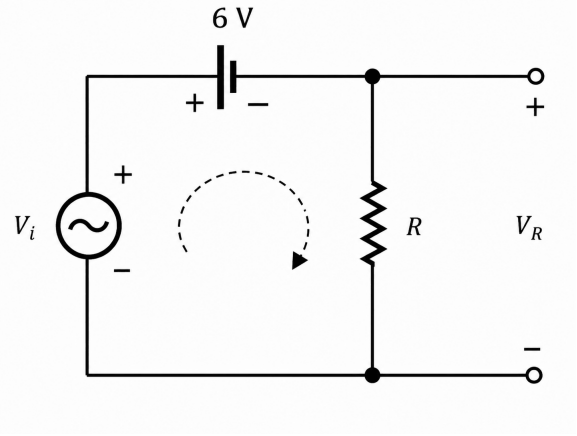

Fig. Solution Q1.06.1

Applying KVL in above figure,

$$-V_i + 6 + V_R = 0$$

$$V_R = V_i - 6$$

Hence, from case 1(a) and case 1(b),

$$V_R = \begin{cases} 0\,\text{V} & ; \quad 0\,\text{V} < V_i < 6\,\text{V} \\ V_i - 6 & ; \quad V_i \geq 6\,\text{V} \end{cases}$$

Case 2 : During negative half cycle, Zener diode will be forward bias and hence it will work as a normal diode. Therefore, Zener diode can be replaced by short circuit.

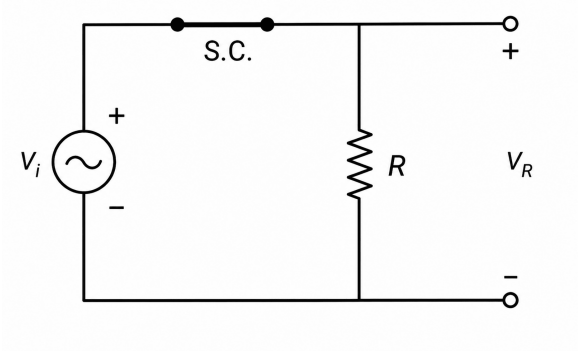

Fig. Solution Q1.06.1

From figure, $V_R = V_i$

Hence, from case 1 and case 2, the expression of output voltage can be summarized as given below,

$$V_R = \begin{cases} V_i & ; \quad V_i < 0\,\text{V} \\ 0\,\text{V} & ; \quad 0\,\text{V} < V_i < 6\,\text{V} \\ V_i - 6 & ; \quad V_i \geq 6\,\text{V} \end{cases}$$

Hence the waveform observed across R is shown below,

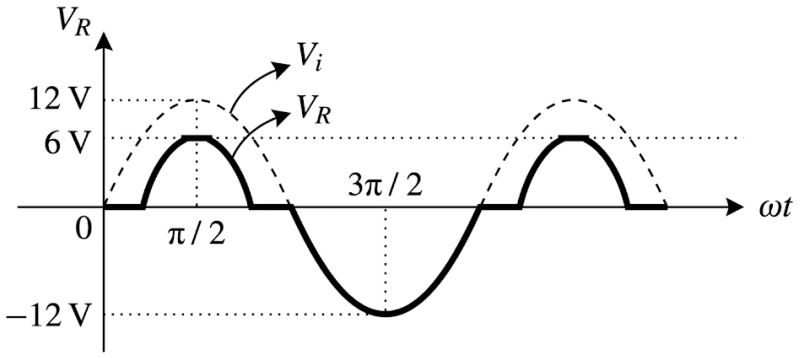

Fig. Solution Q1.06.1

Hence, the correct option is (B)

Q1.05.1 MCQ 2005 2M Ans: C

In a Zener diode voltage regulator circuit, the Zener diode maintains a constant voltage (V_z) across the load. For the Zener diode to function properly, it must maintain at least its knee current I_{zk} (minimum current to keep the Zener breakdown region active).

The total current through the series resistor (I_R) is supplied by the input voltage and is divided between the load current (I_{load}) and the Zener diode current (I_z).

$$I_R = I_{\text{load}} + I_z$$

To ensure the Zener diode is always in regulation, its current must be at least the knee current I_{zk} even at the highest load current and lowest input voltage. Hence, for proper operation:

$$I_R \geq I_{\text{load}} + I_{zk}$$

where,

$$I_R = \frac{V_i - V_z}{R}$$

Step 1: Calculate the resistor current at minimum input voltage Considering the worst case to calculate maximum load current, we use the minimum input voltage ($V_{i(\min)} = 20\,\text{V}$) which gives the minimum resistor current:

$$I_{R(\min)} = \frac{V_{i(\min)} - V_z}{R}$$

$$I_{R(\min)} = \frac{20 - 5.8}{1000} = \frac{14.2\,\text{V}}{1000\,\Omega} = 0.0142\,\text{A} = 14.2\,\text{mA}$$

Step 2: Calculate maximum load current Since the Zener diode must have at least 0.5 mA current to regulate properly, the remaining current can be drawn by the load.

$$I_{\text{load(max)}} = I_{R(\min)} - I_{zk}$$

$$I_{\text{load(max)}} = 14.2\,\text{mA} - 0.5\,\text{mA} = 13.7\,\text{mA}$$

(C) 13.7 mA

Key Points

- To find the **maximum load current** ($I_{\text{load(max)}}$), analyze the circuit under the **worst-case condition** where the available source current is minimum. This occurs when the input voltage is at its minimum ($V_{i(\min)}$).

Mistakes to be avoided

- Don't consider maximum input voltage for maximum load current.

Q1.04.1 MCQ 2004 2M Ans: B ● ○ ○

The output of a centre-tapped full wave rectifier is shown below.

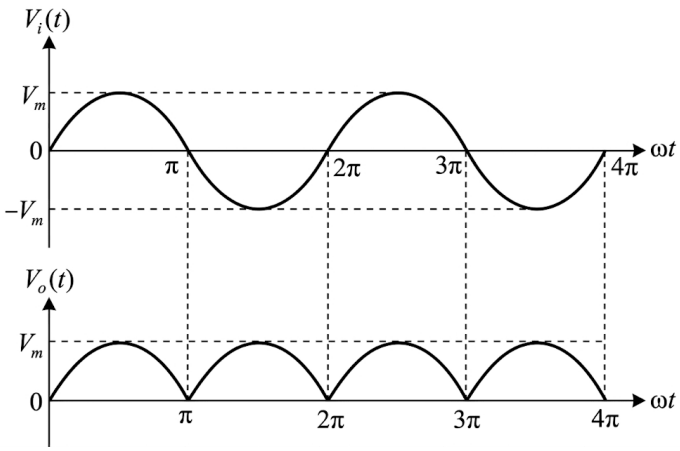

Fig. Solution Q1.04.1

(i) DC or average value of output voltage (V_{dc}) :

DC or average value of output voltage is given by,

$$V_{dc} = \frac{1}{T}\int_0^T V_0(t)\,d(\omega t)$$

$$V_{dc} = \frac{1}{\pi}\int_0^{\pi} V_m \sin\omega t\; d(\omega t)$$

$$V_{dc} = \frac{1}{\pi}V_m\Big[-\cos\omega t\Big]_0^{\pi}$$

$$V_{dc} = \frac{2V_m}{\pi}$$

(ii) Peak inverse voltage (PIV) :

Peak inverse voltage (PIV) or peak reverse voltage (PRV) is the maximum value of reverse voltage which occurs at the peak of the input cycle when the diode is in reverse-bias.

OR

PIV represents maximum reverse bias voltage across diode for which diode can operate safely without breakdown.

Case 1 : During positive half cycle,

D_1 will be forward biased and D_2 will be reverse biased.

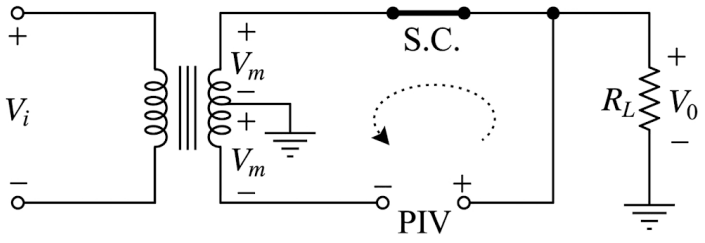

Fig. Solution Q1.04.1

Applying KVL in loop shown in figure,

$$V_m + V_m - \text{PIV} = 0$$

$$\text{PIV} = 2V_m$$

we will get the same value during negative half cycle as well.

(B) $V_{dc} = 2V_m,\ \text{PIV} = 2V_m$

Q1.04.2 MCQ 2004 2M Ans: D ● ○ ○

Given,

(i) $I_{L(\min)} = 100\,\text{mA},\ I_{L(\max)} = 500\,\text{mA}$

(ii) $V_i = 12\,\text{V},\ V_z = 5\,\text{V},\ I_{Z(\min)} = 0\,\text{A}$

For variable load, input current I is given by,

$$I = I_{Z(\min)} + I_{L(\max)} \quad \ldots (\text{i})$$

$$I = I_{Z(\max)} + I_{L(\min)} \quad \ldots (\text{ii})$$

Current through resistor R is given by,

$$I = \frac{12 - V_z}{R} = \frac{12 - 5}{R} = \frac{7}{R} \quad \ldots (\text{iii})$$

Since nothing has been given about the maximum zener current or the power rating we will take the min zener current case to ensure that the device goes into breakdown. So from equation (i),

$$I = 0 + 500 = 500\,\text{mA} = 0.5\,\text{A}$$

From equation (iii),

$$0.5 = \frac{7}{R}$$

$$R = 14\,\Omega$$

(D) $14\,\Omega$

Q1.03.1 MCQ 2003 1M Ans: D ●●○

Given circuit is shown below,

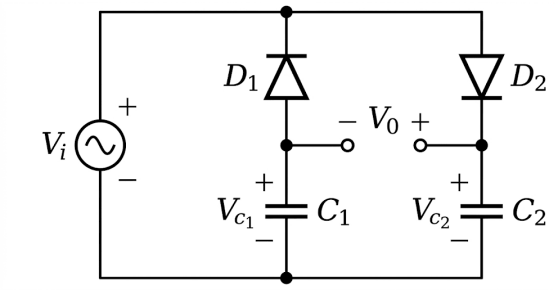

Fig. Solution Q1.03.1

Output voltage V_0 is given by,

$$V_0 = V_{c_2} - V_{c_1}$$

Case 1 : During 1st positive half cycle,

Diode D_1 is in reverse bias, hence OFF.

Diode D_2 is in forward bias, hence ON.

Capacitor C_2 will start charging from zero volt to maximum of input voltage i.e. V_m.

In this case, charging time constant, $\tau_C = 0$ sec

Therefore, $V_{c_2} = V_m$

Case 2 : During 1st negative half cycle,

Diode D_1 is in forward bias, hence ON.

Diode D_2 is in reverse bias, hence OFF.

Capacitor C_1 starts charging in negative direction from zero volt to minimum of input voltage i.e. $-V_m$.

In this case, charging time constant, $\tau_C = 0$ sec

Therefore, $V_{c_1} = -V_m$

Case 3 : During 2nd positive half cycle,

The positive terminal of diode D_1 is at potential $-V_m$. Diode D_1 will be forward biased when the potential of negative terminal of D_1 will be lower or more negative than $-V_m$. Since, the potential of negative terminal of D_1 can not be less than $-V_m$, hence D_1 will be always reverse biased.

The negative terminal of diode D_2 is at potential V_m. Diode D_2 will be forward biased when the potential of positive terminal of D_2 will be higher than V_m. Since, the potential of positive terminal of D_2 can not be higher than V_m, hence D_2 will be always reverse biased.

At steady state, diodes will be OFF and capacitors will not discharge and the circuit becomes,

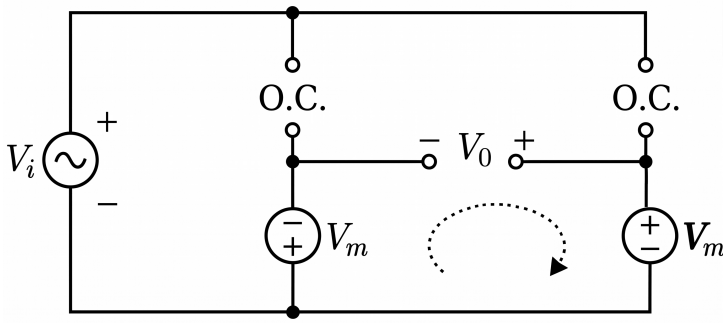

Fig. Solution Q1.03.1

From figure, $V_0 = V_{c_2} - V_{c_1}$

$$V_0 = V_m - (-V_m) = 2V_m$$

Since the output voltage is twice the peak input voltage, the circuit is a voltage doubler.

(D) voltage doubler

Q1.02.1 MCQ 2002 1M Ans: B ● ○ ○

For a silicon diode, the forward voltage decreases approximately by 2 mV/°C with increase in temperature.
Given: $V_D(20°C) = 700\,mV$
Temperature rise: $\Delta T = 40°C - 20°C = 20°C$
Voltage reduction: $\Delta V_D = 20 \times 2\,mV = 40\,mV$
Therefore, $V_D(40°C) = 700\,mV - 40\,mV$

$$V_D(40°C) = 660\,mV$$

(B) 660 mV

Key Points

- Temperature coefficient of a silicon diode is approximately −2 mV/°C.
- At constant current, increase in temperature reduces barrier potential.

Q1.02.2 MCQ 2002 2M Ans: A ● ● ○

Given $V_z = 10\,V$, $I_L = 10\,mA$, $I_{zk} = 1\,mA$ and $30\,V \le V_{in} \le 50\,V$.
For proper zener regulation, the zener current must always remain greater than or equal to the knee current.
The worst case for maintaining regulation occurs at minimum input voltage since minimum current flows through zener diode at that point i.e, $V_{in(min)} = 30\,V$
At this condition,

$$I_{R(min)} = I_L + I_{zk}$$

$$I_{R(min)} = 10\,mA + 1\,mA$$

$$I_{R(min)} = 11\,mA$$

Voltage across series resistor:

$$V_R = V_{in(min)} - V_z$$

$$V_R = 30\,V - 10\,V$$

$$V_R = 20\,V$$

Therefore,

$$R \le \frac{20\,V}{11\,mA}$$

$$R \le 1818\,\Omega$$

Thus, for satisfactory operation,

(A) $R \le 1800\,\Omega$

Mistakes to be avoided

- Do not use maximum input voltage while calculating minimum required current.

Q1.02.3 Descriptive | 2002 | 5M | Ans: -

For ideal diodes, the 5 V divider holds the output at 3 V while $v_i \le 3$ V. D1 then conducts when the input exceeds that level. Before D2 conducts, KCL gives

$$\frac{v_o - v_i}{0.5} + \frac{v_o - 5}{1} + \frac{v_o}{1.5} = 0,$$

where resistances are in kΩ. Thus $v_o = (6v_i + 15)/11$. D2 clamps the input-side diode node at 7 V when both diodes conduct. The onset occurs at $v_i = 31/3$ V, where $v_o = 7$ V.

$$v_o = 3\,\text{V} \quad (v_i \le 3\,\text{V}),$$

$$v_o = (6v_i + 15)/11 \quad (3 < v_i < 31/3),$$

$$v_o = 7\,\text{V} \quad (v_i \ge 31/3\,\text{V}).$$

All thresholds and intermediate voltages are in volts; the pieces meet continuously.

Q1.02.4 Descriptive | 2002 | 5M | Ans: -

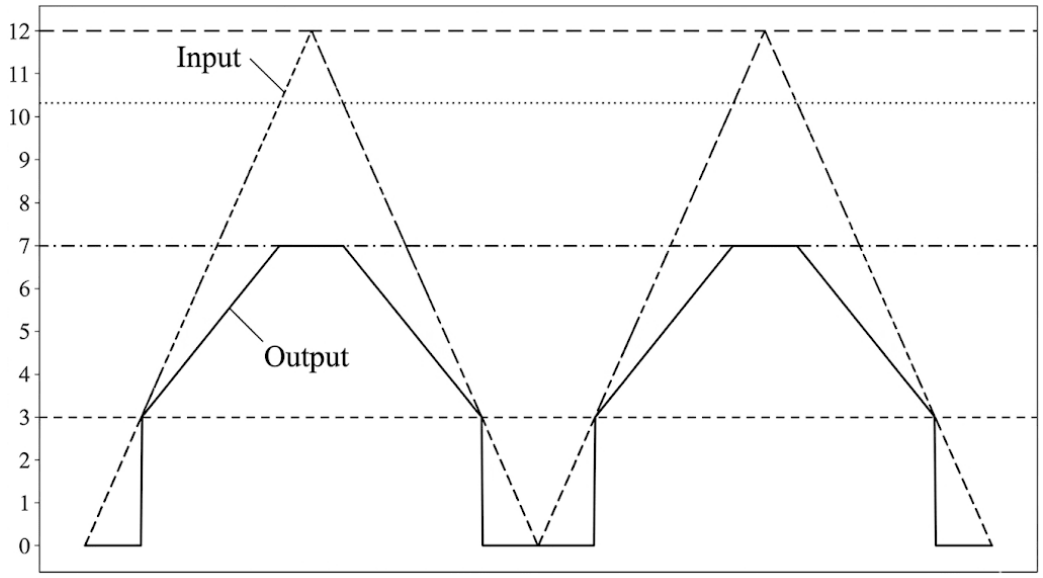

Fig. Solution Q1.02.4

Q1.01.1 Descriptive | 2001 | 2M | Ans: -

The diagram labels both diode voltages from anode to cathode. Thus the reverse-biased diode has $V_r < 0$, and KVL is

$$V_f - V_r = 50\,\text{mV}.$$

Use the full identical-diode Shockley equations; at only two thermal voltages of total bias, approximating reverse current as exactly I_s is not accurate:

$$I/I_s = e^{V_f/V_T} - 1,$$

$$-I/I_s = e^{V_r/V_T} - 1.$$

Let $u = e^{V_f/V_T}$. Then $e^{V_r/V_T} = ue^{-2}$ and the equal-current condition gives

$$u(1 + e^{-2}) = 2.$$

Consequently

$$\boxed{V_f = 14.156\,\text{mV}},$$

$$\boxed{V_r = -35.844\,\text{mV}}.$$

The current is $I/I_s = \tanh(1) = 0.761594$.

Q1.01.2 NAT 2001 2M Ans: 0.762 ●●○

For the oppositely oriented identical diodes, let $j = I/I_s$. The two anode-to-cathode voltages obey

$$V_f = V_T \ln(1 + j), \qquad V_r = V_T \ln(1 - j).$$

KVL from the labelled circuit gives $V_f - V_r = 50$ mV. Since $V_T = 25$ mV,

$$\frac{1 + j}{1 - j} = e^2, \qquad j = \frac{e^2 - 1}{e^2 + 1}.$$

$$I = I_s \tanh(1) = 0.761594\,\text{pA}.$$

$$\boxed{0.762\,\text{pA (three decimals)}}$$

Full reverse saturation is not a valid approximation at this small applied voltage.

Q1.00.1 MCQ 2000 2M Ans: D ●○○

Short Circuit Test :

(i) Assume diode is in forward bias i.e. ON.

(ii) For diode to be ON, current I should be greater than zero ($I_D > 0$ from p to n).

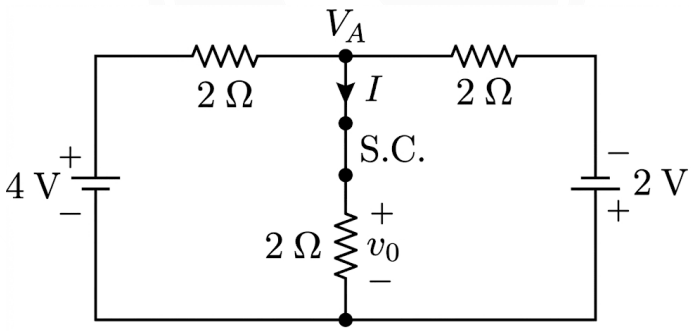

Fig. Solution Q1.00.1

Applying KCL at node V_A,

$$\frac{V_A}{2} + \frac{V_A + 2}{2} + \frac{V_A - 4}{2} = 0$$

$$4 = 3V_A + 2$$

$$V_A = \frac{2}{3}\,\text{V}$$

From figure, $I = \frac{V_A}{2} = \frac{1}{3}$ A

Since, I is positive, so our assumption is correct i.e. diode is ON.

From figure,

$$v_0 = V_A = 2I = \frac{2}{3}\,\text{V}$$

$$\boxed{\text{(D)} \quad \text{None of these}}$$

Key Points

- We can also assume the diode as open circuit initially and find the open circuit voltage, if it is above zero then our assumption is wrong else our assumption is correct.

Q1.99.1 MCQ 1999 1M Ans: A ● ○ ○

Given,
(i) $V_{\text{No load}} = 30\text{ V}$ and $V_{\text{Full load}} = 25\text{ V}$
(ii) $I_{\text{Full load}} = I_{DC} = 1\text{ A}$
Voltage Regulation (in %) is given by,

$$\%\text{ regulation} = \frac{V_{\text{No load}} - V_{\text{Full load}}}{V_{\text{Full load}}} \times 100\%$$

$$\%\text{ regulation} = \frac{30 - 25}{25} \times 100\%$$

$$\%\text{ regulation} = \frac{1}{5} \times 100\% = 20\%$$

Output resistance,

$$R = \frac{V_{\text{No load}} - V_{\text{Full load}}}{I_{DC}}$$

$$R = \frac{30 - 25}{1} = 5\,\Omega$$

(A) 5 Ω and 20%

Key Points

- **Voltage Regulation** indicates how well a power supply maintains its output voltage under varying load conditions. It is the ratio of the change in voltage (from no-load to full-load) to the full-load voltage.

Q1.98.1 MCQ 1998 1M Ans: D ● ○ ○

A four diode bridge rectifier uses a smaller size of transformer, which is less expensive transformer and these rectifiers are suitable for higher voltage applications because of low peak inverse voltage (PIV) rating requirement of each diode.

(D) (1), (2), as well as (3) are true

Q1.98.2 MCQ 1998 1M Ans: D ● ○ ○

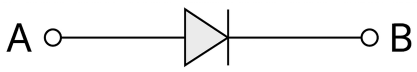

Fig. Solution Q1.98.2

Equivalent circuit representation of above practical diode is series combination of a forward resistance and a cut-in voltage, as shown below,

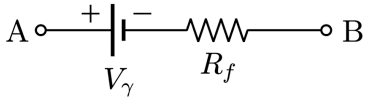

Fig. Solution Q1.98.2

In small signal ac analysis, dc voltage source is replaced by short circuit. Therefore in ac analysis any practical diode is represented by a forward resistance only.

(D) a resistance

Q1.97.1 MCQ | 1997 | 1M | Ans: C | ● ○ ○

PrepFusion

Method 1

Open circuit test:

Assume the diode is OFF. Then the circuit becomes

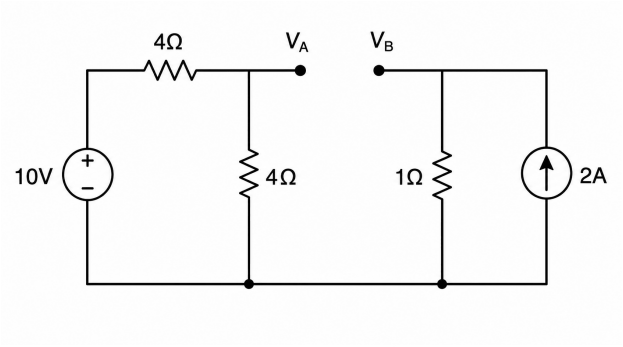

Fig. Solution Q1.97.1

$$V_A = 10 \times \left(\frac{4}{4+4}\right) \quad \text{(By VDR)}$$

$$V_A = 10 \times \frac{1}{2} = 5\,\text{V}$$

$$V_B = 1 \times 2 = 2\,\text{V}$$

Since $V_A > V_B$, our assumption is not correct and the diode is ON. The circuit becomes

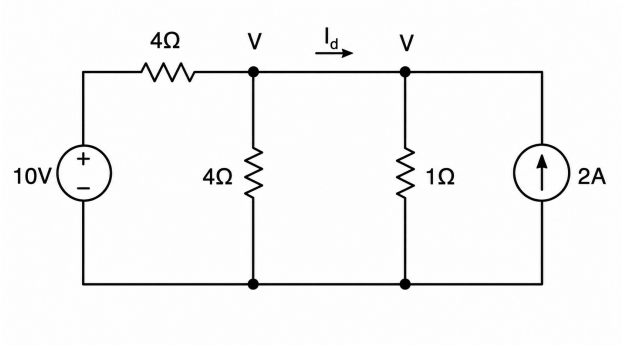

Fig. Solution Q1.97.1

Apply KCL at node V,

$$\frac{V-10}{4} + \frac{V}{4} + \frac{V}{1} - 2 = 0$$

$$6V - 10 - 8 = 0$$

$$6V = 18 \implies V = 3\,\text{V}$$

Now apply KCL at only left node 'V':

$$\frac{V-10}{4} + \frac{V}{4} + I_D = 0$$

$$\frac{3-10}{4} + \frac{3}{4} + I_D = 0 \implies 4I_D = 4 \implies I_D = 1\,\text{A}$$

(C) 1 A

Method 2

Short circuit test:

Assume the diode is ON. Then the circuit becomes

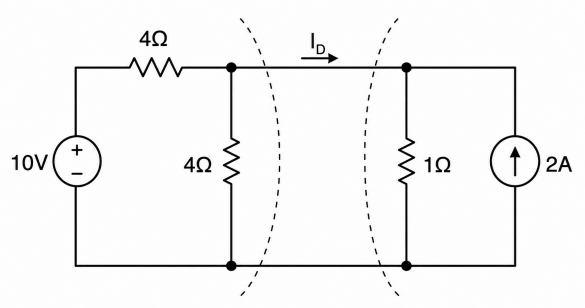

Fig. Solution Q1.97.1

Replace the left and right circuits with their Thevenin equivalents and the circuit becomes:

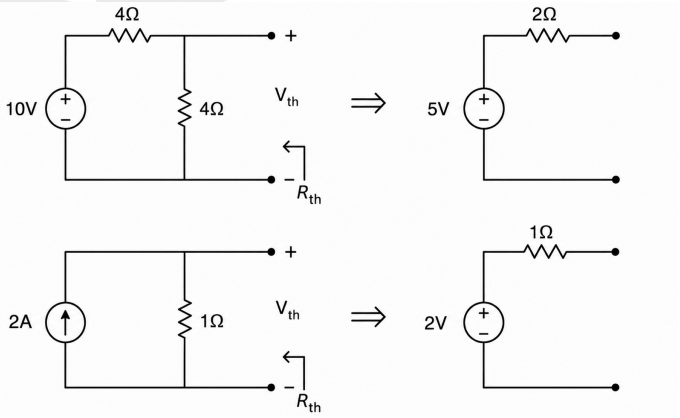

Fig. Solution Q1.97.1

Replacing the Thevenin equivalents,

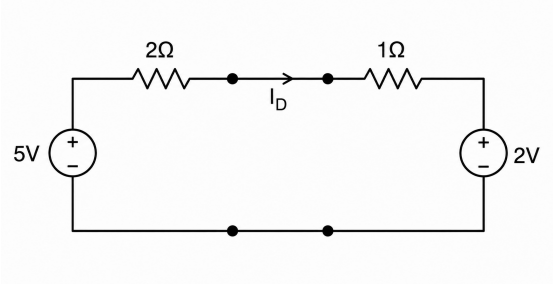

Fig. Solution Q1.97.1

Apply KVL, $-5 + 2I_D + I_D + 2 = 0$

$$3I_D = 3 \implies I_D = 1\,\text{A}$$

Since I_D is positive our assumption is correct and diode is ON.

Hence current through diode = 1 A.

(C) 1 A

Q1.96.1 MCQ 1996 1M Ans: D ● ○ ○

Given circuit is

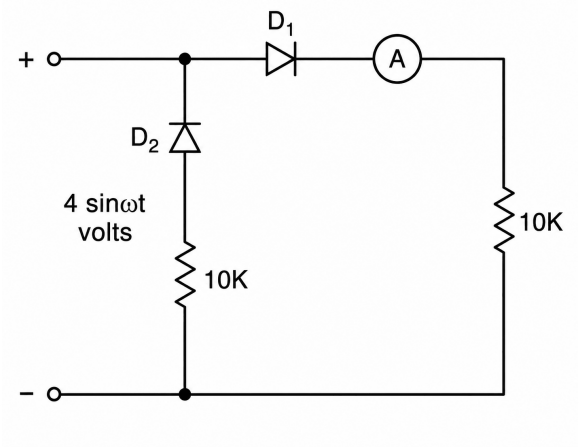

Fig. Solution Q1.96.1

Given input signal, $v_i(t) = 4\sin\omega t$
During the positive half cycle, diode D_1 conducts and diode D_2 remains reverse biased.
Hence the current through the ammeter is

$$i(t) = \frac{4\sin\omega t}{10\,k\Omega}$$
$$i(t) = 0.4\sin\omega t\ \text{mA}$$

Therefore, the peak current is $I_m = 0.4\,\text{mA}$
During the negative half cycle, diode D_1 is OFF and no current flows through the ammeter.
Thus the ammeter current is a half-wave rectified sine wave.
Average value of a half-wave rectified sine wave is

$$I_{avg} = \frac{I_m}{\pi} = \frac{0.4}{\pi}\ \text{mA}$$

(D) $\frac{0.4}{\pi}$ A

Key Points

- Ammeter reads the average value of current in this problem.

Q1.96.2 MCQ 1996 2M Ans: C ● ● ○

Given circuit is

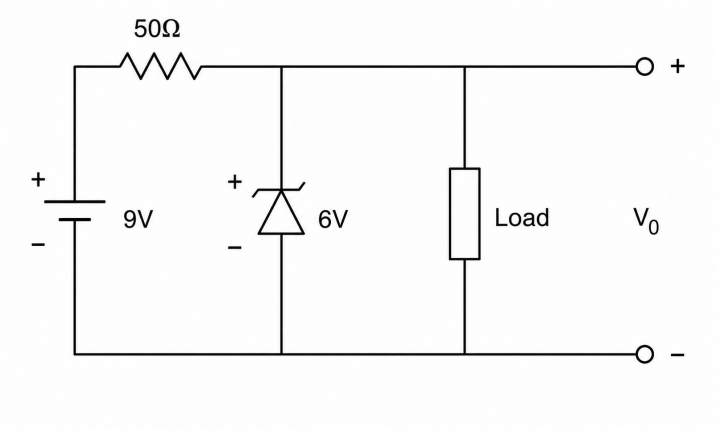

Fig. Solution Q1.96.2

Given, $V_S = 9\,\text{V}$, $V_Z = 6\,\text{V}$, $R = 50\,\Omega$.
Zener knee current, $I_{Z(\min)} = 5\,\text{mA}$.
Maximum zener power dissipation, $P_{Z(\max)} = 300\,\text{mW}$.

Current through the series resistor is

$$I = \frac{V_S - V_Z}{R}$$

$$I = \frac{9 - 6}{50}$$

$$I = 60\,\text{mA}$$

This current divides as $I = I_L + I_Z$

Case 1: For maximum load current, zener current must be minimum:

$$I_Z = I_{Z(\min)} = 5\,\text{mA}$$

$$\therefore I_{L(\max)} = I - I_{Z(\min)} = 60 - 5$$

$$I_{L(\max)} = 55\,\text{mA}$$

Case 2: For minimum load current, zener current becomes maximum.
Maximum permissible zener current is obtained using power rating:

$$P_{Z(\max)} = V_Z I_{Z(\max)}$$

$$300\,\text{mW} = 6\,\text{V} \times I_{Z(\max)}$$

$$I_{Z(\max)} = 50\,\text{mA}$$

Thus,

$$I_{L(\min)} = 60 - 50 = 10\,\text{mA}$$

$$\boxed{\text{(C)} \quad 10\,\text{mA},\ 55\,\text{mA}}$$

Mistakes to be avoided

- Do not assume $I_{L(\min)} = 0$ without checking zener power rating.

Q1.93.1 MCQ | 1993 | 1M | Ans: A

Given source voltage is sinusoidal hence, consider two cases for positive and negative half cycles.

Case 1 : During positive half cycle,

Diode D_1 will be in forward bias, hence it acts as a short circuit.

(a) $0 < V_i < 4.1$ V :

When input voltage is less than breakdown voltage of D_2, D_2 will act as normal diode in reverse bias without breakdown and therefore, D_2 can be replaced by open circuit.

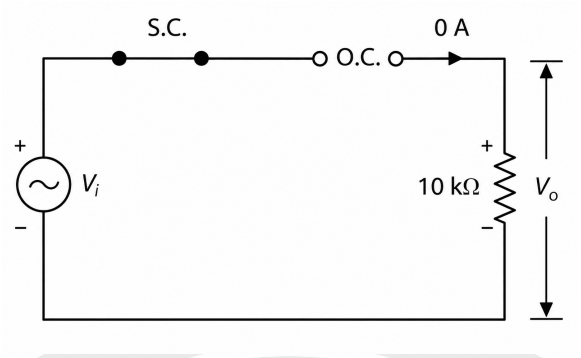

Fig. Solution Q1.93.1

Hence, $V_0 = 0 \times 10 = 0$ V

(b) $V_i > 4.1$ V :

When input voltage will be higher than breakdown voltage of D_2, D_2 will work in reverse bias breakdown region. Therefore, D_2 can be replaced by its breakdown voltage.

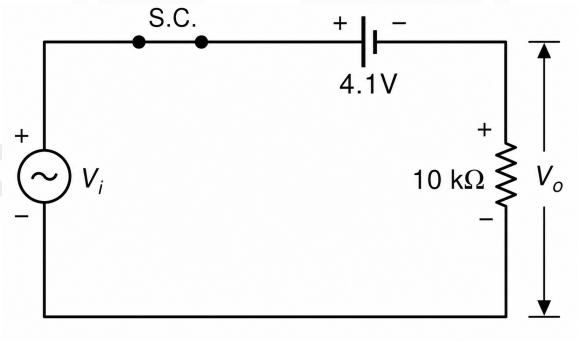

Fig. Solution Q1.93.1

Applying KVL in above figure,

$$-V_i + 4.1 + V_0 = 0$$

$$V_0 = V_i - 4.1 = 10 \sin \omega t - 4.1$$

Case 2 : During negative half cycle,

Diode D_2 will be in forward bias, hence acts as a short circuit.

(a) -4.1 V $< V_i < 0$ V :

When input voltage will be less than breakdown voltage of D_1, D_1 will act as a normal diode in reverse bias. Therefore, D_1 can be replaced by open circuit.

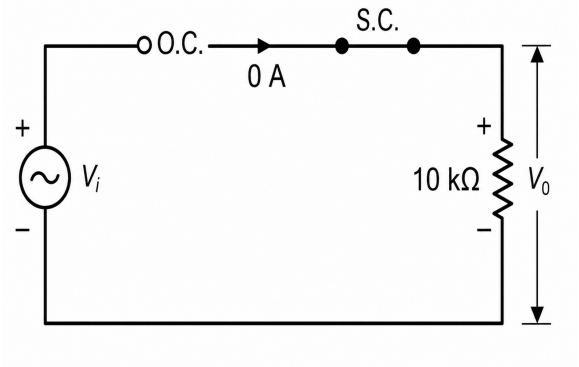

Fig. Solution Q1.93.1

Hence, $V_0 = 0 \times 10 = 0\,\text{V}$

(b) $\mathbf{V_i < -4.1\,V}$:

When input voltage will be higher than breakdown voltage of D_1, D_1 will work in reverse bias breakdown region. Therefore, D_1 can be replaced by its breakdown voltage.

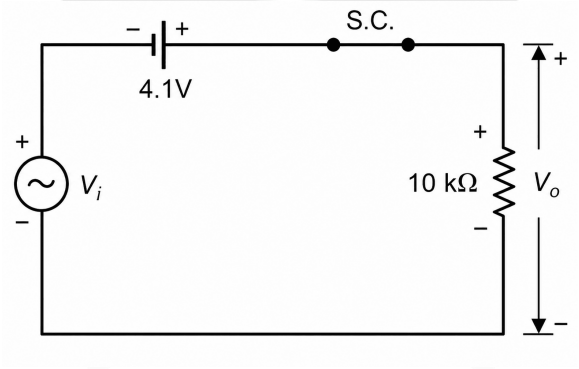

Fig. Solution Q1.93.1

Applying KVL in above figure,

$$-V_i - 4.1 + V_0 = 0$$

$$V_0 = V_i + 4.1 = 10\sin\omega t + 4.1$$

Hence, from case (1) and case (2), expression of output voltage can be written as given below,

$$V_0 = \begin{cases} 0\,\text{V} & ; \quad 0\,\text{V} < V_i < 4.1\,\text{V} \\ V_i - 4.1 & ; \quad V_i > 4.1\,\text{V} \\ 0\,\text{V} & ; \quad -4.1\,\text{V} < V_i < 0\,\text{V} \\ V_i + 4.1 & ; \quad V_i < -4.1\,\text{V} \end{cases}$$

Hence the output waveform is given by,

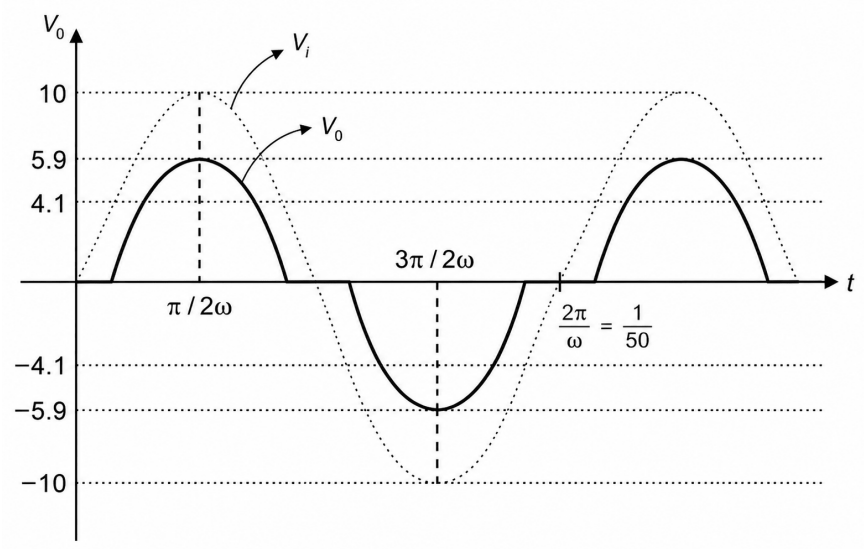

Fig. Solution Q1.93.1

Key Points

- A Zener diode in forward bias operates identically to a standard PN junction diode.
- In reverse bias below the Zener breakdown voltage (V_Z), the diode acts as an open circuit.
- When the reverse bias voltage exceeds V_Z, the Zener diode enters the breakdown region, acting as a constant voltage source equal to V_Z in opposition to the applied voltage.

Q1.93.2 MCQ 1993 1M Ans: B ● ○ ○

Given circuit is,

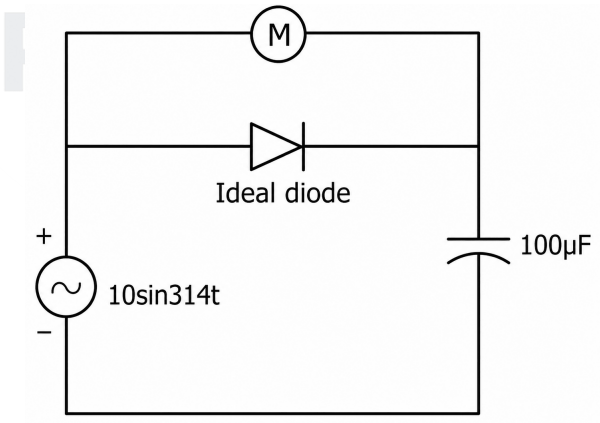

Fig. Solution Q1.93.2

Given input voltage, $v_i(t) = 10 \sin 314t$
Hence the peak value is $V_m = 10\,\text{V}$
During the positive half cycle, the diode conducts and charges the capacitor to the peak value of the input voltage.
Therefore, $V_C = 10\,\text{V}$ and the voltage across voltmeter is zero.
During the negative half cycle, the diode becomes reverse biased and the capacitor retains its charge because the voltmeter draws negligible current.

Hence the voltage across voltmeter is

$$V_i(t) - V_C = (10 \sin 314t - 10)\ \text{V}$$

Therefore the voltage across the voltmeter (diode) is a half-wave waveform whose voltage varies from 0 V to −20 V
An ideal moving iron voltmeter reads the RMS value of the applied voltage.
RMS value of this waveform is

$$V_{\text{RMS}} = \sqrt{\frac{1}{2\pi}\int_{\pi}^{2\pi} (10 \sin\theta - 10)^2\, d\theta} \qquad \text{where } \theta = 314t$$

$$= \sqrt{\frac{100}{2\pi}\int_{\pi}^{2\pi} (\sin\theta - 1)^2\, d\theta}$$

$$= \sqrt{\frac{100}{2\pi}\int_{\pi}^{2\pi} \left(\sin^2\theta - 2\sin\theta + 1\right) d\theta}$$

Using,

$$\int_{\pi}^{2\pi} \sin^2\theta\, d\theta = \frac{\pi}{2}, \int_{\pi}^{2\pi} \sin\theta\, d\theta = -2, \int_{\pi}^{2\pi} 1\, d\theta = \pi$$

Therefore,

$$V_{\text{RMS}} = \sqrt{\frac{100}{2\pi}\left(\frac{\pi}{2} + 4 + \pi\right)}$$

$$= \sqrt{\frac{100}{2\pi}\left(\frac{3\pi}{2} + 4\right)}$$

$$= 10\sqrt{\frac{1}{2\pi}\left(\frac{3\pi}{2} + 4\right)}$$

$$V_{\text{RMS}} \approx 12.24\ \text{V}$$

(B) 12.24 V

Key Points

- Moving iron instruments respond to RMS value.

Q1.92.1 MCQ | 1992 | 2M | Ans: B | ● ○ ○

Given $V_S = 10\,\text{V}$, $R_S = 50\,\Omega$, $V_Z = 6\,\text{V}$, $I_{Z(\min)} = 5\,\text{mA}$

For the voltage across R not to fall below 6 V, the zener diode must remain in breakdown region.
Hence the current through the zener diode should satisfy

$$I_Z \geq 5\,\text{mA}$$

Current through the series resistor is

$$I = \frac{V_S - V_Z}{R_S} = \frac{10 - 6}{50} = 80\,\text{mA}$$

This current divides into load current and zener current:

$$I = I_R + I_Z$$

For minimum permissible value of R, $I_Z = 5\,\text{mA}$
Therefore,

$$I_R = 80 - 5$$

$$I_R = 75\,\text{mA}$$

Now,

$$R_{\min} = \frac{V_Z}{I_R}$$

$$R_{\min} = \frac{6}{75 \times 10^{-3}} = 80\,\Omega$$

(B) $80\,\Omega$

Q1.91.1 NAT 1991 1M Ans: 0

Given circuit and input are shown below.

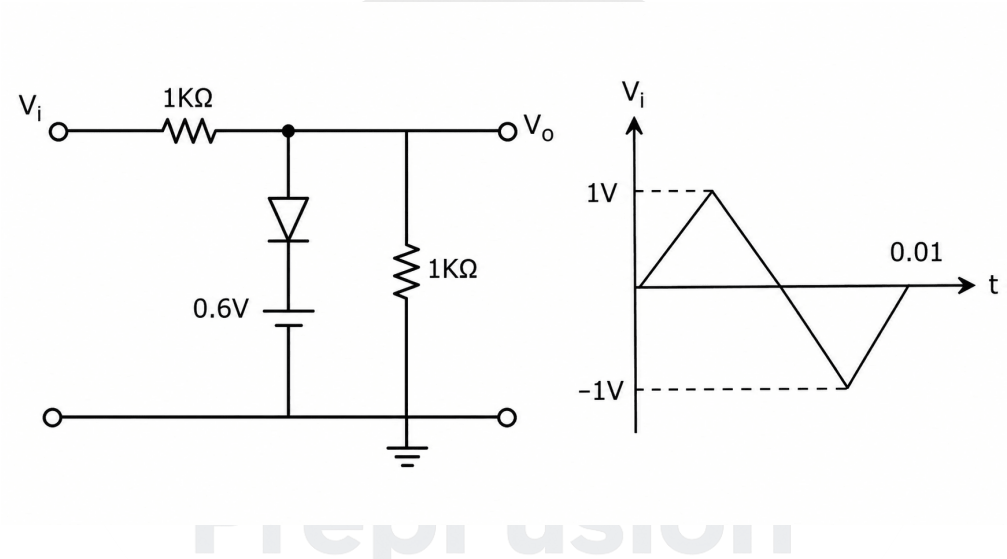

Fig. Solution Q1.91.1

Case 1: Assume the diode is OFF. The circuit acts as a simple voltage divider consisting of the two 1 kΩ resistors. The output voltage V_o without the diode conducting is:

$$V_o = V_i \left(\frac{1\,\text{k}\Omega}{1\,\text{k}\Omega + 1\,\text{k}\Omega} \right) = \frac{V_i}{2}$$

Case 2: For the ideal diode to conduct, the open-circuit voltage at its anode must exceed the voltage at its cathode (0.6 V).

$$V_{\text{anode}} > V_{\text{cathode}}$$

$$\frac{V_i}{2} > 0.6\,\text{V} \implies V_i > 1.2\,\text{V}$$

Since the maximum possible value of the input voltage V_i is 1 V, the condition $V_i > 1.2\,\text{V}$ is never satisfied. Therefore, the diode remains in the OFF state (reverse biased) throughout the entire input cycle.
The output voltage waveform is simply a scaled-down version of the input waveform for all time t:

$$V_o(t) = \frac{V_i(t)}{2}$$

Since the input triangular wave $V_i(t)$ has an average value of 0 V, the average value of the output $V_o(t)$ is also zero.

$$V_{o(\text{avg})} = \frac{0}{2} = 0\,\text{V}$$

Q1.90.1 MSQ 1990 1M Ans: A,D ● ○ ○

For the same a.c. input voltage and load impedance:
Average load current for a half-wave rectifier is

$$I_{\text{dc(HWR)}} = \frac{I_m}{\pi}$$

Average load current for a full-wave rectifier is

$$I_{\text{dc(FWR)}} = \frac{2I_m}{\pi}$$

Therefore, $I_{\text{dc(FWR)}} = 2I_{\text{dc(HWR)}}$
Hence statement (a) is correct.
Full wave rectifier generally means Center tapped full wave rectifier which uses center tapped transformer which is bigger than a normal transformer.
Hence statement (d) is correct.

(A)	The average load current in a full wave rectifier is twice than in a half wave rectifier
(D)	Half wave rectifier will have a small sized transformer compared to full wave rectifier

Q1.90.2 Descriptive 1990 1M Ans: - ● ○ ○

For an ideal diode with ideality factor $\eta = 1$ the diode equation is

$$I = I_S\left(e^{\frac{eV}{KT}} - 1\right)$$

where V = voltage across the diode
Since the two diodes are connected in series, the same current flows through both diodes.
Let the voltage drops across D_1 and D_2 be V_1 and V_2, respectively.
For diode D_1,

$$I = I_S\left(e^{\frac{eV_1}{KT}} - 1\right)$$

$$\frac{I}{I_S} + 1 = e^{\frac{eV_1}{KT}}$$

$$e^{\frac{eV_1}{KT}} = \frac{I}{I_S} + 1$$

For diode D_2, the current direction is opposite to its forward direction. Hence,

$$I = I_S\left(1 - e^{-\frac{eV_2}{KT}}\right)$$

$$e^{-\frac{eV_2}{KT}} = 1 - \frac{I}{I_S}$$

Adding the two expressions,

$$e^{\frac{eV_1}{KT}} + e^{-\frac{eV_2}{KT}} = \left(\frac{I}{I_S} + 1\right) + \left(1 - \frac{I}{I_S}\right)$$

$$\boxed{e^{\frac{eV_1}{KT}} + e^{-\frac{eV_2}{KT}} = 2}$$

Mistakes to be avoided

- Do not use the same polarity expression for both diodes as they are conneted in opposite directions.

PrepFusion

SOLUTIONS

II

MOSFETs

Topics

1. Introduction to MOS - Depletion and Enhancement Type MOSFETs
2. Important Concepts of MOS Physics
3. Biasing a MOSFET - Transconductance and Output Impedance
4. Biasing a MOSFET - Amplification & Small Signal Analysis
5. Small Signal Model of MOSFET and Port Impedance
6. Common Source Amplifier - With and Without Degeneration
7. Common Source Amplifier with Different Biasing
8. Miller's Theorem
9. Common Gate and Common Drain Amplifiers
10. GmRout Method for Gain
11. PMOS Amplifiers
12. Cascode Current Source and Amplifiers
13. Current Mirror Circuits
14. Multistage Amplifiers and Pass Transistor Logic
15. Frequency Response of MOSFET Amplifiers
16. Frequency Response of Common Gate Amplifiers
17. Introduction to Differential Amplifiers
18. Current Source, Voltage Source as Tail
19. Large Signal Analysis of Differential Amplifiers
20. Concept of Half Circuit
21. Building a Two-Stage Op Amp
22. Types and Effects of Feedback
23. Feedback Topologies - Voltage-Voltage, Voltage-Current, Current-Voltage, Current-Current
24. MOS Amplifiers with Body Effect

Unit Snapshot*

Stream: EC	**Questions:** 74	**MCQ:** 49	**MSQ:** 2	**NAT:** 23
1M: 28	**2M:** 44	**Descriptive:** 2	**Avg Weightage** ~4M	**Range:** 2M(2026)–8M(2019)

**See preface for how Avg Weightage and Range are calculated.*

PrepFusion

Q1.26.1 NAT | 2026 | 2M | Ans: 2.56 | ● ○ ○

Given circuit is,

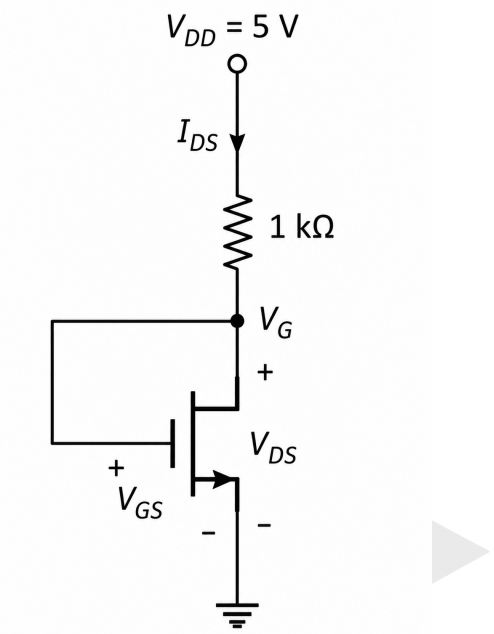

Fig. Solution Q1.26.1

To find the gate voltage V_G of the n-channel MOSFET, we start by analyzing the circuit. The MOSFET is in saturation since it's connected as a source follower with $V_{DD} > V_{TH}$. The current equation in saturation is:

$$I_{DS} = \frac{1}{2}\mu C_{ox}\left(\frac{W}{L}\right)(V_{GS} - V_{TH})^2$$

Here, the drain current (I_D) through the 1 kΩ resistor can be calculated as:

$$I_D = \frac{V_{DD} - V_G}{1\,\text{k}\Omega}$$

Equating both expressions for I_D:

$$\frac{V_{DD} - V_G}{1\,\text{k}\Omega} = \frac{1}{2} \cdot 2\,\text{mA/V}^2 \cdot (V_G - V_{TH})^2$$

$$V_{DD} - V_G = (V_G - V_{TH})^2$$

Substitute $V_{DD} = 5\,\text{V}$ and $V_{TH} = 1\,\text{V}$:

$$5 - V_G = (V_G - 1)^2$$

$$5 - V_G = V_G^2 - 2V_G + 1$$

$$V_G^2 - V_G - 4 = 0$$

Using the quadratic formula $V_G = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, where $a = 1, b = -1, c = -4$:

$$V_G = \frac{1 \pm \sqrt{1 + 16}}{2}$$

$$V_G = \frac{1 \pm \sqrt{17}}{2}$$

$$V_G = \frac{1 \pm 4.12}{2}$$

Calculating the positive root (ensures $V_{GS} > V_{TH}$):

$$\boxed{V_G = \frac{5.12}{2} = 2.56\,\text{V}}$$

Key Points

- When the gate of an N-MOSFET is tied to its drain then it is always in saturation as V_{DS} is always grater than $V_{GS} - V_{TH} = V_{DS} - V_{TH}$.
- Always select the positive root of the quadratic equation that satisfies the strong inversion condition $V_{GS} > V_{TH}$.

Q2.25.1 MSQ | 2025 | 1M | Ans: B,D | ● ○ ○

From the small-signal characteristics of MOSFET amplifiers:

(i) Common gate amplifier

Input resistance of common gate amplifier is

$$R_i = \frac{1}{g_m}$$

Hence statement (i) is incorrect.

(ii) Common source amplifier

Since gate current is ideally zero,

$$R_i = \infty$$

Hence statement (ii) is correct.

(iii) Common source amplifier phase relation

Voltage gain of common source amplifier is

$$A_v = -g_m R_L'$$

Negative sign indicates a phase shift of 180°. Hence input and output voltages are 180° out of phase.
Therefore statement (iii) is incorrect.

(iv) Common drain amplifier phase relation

Voltage gain of common drain amplifier is

$$A_v = \frac{g_m R_L'}{1 + g_m R_L'} \approx 1$$

Since the gain is positive, there is no phase inversion. Hence input and output voltages are in phase.
Therefore statement (iv) is correct.

(B)	The common-source amplifier has an infinite input resistance
(D)	The input and output voltages of the common-drain amplifier are in phase

Key Points

- Common source amplifier provides 180° phase shift.
- Common drain amplifier is also called source follower.
- Common gate amplifier has low input resistance equal to $\frac{1}{g_m}$.

Q2.25.2 MCQ | 2025 | 2M | Ans: D | ● ○ ○

Given : Identical MOSFET M_1 and M_2

$$g_m = 5\,\text{mS}$$

$$V_1 = 2.5 + 0.01\sin\omega t$$

$$V_2 = 2.5 - 0.01\sin\omega t$$

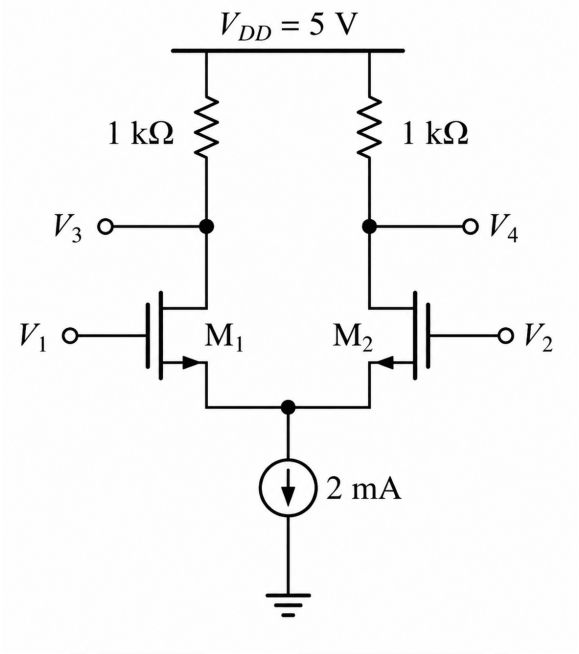

Fig. Solution Q2.25.2

From the given circuit

$$I_{DS_1} = I_{DS_2} = \frac{2\,\text{mA}}{2} = 1\,\text{mA}$$

i.e.,

$$I_{DS} = 1\,\text{mA}$$

And differential mode voltage

$$\frac{V_d}{2} = 0.01\sin\omega t$$

$$V_d = 0.02\sin\omega t$$

Common mode voltage, $V_{CM} = 2.5\,\text{V}$
Differential mode gain is given by,

$$A_{DM} = \frac{V_0}{V_d} = \frac{-g_m \times R_D}{2}$$

$$\frac{V_0}{V_d} = \frac{-5 \times 1}{2} = -2.5$$

$$V_0 = -2.5 \times V_d$$

$$V_0 = -2.5 \times 0.02\sin\omega t = -0.05\sin\omega t$$

AC output at V_3 ,

$$V_{3_{AC}} = -0.05\sin\omega t$$

DC output voltage at V_3

$$V_{3_{DC}} = V_{DD} - I_{DS}(R_D) = 5 - 1(1) = 4\,\text{V}$$

The effect of common mode input is cancelled in given circuit i.e. $A_{CM} = 0$.
Hence total output voltage

$$V_3 = V_{3_{AC}} + V_{3_{DC}} = 4 - 0.05 \sin \omega t$$

(D) $4 - 0.05 \sin \omega t$

Key Points

- The total output voltage in a given amplifier node is the superposition of its DC bias voltage and the AC small-signal variation: $V_{out} = V_{DC} + V_{AC}$.
- For a perfectly symmetrical differential pair, the single-ended AC differential gain is evaluated as $A_{DM} = \frac{-g_m R_D}{2}$.
- An ideal constant tail current source provides infinite common-mode rejection, rendering the common mode gain $A_{CM} = 0$.

Q2.24.1 MCQ 2024 1M Ans: A

Replace the MOSFET in the given circuit using its small signal model and connect a voltage source of V_i to calculate Z_{in} and the circuit becmoes,

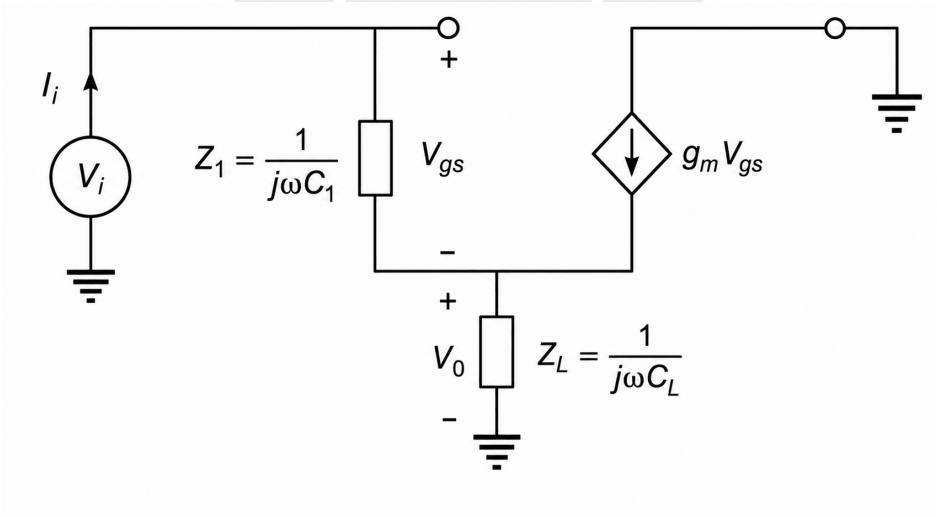

Fig. Solution Q2.24.1

$$I_i = \frac{V_i - V_0}{z_1} = \frac{V_i}{z_1} - \frac{V_0}{z_1}$$

Where,

$$V_0 = (I_i + g_m V_{gs}) z_L$$

$$V_0 = I_i z_L + g_m V_{gs} z_L$$

$$V_0 = I_i (z_L + g_m z_1 z_L)$$

$$\therefore I_i = \frac{V_i}{z_1} - \frac{I_i (z_L + g_m z_1 z_L)}{z_1}$$

$$I_i \frac{[z_1 + z_L + g_m z_1 z_L]}{z_1} = \frac{V_i}{z_1}$$

$$z_{in} = \frac{V_i}{I_i} = z_1 + z_L + g_m z_1 z_L$$

(A) $\dfrac{-g_m}{C_1C_L\omega^2}+\dfrac{1}{j\omega C_1}+\dfrac{1}{j\omega C_L}$

Q2.24.2 MCQ | 2024 | 1M | Ans: A | ● ○ ○ | CLICK HERE FOR VIDEO SOLUTION

In the given amplifier circuit, we need to find the closed loop low frequency small signal voltage gain.
This circuit represents a differential amplifier with an active current mirror load, in which the output V_{out} is connected to V_- forming a negative feedback.

Concept: Why it's a negative feedback ?

The gate of the right NMOS transistor is directly connected to the output node, therefore

$$V_{GS,\text{right}} = V_{out} - V_s$$

where V_s is the common source voltage of the differential pair.
When V_{in} increases, the left NMOS transistor conducts more current. Since the tail current source I_{SS} is ideal and constant,

$$I_{Left} + I_{Right} = I_{SS}$$

hence an increase in the left branch current causes the right branch current to decrease:

$$I_{Left} \uparrow \Rightarrow I_{Right} \downarrow$$

The left PMOS transistor is diode-connected and forms a current mirror with the right PMOS transistor. Therefore, when the left branch current increases, the mirrored PMOS current in the right branch also increases:

$$I_{PMOS,\text{right}} \uparrow$$

while simultaneously,

$$I_{NMOS,\text{right}} \downarrow$$

At the output node, the PMOS transistor supplies current from the top whereas the right NMOS transistor sinks current downward. Since the PMOS current becomes larger than the NMOS current, excess current flows into the output node and charges the output capacitance. Hence,

$$V_{out} \uparrow$$

Since the gate of the right NMOS transistor is connected directly to V_{out}, any increase in V_{out} increases its gate voltage, thereby increasing $V_{GS,\text{right}}$
As a result, the right NMOS transistor conducts more current:

$$I_{NMOS,\text{right}} \uparrow$$

This increased downward current tends to pull the output voltage downward, opposing the original rise in V_{out}. Therefore,

$$V_{out} \uparrow \Rightarrow I_{NMOS,\text{right}} \uparrow \Rightarrow V_{out} \downarrow$$

Thus, the fed back signal opposes the original change in output and hence the circuit exhibits negative feedback.
That is also the reason why left NMOS gate acts as non-inverting terminal and right NMOS gate acts as inverting terminal.

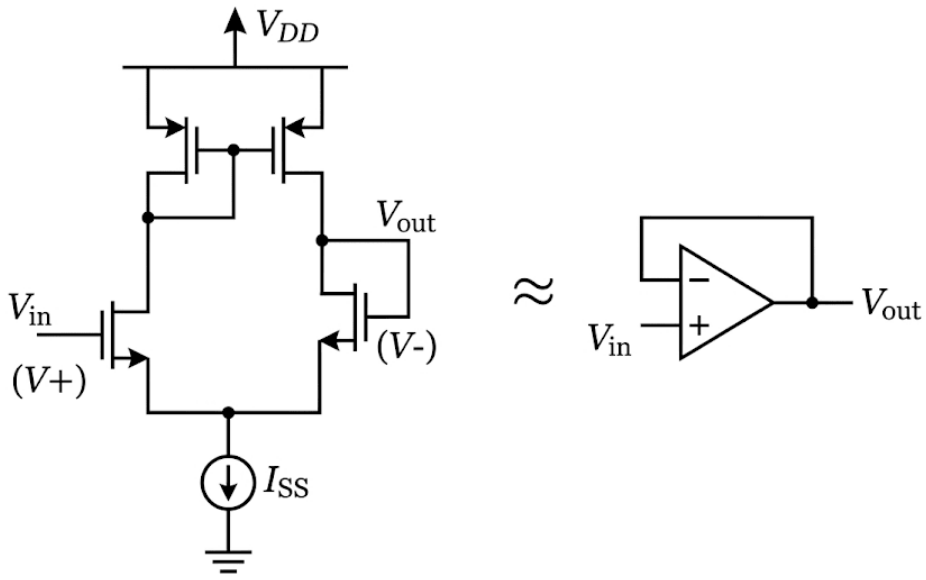

Fig. Solution Q2.24.2

Since there is no external feedback network dividing the output voltage, the entire output is fed back to the inverting terminal. Therefore, the feedback factor is:

$$\beta = 1$$

The formula for the closed-loop gain in a negative feedback amplifier is:

$$A_{CL} = \frac{A_{OL}}{1 + A_{OL}\beta}$$

Given the open-loop gain $A_{OL} = 50$ and $\beta = 1$, we can calculate the closed-loop gain A_{CL} as :

$$A_{CL} = \frac{40}{1 + 40 \times 1}$$

$$A_{CL} = \frac{40}{41}$$

$$A_{CL} \approx 0.976$$

$$\boxed{\text{(A)} \quad 0.976}$$

Key Points

- The feedback factor β represents the fraction of the output voltage fed back to the input. For a direct connection, $\beta = 1$.
- The above circuit is known as voltage follower when open loop gain is ∞.

Mistakes to be avoided

- Assuming $A_{CL} = 1$ (voltage follower) without checking if the open-loop gain A_{OL} is finite. The approximation only holds when $A_{OL} \gg 1$.

Q2.24.3 MCQ | 2024 | 2M | Ans: B

Given,
(i) Transistors M_1 and M_2 are biased in saturation.
(ii) Small signal transconductance for M_1 and M_2 is g_{m1} and g_{m2}.
(iii) Body effect, channel length modulation and intrinsic device capacitances are negligible.
Small signal equivalent circuit is shown below,

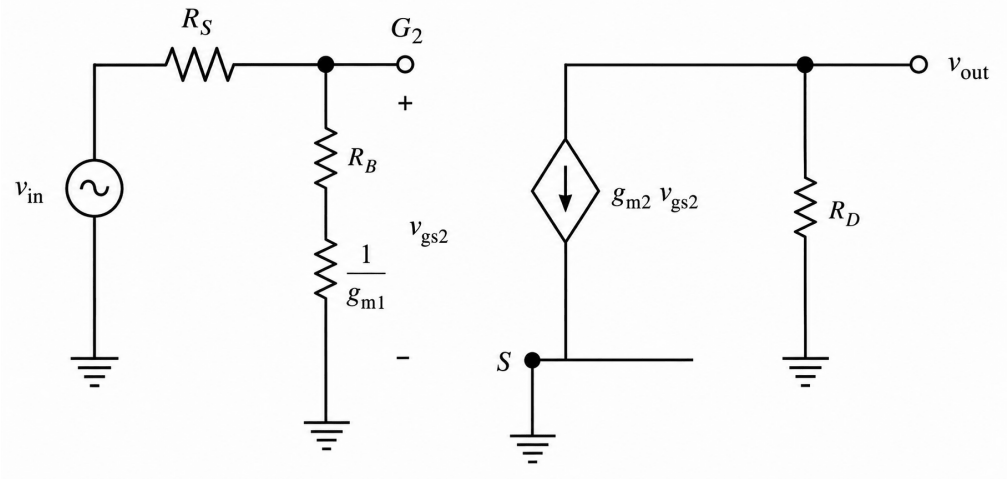

Fig. Solution Q2.24.3

From figure,

$$v_{out} = -g_{m2} v_{gs2} R_D \quad \ldots (i)$$

Apply voltage division rule at input

$$v_{gs2} = \frac{v_{in}\left(R_B + \frac{1}{g_{m1}}\right)}{R_S + R_B + \frac{1}{g_{m1}}} \quad \ldots (ii)$$

From equation (ii),

$$v_{out} = -g_{m2} R_D v_{in} \frac{\left(R_B + \frac{1}{g_{m1}}\right)}{R_B + R_S + \frac{1}{g_{m1}}}$$

$$\frac{v_{out}}{v_{in}} = -g_{m2} R_D \frac{\left(R_B + \frac{1}{g_{m1}}\right)}{\left(R_B + \frac{1}{g_{m1}} + R_S\right)}$$

Magnitude of small signal voltage gain,

$$\left|\frac{v_{out}}{v_{in}}\right| = g_{m2} R_D \frac{\left(R_B + \frac{1}{g_{m1}}\right)}{\left(R_B + \frac{1}{g_{m1}} + R_S\right)}$$

$$\boxed{\text{(B)} \quad \frac{g_{m2} R_D \left(R_B + \frac{1}{g_{m1}}\right)}{R_B + \frac{1}{g_{m1}} + R_S}}$$

Key Points

- The small-signal resistance looking into the source of a diode-connected MOSFET (M_1) is evaluated as $\frac{1}{g_{m1}}$ when body effect and channel length modulation are neglected.
- The input signal v_{in} experiences attenuation before reaching the gate of M_2 due to the voltage divider formed by source resistance R_S and the series combination $(R_B + \frac{1}{g_{m1}})$.

Q2.23.1 MCQ 2023 1M Ans: D ●○○

In the given circuit, M_1 is common-gate amplifier and M_2 behaves as an active load.
By using properties of common gate (CG) amplifier,
Input impedance (R_i) is low $\approx 1/g_m$
Output impedance (R_0) is high $\approx r_0$
As the input impedance is low it is easy for the circuit to accept input current and as the output impedance is high it is easy for the circuit to transfer the entire current to load.
So it is a current controlled current source (current amplifier).

(D) current controlled current source

Key Points

- A Common-Gate (CG) amplifier provides a low input impedance ($\approx \frac{1}{g_m}$) at the source terminal.
- It provides a high output impedance at the drain terminal, making it suitable for driving current into a load.
- An amplifier with low input impedance (to perfectly sample input current) and high output impedance (to perfectly deliver output current) is ideally modeled as a Current Controlled Current Source (CCCS).

Q2.23.2 MCQ 2023 1M Ans: D ●○○

Note : This question has been modified. In the official GATE question, the options are given as closed loop gain which is not possible.
Hence MTA was given. Here We have modified the options and the solution is done on the basis of modified options.
Concept:
The Barkhausen Stability Criterion is a mathematical condition to determine when a linear electronic circuit will oscillate. It states that a circuit will oscillate if it meets these two conditions at the same frequency:
1) **The total phase shift around a loop is 0 degrees or an integral multiple of 360 degrees**
2) **The total gain around the loop is equal to or greater than 1.**
When we apply the Barkhausen Stability Criterion to the options given, the answer becomes clear.
From option - 4, the closed loop gain is greater than 1 and the phase shift is greater than 180°. The common source stage provides a phase shift of 180°. So overall phase shift is grater than 360° This condition can cause oscillation. Although having the phase shift exactly 360° is the ideal case for sustained oscillation, a phase shift greater than 360° might also lead to instability which may result in oscillations.

(D) the loop gain is greater than 1 and the phase shift is greater than 180°

Key Points

- The Barkhausen Stability Criterion dictates the mathematical requirements for a linear circuit to oscillate: a loop gain magnitude equal to or greater than 1 ($|A\beta| \geq 1$).
- The total phase shift around the closed loop must be 0° or an integer multiple of 360° (i.e., $\angle A\beta = 360° \times n$).

Q2.23.3 NAT | 2023 | 2M | Ans: 2 | ● ○ ○

Given circuit is,

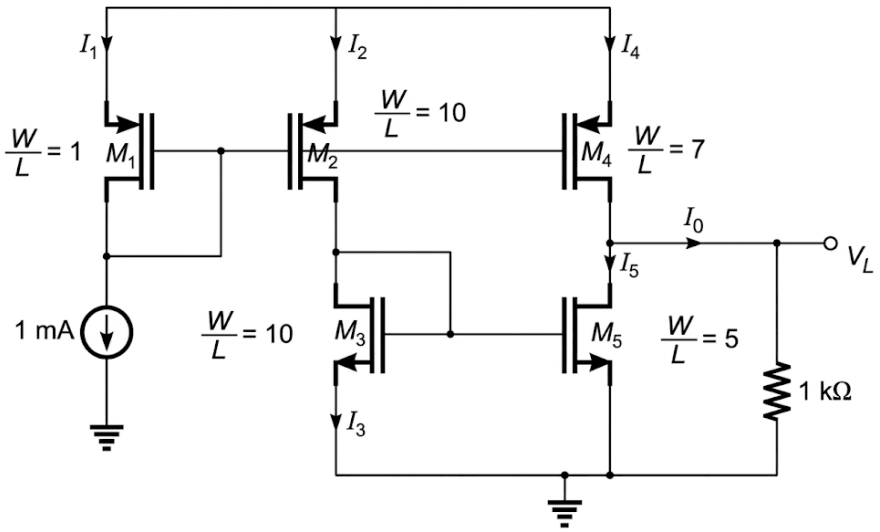

Fig. Solution Q2.23.3

Here M_1, M_2, M_4 form a current mirror whereas M_3, M_5 form another current mirror.
We know, in a current mirror drain current

$$I_D \propto \left(\frac{W}{L}\right)$$

$$I_1 = 1\,\text{mA}$$

$$I_2 = \frac{\left(\frac{W}{L}\right)_2}{\left(\frac{W}{L}\right)_1} \times I_1 = \frac{10}{1} \times 1 = 10\,\text{mA}$$

$$I_4 = \frac{\left(\frac{W}{L}\right)_4}{\left(\frac{W}{L}\right)_1} \times I_1 = \frac{7}{1} \times 1 = 7\,\text{mA}$$

$$I_3 = I_2 = 10\,\text{mA}$$

$$I_5 = \frac{\left(\frac{W}{L}\right)_5}{\left(\frac{W}{L}\right)_3} \times I_3 = \frac{5}{10} \times 10 = 5\,\text{mA}$$

$$I_0 = I_4 - I_5 = 7 - 5 = 2\,\text{mA}$$

$$\boxed{V_L = 2 \times 1 = 2\,\text{V}}$$

Key Points

- In a standard MOSFET current mirror, the mirrored drain current I_D is directly proportional to the aspect ratio $\left(\frac{W}{L}\right)$ of the transistors, assuming they share the same V_{GS} and operate in saturation.

Q2.22.1 MCQ | 2022 | 1M | Ans: C | ● ○ ○

Concept:
MOSFET is in saturation region when

$$V_{GS} > V_{th}$$

$$V_{DS} > V_{GS} - V_{th}$$

The current equation for a MOSFET in the saturation region is given by:

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{th})^2$$

$$I_D = \frac{K_n}{2}(V_{GS} - V_{th})^2$$

Calculation:

Both MOSFETs are in saturation because drain is shorted to Gate and they both are in series so same current flows through them.

$$I_{DSN} = I_{SDP}$$

$$\frac{\mu_n C_{ox}}{2} \times \left(\frac{W}{L}\right)_N (V_{GSN} - V_{TN})^2 = \frac{\mu_p C_{ox}}{2}\left(\frac{W}{L}\right)_P (V_{SGP} - |V_{TP}|)^2$$

$$300 \times 1(V_0 - 1)^2 = 40 \times 5(4 - V_0 - 1)^2$$

$$3(V_0^2 + 1 - 2V_0) = 2(9 + V_0^2 - 6V_0)$$

$$\Rightarrow V_0^2 + 6V_0 - 15 = 0$$

$$V_0 = \frac{-6 \pm \sqrt{36 + 4 \times 15}}{2} = \frac{-6 \pm \sqrt{96}}{2}$$

$$V_0 = 1.898\,\text{V}, -7.89\,\text{V}$$

V_0 cannot be negative because V_0 should lie between 0 and 4 V. So, $V_0 = 1.898$ V (Less than 2 V)

(C) less than 2 V

Key Points

- A diode-connected MOSFET (Drain shorted to Gate) is always in saturation if $V_{GS} > V_{th}$, because $V_{DS} = V_{GS}$, which strictly satisfies the saturation condition $V_{DS} \geq V_{GS} - V_{th}$.

Q2.22.2 NAT | 2022 | 2M | Ans: -2 | ● ● ○

AC equivalent circuit of the given figure by

(i) All capacitors are short circuited.

(ii) All DC voltage sources are replaced by short circuit.

(iii) All DC current sources are replaced by open circuit.

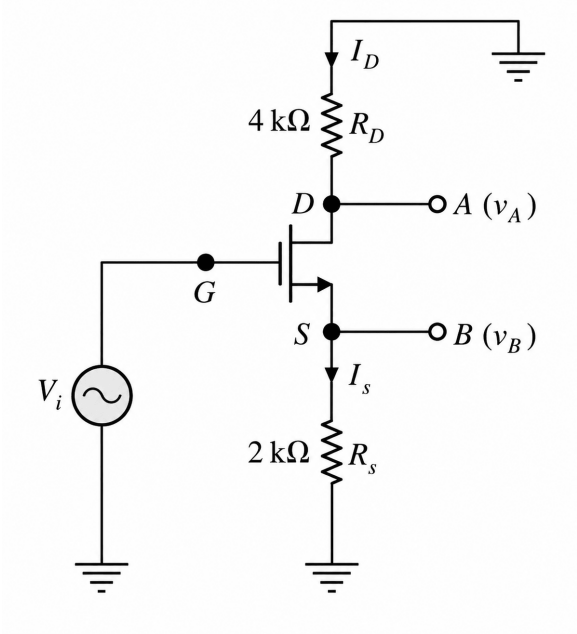

Fig. Solution Q2.22.2

Small signal equivalent circuit is shown below,

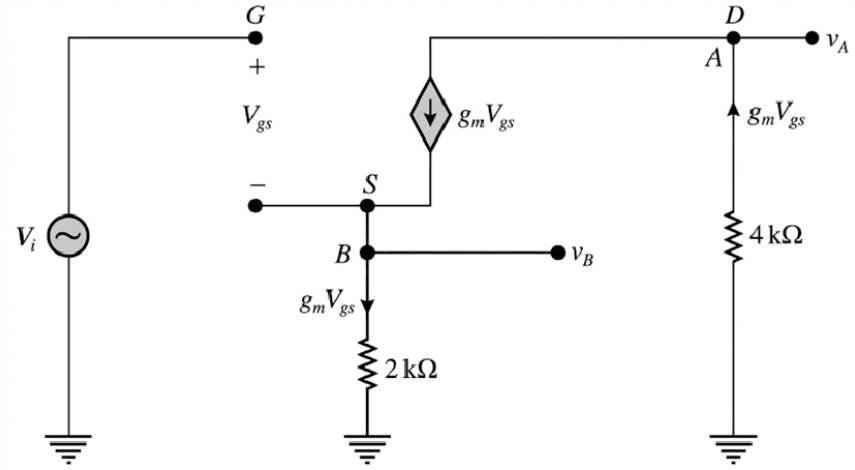

Fig. Solution Q2.22.2

Voltage at A,

$$v_A = -R_D g_m V_{gs} \quad \ldots (i)$$

Voltage at B,

$$v_B = R_s g_m V_{gs} \quad \ldots (ii)$$

Thus,

$$\frac{v_A}{v_B} = \frac{-R_D g_m V_{gs}}{R_s g_m V_{gs}} = \frac{-R_D}{R_s} = \frac{-4}{2} = -2$$

$$\boxed{\frac{v_A}{v_B} = -2}$$

Q2.22.3 MCQ 2022 2M Ans: C ● ○ ○

Given,

$$L = 10\,\mu\text{m}, \quad W = 100\,\mu\text{m}, \quad \mu_n C_{ox} = 1\,\text{mA/V}^2, \quad V_T = 1\,\text{V}, \quad V_{DS} = 1\,\text{V}, \quad V_{GS} = [2 - \sin(2t)]\,\text{V}$$

The maximum and minimum values of gate-to-source voltage are:

$$V_{GS(\max)} = 3\,\text{V} \text{ and } V_{GS(\min)} = 1\,\text{V}$$

To find the maximum drain current, we use $V_{GS(\max)} = 3\,\text{V}$. Let's check the region of operation at maximum V_{GS}:

$$V_{GS(\max)} - V_T = 3 - 1 = 2\,\text{V}$$

Since $V_{DS} = 1\,\text{V}$, we have $V_{DS} < (V_{GS(\max)} - V_T)$. Here, $V_{DS} > (V_{GS} - V_T)$ condition will not be satisfied.
Hence, nMOSFET is in Linear region.
The maximum drain current is given by the linear region equation:

$$I_{D_{\max}} = \mu_n C_{ox} \frac{W}{L}\left[(V_{GS(\max)} - V_T)V_{DS} - \frac{1}{2}V_{DS}^2\right]$$

$$I_{D_{\max}} = 1\,\text{mA/V}^2 \times 10\left[(3-1)1 - \frac{1}{2}\times 1^2\right]$$

$$I_{D_{\max}} = 10[1.5]\,\text{mA} = 15\,\text{mA}$$

(C) 15 mA

Key Points

- Before calculating the current, it is crucial to verify the region of operation using the condition $V_{DS} < V_{GS} - V_T$ for the linear (triode) region and $V_{DS} \geq V_{GS} - V_T$ for the saturation region.
- The drain current equation for an nMOSFET operating in the linear region is $I_D = \mu_n C_{ox}\frac{W}{L}\left[(V_{GS} - V_T)V_{DS} - \frac{1}{2}V_{DS}^2\right]$.

Q2.21.1 MCQ 2021 1M Ans: D ● ○ ○

Method 1

Channel length modulation coefficient $\lambda \neq 0$, it means internal resistance of MOSFET M_1 and M_2 are not zero. That is $r_{o1} \neq 0, r_{o2} \neq 0$
The AC equivalent circuit is given by shorting DC voltage sources.

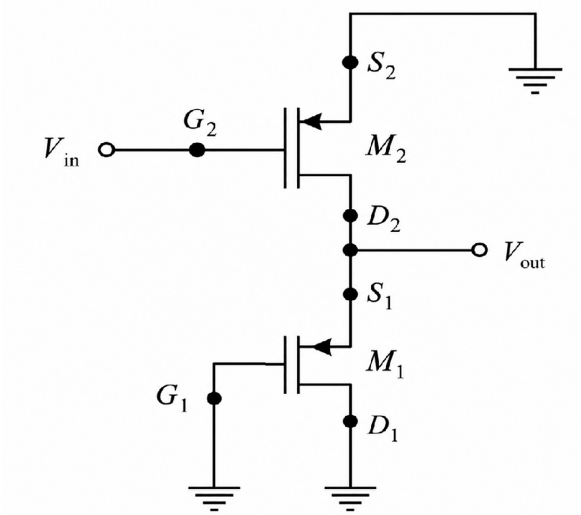

Fig. Solution Q2.21.1

From figure $V_{g_1} = 0\,\text{V}$, $V_{s_2} = 0\,\text{V}$ and $V_{d_1} = 0\,\text{V}$, $V_{s_1} = V_{d_2} = V_{out}$, $V_{g_2} = V_{in}$. So,

$$V_{gs_1} = V_{g_1} - V_{s_1} = 0 - V_{out} = -V_{out}$$

$$V_{gs_2} = V_{g_2} - V_{s_2} = V_i - 0 = V_i$$

The AC equivalent model of given circuit is shown below,

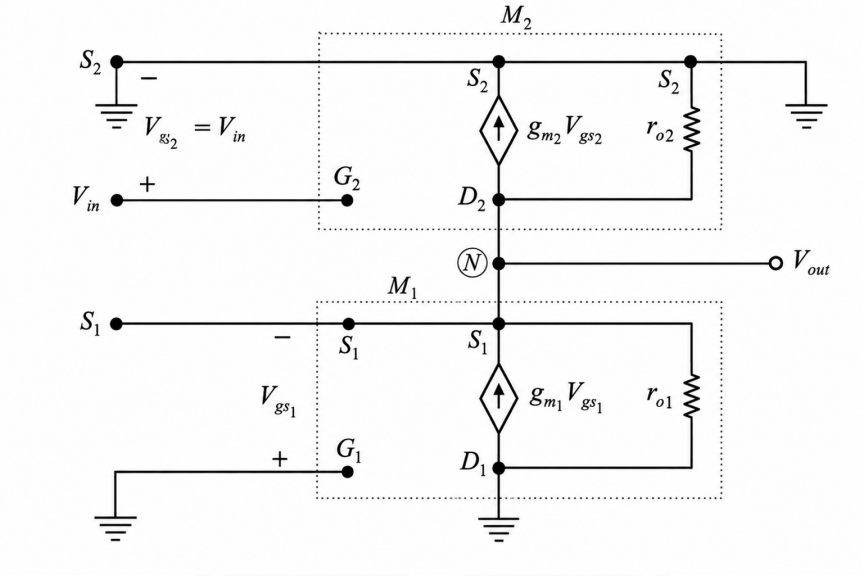

Fig. Solution Q2.21.1

Applying KCL at node N,

$$\frac{V_{out}}{r_{o1}} - g_{m_1}V_{gs_1} + g_{m_2}V_{gs_2} + \frac{V_{out}}{r_{o2}} = 0 \quad \ldots (\text{i})$$

Here,

$$V_{gs_1} = -V_{out} \quad \ldots (\text{ii})$$

$$V_{gs_2} = V_{in} \quad \ldots (\text{iii})$$

So, put equation (ii) and (iii) into equation (i),

$$V_{out}\left[\frac{1}{r_{o1}} + \frac{1}{r_{o2}} + g_{m_1}\right] = -g_{m_2}V_i$$

$$\frac{V_{out}}{V_i} = -g_{m_2}\left[\frac{1}{g_{m_1}}||r_{o1}||r_{o2}\right] \implies \frac{\partial V_{out}}{\partial V_i} = -g_{m_2}\left[\frac{1}{g_{m_1}}||r_{o1}||r_{o2}\right]$$

$$\boxed{\text{(D)} \quad -g_{m2}\left(\frac{1}{g_{m1}} \parallel r_{o1} \parallel r_{o2}\right)}$$

Method 2

Channel length modulation coefficient $\lambda \neq 0$, it means internal resistance of MOSFET M_1 and M_2 are not zero i.e., $r_{o1} \neq 0$, $r_{o2} \neq 0$.

Here, drain (D_1) and gate (G_1) of MOSFET (M_1) are grounded, so it can be drawn as,

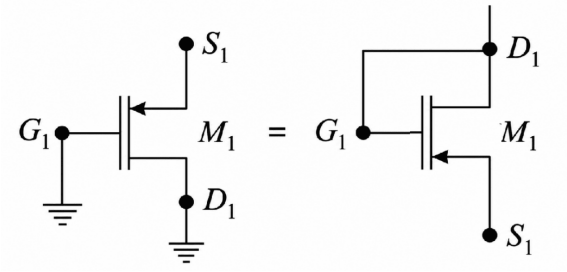

Fig. Solution Q2.21.1

When drain and gate of MOSFET M_1 are shorted to each other then it can be replaced by resistance of value $\left(\frac{1}{g_{m1}}||r_{o1}\right)$ as shown below,

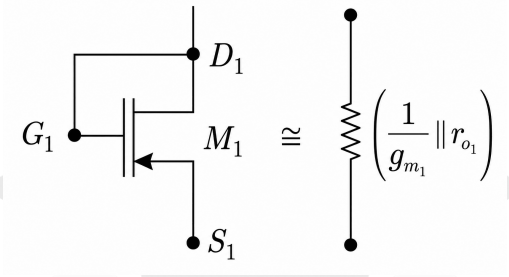

Fig. Solution Q2.21.1

So, given circuit becomes as,

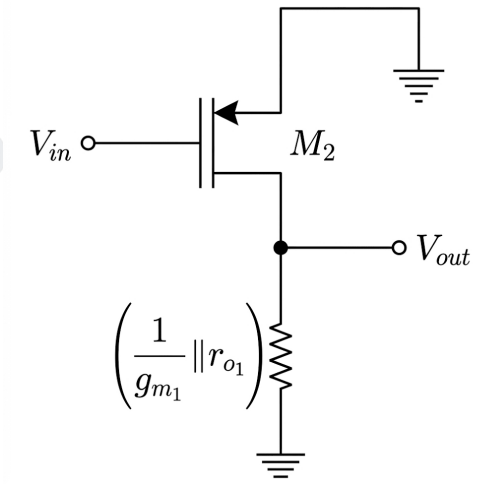

Fig. Solution Q2.21.1

Here, $\frac{1}{g_{m1}}||r_{o1} = R_L$ work as load for MOS (M_2), so small signal voltage gain of MOS (M_2) is

$$(A_V)_{M_2} = -g_{m2}(R_L||r_{o2})$$

$$(A_V)_{M_2} = -g_{m2}\left[\left(\frac{1}{g_{m1}}||r_{o1}\right)||r_{o2}\right]$$

$$\boxed{\text{(D)} \quad -g_{m2}\left(\frac{1}{g_{m1}} \parallel r_{o1} \parallel r_{o2}\right)}$$

Key Points

- The internal output resistance of a MOSFET is finite ($r_o \neq \infty$) when the channel length modulation effect is considered ($\lambda \neq 0$).

Q2.21.2 NAT | 2021 | 2M | Ans: 0.5 | ● ○ ○

Let threshold voltage of MOSFET is V_{th}

Given:

$$\mu_n C_{ox} = 100\,\mu\text{A/V}^2, \quad \frac{W}{L} = 10, \quad V_{GS} = 1\text{ V}$$

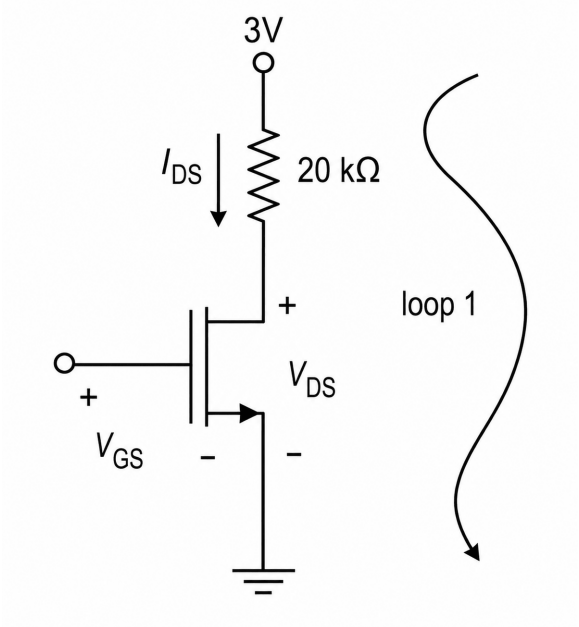

Fig. Solution Q2.21.2

PrepFusion

Drain current:

$$I_{DS} = \frac{\mu_n C_{ox}}{2} \times \frac{W}{L}(V_{GS} - V_T)^2$$

$$I_{DS} = \frac{100}{2} \times 10 \times 10^{-6}(1 - V_{th})^2\text{ A}$$

$$I_{DS} = \frac{1}{2}(1 - V_{th})^2\text{ mA} \quad \ldots(1)$$

Applying KVL at loop 1

$$V_{DS} = 3 - 20I_{DS} \quad \ldots(2)$$

From equation (1) put I_{DS} into equation (2)

$$V_{DS} = 3 - 20 \times \frac{1}{2}(1 - V_{th})^2$$

$$V_{DS} = 3 - 10(1 - V_{th})^2 \quad \ldots(3)$$

Since MOSFET is at edge of saturation

$$V_{DS} = V_{GS} - V_{th}$$

$$1 - V_{th} = 3 - 10(1 - V_{th})^2$$

Assume $1 - V_{th} = a$

$$a = 3 - 10a^2$$

$$10a^2 + a - 3 = 0$$

$$a = \frac{-1 \pm \sqrt{1 + 120}}{20} = \frac{-1 \pm 11}{20}$$

$$a = 0.5\,\text{V} \quad \text{or} \quad -0.6\,\text{V}$$

When $a = 0.5$,

$$1 - V_{th} = 0.5 \implies V_{th} = 0.5\,\text{V}$$

When $a = -0.6\,\text{V}$,

$$1 - V_{th} = -0.6 \implies V_{th} = 1.6\,\text{V}$$

But $V_{GS} > V_{th}$ {to flow drain current I_{DS}} Since $V_{GS} = 1\,\text{V}$, V_{th} must be strictly less than 1 V. So $V_{th} = 0.5\,\text{V}$ is the correct value of threshold voltage.

$$\boxed{V_{th} = 0.5\text{V}}$$

Key Points

- A MOSFET operating at the edge of saturation satisfies the condition $V_{DS} = V_{GS} - V_{th}$.

Q2.20.1 MCQ 2020 2M Ans: B ● ○ ○

The small signal equivalent circuit is shown below,

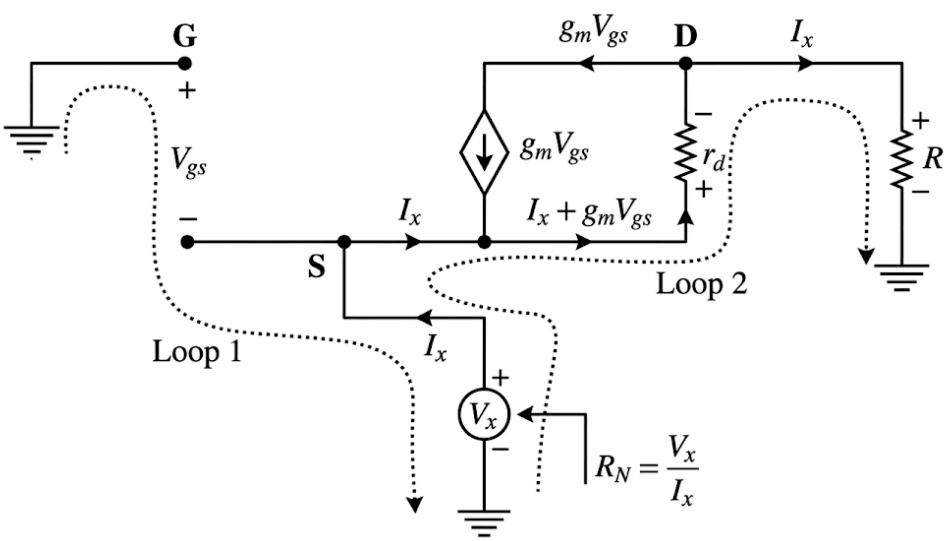

Fig. Solution Q2.20.1

Applying KVL in loop 1,

$$-V_{gs} - V_x = 0$$

$$V_{gs} = -V_x \quad \ldots (i)$$

Applying KVL in loop 2,

$$V_x - r_{ds}(I_x + g_m V_{gs}) - RI_x = 0$$

From equation (i), putting $V_{gs} = -V_x$,

$$+V_x - r_{ds} I_x + g_m V_x r_{ds} - RI_x = 0$$

$$V_x(1 + g_m r_{ds}) = I_x(r_{ds} + R)$$

$$\frac{V_x}{I_x} = R_N = \frac{r_{ds} + R}{1 + g_m r_{ds}}$$

(B) $\dfrac{r_{ds} + R}{1 + g_m r_{ds}}$

Key Points

- To calculate the Norton equivalent resistance (R_N) looking into a specific terminal, apply a test voltage source V_x at that terminal and evaluate the current I_x supplied by the source. The resistance is $R_N = \frac{V_x}{I_x}$.
- For a common-gate topology (looking into the source), the input resistance is not simply $\frac{1}{g_m}$ when r_{ds} is finite and a drain resistor R is present. The general expression becomes $R_{in} = \frac{r_{ds}+R}{1+g_m r_{ds}}$.

Q2.20.2 MCQ 2020 2M Ans: C ● ○ ○

Sample and hold circuit is an analog device that samples the voltage of a continuously varying signal and holds its signal value at a constant level for a specified period.
For proper sampling operation, the MOSFET should be on so that the capacitor can charge upto the input voltage i.e.

$$V_{GS} > V_T$$

For proper holding operation, the MOSFET should be off so that the capacitor can hold its charge i.e.

$$V_{GS} < V_T$$

Given: $V_T = 3\,\text{V}$, $V_{(I)\,\min} = -10\,\text{V}$, $V_{(I)\,\max} = 10\,\text{V}$

The MOSFET should be on for sampling operation, i.e.

$$V_{GS} > V_T$$
$$V_G - V_{(I)\,\max} > V_T$$
$$V_G - 10 > 3$$
$$V_G > 13\,\text{V}$$

The MOSFET should be off for holding operation, i.e.

$$V_{GS} < V_T$$
$$V_G - V_{(I)\,\min} < V_T$$
$$V_G - (-10) < 3$$
$$V_G < -7\,\text{V}$$

Hence, the minimum and the maximum values of V_G required for proper sampling and holding respectively, are 13 V and −7 V respectively.

(C) 13 V and − 7 V

Key Points

- During sampling phase, MOSFET must be fully turned ON, satisfying the condition $V_{GS} > V_T$ even for the most positive possible input voltage level ($V_{(I)\,\max}$).
- During holding phase, MOSFET must be fully turned OFF, satisfying the condition $V_{GS} < V_T$ even for the most negative possible input voltage level ($V_{(I)\,\min}$).

Q2.19.1 MCQ | 2019 | 2M | Ans: C | ● ○ ○

Method 1

Given, threshold voltage of each NMOS transistor is 0.6 V and $V_G = 3$ V for all NMOS transistors.

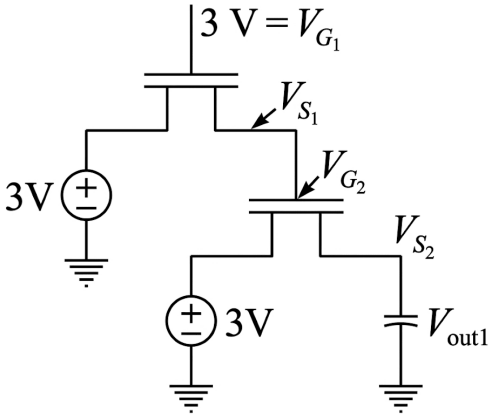

Fig. Solution Q2.19.1

Method 2

Given circuit (I) is shown in below figure and the n-MOS used have threshold voltage as 0.6 V. For pass transistor logic V_{GS} is maintained as V_{TH}.

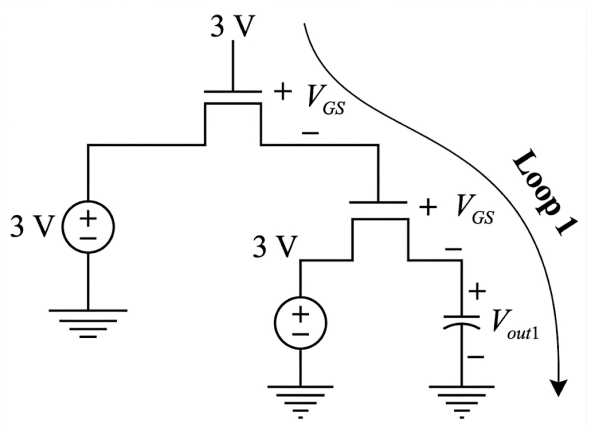

Fig. Solution Q2.19.1

In a pass transistor logic, V_{GS} is automatically maintained at V_{TH}.
If V_G is fixed, V_S should be adjusted so that

$$V_G - V_S = V_{TH} \implies V_S = V_G - V_{TH}$$

From above figure,

$$V_{s_1} = V_{G_1} - V_{TH} = 3 - 0.6 = 2.4\,\text{V}$$

$$V_{s_2} = V_{G_2} - V_{TH} = 2.4 - 0.6 = 1.8\,\text{V} = V_{out1}$$

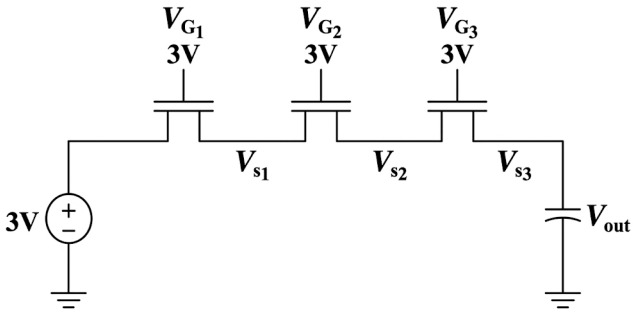

Fig. Solution Q2.19.1

Here, $V_{s_1} = V_{G_1} - V_{TH} = 3 - 0.6 = 2.4\,\text{V}$

$V_{s_2} = V_{G_2} - V_{TH} = 3 - 0.6 = 2.4\,\text{V}$

$V_{s_3} = V_{G_3} - V_{TH} = 3 - 0.6 = 2.4\,\text{V} = V_{out2}$

(C) 1.8 and 2.4

Applying KVL in loop (1), we get

$$-3 + V_{GS} + V_{GS} + V_{out_1} = 0$$

$$V_{out_1} = 3 - 2V_{GS} = 3 - 2 \times 0.6$$

$$V_{out_1} = 1.8\,\text{V}$$

Similarly from circuit (II),

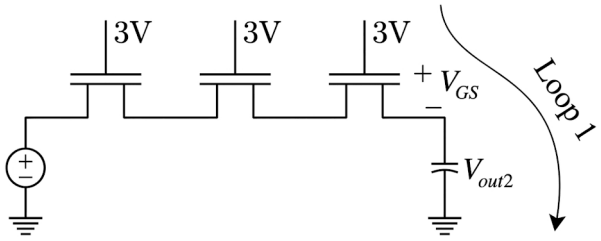

Fig. Solution Q2.19.1

Applying kVL in loop (1)

$$-3 + V_{GS} + V_{out_2} = 0$$

$$V_{out_2} = 3 - V_{GS} = 3 - 0.6$$

$$V_{out_2} = 2.4 \text{ volt}$$

Key Points

- An nMOS transistor can pass a logic HIGH only up to $V_{gate} - V_{th}$ at its source terminal.

Q2.19.2 NAT | 2019 | 2M | Ans: -900 | ● ● ○

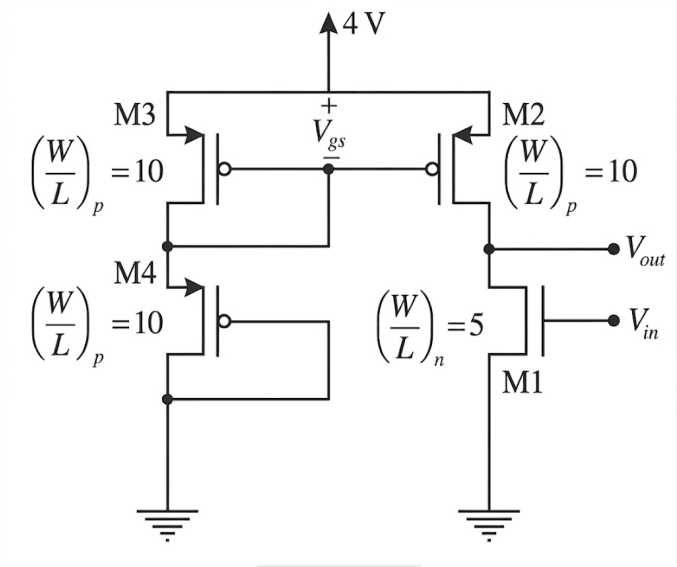

Fig. Solution Q2.19.2

Given,
For N-MOS, $\mu_n C_{ox} = 60\,\mu\text{A/V}^2 \quad W/L = 5$
For P-MOS, $\mu_p C_{ox} = 30\,\mu\text{A/V}^2 \quad W/L = 10$
Output resistance, $r_{ds} = 6\,\text{M}\Omega$
For both MOSFET, $|V_t| = 1\,\text{V}$
From above figure, the M_3 and M_4 have same V_{SG} so $V_{SG} = V_{DD}/2 = 4/2 = 2\,\text{V}$
and

$$I_D = \frac{\mu_n C_{ox}}{2}\frac{W}{L}\left[V_{GS} - |V_t|\right]^2$$

$$I_D = \frac{\mu_p C_{ox}}{2}\frac{W}{L}\left[V_{GS} - |V_t|\right]^2$$

$$I_D = \frac{1}{2} \times 30 \times 10 \times (2-1)^2$$

$$I_D = 150\,\mu\text{A}$$

$$g_{mn} = \sqrt{2I_D \cdot \left(\frac{W}{L}\right)_n \cdot \mu_n C_{ox}}$$

$$g_{mn} = \sqrt{2 \times 150 \times 5 \times 60}$$

$$g_{mn} = 300\,\mu\text{A/V}$$

Here M_2,M_3,M_4 form active load for M_1 and the resistance seen through the drain of M_1 looking up is r_{ds} of M_2. So the AC small signal equivalent is given by,

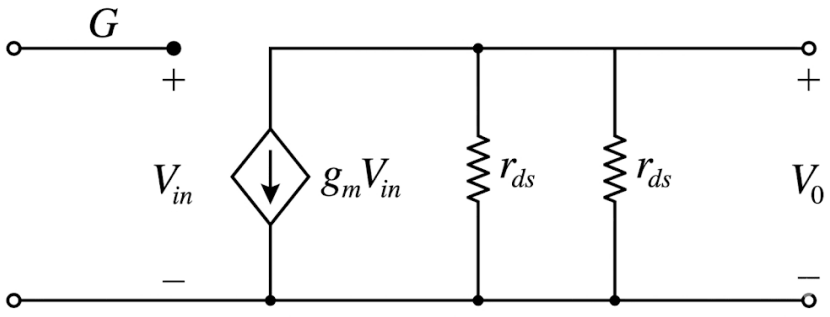

Fig. Solution Q2.19.2

$$\text{Gain} = \frac{V_0}{V_{in}}$$

$$V_0 = -g_{mn}V_{in}[r_{ds}||r_{ds}]$$

$$V_0 = \frac{-g_{mn}V_{in}r_{ds}}{2} = \frac{-300 \times 10^{-6} \times 6 \times 10^6}{2}$$

$$\therefore \quad \text{Gain} = \frac{-g_{mn}V_{in}r_{ds}}{2V_{in}}$$

$$\boxed{\text{Gain} = -900}$$

Key Points

- In the biasing branch, two identical diode-connected PMOS transistors in series divide the supply voltage equally, establishing $V_{SG} = \frac{V_{DD}}{2}$.
- The drain current I_D is determined by this V_{SG} and is mirrored to the active load PMOS.
- The transconductance of the amplifying NMOS is calculated using $g_{mn} = \sqrt{2I_D\mu_n C_{ox}(W/L)_n}$.
- The voltage gain of a common-source amplifier with an active load is given by $-g_{mn}(r_{ds_n}||r_{ds_p})$.

Q2.19.3 NAT | 2019 | 2M | Ans: 6 | ● ○ ○

Given circuit is shown below,

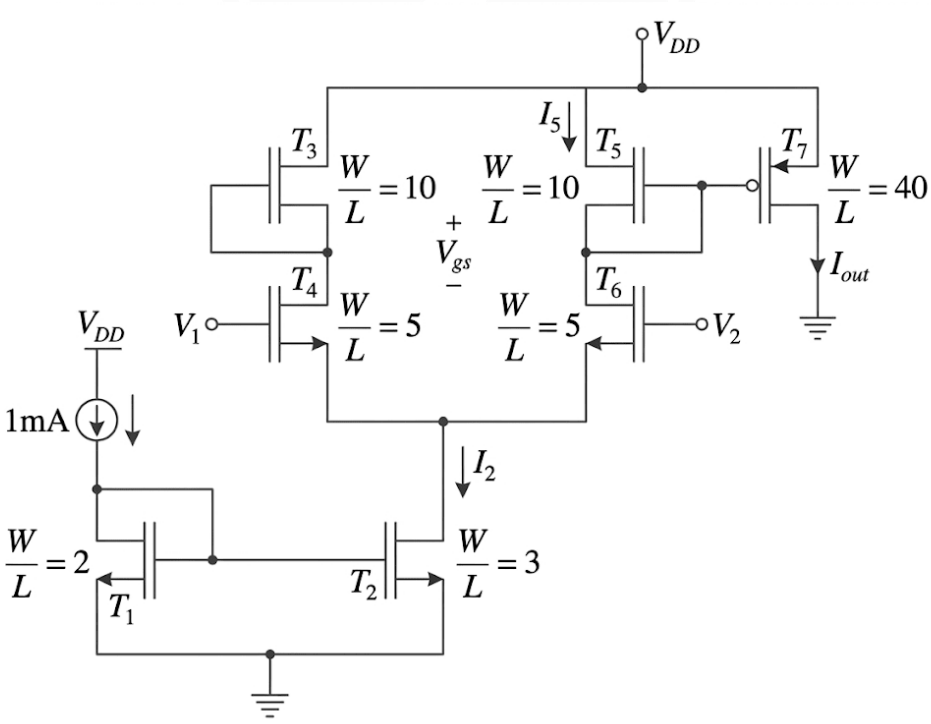

Fig. Solution Q2.19.3

I_D (drain current) is given as

$$I_D = \frac{\mu_n C_{ox}}{2}\left(\frac{W}{L}\right)(V_{GS} - V_T)^2$$

$$\therefore \quad I_D \propto \left(\frac{W}{L}\right)$$

Here, transistor T_1 and T_2 form a current mirror circuit,
Therefore,

$$\frac{I_{D_1}}{I_{D_2}} = \frac{\left(\frac{W}{L}\right)_1}{\left(\frac{W}{L}\right)_2}$$

$$\frac{I_{D_1}}{I_{D_2}} = \frac{2}{3} \quad \ldots (i)$$

$$I_{D_2} = \frac{3}{2} I_{D_1} = 1.5\,\text{mA}$$

Given that $V_1 = 0\,\text{V}$,
So, transistor T_4 will be off because $V_G = 0$ ($V_{GS} < V_t$). So, we get $I_{D_3} = I_{D_4} = 0\,\text{A}$ Which gives, $I_{D_2} = I_{D_6}$
T_5 and T_6 are in series and T_7 and T_5 form another current mirror, therefore we get,

$$I_{D_5} = I_{D_6} = I_{D_2} \quad \ldots (ii)$$

$$\frac{I_{D_7}}{I_{D_5}} = \frac{\left(\frac{W}{L}\right)_7}{\left(\frac{W}{L}\right)_5}$$

$$I_{out} = I_{D_7} = I_{D_5}\left[\frac{\left(\frac{W}{L}\right)_7}{\left(\frac{W}{L}\right)_5}\right] = I_{D_5}\left(\frac{40}{10}\right)$$

$$I_{out} = 4I_{D_5} = 4I_{D_2} \quad \text{[From equation (ii)]}$$

$$I_{out} = 4 \times 1.5\,\text{mA} \quad \text{[From equation (i)]}$$

$$\boxed{I_{out} = 6\,\text{mA}}$$

Q2.19.4 NAT | 2019 | 2M | Ans: 25.4 to 25.6 | ● ○ ○

For the intended source-reference bias, $V_{GS} = 5$ V and overdrive is 4.3 V. The square-law saturation current is

$$I_D = \frac{\mu_n C_{ox}}{2} \frac{W}{L} (V_{GS} - V_T)^2.$$

Here $\mu_n C_{ox}(W/L) = 2.76\,\text{mA/V}^2$, so

$$I_D = 1.38(4.3)^2 = 25.5162\,\text{mA}.$$

$$\boxed{25.52\,\text{mA (two decimals)}}$$

The third decimal digit is 6, so 25.5162 rounds to 25.52. A long-channel MOSFET does not inherently ground its source: the calculation uses the source-reference voltage intended by this question.

Q2.18.1 MCQ | 2018 | 1M | Ans: C | ● ○ ○

Given :
(i) Both NMOS transistor, M_1 and M_2 are identical i.e., $I_D = I_{D1} = I_{D2}$
(ii) Both transistors are in saturation region.
(iii) Input is applied between terminals G and S of transistor M_1.
(iv) Output is taken between terminals D and S.

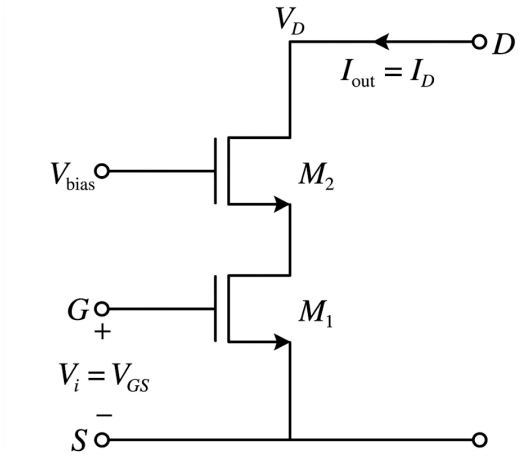

Fig. Solution Q2.18.1

Overall transconductance is given by,

$$g_m = \frac{\partial I_{out}}{\partial V_{in}} = \frac{\partial I_D}{\partial V_{GS}} \quad \ldots (i)$$

Transconductance of transistor M_1 is given by,

$$g_{m1} = \frac{\partial I_{D1}}{\partial V_{GS1}} = \frac{\partial I_D}{\partial V_{GS}}$$

[From above figure, $V_{GS1} = V_{GS}$]

From equation (i),

$$g_m = g_{m1}$$

AC analysis (calculation of r_o) :

For calculation of output resistance, input bias DC voltage should be short circuited.

AC equivalent circuit is shown below,

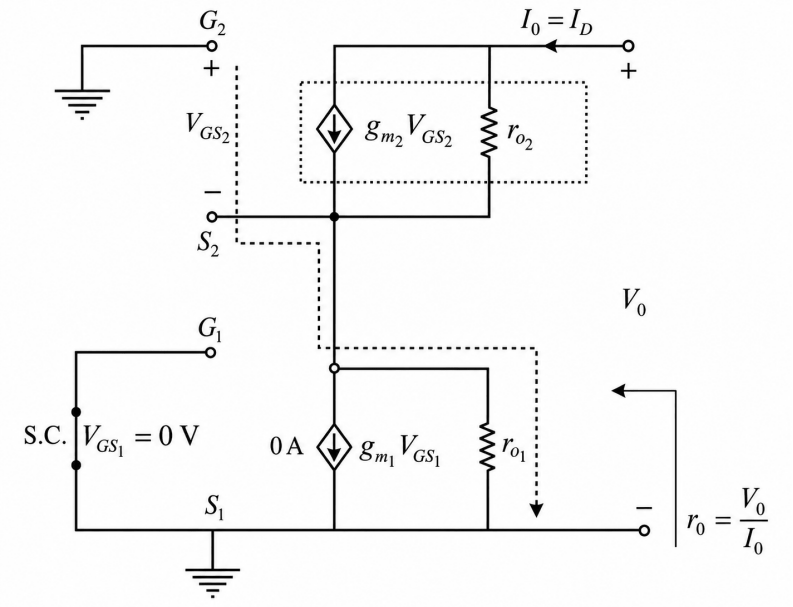

Fig. Solution Q2.18.1

Modified circuit diagram is shown below,

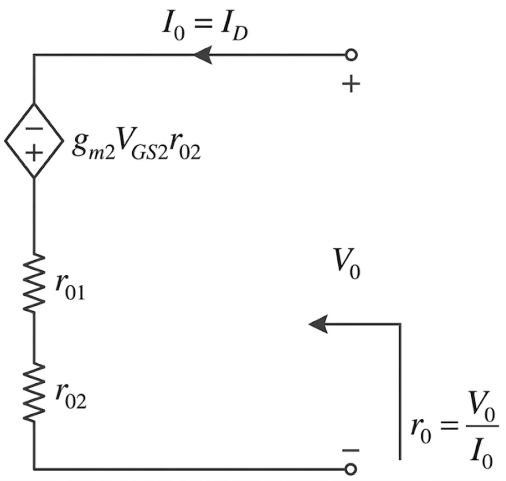

Fig. Solution Q2.18.1

Applying KVL in above figure,

$$-V_0 - g_{m2}V_{GS2}r_{o2} + (r_{o1} + r_{o2})I_0 = 0$$

$$V_0 = (r_{o1} + r_{o2})I_0 - g_{m2}V_{GS2}r_{o2} \quad \ldots(\text{ii})$$

Applying KVL in figure (b),

$$+V_{GS2} + r_{o1}I_0 = 0$$

$$V_{GS2} = -I_0 r_{o1} \quad \ldots(\text{iii})$$

From equation (ii) and (iii),

$$V_0 = (r_{o1} + r_{o2})I_0 + g_{m2}r_{o1}r_{o2}I_0$$

Since, r_{o1} and r_{o2} have very high values i.e. $r_{o1}r_{o2} > (r_{o1} + r_{o2})$ therefore $(r_{o1} + r_{o2})$ can be neglected.

$$V_0 \approx g_{m2}r_{o1}r_{o2}I_0$$

$$r_o = \frac{V_0}{I_0} \approx g_{m2}r_{o1}r_{o2}$$

(C) $g_m \approx g_{m1}$ and $r_o \approx r_{o1} \cdot g_{m2} \cdot r_{o2}$

Key Points

- The overall transconductance (g_m) of a cascode configuration is approximately equal to the transconductance of the input common-source transistor (g_{m1}).
- The overall output resistance (r_o) of a cascode amplifier is greatly increased by the common-gate transistor (M_2), acting as a current buffer. It is approximately the intrinsic gain of M_2 multiplied by the output resistance of M_1: $r_o \approx g_{m2}r_{o2}r_{o1}$.

Q2.18.2 MCQ 2018 1M Ans: D ●○○

A transimpedance amplifier (TIA) is a specialized electronic amplifier that converts an input current signal into an output voltage signal. A good TIA should have **Low input impedance and low output impedance**

Reason: The generic diagram of a TIA is shown below.

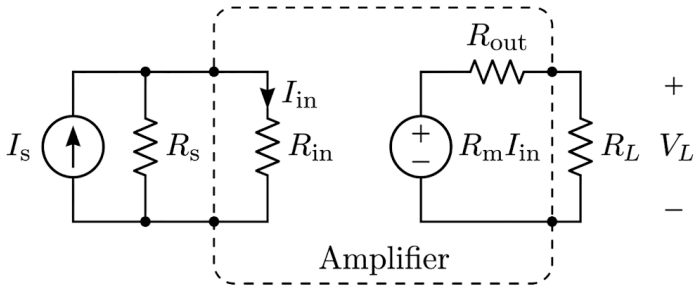

Fig. Solution Q2.18.2

Importance of Low Input Impedance:
Efficient Current Transfer: To efficiently capture the input current most of the input current should flow into the amplifier(R_{in}) for which R_{in} should be as low as possible(so that using current division rule most of the current goes into R_{in}).
Importance of Low Output Impedance
Driving loads effectively: In order to drive the load efficiently most of the output voltage should drop across load for which R_{out} should be as low as possible (so that using voltage division rule most of the voltage drops across load).

(D) low input impedance and low output impedance

Key Points

- **Voltage Amplifier:** Input = Voltage, Output = Voltage → High Z_{in}, Low Z_{out}.
- **Current Amplifier:** Input = Current, Output = Current → Low Z_{in}, High Z_{out}.
- **Transconductance Amplifier:** Input = Voltage, Output = Current → High Z_{in}, High Z_{out}.
- **Transimpedance Amplifier:** Input = Current, Output = Voltage → Low Z_{in}, Low Z_{out}.

Q2.18.3 NAT 2018 2M Ans: 0.42 ●●○

Given circuit is,

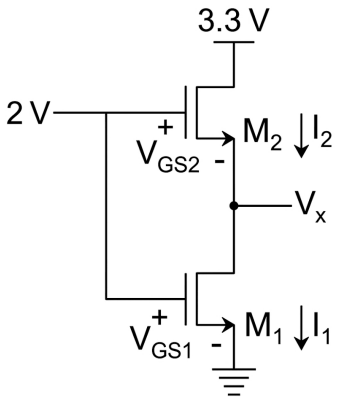

Fig. Solution Q2.18.3

Since both MOSFETS are in series same current flows through them.
To equate currents first we need to find the region of operation of two MOSFETs.
For M_2 : Check for saturation:

$$V_{DS} > V_{GS} - V_T \implies 3.3 > 2 - 1 \implies 3.3 > 1$$

M_2 is in saturation

For M_1 : Check for saturation:

$$V_{DS} > V_{GS} - V_{TH} \implies V_x > 2 - 1 \implies V_x > 1\,\text{V}$$

But if $V_x > 1$ then V_{GS2} will be less than 1 which will turn off M_2. Hence, M_1 is not in saturation, it is in the triode region.

$$I_1 = I_2$$

$$I_1 = \mu_n c_{ox} \left(\frac{W}{L}\right)_1 \left([V_{GS1} - V_{TH}]V_{DS1} - \frac{1}{2}V_{DS1}^2\right)$$

$$I_2 = \frac{1}{2}u_n C_{ox} \left(\frac{W}{L}\right)_2 (V_{GS2} - V_{th})^2$$

Also, $\left(\frac{W}{L}\right)_2 = 2\left(\frac{W}{L}\right)_1$

$$V_x - \frac{1}{2}V_x^2 = (1 - V_x)^2$$

$$3V_x^2 - 6V_x + 2 = 0$$

$$V_x = 0.42\,\text{V}, -1.58\,\text{V}$$

V_x cannot be negative Hence, $\boxed{V_x = 0.42\,\text{V}}$

Key Points

- To analyze a circuit with multiple MOSFETs, first assume a region of operation (usually saturation for amplifying stages) and verify it using the calculated voltages.
- A MOSFET acts as an active load when its drain and gate are not tied together but it remains in saturation. Here M_2 acts as a source follower type load.
- The current flowing through series-connected MOSFETs must be equal. Equating their drain current expressions allows solving for the intermediate node voltage.

Q2.17.1 NAT 2017 2M Ans: 2 ●○○

Given circuit is,

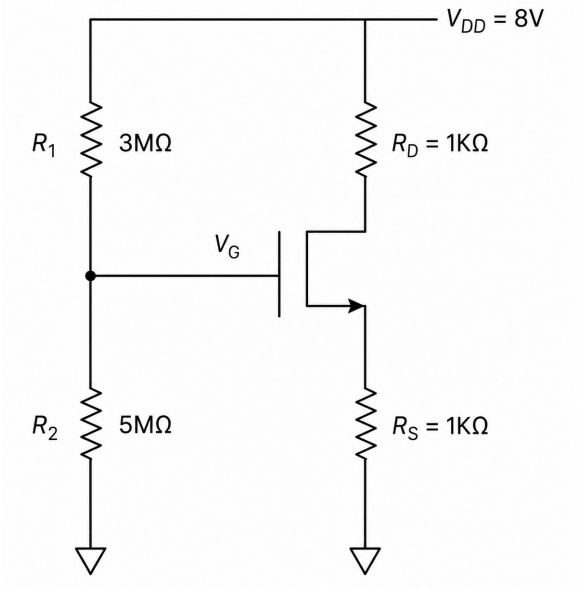

Fig. Solution Q2.17.1

The current in a MOSFET working in saturation is given by:

$$I_D = \frac{1}{2}\mu_n C_{ox} \times \frac{W}{L}(V_{gs} - V_{tn})^2$$

where

$$V_{gs} = V_g - V_s \quad \ldots(\text{i})$$

$$V_g = 8 \times \frac{5}{3+5} = 5\,\text{V} \quad \ldots(\text{ii})$$

Note : We used voltage division rule above because same current flows through R_1 and R_2 as MOSFET's gate terminal draws no current.

Applying KVL from the source terminal to ground, we get:

$$V_s = I_D R_S$$

$$V_s = I_S(1\,\text{k})$$

$$V_{gs} = 5 - I_D(1\,\text{k})$$

$$V_{gs} - V_t = 5 - I_D(1\,\text{k}) - 1$$

$$V_{gs} - V_t = 4 - I_D(1\,\text{k})$$

Putting on the respective values, we get:

$$I_D = \frac{1}{2} \times 10^{-3}(4 - I_D \cdot 1\,\text{k})^2$$

$$10^6 I_D^2 - 10^4 I_D + 16 = 0$$

$$I_D = 8\,\text{mA or } 2\,\text{mA}$$

For, $I_D = 8\,\text{mA}$ $V_S = 8\,\text{V}$ (not possible since the $V_D D$ itself is 8V)

For $I_D = 2\,\text{mA}$ $V_S = 2\,\text{V}$ (possible)

$$\boxed{\text{Hence,} I_D = 2\,\text{mA}}$$

Key Points

- The gate voltage is established by the voltage divider network: $V_G = V_{DD}\left(\frac{R_2}{R_1+R_2}\right)$ and it is only valid when same current is flowing through R_1 and R_2.

Q2.17.2 MCQ 2017 1M Ans: C ●○○

Ideal enhancement MOSFET is a MOSFET which saturates when $V_{DS} \geq V_{GS} - V_T$ and allows a constant current to flow across it even after a further increase in V_{DS}. So it acts as a constant current source with infinte output impedance.

But for a non-ideal MOSFET (i.e. MOSFET in which channel length modulation effect is significant), a change in V_{DS} beyond $V_{GS} - V_T$, will result in a further increase in current beyond saturation. Hence it behaves as a constant current source with finite output impedance as current is changing based on V_{DS}

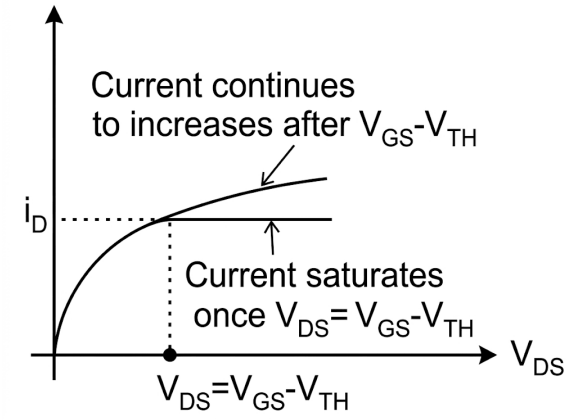

Fig. Solution Q2.17.2

(C) current source with finite output impedance

Key Points

- In the saturation region, an ideal MOSFET acts as a voltage-controlled current source with infinite output impedance, as I_D is completely independent of V_{DS}.
- When the channel length modulation effect is significant, the effective channel length decreases as V_{DS} increases, causing a proportional increase in the drain current I_D.

Q2.17.3 NAT 2017 1M Ans: 0.5 ●○○

Given $V_{DS} = 0.1\,\text{V}$, $V_{GS} = 0.7\,\text{V}$, $V_{th} = 0.3\,\text{V}$.

$$V_{GS} - V_{th} = 0.7 - 0.3 = 0.4\,\text{V} > V_{DS}$$

Since $V_{DS} < V_{GS} - V_{th}$, MOSFET is in the triode region.

$$\therefore I_{DS} = \mu_n C_{ox} \frac{W}{L}\left((V_{GS} - V_{th})V_{DS} - \frac{1}{2}V_{DS}^2\right)$$

$$g_m = \frac{\partial I_D}{\partial V_{GS}} = \mu_n C_{ox} \frac{W}{L}(V_{DS})$$

$$\therefore g_m = 50 \times 100\,\mu\text{A/V}^2 \times 0.1\,\text{V}$$

$$g_m = 500\,\mu\text{A/V} = 0.5\,\text{mA/V}$$

The transconductance $g_m = 0.5\,\text{mA/V}$

Key Points

- Always verify the region of operation first. Since $V_{DS} < V_{GS} - V_{TH}$, the MOSFET operates in the linear/triode region.
- Transconductance is defined as the change in drain current for a given change in gate-source voltage at a constant drain-source voltage: $g_m = \frac{\partial I_D}{\partial V_{GS}}$ (V_{DS} is taken as constant.

Q2.17.4 MCQ 2017 2M Ans: B ● ○ ○

n-channel MOSFET biased in saturation region will result in a current given by:

$$\Rightarrow I_D = \frac{W\mu C_{ox}}{2L}(V_{GS} - V_{th})^2$$

And the transconductance is defined as $g_m = \frac{\partial I_D}{\partial V_{GS}}$

$$\Rightarrow g_m = \frac{W\mu C_{ox}}{L}(V_{GS} - V_{th})$$

Calculation:

Given, two transistors T_1 and T_2 which are identical in all respects except for width and gate overdrive voltage $(V_{GS} - V_{th})$.

For T_1 MOSFET,

$$g_{m1} \propto W_1(V_{GS} - V_{th})_1 \quad \text{and} \quad I_{D1} \propto W_1(V_{GS} - V_{th})_1^2$$

Similarly for T_2,

$$g_{m2} \propto W_2(V_{GS} - V_{th})_2 \quad \text{and} \quad I_{D2} \propto W_2(V_{GS} - V_{th})_2^2$$

$$\Rightarrow \frac{I_{D1}}{I_{D2}} = \frac{W_1(V_{GS} - V_{th})_1^2}{W_2(V_{GS} - V_{th})_2^2} \quad \ldots(1)$$

Given $W_2 = 2W_1$, and $(V_{GS} - V_{th})_2 = 2(V_{GS} - V_{th})_1$

Putting these in Equation (1), we get,

$$\Rightarrow \frac{I_{D1}}{I_{D2}} = \frac{W_1(V_{GS} - V_{th})_1^2}{2W_1(2)^2(V_{GS} - V_{th})_1^2}$$

$$\Rightarrow \frac{I_{D1}}{I_{D2}} = \frac{W_1(V_{GS} - V_{th})_1^2}{8W_1(V_{GS} - V_{th})_1^2} \Rightarrow I_{D2} = 8I_{D1}$$

Similarly,

$$\frac{g_{m1}}{g_{m2}} = \frac{W_1(V_{GS} - V_{th})_1}{W_2(V_{GS} - V_{th})_2}$$

$$\frac{g_{m1}}{g_{m2}} = \frac{W_1(V_{GS} - V_{th})_1}{2W_1(2)(V_{GS} - V_{th})_1}$$

$$g_{m2} = 4g_{m1}$$

(B) $8I_{D1}$ and $4g_{m1}$

Q2.17.5 MCQ 2017 2M Ans: C ● ● ○

For the given figure, let x be the voltage at node A.

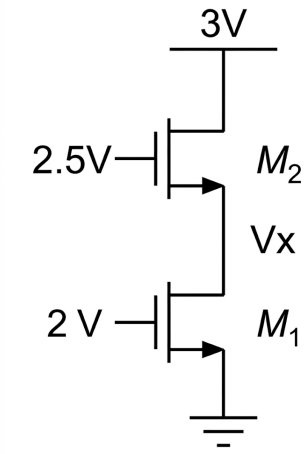

Fig. Solution Q2.17.5

$$V_{DS_2} = 3 - x$$

$$V_{GS_2} - V_t = 2.5 - x - 1 = 1.5 - x$$

We observe that, V_{DS_2} is always $> V_{GS_2} - V_t$ So, M_2 will always be in saturation.
Assuming M_1 to be in saturation, we calculate the saturation current as shown.

$$(I_{DSat})_1 = (I_{DSat})_2$$

$$\Rightarrow K_n(V_{GS} - V_t)_1^2 = K_n(V_{GS} - V_t)_2^2$$

$$\Rightarrow (2-1)^2 = (2.5 - x - 1)^2$$

$$\Rightarrow 1 = 1.5 - x$$

$$x = 0.5\,\text{V}$$

If 'x' were to be 0.5 V then for M_1,

$$V_{DS} = x - 0 = 0.5\,\text{V}$$

and $V_{GS} - V_t = 1\,\text{V}$. So, $V_{DS} < V_{GS} - V_t$. This shows that our assumption is wrong, and M_1 cannot be in saturation. So, M_1 is in linear.

(C) Linear, Saturation

Key Points

- To determine the region of operation for transistors in series, start by assuming both are in the saturation region.
- Equate their drain currents ($I_{D1} = I_{D2}$) to find the intermediate node voltages.
- Verify the saturation assumption by checking if $V_{DS} \geq V_{GS} - V_T$ for each transistor using the calculated node voltage.

Q2.16.1 NAT 2016 2M Ans: 1.2 ●○○

Given $g_m = 0.5\,\mu\text{A/V}$ for $V_{ds} = 50\,\text{mV}$, $V_{GS} = 2\,\text{V}$,
$g_d = 8\,\mu\text{A/V}$ for $V_{GS} = 2\,\text{V}$, $V_{DS} = 0\,\text{V}$
Since, $V_{GS} > V_{DS}$, MOSFET is in linear region, so

$$I_D = K_n'(V_{GS} - V_T)V_{DS}$$

$$\frac{\partial I_D}{\partial V_{GS}} = K_n' V_{DS}$$

$$g_m = K_n' V_{DS}$$

$$0.5 \times 10^{-6} = K_n'[50 \times 10^{-3}]$$

$$\Rightarrow K_n' = 10^{-5}$$

$$I_D = K_n'(V_{GS} - V_T)V_{DS}$$

$$\frac{\partial I_D}{\partial V_{DS}} = K_n'[V_{GS} - V_T]$$

$$g_d = K_n'[V_{GS} - V_T]$$

$$8 \times 10^{-6} = 10^{-5}[2 - V_T]$$

$$V_T = 1.2\,\text{V}$$

Key Points

- For very small values of V_{DS} (deep triode region), the quadratic term $\frac{1}{2}V_{DS}^2$ can be neglected, and the drain current simplifies to $I_D \approx \mu_n C_{ox}\frac{W}{L}(V_{GS} - V_T)V_{DS}$.
- The drain conductance (or channel conductance) is defined as $g_d = \frac{\partial I_D}{\partial V_{DS}} = \mu_n C_{ox}\frac{W}{L}(V_{GS} - V_T)$. It represents the inverse of the small-signal channel resistance (r_{ds}).

Q2.16.2 MCQ | 2016 | 2M | Ans: C | ● ○ ○

Method 1

Given the channel length modulation coefficient(λ) is non-zero which means all MOSFETs have finite r_0
The AC equivalent of the given circuit is obtained by short circuiting DC voltage sources.

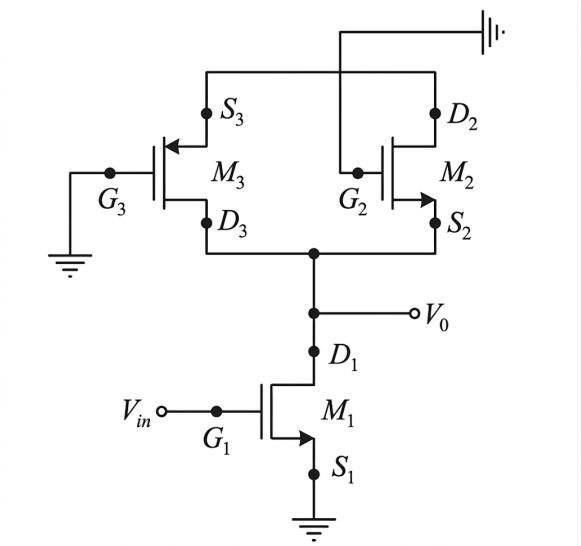

Fig. Solution Q2.16.2

From figure,

$$V_{gs3} = V_{g3} - V_{s3} = 0 - 0 = 0\,\text{V}$$

$$V_{in} = V_{gs1}$$

$$V_{o1} = V_0 = -V_{gs2}$$

The small signal model is given below.
Applying KCL equation at D_1 i.e. D_3,

$$g_{m1}V_{gs1} + \frac{V_0}{r_{o1}} + \frac{V_0}{r_{o2}} - g_{m2}V_{gs2} - g_{m3}V_{gs3} + \frac{V_0}{r_{o3}} = 0$$

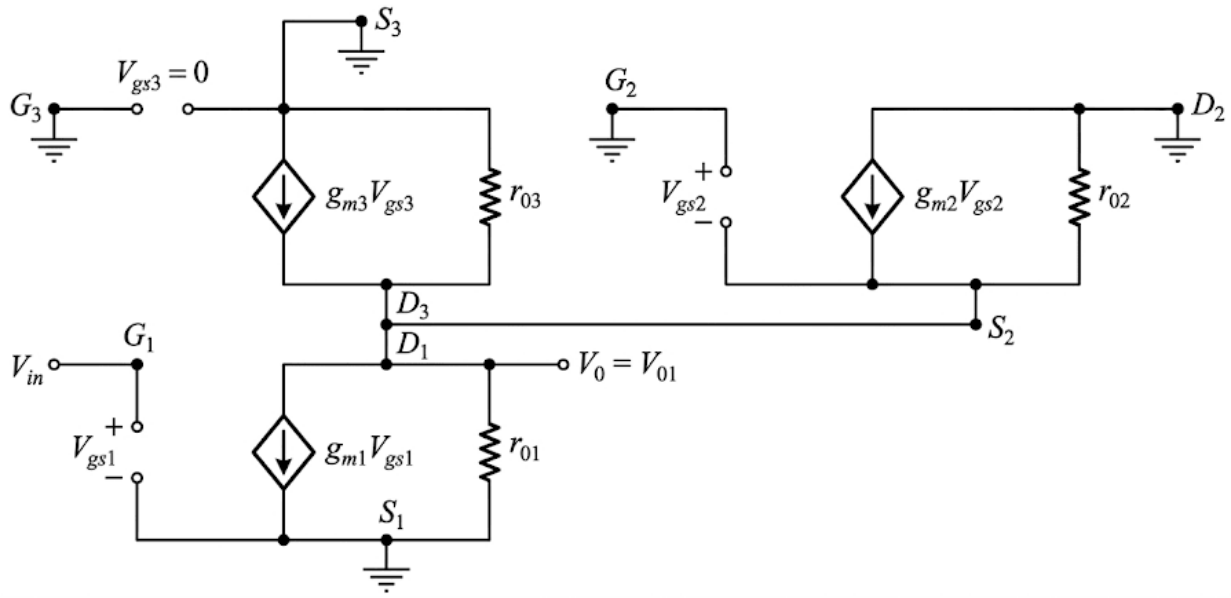

Fig. Solution Q2.16.2

From figure,

$$V_{gs1} = V_{in}, \quad V_{gs2} = -V_0, \quad V_{gs3} = 0\,\text{V}$$

Substituting these values into the KCL equation:

$$V_0\left(\frac{1}{r_{o1}} + \frac{1}{r_{o2}} + g_{m2} + \frac{1}{r_{o3}}\right) = -g_{m1}V_{in}$$

$$A_V = \frac{V_0}{V_{in}} = -g_{m1}\left[r_{o1} \| \left(r_{o2} \| \frac{1}{g_{m2}}\right) \| r_{o3}\right]$$

Hence, the correct option is (C).

Method 2

Transistor M_3 has its gate connected to a DC bias V_G, which acts as an AC ground. The resistance seen looking into the drain terminal is:

$$R_{in1} \approx r_{o3}$$

For transistor M_2, both the gate and drain are tied to V_{DD} (AC ground). Looking into the source terminal, the resistance is:

$$R_{in2} \approx \frac{1}{g_{m2}} \| r_{o2}$$

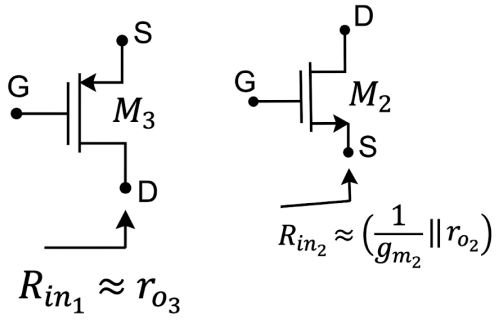

Fig. Solution Q2.16.2

The circuit simplifies to a common-source amplifier driven by M_1. The driver transconductance is $G_m = g_{m1}$. The total output resistance R_{out} at node V_o is the parallel combination of the driver's output resistance (r_{o1}) and the input resistances of the active loads:

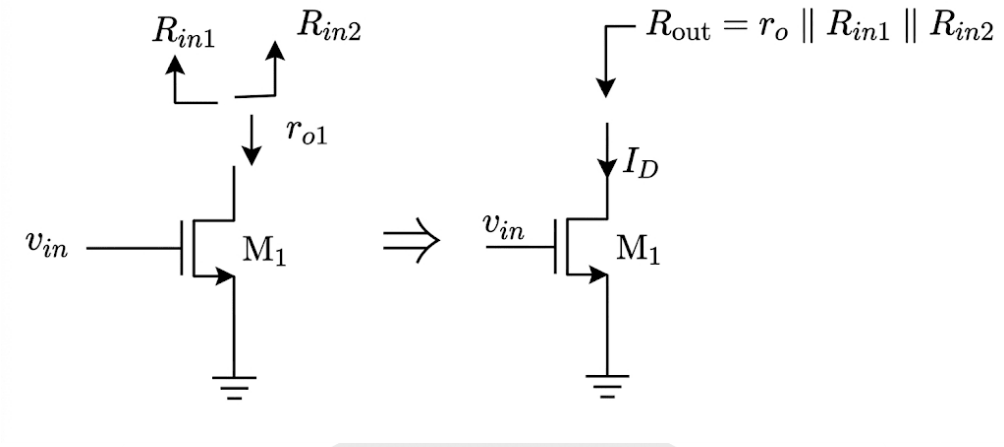

Fig. Solution Q2.16.2

$$R_{out} = r_{o1}||R_{in1}||R_{in2}$$

$$R_{out} = r_{o1}||r_{o3}||\left(\frac{1}{g_{m2}}||r_{o2}\right)$$

The small-signal voltage gain is calculated as:

$$\frac{V_{out}}{V_{in}} = -G_m R_{out} = -g_{m1}\left[r_{o1}||\left(\frac{1}{g_{m2}}||r_{o2}\right)||r_{o3}\right]$$

$$\boxed{\text{(C)} \quad -g_{m1}\left(r_{o1} \parallel \left(\frac{1}{g_{m2}} \parallel r_{o2}\right) \parallel r_{o3}\right)}$$

Key Points

- For small-signal AC analysis, independent DC voltage sources are deactivated (grounded), and DC current sources are open-circuited.
- Transistor M_2 is in a diode-connected-like configuration from the perspective of the output node, effectively contributing a conductance of g_{m2} and its output resistance r_{o2} to the total output impedance.
- Transistor M_3 acts as an active load with its gate grounded, contributing only its output resistance r_{o3} because its V_{gs3} is zero.

Q2.16.3 NAT 2016 2M Ans: 57.87 ●●○

Given, $\lambda = 0$, $g_m = 0.01S$ and M1 is in saturation.
The AC equivalent circuit is given by

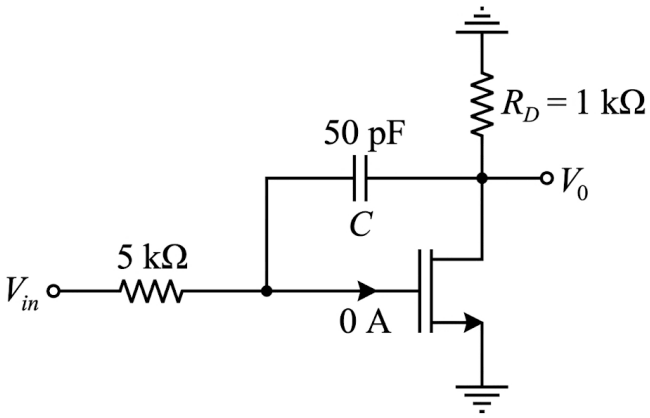

Fig. Solution Q2.16.3

Applying Miller's theorem across feedback capacitor 50 pF.

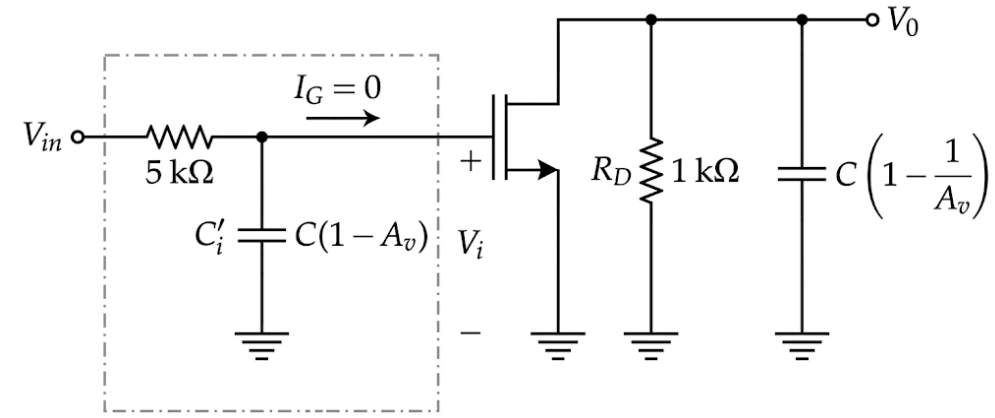

Fig. Solution Q2.16.3

Calculation of A_V (Internal voltage gain) :
(i) 50 pF (small value internal capacitor) is replaced by open circuit.
(ii) It is a common source without R_S amplifier, hence

$$A_V = -g_m R_L'; \text{ where, } R_L' = r_0 || R_D$$

Note : If we are using standard results then no need of AC model.
Internal output resistance of MOSFET is given by,

$$r_0 = \left|\frac{1}{\lambda I_D}\right|$$

where, λ = Channel length modulation If $\lambda = 0$ (given) then $r_0 = \infty$

$$R_L' = \infty || 1 = 1\ \text{k}\Omega$$

$$A_V = -g_m R_L'$$

$$A_V = -0.01 \times 1000 = -10$$

Calculation of 3-dB frequency :
From figure,

$$\frac{V_0}{V_{in}} = \frac{V_0}{V_1} \times \frac{V_1}{V_{in}} = A_V \times \frac{V_1}{V_{in}}$$

$$\frac{V_0}{V_{in}} = -g_m R'_L \times \frac{V_1}{V_{in}} \quad \ldots (i)$$

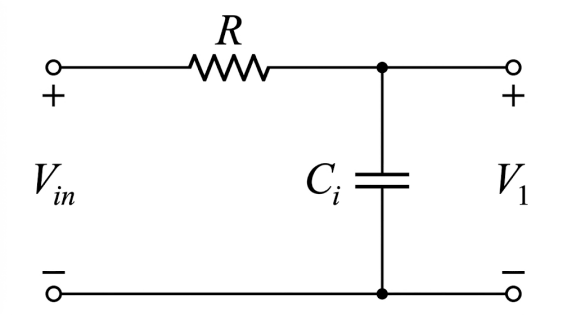

Fig. Solution Q2.16.3

$$\frac{V_1}{V_{in}} = \frac{1/C_i s}{R + \frac{1}{C_i s}} = \frac{1}{1 + RC_i s}$$

$$\frac{V_1}{V_{in}} = \frac{1}{1 + j\omega RC_i}$$

From equation (i),

$$\frac{V_0}{V_{in}} = \frac{-g_m R'_L}{1 + j\omega RC_i}$$

$$A'_V = \frac{-g_m R'_L}{1 + j2\pi f RC_i} \quad \ldots (ii)$$

Standard form of frequency response of first order LPF is given by,

$$A'_V = \frac{V_0}{V_{in}} = \frac{A_{V_0}}{1 + j\frac{f}{f_{Hi}}} \quad \ldots (iii)$$

Comparing equations (ii) and (iii), Small signal input pole frequency or input higher 3 dB frequency is given by,

$$f_{Hi} = \frac{1}{2\pi RC_i}$$

where, $C_i = C(1 - A_V) = 50(1 + 10)$

$$C_i = 550\,\text{pF}$$

$$f_{Hi} = \frac{1}{2\pi \times 5 \times 10^3 \times 550 \times 10^{-12}}$$

$$f_{Hi} = 57.87\,\text{kHz}$$

The small signal input pole frequency is **57.87** kHz.

Key Points

- The effective input capacitance due to the Miller effect is $C_{in} = C_f(1 - A_v)$, where A_v is the inverting voltage gain of the amplifier. A large negative gain significantly amplifies the apparent input capacitance.
- The dominant high-frequency pole (input pole) is formed by the source resistance and this magnified equivalent input capacitance: $f_p = \frac{1}{2\pi R_{in} C_{in}}$.

Q2.16.4 NAT 2016 2M Ans: 28.35 ● ○ ○

For a MOSFET in saturation, the current is given by:

$$I_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)(V_{GS} - V_T)^2(1 + \lambda V_{DS})$$

The drain conductance (g_d) is calculated as the rate of change of drain current with respect to the Drain to source voltage, i.e.

$$g_d = \frac{\partial I_D}{\partial V_{DS}}$$

$$g_d = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)(V_{GS} - V_T)^2\lambda$$

Putting on the respective values, we get:

$$g_d = \frac{1}{2} \times 70 \times 10^{-6}(4)(1.8 - 0.3)^2 \times 0.09$$

$$\boxed{g_d = 28.35\,\mu \text{ Seimens}}$$

Q2.16.5 MCQ 2016 1M Ans: D ● ○ ○

The current in an NMOS in the linear region is:

$$I_D = \frac{W\mu_n C_{ox}}{2L}\left[2(V_{GS} - V_T)V_{DS} - V_{DS}^2\right]$$

The transistor operates in the linear region for small values of V_{DS}, where the V_{DS}^2 term can be neglected:

$$I_D \approx \frac{W\mu_n C_{ox}}{L}\left[(V_{GS} - V_T)V_{DS}\right]$$

The channel resistance R_d is given by:

$$R_d = \frac{V_{DS}}{I_{DS}} = \frac{1}{\frac{W\mu_n C_{ox}}{L}(V_{GS} - V_T)}$$

Observations:

As W increases, the resistance decreases.
As V_T decreases, the resistance decreases.
As L increases, the resistance increases.
However, as V_{GS} increases, the resistance decreases. So, Option D is the required incorrect statement.

$$\boxed{\text{(D)} \quad \text{If } V_{GS} \text{ is increased, the resistance increases}}$$

Q2.15.1 MCQ 2015 2M Ans: A ● ○ ○

In the given figure, since gate is shorted to drain ,the MOSFET is in saturation.
Current in saturation is given by

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{TH})^2$$

We have $V_{GS} = V_{DD} - V_{SS}$ and let $\frac{1}{2}\mu_n C_{ox}\frac{W}{L} = K_n$, then

$$I_D = K_n(V_{DD} - V_{TH} - V_{SS})^2$$

Let $V_{DD} - V_{TH} = A$ = constant

$$\Rightarrow I_D = K_n(A - V_{SS})^2$$

This is equation of parabola
For $V_{SS} = 0, I_D = K_n A$(maximum) and for $V_{SS} = V_{DD} - V_{TH}, I_D = 0$

$$\boxed{\text{Option (A) is correct.}}$$

Q2.15.2 NAT 2015 2M Ans: 3 ● ○ ○

Let the common node between the transistors be x. The lower transistor has $V_{GS1} = 2$ V, so its overdrive is 1 V. Saturation requires $x \geq 1$ V, and then its drain current is 0.5 mA. The upper NMOS is diode connected. To carry the same positive current it must be ON:

$$V_{GS2} - V_T = V_{DD} - x - 1 > 0.$$

Its current equation is

$$0.5 = 0.5(V_{DD} - x - 1)^2$$

with current in mA. The square root is therefore the positive overdrive,+1 V. The algebraic negative root would imply zero gate-source voltage, below threshold, so the assumed positive saturation current would not exist.

$$V_{DD} = x + 2 \geq 3\,\text{V}.$$

$$\boxed{V_{DD,\text{min}} = 3\,\text{V}}$$

Checking only M1 does not validate the other algebraic root: M2 must satisfy its own ON condition too.

Q2.15.3 NAT 2015 2M Ans: 20 ● ○ ○

Given: $I_D = 1$ mA Channel length modulation coefficient = $0.05\,\text{V}^{-1}$
The drain current when the MOSFET is in saturation is given by:

$$I_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right) \times (V_{GS} - V_T)^2$$

The output resistance (r_o) is given by:

$$r_o = \frac{V_A}{I_D}$$

$$V_A = \frac{1}{\lambda}$$

where λ = Channel length parameter

$$V_A = \frac{1}{\lambda} = 20$$

$$r_o = \frac{V_A}{I_D} = \frac{20}{1\,\text{mA}} = 20\,\text{k}\Omega$$

$$\boxed{\text{The output impedance is } 20\,\text{k}\Omega}$$

Key Points

- It is mathematically defined as $r_o = \frac{1}{\lambda I_D}$, where λ is the channel length modulation coefficient.
- By introducing the Early voltage $V_A = \frac{1}{\lambda}$, the equation simplifies to $r_o = \frac{V_A}{I_D}$, which makes it analogous to BJT output resistance calculations.

Q2.15.4 NAT 2015 2M Ans: 0.022 ● ○ ○

At fixed gate-source bias, the saturation model is $I_D = I_0(1 + \lambda V_{DS})$. Taking the ratio of the two measured currents cancels I_0:

$$1.02 = \frac{1 + 6\lambda}{1 + 5\lambda}.$$

Thus $0.9\lambda = 0.02$ and

$$\lambda = 0.022222\ldots\ \text{V}^{-1}.$$

$$\boxed{0.022\,\text{V}^{-1} \text{ (three decimals)}}$$

Q2.14.1 MCQ 2014 1M Ans: B ● ○ ○

A current buffer is a specialized electronic amplifier that converts an input current signal into an output current signal. A good current buffer should have **Low input impedance and high output impedance**

Reason: The generic diagram of a current buffer is shown below.

Importance of Low Input Impedance:

Efficient Current Transfer: To efficiently capture the input current most of the input current should flow into the amplifier(R_i) for which R_i should be as low as possible(so that using current division rule most of the current goes into R_i).

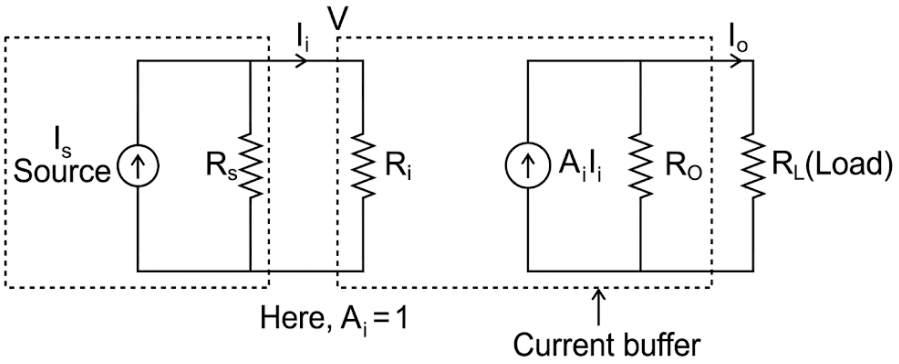

Fig. Solution Q2.14.1

Importance of High Output Impedance

Driving loads effectively: In order to drive the load efficiently most of the output current should flow into load for which R_0 should be as high as possible (so that using current division rule most of the current flows into load).

(B) low input impedance and high output impedance

Key Points

- **Current Amplifier:** Input = Current, Output = Current → Low Z_{in}, High Z_{out}.

Q2.14.2 MCQ 2014 1M Ans: B ● ○ ○

Given circui is,

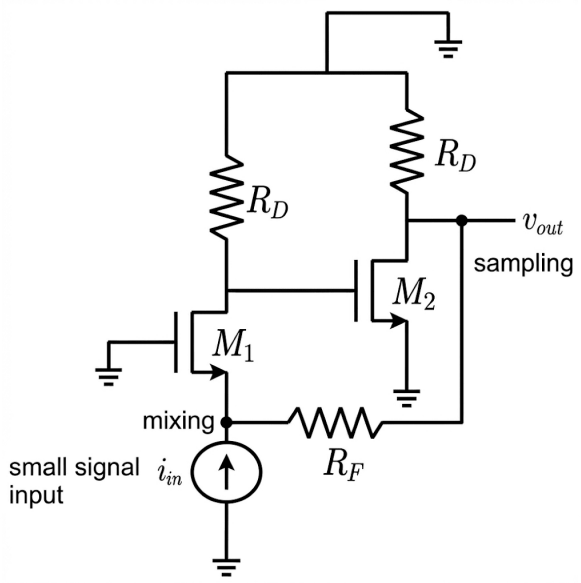

Fig. Solution Q2.14.2

Since the input signal is an input current i_{in}, the feedback signal mixes in shunt at the input. Hence, the input mixing is current mixing.
Also, the feedback resistor R_F is directly connected from the output node back to the input node, i.e. the sampled quantity is output voltage.
Hence, the feedback topology is voltage shunt feedback or voltage current feedback.

(B) voltage-current feedback

Key Points

- Shunt connection at input ⇒ current mixing.
- Feedback element directly connected to output node ⇒ Voltage sampling.

Mistakes to be avoided

- R_F connected across the output node implies voltage sensing, not current sensing.
- Do not reverse the answer. Output topology comes first in the answer.

Q2.14.3 NAT 2014 2M Ans: 500 ●○○

Region qualification: the stated bias is $V_{DS} = 5$ V. With the stated gate and threshold voltages, the overdrive is 2.5 V, so that bias is in saturation, not the asserted linear region. The correct triode condition is $0 \le V_{DS} < V_{GS} - V_{TH}$. Write $k = \mu_n C_{ox}(W/L) = 0.0008$ A/V^2. In that region,

$$I_D = k\left(2.5V_{DS} - V_{DS}^2/2\right).$$

For a voltage-controlled resistor at sufficiently small drain voltage, the zero-bias differential resistance is

$$r_{ds}(0) = \frac{1}{k(2.5)} = 500\,\Omega.$$

This explains the intended key, but it does not validate using the triode equation at 5 V. At the printed 5 V bias, the ideal zero-channel-modulation saturation model gives 2.5 mA: static $V/I = 2000$ ohms and infinite differential output resistance. Taking an absolute value of an invalid negative triode derivative is not a physical correction.

Q2.14.4 MCQ 2014 2M Ans: C ● ○ ○

Given $V_{Th} = 0.8\,\text{V}, V_{D_1} = 1.6\,\text{V}, \quad I_{D_1} = 0.5\,\text{mA}, V_{D_2} = 2\,\text{V}, \quad I_{D_2} = ?,$
Since the drain and gate of the MOSFET are shorted, it is in saturation. The saturation current equation neglecting channel length modulation is given by,

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_T)^2$$

$$I_D = \frac{1}{2}K_N\frac{W}{L}(V_{GS} - V_T)^2$$

For $V_D = 1.6v$

$$I_{D_1} = k_n(V_{GS_1} - V_T)^2$$

$$k_n = \frac{I_{D_1}}{(1.6 - 0.8)^2} = \frac{0.5}{(0.8)^2}$$

$$k_n = 0.78\,\text{mA/v}$$

For $V_{D_2} = V_{GS_2} = 2\,\text{V}$,

$$I_{D_2} = k_n(V_{GS_2} - V_T)^2$$

$$I_{D_2} = 0.78(2 - 0.8)^2$$

$$I_{D_2} = 1.125\,\text{mA}$$

$$\boxed{\text{(C)} \quad 1.125}$$

Q2.14.5 NAT 2014 2M Ans: 0.9 ● ○ ○

The PMOS gate is at 6 V and its source at 12 V, giving overdrive 4 V. The NMOS gate is 0 V and its source is -5 V, giving overdrive 3 V. If both were saturated their currents would be 1.6 and 0.9 mA, respectively, so that assumption is inconsistent with their series connection. Assume instead the NMOS is saturated and PMOS is in triode. Then $I_D = 0.1(3)^2 = 0.9$ mA. Let $y = V_{SD,p} = 12 - V_x$. The PMOS triode equation is

$$0.9 = 0.1(8y - y^2).$$

Its triode-valid root is $y = 4 - \sqrt{7} = 1.35425$ V; the other root exceeds its 4 V overdrive. Therefore

$$V_x = 8 + \sqrt{7} = 10.64575\,\text{V}.$$

The PMOS indeed has $V_{SD,p} < 4$ V. The NMOS has $V_{DS,n} = V_x + 5 > 3$ V and is saturated. Its node condition is $V_x \geq -2$ V, not $V_x \geq 3$ V.

$$\boxed{I_D = 0.9\,\text{mA}}$$

Q2.14.6 MCQ 2014 1M Ans: A ● ○ ○

A transconductance amplifier is a specialized electronic amplifier that converts an input voltage signal into an output current signal. A good transconductance amplifier should have **high input impedance and high output impedance**
Reason: The generic diagram of a transconductance amplifier is shown below.

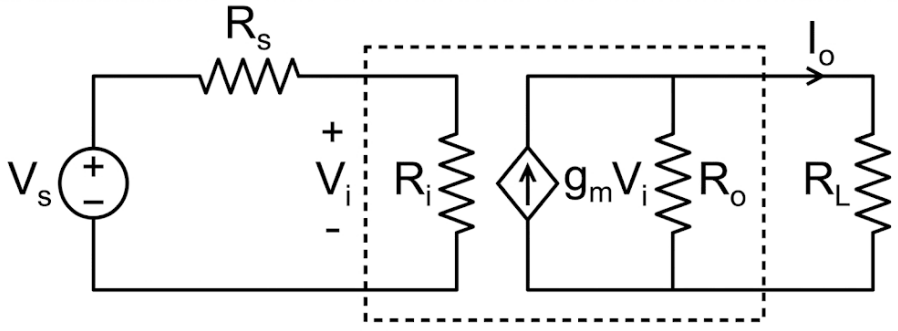

Fig. Solution Q2.14.6

Importance of High Input Impedance:
Efficient Voltage Transfer: To efficiently capture the input voltage most of the input voltage should be dropped acorss R_i for which R_i should be as high as possible(so that using voltage division rule most of the voltage drops across R_i). **Importance of High Output Impedance**
Driving loads effectively: In order to drive the load efficiently most of the output current should flow into load for which R_0 should be as high as possible (so that using current division rule most of the current flows into load).

(A) high input resistance and high output resistance

Key Points

- **Transconductance Amplifier:** Input = Voltage, Output = Current → High Z_{in}, High Z_{out}.

Q2.14.7 NAT 2014 2M Ans: 0.0707

Given

$$\frac{\partial I_{DS}}{\partial V_{GS}} = 10^{-3}$$

Also, the MOSFET is operating in the linear region with $V_{DS} = 0.1\,\text{V}$
The current in linear region neglecting channel length modulation is given by,

$$I_{DS} = \mu_n C_0 \times \frac{W}{L}\left[(V_{GS} - V_t)V_{DS} - \frac{V_{DS}^2}{2}\right]$$

$$I_{DS} = K_n\left[(V_{GS} - V_t)V_{DS} - \frac{V_{DS}^2}{2}\right]$$

$$\frac{\partial I_{DS}}{\partial V_{GS}} = K_n V_{DS}$$

$$10^{-3} = K_n(0.1)$$

$$K_n = 10^{-2}$$

Now, the current in saturation region is given by,

$$I_{DS} = \frac{K_n}{2}[V_{GS} - V_t]^2$$

$$\sqrt{I_{DS}} = \sqrt{\frac{K_n}{2}}(V_{GS} - V_t)$$

Slope is nothing but the rate of change or the derivative, so

$$\frac{\partial\sqrt{I_{DS}}}{\partial V_{GS}} = \sqrt{\frac{K_n}{2}}$$

Putting on the respective values, we get:

$$\text{Slope} = \frac{\partial\sqrt{I_{DS}}}{\partial V_{GS}} = \sqrt{\frac{10^{-2}}{2}} = 0.0707\ \sqrt{\text{A}}/\text{V}$$

$$\boxed{\text{Slope} = 0.0707\ \sqrt{\text{A}}/\text{V}}$$

Key Points

- To find slope we have to find the derivative of current with respect to voltage.

Q2.14.8 NAT | 2014 | 2M | Ans: 1.5 ● ○ ○

Given $\frac{W}{L} = 2$, $\mu_n C_{ox} = 100\,\mu\text{A/V}^2$, $V_{th} = 0.5\,\text{V}$

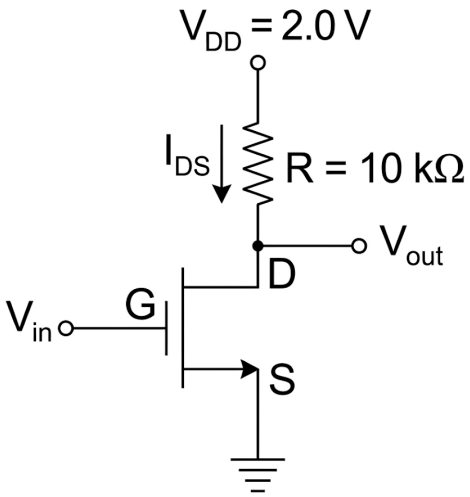

Fig. Solution Q2.14.8

For transition from saturation to linear region,

$$V_{DS} = V_{GS} - V_T$$

Since, $V_{GS} = V_{in}$, $V_{DS} = V_{out}$,

$$V_{out} = V_{in} - 0.5$$

Current through the resistor is

$$I_D = \frac{V_{DD} - V_{out}}{R}$$

$$I_D = \frac{2 - (V_{in} - 0.5)}{10^4}$$

$$I_D = \frac{2.5 - V_{in}}{10^4}$$

At the boundary condition,

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_T)^2$$

$$I_D = \frac{1}{2}(100 \times 10^{-6})(2)(V_{in} - 0.5)^2$$

$$I_D = 100 \times 10^{-6}(V_{in} - 0.5)^2$$

Equating the two expressions of current,

$$100 \times 10^{-6}(V_{in} - 0.5)^2 = \frac{2.5 - V_{in}}{10^4}$$

$$(V_{in} - 0.5)^2 = 2.5 - V_{in}$$

$$V_{in}^2 - V_{in} + 0.25 = 2.5 - V_{in}$$

$$V_{in}^2 + 0.25 = 2.5$$

$$V_{in}^2 = 2.25$$

$$\boxed{V_{in} = 1.5\ V}$$

Key Points

- At the boundary of saturation and linear region, $V_{DS} = V_{GS} - V_{th}$.

Q2.13.1 MCQ | 2013 | 1M | Ans: A | ● ○ ○

Given circuit is,

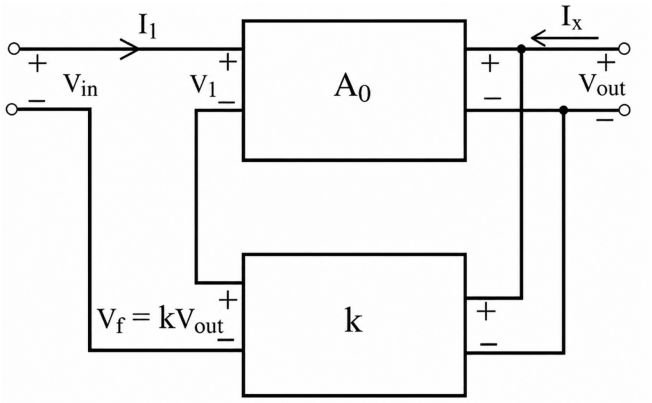

Fig. Solution Q2.13.1

Here A_0 is the amplifier and assume R_{in},R_{in_f} are the input resistances without and with feedback respectively, are given by

$$R_{in} = \frac{V_1}{I_1} \quad R_{in_f} = \frac{V_{in}}{I_1}$$

Applying KVL in the input loop, we get input voltage as,

$$V_{in} = V_1 + V_f$$

$$V_{in} = V_1 + kV_{out}$$

$$V_{in} = V_1 + kA_oV_1 \quad (V_{out} = A_oV_1)$$

$$V_{in} = V_1(1 + kA_o)$$

$$\therefore R_{in_f} = \frac{V_1(1 + kA_o)}{I_1} = (1 + kA_0)R_{in}$$

Therefore input impedance is increased with feedback.

Now, assume R_{out}, R_{out_f} are the output resistances without and with feedback respectively. To find output resistance, set $V_{in} = 0$ and inject a current from output node, then

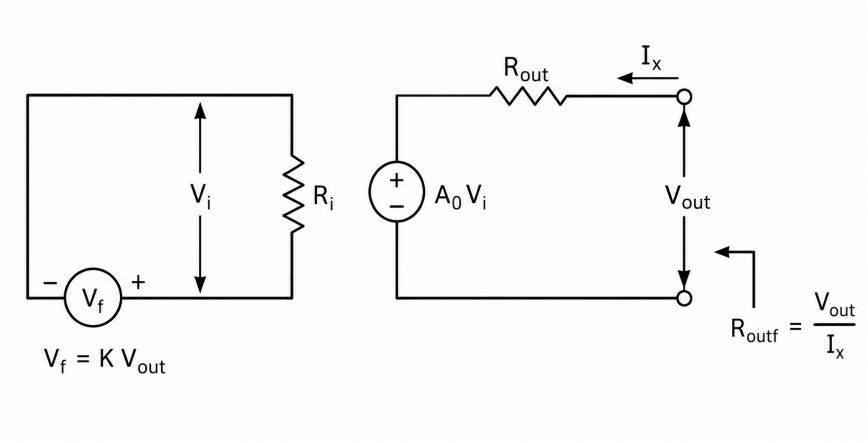

Fig. Solution Q2.13.1

Applying KVL to the output side we get,

$$A_0V_i + I_xR_{out} - V_{out} = 0$$

$$\therefore \quad I_x = \frac{V_{out} - A_0V_i}{R_{out}} \tag{1}$$

The input voltage is given as

$$V_i = -V_f = -KV_{out} \qquad \because V_{in} = 0 \tag{2}$$

Substituting the V_i from equation (2) in equation (1) we get,

$$I_x = \frac{V_{out} + A_0KV_{out}}{R_{out}} = \frac{V_{out}(1 + KA_0)}{R_{out}}$$

$$\therefore \quad R_{out_f} = \frac{V_{out}}{I_x} \implies R_{out_f} = \frac{R_{out}}{(1 + KA_0)}$$

Hence output impedance with feedback decreases.

(A) The input impedance increases and output impedance decreases

Key Points

- For voltage-voltage (series-shunt) feedback, the input impedance increases by a factor of $(1 + A\beta)$ or $(1 + A_ok)$.
- Shunt sampling at the output decreases the output impedance by a factor of $(1 + A_ok)$.

Q2.13.2 MCQ 2013 1M Ans: D ●○○

Because of channel length modulation effect, the drain current equation in saturation becomes,

$$I_D = I_{Dsat}(1 + \lambda V_{DS})$$

where λ is the channel length modulation parameter.
Due to channel length modulation, the drain current increases with V_{DS}, causing the output characteristics to have a finite slope. The output resistance is given by

$$r_o = \frac{1}{\lambda I_D}$$

Hence, due to channel length modulation, $r_o \downarrow$
The gate-source capacitance depends on device geometry and independent on channel length modulation. So option A is incorrect.

The transconductance increases as the drain current is increased because of channel length modulation. So option B is incorrect.
Unity gain frequency is directly proportional to g_m so it is increased. So option C is incorrect.

(D) a decrease in the output resistance

Q2.13.3 MCQ 2013 2M Ans: B ● ○ ○

Given circuit is,

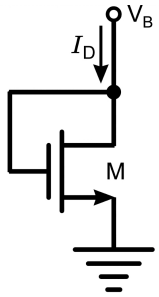

Fig. Solution Q2.13.3

The MOS is operating in saturation region as the drain and gate are shorted.

$$I_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)(V_{GS} - V_T)^2$$

$$I_D = \frac{1}{2} \times 40 \times 10^{-6}(V_B - 1)^2$$

$$I_D = 20 \times 10^{-6}(V_B - 1)^2$$

So,

$$\frac{\partial I_D}{\partial V_B} = 40 \times 10^{-6}(V_B - 1) = 40 \times 10^{-6} \times (2 - 1)$$

$$\frac{\partial I_D}{\partial V_B} = 40 \times 10^{-6}$$

$\therefore$ small signal resistance,

$$r_o = \frac{\partial V_B}{\partial I_D} = \frac{10^6}{40}$$

$$r_o = 25\,\text{k}\Omega$$

(B) 25

Q2.13.4 MCQ 2013 2M Ans: A ● ● ○

Given circuit is,

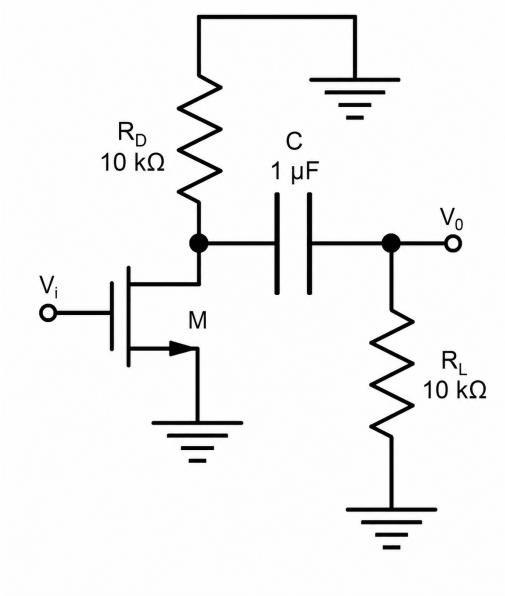

Fig. Solution Q2.13.4

Given that $\lambda = 0$, $r_{out} = \frac{1}{\lambda I_{out}} = \infty$ and $g_m = 1$ mA/V

The AC equivalent model is given as follows.

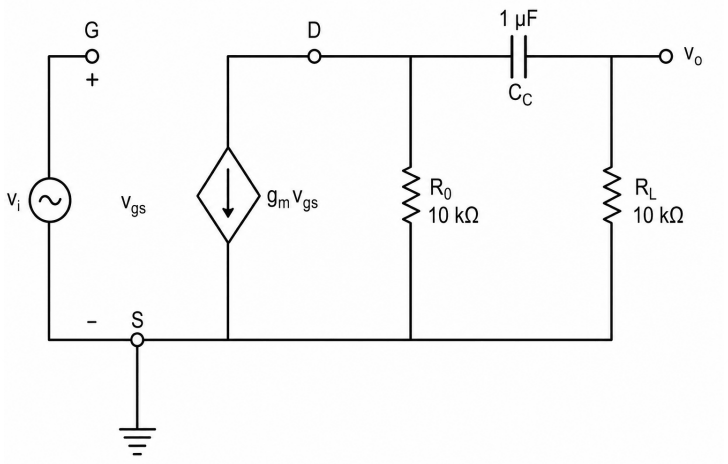

Fig. Solution Q2.13.4

To find cutoff frequency we need to find the equivalent resistance seen through the capacitor, So disable the independent sources and connect a voltage source in place of C_c and find the current through it.

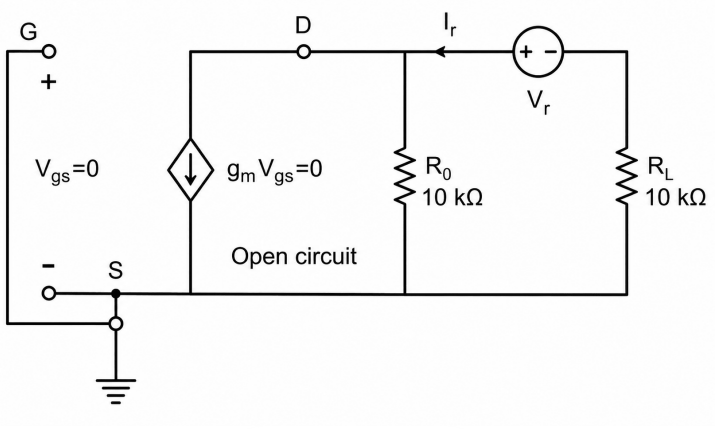

Fig. Solution Q2.13.4

$$R = \frac{V_r}{I_r} = R_D + R_L$$

$$R = 10\,\text{k}\Omega + 10\,\text{k}\Omega = 20\,\text{k}\Omega$$

Lower cut-off frequency will be:

$$f_L = \frac{1}{2\pi R C_c}$$

$$f_L = \frac{1}{2\pi \times 20 \times 10^3 \times 10^{-6}}$$

$$f_L \approx 8\,\text{Hz}$$

(A) 8

Key Points

- When finding the equivalent resistance seen by a coupling capacitor, independent active sources are deactivated (voltage sources replaced by a short circuit).
- The lower cutoff frequency due to a single high-pass RC network is calculated using $f_L = \frac{1}{2\pi R_{eq} C}$.

Mistakes to be avoided

- The resistances when looking through the capacitor will be in series not in parallel.

Q2.11.1 MCQ 2011 2M Ans: C ● ○ ○

Given circuit is,

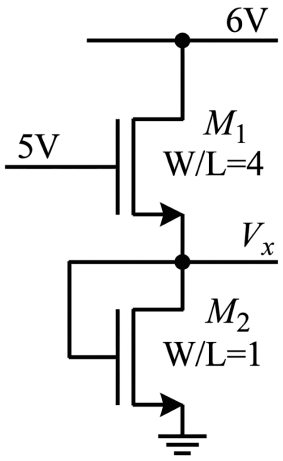

Fig. Solution Q2.11.1

Since the Gate and Drain of M_2 transistor is shorted, it is operated in saturation region. Let the M_1 transistor is in saturation region. Since both MOSFETs are in series same current flows through them. Hence,

$$I_{D1} = I_{D2}$$

$$\frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)_1 (V_{GS1} - V_{T1})^2 = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)_2 (V_{GS2} - V_{T2})^2$$

$$4[5 - V_x - 1]^2 = 1[V_x - 1]^2$$

$$2(4 - V_x) = V_x - 1$$

$$9 = 3V_x \rightarrow V_x = 3\,\text{V}$$

$$V_{GS1} = 5 - 3 = 2, \quad V_{DS1} = 6 - 3 = 3\,\text{V}$$

$$\therefore V_{DS1} = (V_{GS1} - V_{T1})$$

So our assumption is correct and M_1 is in saturation region. Hence $V_x = 3\,\text{V}$

(C) 3 V

Q2.09.1 MCQ 2009 2M Ans: D ● ○ ○

Given circuit is,

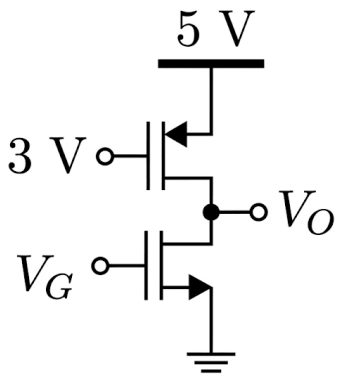

Fig. Solution Q2.09.1

Here, for the pMOS transistor, $V_{SG,p} = 5 - 3 = 2V$
Since, $V_{SG,p} > |V_T|$, the pMOS is already ON strongly. Initially NMOS is OFF, so output node is charged to 5V through PMOS.
The nMOS turns ON when $V_{GS,n} = V_T = 1V$
The question states that V_G is increased slightly beyond $1V$. Hence the nMOS has just started conducting and its current is very small.
Since the pMOS is strongly ON while the nMOS current is still very small, the output voltage remains close to $5V$.
For nMOS saturation condition,

$$V_{DS,n} \geq V_{GS,n} - V_T$$

Now, $V_{GS,n} - V_T$ is very small because V_G is only slightly greater than $1V$. Also, $V_{DS,n} = V_0 \approx 5V$. Thus,

$$V_{DS} \gg (V_{GS} - V_T)$$

Hence the nMOS operates in saturation region.
Now, for pMOS, $V_{SG,p} - |V_T| = 2 - 1 = 1V$, also, $V_{SD,p} = 5 - V_0$
Since $V_0 \approx 5V$, $V_{SD,p} \approx 0$. Thus,

$$V_{SD,p} < V_{SG,p} - |V_T|$$

Hence the pMOS operates in triode region.

(D) n-MOSFET is in saturation and p-MOSFET is in triode region

Q2.09.2 MCQ 2009 2M Ans: D ●○○

From the previous question, nMOS is in saturation region and pMOS is in triode region.
Given, $V_G = 1.5V$ and $\mu C_{ox}(W/L) = 1$ mA/V^2
Since both MOSFETs are in series same current flows through them.

$$I_{Dn} = I_{Dp}$$

$$\frac{1}{2}\mu C_{ox}\frac{W}{L}(V_{GS} - V_T)^2 = \mu C_{ox}\frac{W}{L}\left[(V_{SG} - |V_T|)V_{SD} - \frac{V_{SD}^2}{2}\right]$$

$$\frac{1}{2}(1)(0.5)^2 = 1\left[(2-1)(5-V_0) - \frac{(5-V_0)^2}{2}\right]$$

$$0.125 = (5 - V_0) - \frac{(5-V_0)^2}{2}$$

Let,$x = 5 - V_0$. Then,

$$0.125 = x - \frac{x^2}{2}$$

$$0.25 = 2x - x^2$$

$$x^2 - 2x + 0.25 = 0$$

$$x = 1 \pm \frac{\sqrt{3}}{2}$$

Since pMOS is in triode region, $V_{SD} < 1$. Hence, $x < 1$.

$$\therefore x = 1 - \frac{\sqrt{3}}{2}$$

$$5 - V_0 = 1 - \frac{\sqrt{3}}{2}$$

$$V_0 = 4 + \frac{\sqrt{3}}{2}$$

$$\boxed{\text{(D)} \quad 4 + \frac{\sqrt{3}}{2}\ \text{V}}$$

Q2.08.1 MCQ 2008 1M Ans: B ● ○ ○

Given,

$$I_D = K(V_{GS} - V_T)^2$$

The transconductance is defined as

$$g_m = \frac{\partial I_D}{\partial V_{GS}}$$

Differentiating I_D with respect to V_{GS},

$$g_m = \frac{\partial}{\partial V_{GS}}\left[K(V_{GS} - V_T)^2\right]$$
$$= 2K(V_{GS} - V_T)$$

Hence,

$$g_m = 2K(V_{GS} - V_T)$$

$$\boxed{\text{(B)} \quad 2K(V_{GS} - V_T)}$$

Q2.08.2 MCQ 2008 2M Ans: B ● ○ ○

Given circuit is,

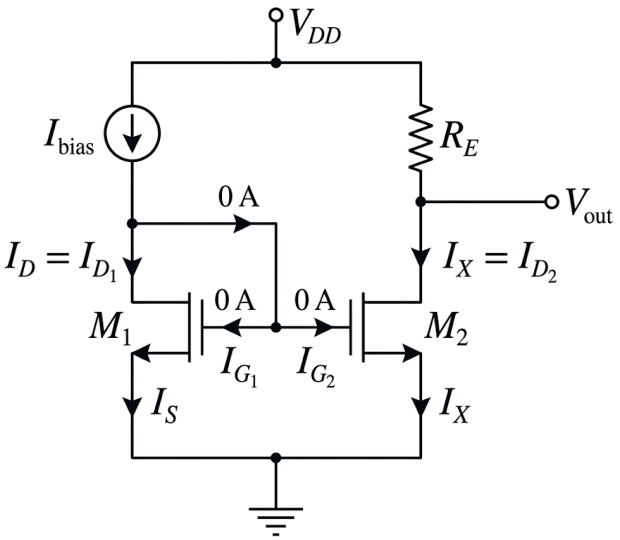

Fig. Solution Q2.08.2

The above circuit is a MOSFET based current mirror circuit.
From the figure,

$$I_{bias} = I_{G_1} + I_{G_2} + I_D$$

As shown in the figure, the gate current of both MOSFET will be zero. i.e.

$$I_{G_1} = I_{G_2} = 0\,\text{A}$$

Hence,

$$I_{bias} = I_D = I_{D_1} \quad \ldots (i)$$

Since, both MOSFET are identical, hence

$$I_{D_1} = I_{D_2} = I_D$$

$$I_D = I_X \quad \ldots (ii)$$

From equation (i) and (ii),

$$I_X = I_{bias}$$

(B) $I_x = i_{bias}$

Key Points

- The DC gate current of a MOSFET is ideally zero ($I_G = 0$).
- In a basic MOSFET current mirror with identical transistors, the output current mirrors the reference (bias) current exactly, assuming no channel-length modulation ($\lambda = 0$).

Q2.08.3 MCQ 2008 2M Ans: *

Given that NMOS is in linear region, so drain current I_D in the linear region is given by

$$I_D = K_n \left[(V_{GS} - V_T)V_{DS} - \frac{V_{DS}^2}{2} \right]$$

$$\text{and} \quad g_m = \frac{\partial I_D}{\partial V_{GS}} = K_n V_{DS}$$

So it is clear that g_m is independent on V_{GS}.
Hence none of the options match.

Key Points

- In the linear region, g_m is independent on V_{GS}.

Q2.08.4 MCQ 2008 2M Ans: C

Given circuit is,

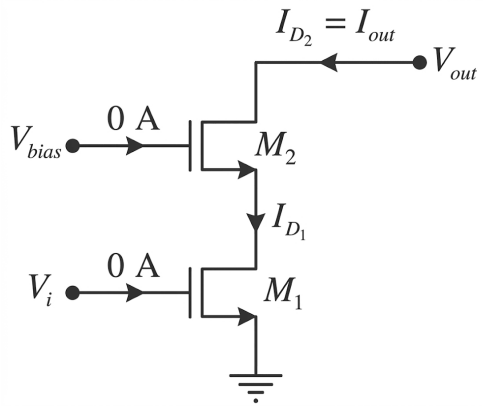

Fig. Solution Q2.08.4

Since, gate current of both NMOS is zero,
i.e. $I_{G_1} = I_{G_2} = 0$

So, $I_{D_2} = I_{D_1} = I_{out}$ ... (i)
Equivalent tranconductance g_m is given by,

$$g_m = \frac{I_{out}}{V_i} \quad \text{... (ii)}$$

Transconductance of M_2 is given by,

$$g_{m_2} = \frac{I_{D_2}}{V_{bias}}$$

Transconductance of M_1 is given by,

$$g_{m_1} = \frac{I_{D_1}}{V_i} \quad \text{... (iii)}$$

From equation (i), (ii) and (iii),

$$g_m = g_{m_1}$$

(C) nearly equal to the g_m of M1

Key Points

- For a cascode configuration (transistors in series), the effective transconductance of the entire structure is determined by the transconductance of the common-source transistor at the input (in this case, M_1).
- M_2 acts as a common-gate stage, which passes the current I_{D1} to the output but does not change the transconductance with respect to the input voltage V_i.

Q2.06.1 MCQ 2006 1M Ans: D ●○○

Given MOSFET is a depletion NMOS which means channel is already present and even at negative gate voltages I_D can exist and
Given $V_{GS} = 0$ at $I_D = 12$ mA and $V_{GS} = -6$ Volts at $I_D = 0$

$$\textbf{Transconductance } (g_m) = \frac{\partial I_D}{\partial V_{GS}} \textbf{ for } V_{DS} = \textbf{constant}$$

The I_D vs V_{GS} characteristics are plotted below.

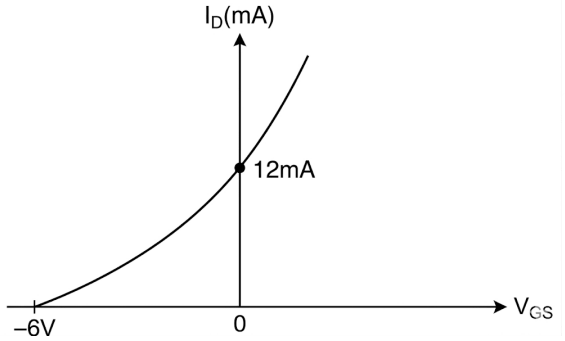

Fig. Solution Q2.06.1

If we observe, the slope of the characteristics is increasing with increase in V_{GS} and the slope is nothing but g_m. Hence g_m will be maximum at maximum value of V_{GS}(i.e,maximum value in the options).

(D) $V_{GS} = 3$ V

Key Points

- The transconductance g_m is the slope of the transfer characteristic curve (I_D vs. V_{GS}).
- For an n-channel depletion MOSFET, the slope increases as V_{GS} becomes more positive. Therefore, the most positive V_{GS} among the given options provides the highest transconductance.

Q2.05.1 MCQ 2005 2M Ans: C

Given circuit is,

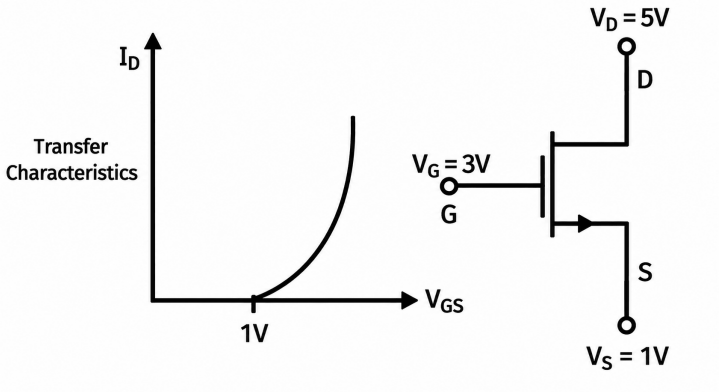

Fig. Solution Q2.05.1

From the transfer characteristics,

$$I_D = 0 \quad \text{at} \quad V_{GS} = 1V$$

Hence the threshold voltage is $V_T = 1V$
Now, from the circuit:

$$V_G = 3V, \qquad V_S = 1V \qquad V_D = 5V$$

Therefore,

$$V_{GS} = V_G - V_S$$

$$V_{GS} = 3 - 1 = 2V$$

And

$$V_{DS} = V_D - V_S$$

$$V_{DS} = 5 - 1 = 4V$$

For an nMOS transistor to operate in saturation region,

$$V_{DS} \geq V_{GS} - V_T$$

$$4 \geq 2 - 1$$

$$4 \geq 1$$

which is true. Hence the MOSFET operates in saturation region.

(C) 1 V and the device is in saturation region

Q2.05.2 MCQ 2005 2M Ans: C ● ○ ○

Given circuit is,

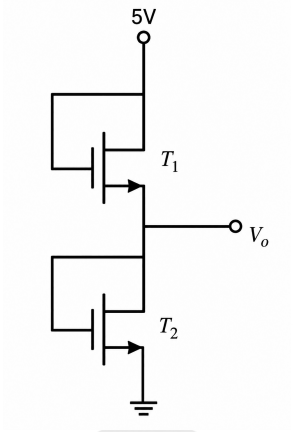

Fig. Solution Q2.05.2

Since both the T_1 and T_2 gate and drain are shorted , they are in saturation region.
Since both MOSFETs are in series, same current flows through them.

$$I_{D1} = I_{D2}$$

$$K_1(V_{GS1} - V_T)^2 = K_2(V_{GS2} - V_T)^2$$

$$36[(5 - V_0) - 1]^2 = 9[(V_0 - 0) - 1]^2$$

$$36(4 - V_0)^2 = 9(V_0 - 1)^2$$

$$4(4 - V_0)^2 = (V_0 - 1)^2$$

$$2(4 - V_0) = V_0 - 1$$

$$8 - 2V_0 = V_0 - 1$$

$$9 = 3V_0$$

$$V_0 = 3V$$

$$\boxed{\text{(C)} \quad 3\,\text{V}}$$

Q2.05.3 MCQ 2005 1M Ans: D ● ○ ○

The generic diagram of a current shunt feedback amplifier is given by,

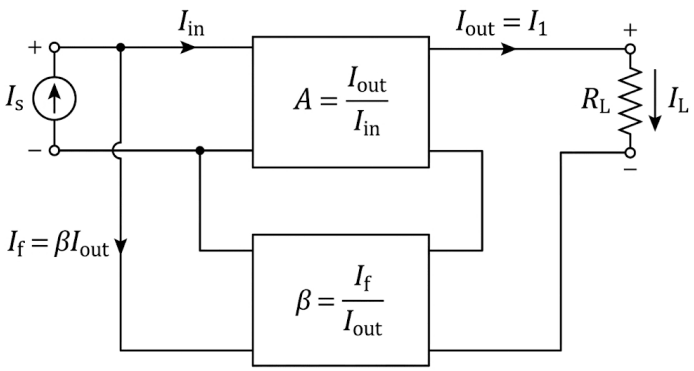

Fig. Solution Q2.05.3

Expression for Input resistance(R_{if}) : The equivalent circuit of the current shunt feedback amplifier is,

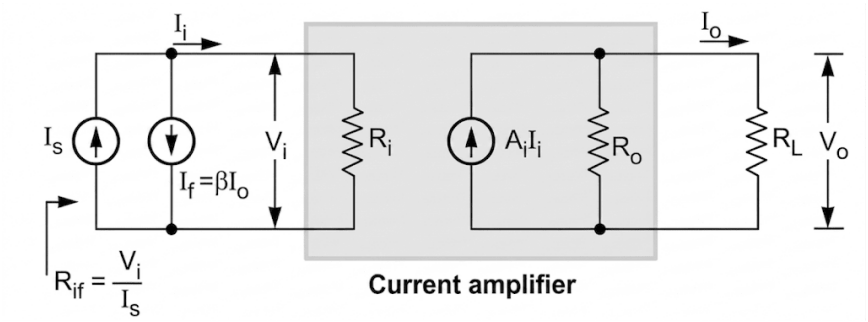

Fig. Solution Q2.05.3

Without feedback the input resistance is, $R_i = \frac{V_i}{I_i}$. With feedback the input resistance is $R_{if} = \frac{V_i}{I_s}$.
Applying KCL at input node,

$$I_s = I_i + I_f = I_i + \beta I_0$$

Here, $A_i = \frac{I_0}{I_i}$

$$I_s = I_i + \beta A_I I_i = I_i(1 + \beta A_I)$$

The input resistance with feedback is given as

$$R_{if} = \frac{V_i}{I_s} = \frac{V_i}{I_i(1 + \beta A_I)}$$

$$\therefore \quad \boxed{R_{if} = \frac{R_i}{(1 + \beta A_I)}} \qquad \because R_i = \frac{V_i}{I_i}$$

Therefore input resistance decreased.
Expression for Output resistance(R_{of}) : The equivalent circuit to find output resistance is given by,

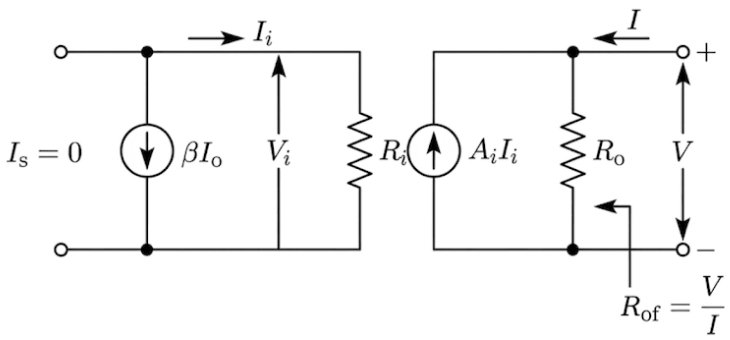

Fig. Solution Q2.05.3

To find the output resistance, deactivate the input source I_s , connect test voltage source at output terminals with R_L disconnected.
Applying the KCL to the output node we get,

$$I = \frac{V}{R_o} - A_i I_i \qquad \ldots (1)$$

The input current is given as

$$I_i = -I_f = -\beta I_o \qquad \because I_s = 0$$
$$= \beta I \qquad \because I = -I_o \qquad \ldots (2)$$

Substituting value of I_i from equation (2) in equation (1) we get,

$$I = \frac{V}{R_o} - A_i\beta I$$

$$\therefore \quad \frac{V}{R_o} = I + A_i\beta I = I(1 + \beta A_i)$$

$$\boxed{R_{of} = \frac{V}{I} = R_o(1 + \beta A_i)}$$

Hence output resistance increased.

(D) decrease the input resistance and increase the output resistance

Key Points

- For current shunt feedback, the input impedance decreases by a factor of $(1 + A\beta)$ or $(1 + A_i k)$.
- Series sampling at the output increases the output impedance by a factor of $(1 + A_o k)$.

Q2.04.1 MCQ 2004 2M Ans: D ● ○ ○

A diode-connected nMOS transistor always operates in saturation region whenever $V_{GS} > V_T$
The drain current in saturation is given by

$$I_D = K(V_{GS} - V_T)^2$$

Given, $V_T = 1\,\text{V}$ and When $V_{GS} = 2\,\text{V}$, $I_D = 1\,\text{mA}$. Thus,

$$1\,\text{mA} = K(2 - 1)^2 \implies K = 1\,\text{mA/V}^2$$

Now for $V_{GS} = 3\,\text{V}$

$$I_D = K(3 - 1)^2 = 1 \times (2)^2 = 4\,\text{mA}$$

(D) 4 mA

Q2.03.1 MCQ 2003 2M Ans: A ● ○ ○

Feedback-definition warning. A current-shunt amplifier mixes currents at its input, so $R_{if} = R_i/(1 + T)$ must use the appropriate dimensionless loop gain T. The question labels 50 as a *voltage* gain but supplies a feedback factor without defining its current/voltage port normalization. Using 50 directly as the forward gain gives $T = 50(0.2) = 10$ and answer (A),1/11 kΩ. That is the conventional intended calculation, not a proof that voltage and current gains are interchangeable. If 50 is instead the open-circuit voltage gain of a unilateral two-port with the stated input/output resistances, its short-circuit current gain would be

$$A_{i,sc} = 50\frac{1}{2.5} = 20.$$

If 0.2 is then explicitly a current-feedback ratio, $T = 4$ and the result is B,1/5 kΩ. A loaded current gain would additionally depend on the load. The source does not establish those port definitions. The numerical result depends on whether the stated forward gain is a voltage gain or a current gain; these definitions are not interchangeable.

Q2.03.2 MCQ 2003 2M Ans: C ●○○

Given, voltage gain $A_V = 50$, input resistance $R_i = 1000\Omega$, output resistance $R_0 = 250\Omega$. The equivalent circuit of the cascaded configuration is,

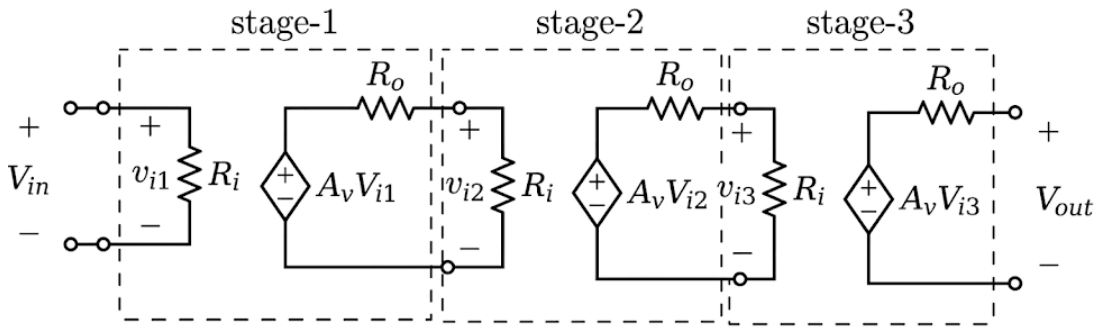

Fig. Solution Q2.03.2

Here,

$$V_{i1} = V_{in} \implies \frac{V_{i1}}{V_{in}} = 1$$

$$V_{i2} = A_V V_{i1}\left(\frac{R_i}{R_i + R_0}\right) \implies \frac{V_{i2}}{V_{i1}} = A_V\left(\frac{R_i}{R_i + R_0}\right)$$

$$V_{i3} = A_V V_{i2}\left(\frac{R_i}{R_i + R_0}\right) \implies \frac{V_{i3}}{V_{i2}} = A_V\left(\frac{R_i}{R_i + R_0}\right)$$

$$V_{out} = A_V V_{i3} \implies \frac{V_{out}}{V_{i3}} = A_V$$

Overall Voltage gain is,

$$A_v = \frac{V_{out}}{V_{in}} = \frac{V_{out}}{V_{i3}} \times \frac{V_{i3}}{V_{i2}} \times \frac{V_{i2}}{V_{i1}} \times \frac{V_{i1}}{V_{in}}$$

$$= 50\left(\frac{1000}{1000 + 250}\right) \times 50\left(\frac{1000}{1000 + 250}\right) \times 50$$

$$A_v = 50\left(\frac{1000}{1250}\right) \times 50\left(\frac{1000}{1250}\right) \times 50$$

$$A_v = 80000$$

Voltage gain in dB is

$$G_v = 20\log_{10}(80000)$$

$$\approx 98\,\text{dB}$$

(C) 98 dB

Q2.01.1 MCQ 2001 1M Ans: B ●○○

In saturation region, increasing drain voltage increases the drain depletion region near the drain terminal.
This causes the pinch-off point to move towards the source, thereby reducing the effective channel length. This effect is called channel length modulation.
Hence the effective channel length decreases with increase in drain voltage.

(B) drain voltage

Q2.99.1 MCQ 1999 1M Ans: A ● ○ ○

In negative feedback, a part of the output signal is fed back to the input in opposite phase. Hence the effective input becomes

$$V_i' = V_i - \beta V_o$$

where β is the feedback factor. The output voltage is

$$V_o = AV_i'$$

$$V_o = A(V_i - \beta V_o)$$

$$V_o = AV_i - A\beta V_o$$

$$V_o(1 + A\beta) = AV_i$$

Thus the closed-loop gain is

$$A_f = \frac{V_o}{V_i} = \frac{A}{1 + A\beta}$$

Since, $1 + A\beta > 1$ therefore, $A_f < A$

Hence negative feedback reduces the amplifier gain but improves stability, bandwidth and reduces distortion and noise.

(A) reduces gain

Q2.99.2 MCQ 1999 1M Ans: C ● ○ ○

For voltage-series feedback,

$$R_{if} = R_i(1 + A\beta) \qquad R_{of} = \frac{R_o}{1 + A\beta}$$

Given, $A = 100$, $\beta = 0.99$ $R_i = 1\text{K}\Omega$ $R_0 = 100\Omega$. Thus,

$$1 + A\beta = 1 + 100(0.99) = 100$$

Input impedance becomes

$$R_{if} = 1\,\text{k}\Omega \times 100 = 100\,\text{k}\Omega$$

Output impedance becomes

$$R_{of} = \frac{100\,\Omega}{100} = 1\,\Omega$$

(C) 100 kΩ and 1 Ω

Q2.99.3 MCQ 1999 1M Ans: B ● ○ ○

For a single-pole system, the unit step response is of the form,

$$v_o(t) = 1 - e^{-t/\tau}$$

Rise time is defined from 10% to 90% response.

$$t_r = -\tau \ln(0.1) - (-\tau \ln(0.9)$$

$$t_r \approx 2.2\tau$$

Also, upper cutoff frequency is

$$f_H = \frac{1}{2\pi\tau} = \frac{1}{2\pi(t_r/2.2)} = \frac{2.2}{2\pi t_r} \approx \frac{0.35}{t_r}$$

$\therefore$ For a single-pole system, the upper cutoff frequency is given by,

$$f_H = \frac{0.35}{t_r}$$

Given, $t_r = 35\,\text{ns} = 35 \times 10^{-9}\,\text{s}$

$$\therefore f_H = \frac{0.35}{35 \times 10^{-9}} = 10 \times 10^6\,\text{Hz} = 10\,\text{MHz}$$

(B) 10

Q2.98.1 MCQ | 1998 | 1M | Ans: C

The composite FET (Two FET connected in parallel) is shown below.

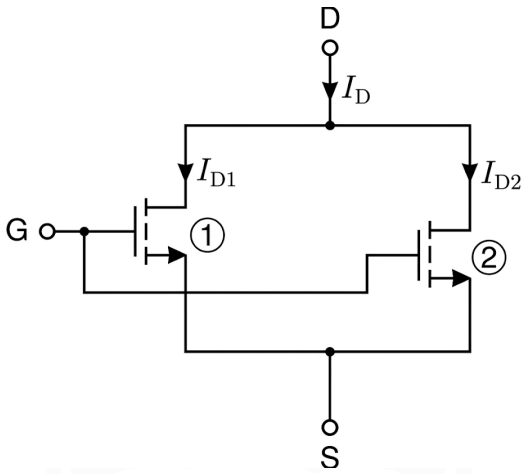

Fig. Solution Q2.98.1

$$I_D = I_{D_1} + I_{D_2}$$

$$g_m = \frac{I_D}{v_{gs}} = \frac{I_{D_1}}{v_{gs}} + \frac{I_{D_2}}{v_{gs}}$$

$$= g_{m_1} + g_{m_2}$$

Since both FET are identical ($g_{m_1} = g_{m_2}$). Therefore overall $g_m = 2g_m$

Now,

$$r_d = \frac{v_{ds}}{I_d} = \frac{v_{ds}}{I_{d_1} + I_{d_2}} = \frac{1}{\left(\frac{I_{d_1}}{v_{ds}}\right) + \left(\frac{I_{d_2}}{v_{ds}}\right)}$$

$$= \frac{1}{\left(\frac{1}{r_{d_1}} + \frac{1}{r_{d_2}}\right)}$$

$$r_d = r_{d_1} \parallel r_{d_2}$$

Since both FET are identical ($r_{d_1} = r_{d_2}$)

$$r_d = \left(\frac{r_d}{2}\right)$$

(C) $2g_m$ and $\frac{r_d}{2}$

Key Points

- When two identical FETs are connected in parallel, their drain currents sum up, effectively doubling the transconductance ($g_{m,eq} = 2g_m$).
- The internal drain resistances appear in parallel, halving the equivalent drain resistance ($r_{d,eq} = \frac{r_d}{2}$).

Q2.98.2 MCQ 1998 1M Ans: A

The generic diagram of a shunt shunt feedback(also called voltage shunt feedback) amplifier is given by,

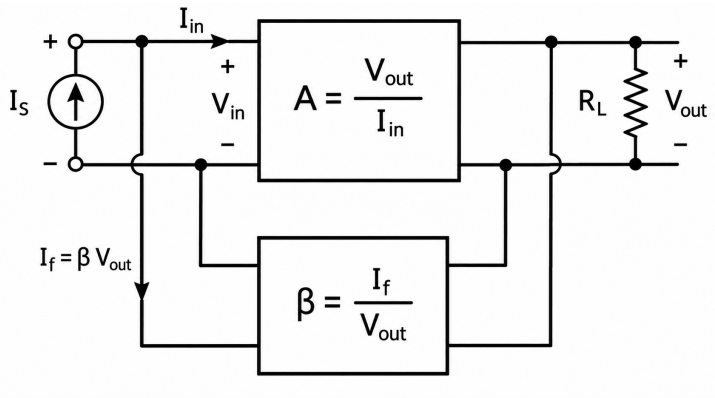

Fig. Solution Q2.98.2

Expression for Input resistance(R_{if}) :
Without feedback the input resistance is, $R_i = \frac{V_{in}}{I_{in}}$. With feedback the input resistance is $R_{if} = \frac{V_{in}}{I_s}$.
Applying KCL at input node,

$$I_s = I_{in} + I_f = I_{in} + \beta V_{out}$$

Here, $A = \frac{V_{out}}{I_{in}}$

$$I_s = I_{in} + \beta A I_{in} = I_{in}(1 + A\beta)$$

The input resistance with feedback is given as

$$R_{if} = \frac{V_{in}}{I_s} = \frac{V_{in}}{I_{in}(1 + A\beta)}$$

$$\therefore \quad \boxed{R_{if} = \frac{R_i}{(1 + A\beta)}} \qquad \because R_i = \frac{V_{in}}{I_{in}}$$

Therefore due to negative feedback, Input resistance decreases by a factor $1 + A\beta$.
Expression for Output resistance(R_{of}) : The equivalent circuit to find output resistance is given by,

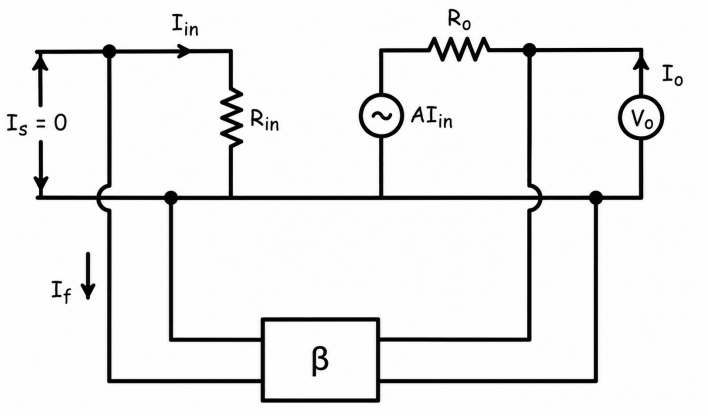

Fig. Solution Q2.98.2

To find the output resistance, deactivate the input source I_s , connect a test voltage source at the output terminals with R_L disconnected.

Apply KVL for negative feedback,

$$V_0 - I_0R_0 - A \times I_{in} = 0$$

But, $I_{in} = -I_f = -\beta V_0$

$$V_0 + A\beta V_0 = I_0R_0$$

$$V_0(1 + A\beta) = I_0R_0$$

$$\frac{V_0}{I_0} = R_{of} = \frac{R_0}{1 + A\beta}$$

Due to negative feedback, output impedance decreases by a factor $1 + A\beta$.

(A) both, input and output impedances decrease

Key Points

- For shunt shunt feedback, the input impedance decreases by a factor of $(1 + A\beta)$ or $(1 + A_ik)$.
- Shunt sampling at the output decreases the output impedance by a factor of $(1 + A_ok)$.

Q2.98.3 MCQ 1998 1M Ans: D ●○○

In the given circuit, short the DC voltage source at the top to ground to find small signal equivalent and current flowing through the MOSFET is I_D.

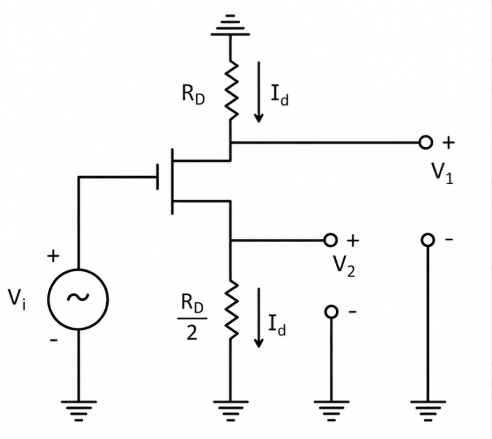

Fig. Solution Q2.98.3

$$V_1 = -i_D R_D \qquad V_2 = i_D \left(\frac{R_D}{2}\right) = \frac{i_D R_D}{2}$$

$$i_D R_D = 2V_2$$

Substituting in V_1,

$$V_1 = -2V_2$$

$$\boxed{\text{(D)} \quad V_1 = -2V_2}$$

Q2.97.1 NAT 1997 5M Ans: *

Given circuit is,

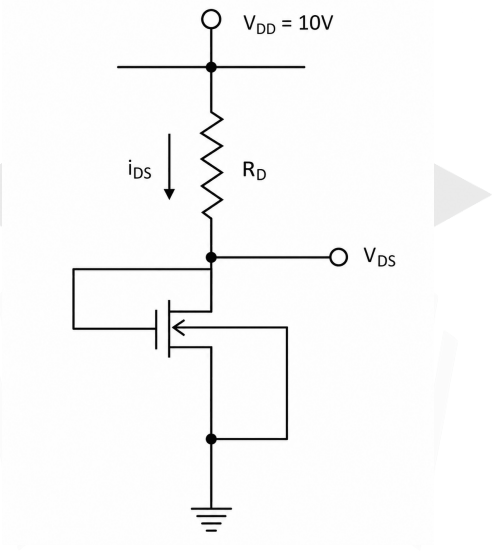

Fig. Solution Q2.97.1

Given: $V_{DD} = 10\,\text{V}, \beta = 10^{-4}\,\text{A/V}^2, V_T = 1\,\text{V}, I_{DS} = 0.5\,\text{mA}$.
Since gate and drain are shorted, $V_{GS} = V_{DS}$ For NMOS in saturation,

$$I_{DS} = \frac{\beta}{2}(V_{GS} - V_T)^2$$

$$0.5 \times 10^{-3} = \frac{10^{-4}}{2}(V_{GS} - 1)^2$$

$$(V_{GS} - 1)^2 = 10$$

$$V_{GS} = 1 + \sqrt{10} \approx 4.16\,\text{V}$$

Hence,

$$\boxed{V_{DS} = 4.16\,\text{V}}$$

Now,

$$R_D = \frac{V_{DD} - V_{DS}}{I_{DS}} = \frac{10 - 4.16}{0.5 \times 10^{-3}}$$

$$\boxed{R_D \approx 11.68\,\text{k}\Omega}$$

Q2.92.1 NAT 1992 8M Ans: 28.77 ●●○

Initially the transistor is OFF and $v_o(0) = 10$ V. After the 4 V gate step its overdrive is 2 V. Taking the device constant as $K = \mu_n C_{ox} W/(2L)$,

$$I_D = K(V_{GS} - V_T)^2 = 20\,\text{mA}.$$

Until the output falls to 5 V it remains above the 2 V saturation boundary, so this constant-current model is valid throughout the requested interval. KCL at the output gives

$$C\frac{dv_o}{dt} = \frac{10 - v_o}{1\,\text{k}\Omega} - 20\,\text{mA}.$$

The time constant is 100 ns. Solving with the initial condition gives

$$v_o(t) = -10 + 20e^{-t/(100\,\text{ns})}.$$

The -10 V term is the extrapolated equilibrium of this saturation-only equation, not the actual final voltage after the transistor leaves saturation. For $v_o = 5$ V,

$$e^{-t/(100\,\text{ns})} = 3/4,$$

$$\boxed{t = 100\ln(4/3) = 28.77\,\text{ns}}.$$

Q2.91.1 NAT 1991 2M Ans: 1 ●○○

Given circuit is,

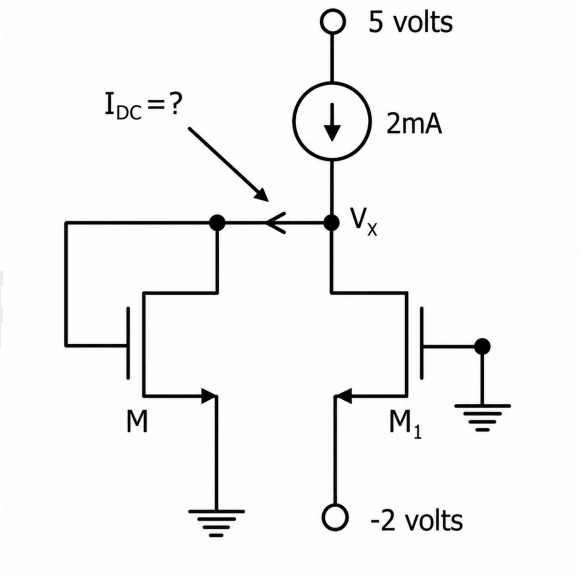

Fig. Solution Q2.91.1

For the left MOSFET, gate and drain are shorted so it is in saturation.

$$V_{GS_M} = V_{DS_M} = V_x$$

For the right MOSFET:

$$V_{GS_1} = 0 - (-2) = 2\,\text{V} \qquad V_{DS_1} = V_x + 2$$

Since

$$V_{DS_1} \geq ((V_{GS_1} - 1) = 1)$$

the right MOSFET operates in saturation. Therefore,

$$I_{D_1} = (V_{GS_1} - 1)^2\,\text{mA}$$

$$I_{D_1} = (2-1)^2 = 1\,\text{mA}$$

Applying KCL at the node V_x:

$$I_{D_1} + I_{DC} = 2\,\text{mA}$$

$$I_{DC} + 1 = 2$$

$$\boxed{I_{DC} = 1\,\text{mA}}$$

Key Points

- We should not assume that since the gate of M_1 is grounded, it is OFF, we have to look at the source potential as well.

Q2.88.1 MSQ | 1988 | 1M | Ans: B,D | ● ○ ○

For MOSFET devices:

- Electron mobility in N-channel MOSFET is higher than hole mobility in P-channel MOSFET.
- Hence N-channel MOSFETs switch faster.
- Due to higher mobility, they provide better current driving capability.
- Noise immunity and TTL compatibility are not the deciding advantages here.

(B) It is faster
(D) It has better drive capability

PrepFusion

SOLUTIONS

III

Operational Amplifier

Topics

1. Basics of Op-Amp, Comparator and Virtual Short
2. Inverting and Non-Inverting Amplifiers
3. Adder, Differential and Instrumentation Amplifiers
4. Log, Anti-Log and Rectifier Using Op-Amp
5. Filters Using Op-Amp
6. Offsets in Op-Amp
7. CMRR, Slew Rate and Frequency Response of Op-Amp
8. Integrator Using Op-Amp
9. Differentiator Using Op-Amp
10. Active Clipper, Clamper and Peak Detector
11. Multivibrators Using Op-Amp
12. Oscillators Using Op-Amp
13. Schmitt Triggers

Unit Snapshot*

Stream: EC	**Questions:** 97	**MCQ:** 61	**MSQ:** 3	**NAT:** 33
1M: 37	**2M:** 56	**Descriptive:** 4	**Avg Weightage** ~3M	**Range:** 1M(2011)–5M(2026)

**See preface for how Avg Weightage and Range are calculated.*

Q3.26.1 MCQ 2026 1M Ans: C ● ○ ○

The circuit after closing the switch is,

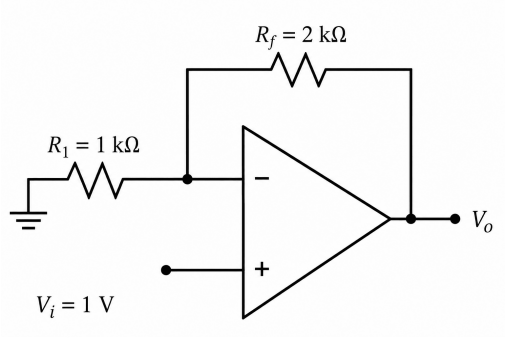

Fig. Solution Q3.26.1

Given, $V_{o1} = x$ (before closing the switch)
When the switch is open, the feedback resistance is the sum of both resistors in the feedback path ($2\,\text{k}\Omega + 1\,\text{k}\Omega = 3\,\text{k}\Omega$).

$$\text{Switch open} \implies V_{o1} = \left(1 + \frac{R_f}{R_1}\right) V_i = \left(1 + \frac{3}{1}\right) \times 1 = 4\,\text{V} = x$$

When the switch is closed, the 1 kΩ resistor is shorted out, leaving only the 2 kΩ resistor as the effective feedback resistance.

$$\text{Switch closed} \implies V_{o2} = \left(1 + \frac{2}{1}\right) \times 1 = 3\,\text{V}$$

Now, taking the ratio of the output voltages:

$$\frac{V_{o2}}{V_{o1}} = \frac{3}{4} \implies V_{o2} = \frac{3}{4}x$$

$$\boxed{\text{(C)} \quad \frac{3}{4}x}$$

Key Points

- The output voltage of a standard non-inverting Op-Amp in negative feedback is given by $V_o = \left(1 + \frac{R_f}{R_1}\right) V_i$.

Mistakes to be avoided

- Always check which type of feedback is present.

Q3.26.2 MCQ 2026 2M Ans: C ● ○ ○

Given circuit is,

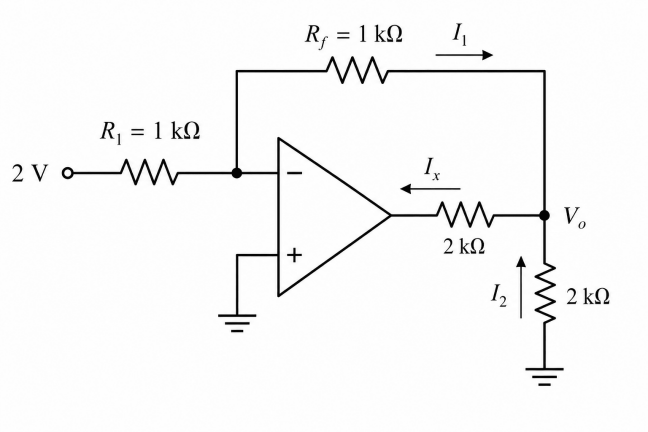

Fig. Solution Q3.26.2

Since the op-amp is connected in negative feedback and it is an inverting amplifier(since input is connected at inverting terminal),

$$V_0 = \frac{-R_f}{R_1} V_i = \frac{-1}{1} \times 2 = -2\ \text{V}$$

Applying KCL at V_0,

$$I_x = I_1 + I_2$$

$$I_x = \frac{0 - V_0}{1} + \frac{0 - V_0}{2}$$

$$I_x = \frac{2}{1} + \frac{2}{2} = 3\ \text{mA}$$

(C) 3.0 mA

Key Points

- **Virtual Ground:** In an ideal op-amp with the non-inverting terminal grounded, the voltage at the inverting terminal is forced to 0 V.
- **Inverting Amplifier Gain:** The output voltage V_0 at the feedback node is determined by the ratio of the feedback resistor to the input resistor: $V_0 = -\left(\frac{R_f}{R_{in}}\right) V_{in}$.

Q3.26.3 NAT 2026 2M Ans: 3

Method 1

Given circuit is,

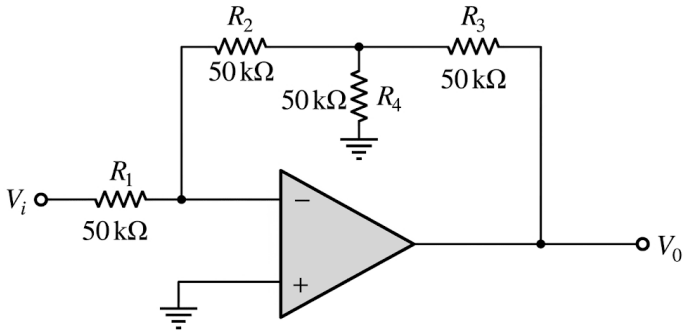

Fig. Solution Q3.26.3

Here, the feeback network is a T-network and its effective resistance is given by,

$$R_f = R_2 + R_3 + \frac{R_2 R_3}{R_4}$$

$$\therefore \text{ Closed loop gain } \frac{V_0}{V_i} = \frac{-R_f}{R_1} = \frac{-(R_2 + R_3 + \frac{R_2 R_3}{R_4})}{R_1}$$

$$\frac{V_0}{V_i} = \frac{-R_2}{R_1}\left[1 + \frac{R_3}{R_2} + \frac{R_3}{R_4}\right]$$

$$= \frac{-50\,\text{k}\Omega}{50\,\text{k}\Omega}\left[1 + \frac{50\,\text{k}\Omega}{50\,\text{k}\Omega} + \frac{50\,\text{k}\Omega}{50\,\text{k}\Omega}\right]$$

$$= -1[1 + 1 + 1] = -3$$

Hence,

$$\left|\frac{V_0}{V_i}\right| = 3$$

Method 2

The circuit is,

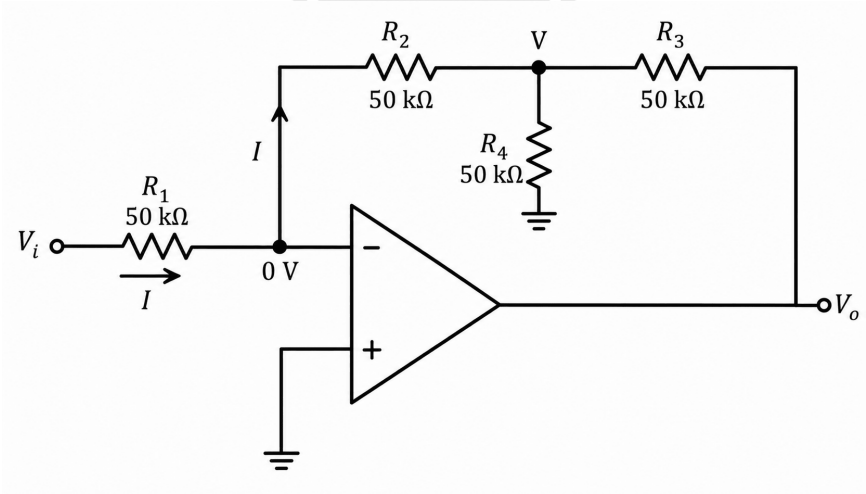

Fig. Solution Q3.26.3

Because of virtual ground due to negative feedback and ideal op-amp, $V_+ = V_- = 0\,\text{V}$
Since no current flows into input terminals of Op-Amp, entire I flows into R_2 and is given by,

$$I = \frac{V_i - 0}{50\,\text{k}} = \frac{V_i}{50k}$$

$$V = 0 - R_2 I = -50\,\text{k}\left(\frac{V_i}{50\,\text{k}}\right) = -V_i$$

Apply KCL at V,

$$\frac{V - 0}{50} + \frac{V}{50} + \frac{V - V_o}{50} = 0$$

$$V + V + V - V_o = 0$$

$$3V - V_o = 0$$

$$V_o = 3V \implies V_o = 3(-V_i)$$

$$\frac{V_o}{V_i} = -3$$

$$\boxed{\left|\frac{V_o}{V_i}\right| = 3}$$

Key Points

- The equivalent feedback resistance of the T-network formed by R_2, R_3, and R_4 is $R_{eq} = R_2 + R_3 + \frac{R_2 R_3}{R_4}$.

Q3.25.1 MSQ | 2025 | 1M | Ans: A,C | ● ○ ○

For an ideal op-amp:

(A) Infinite input resistance $\Longrightarrow$ TRUE

An ideal op-amp should not draw any current from the input terminals. Hence input current must be zero.

$$I_{in} = \frac{V_{in}}{R_{in}}$$

For finite input voltage and zero input current,

$$R_{in} \to \infty$$

(B) Infinite output resistance $\Longrightarrow$ FALSE

An ideal op-amp should behave like an ideal voltage source. An ideal voltage source has zero output resistance so that output voltage remains constant for any load current.

Hence output resistance is zero, not infinite.

(C) Infinite open-loop differential gain $\Longrightarrow$ TRUE

The output voltage of an op-amp is

$$V_o = A_d(V^+ - V^-)$$

For an ideal op-amp, even a very small differential input should produce a large output voltage.

Hence,

$$A_d \to \infty$$

(D) Infinite open-loop common-mode gain $\Longrightarrow$ FALSE

An ideal op-amp should completely reject common-mode signals.

Hence common-mode gain should be zero.

$$A_{cm} = 0$$

(A) Infinite input resistance
(C) Infinite open-loop differential gain

Q3.25.2 NAT | 2025 | 1M | Ans: 50 | ● ○ ○

Given circuit is shown below,

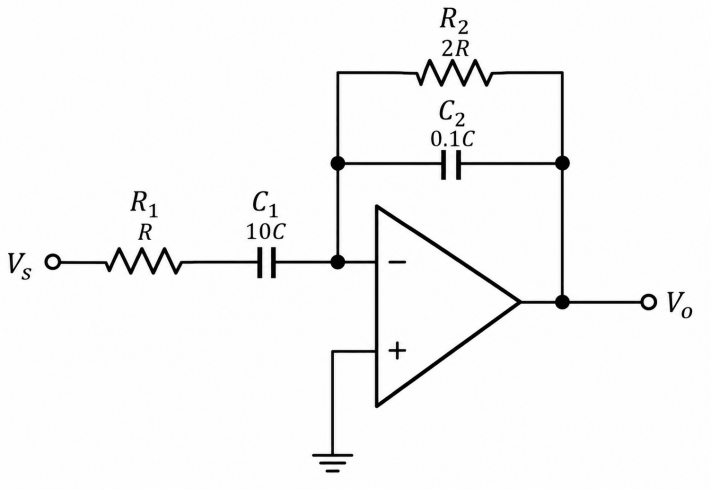

Fig. Solution Q3.25.2

Given, the lower 3-dB frequency, $f_L = 1$ MHz
At very low frequencies, capacitors act as open circuit, so when C_1 gets open circuited, the input is disconnected from the circuit. Hence R_1,C_1 are responsible for deciding lower cutoff frequency.
Similarly at high frequencies, capacitors act as short circuit, so when C_2 is short circuited, the feedback impedance becomes zero, and the gain falls(as gain $\propto (-Z_f/Z_1)$). Hence R_2,C_2 are responsible for deciding higher cutoff frequency.
The Lower 3-dB frequency f_L of a simple RC circuit is,

$$f_L = \frac{1}{2\pi R_1 C_1}$$

$$1 \times 10^6 = \frac{1}{2\pi R \times 10C}$$

$$2\pi \times 10^7 = \frac{1}{RC}$$

The upper 3-dB frequency f_H using component R_2 and C_2.

$$f_H = \frac{1}{2\pi R_2 C_2} = \frac{1}{2\pi \times 2R \times 0.1C} = \frac{1}{0.4\pi RC}$$

Substitute the value of $\frac{1}{RC}$,

$$f_H = \frac{2\pi \times 10^7}{0.4\pi} = 5 \times 10^7 = 50\,\text{MHz}$$

$$\boxed{f_H = 50\,\text{MHz}}$$

Key Points

- In an active bandpass filter formed by an op-amp with input and feedback RC networks, the lower cutoff frequency (f_L) is dictated by the input series RC branch (R_1, C_1).
- The upper cutoff frequency (f_H) is dictated by the feedback parallel RC branch (R_2, C_2).
- The cutoff frequency for each first-order section is determined by the standard formula $f = \frac{1}{2\pi RC}$.

Q3.24.1 MCQ | 2024 | 2M | Ans: D | ● ● ○ | CLICK HERE FOR VIDEO SOLUTION

Given circuit is

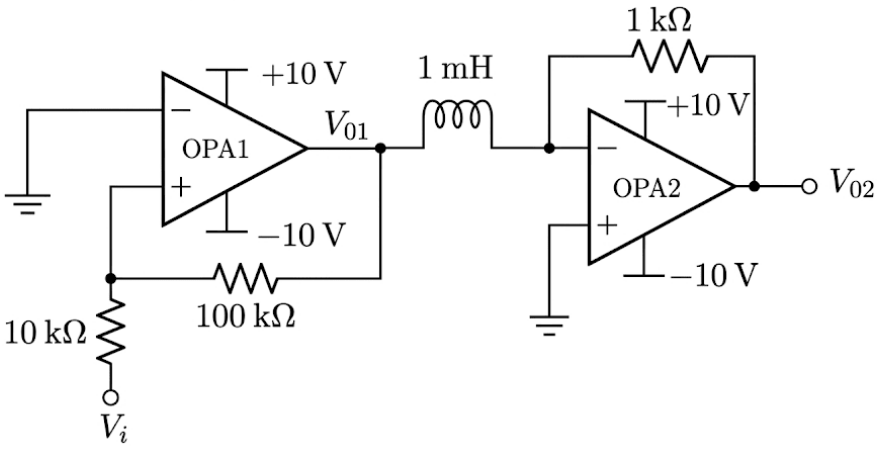

Fig. Solution Q3.24.1

The Op-Amp1 circuit is a non inverting Schmitt trigger circuit. V_{o1} can be either $+V_{\text{sat}}(+10\,\text{V})$ or $-V_{\text{sat}}(-10\,\text{V})$ hence it is a square wave.

(1) Let us assume $V_{o1} = +10\,\text{V}$

$$V_+ = \frac{V_{o1}(10\,\text{k}\Omega) + V_{\text{in}}(100\,\text{k}\Omega)}{110\,\text{k}\Omega}$$

$$V_+ = \frac{V_{o1}}{11} + \frac{10V_{\text{in}}}{11}$$

$$V_+ = \frac{10}{11} + \frac{10V_{\text{in}}}{11}$$

First we have assumed the output to be $+10\,\text{V}$. Now we have to write when it will go to $-V_{\text{sat}}(-10\,\text{V})$.

When $V_+ < V_- \Rightarrow V_+ < 0 \Rightarrow V_{o1} = -10\,\text{V}$

$$\frac{10}{11} + \frac{10V_{in}}{11} < 0$$

$$V_{in} < -1$$

So, when input voltage is less than -1 Volt, output will go from $+V_{sat}$ to $-V_{sat}$.

(2) Let us assume $V_{o1} = -10$ Volt

$$V_+ = \frac{V_{o1}}{11} + \frac{10V_{in}}{11}$$

When $V_+ > V_- \Rightarrow V_+ > 0 \Rightarrow V_{o1} = +10\,\text{V}$

$$\frac{-10}{11} + \frac{10V_{in}}{11} > 0$$

$$V_{in} > 1$$

So, when input voltage is grater than 1 Volt ,output will go from $-V_{sat}$ to $+V_{sat}$.

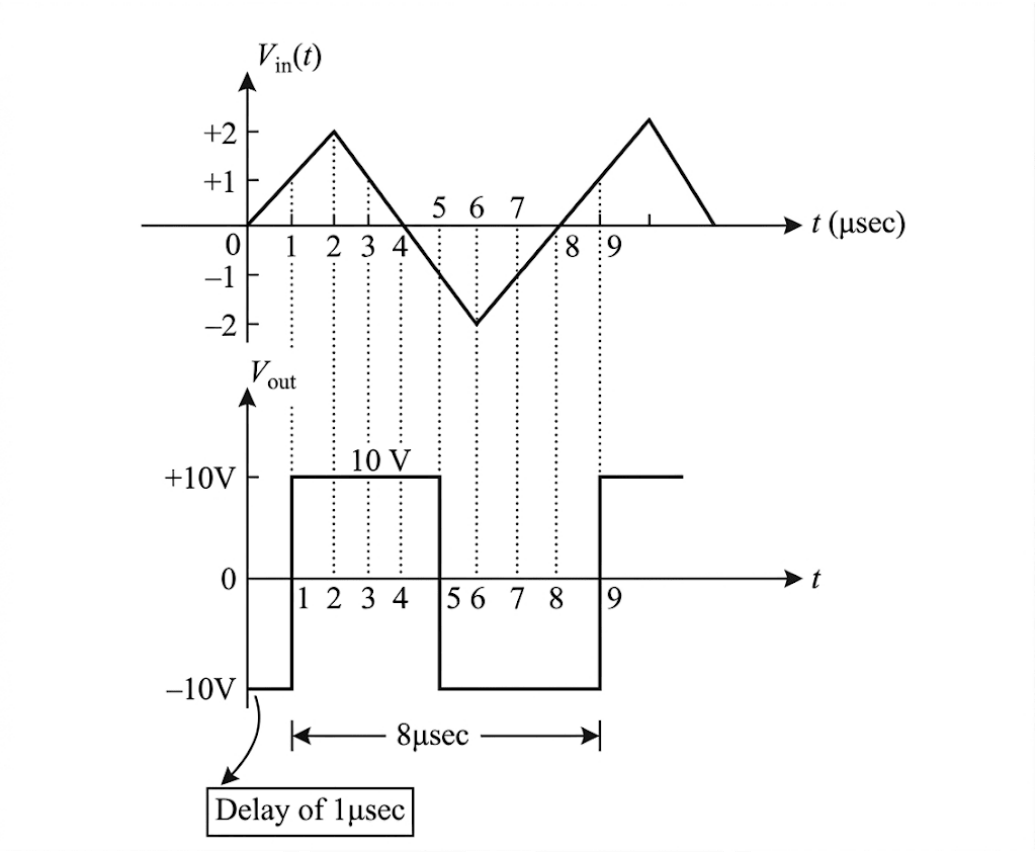

Fig. Solution Q3.24.1

The Op-Amp2 circuit is an integrator and we can write output (V_{O2}).

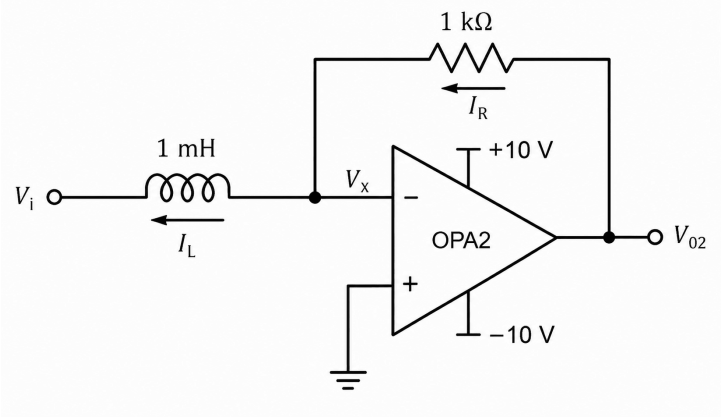

Fig. Solution Q3.24.1

For 0<t<1: $V_i = -10\text{V}$, $V_x = 0$(due to virtual short). The current through inductor is,

$$I_L = \frac{1}{L}\int V_L(t)dt = \frac{1}{1m}\int (0-(-10))dt = \frac{10t}{10^{-3}}$$

The same current flows through the resistor, and output voltage is ,

$$V_{02} = I_R(t) \times R = \frac{10t}{10^{-3}} \times 1K$$

$$V_{02} = 10^7 t \implies V_{02}(t = 1\mu s) = 10\text{ V}$$

So, Op-Amp has reached its saturation voltage which means the concept of virtual short is no longer valid as output is now independent of input. But this assumption is not valid in this case and virtual short concept is still applied.

Why virtual short concept is still applied?

We will analyze what happens if virtual short is not applied.

Since there is no virtual short V_x is no longer fixed at 0 and can vary.

$$I_L(t = 1\mu) = \frac{10 \times 1\mu}{10m} = 10mA \text{ and } V_x(t = 1\mu) = 0V$$

Now at $t = 1\mu s^+$,the input suddenly changes to +10V and the final steady state voltage at V_x will be $V_x(\infty) = 10V$(as the inductor at steady state is shorted and input voltage appears at V_x). Which means the voltage at V_x is rising in the positive direction. But the moment V_x crosses 0 V, the negative terminal voltage becomes grater than positive terminal voltage so Op-Amp output becomes $V_{02} = -10$Vi.e, the output should immediately switch from 10 V to -10 V.

Now, $V_x(1\mu^+) \approx 0$ V and $V_{02}(1\mu^+) = -10$ V. So the current in the resistor instantly reverses its direction but inductor current is still flowing towards left(as shown in figure) as current through inductor can't change instantly.

But this is not possible as 2 currents in the same branch are flowing in opposite directions(note that current through op-amp is always zero). Hence our initial assumption is incorrect and virtual short concept is still valid and output will not suddenly change to -10 V

For t>1 : The current through inductor after $1\mu s$ is ,

$$I_L(t) = 10mA - \frac{10(t - 1\mu s)}{1m} \quad t > 1\mu s$$

$$V_{02}(t) = I_R(t) \times 1K \qquad \because I_R = I_L$$

$$\text{at } t = 3\mu s \quad V_{02}(t) = -10 \text{ V}$$

Now the Op-Amp has reached saturation voltage again, and this time the concept of virtual short can't be applied for real. This time it is valid because,

For $t = 3\mu s^-$, $V_x = 0$ V and at $t = \infty$, $V_x = 10$ V, so V_x is increasing in positive direction so output of op-Amp should be at -10V(as $V_- > V_+$) and it is already at -10 V. So this time V_x can vary which means concept of virtual short is not applied.

So for $3\mu s < t < 5\mu s$, the output stays constant at -10 V which is shown in the waveform below. From the next input cycle onward the same analysis can be applied and virtual short concept isn't applied anymore.

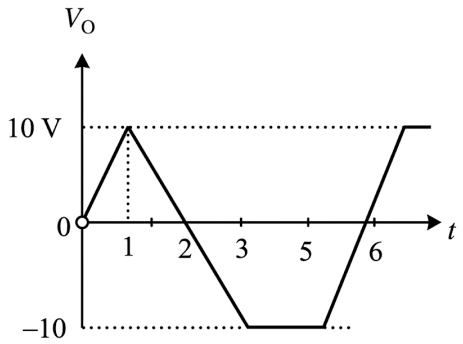

Fig. Solution Q3.24.1

Hence we can say that output is delayed by $1\mu s$ and V_{02} is a trapezoidal waveform.

(D) V_{01} is delayed by 1 μs relative to V_i, and V_{02} is a trapezoidal waveform.

Key Points

- A Schmitt Trigger utilizes positive feedback, causing the op-amp to operate in saturation mode ($+V_{sat}$ or $-V_{sat}$).

Mistakes to be avoided

- Applying the virtual ground concept ($V_+ = V_-$) to find steady-state node voltages. Virtual ground is only valid for negative feedback circuits in the linear region.
- Always check for saturation voltages of Op-Amp and note that output can't go beyond that.

Q3.24.2 MCQ | 2024 | 2M | Ans: B | ● ● ○

Given, circuit is shown below,

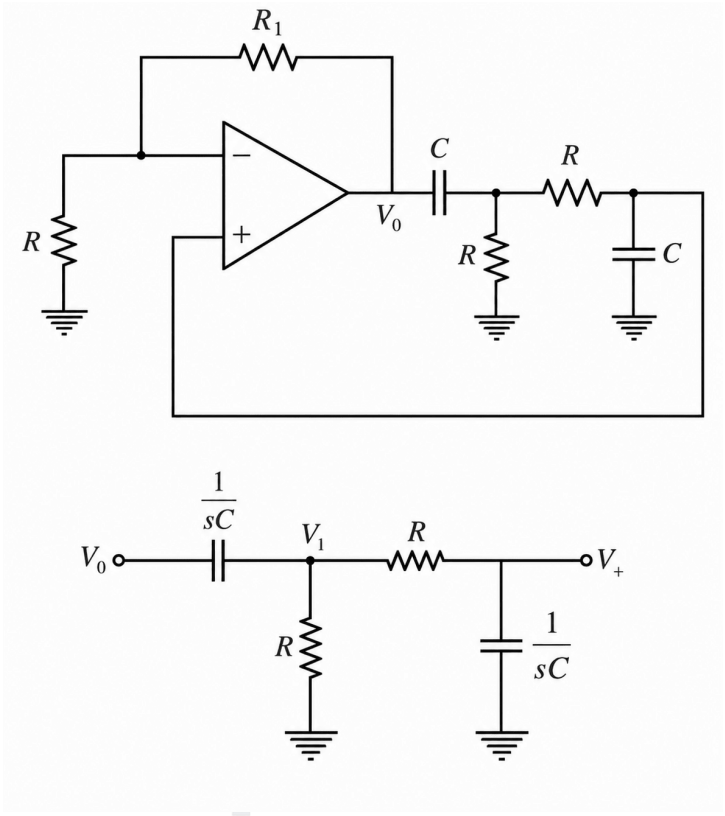

Fig. Solution Q3.24.2

Applying KCL at V_1,

$$\frac{V_1 - V_0}{\left(\frac{1}{sC}\right)} + \frac{V_1}{R} + \frac{V_1 - V_+}{R} = 0$$

$$sC(V_1 - V_0) + \frac{(2V_1 - V_+)}{R} = 0$$

$$sCR(V_1 - V_0) + 2V_1 - V_+ = 0$$

$$V_1(2 + sCR) - V_+ - sCRV_0 = 0 \quad \ldots (i)$$

Applying KCL at V_+,

$$\frac{V_+}{\left(\frac{1}{sC}\right)} + \frac{V_+ - V_1}{R} = 0$$

$$sC(V_+) + \frac{(V_+ - V_1)}{R} = 0$$

$$sCR(V_+) + V_+ - V_1 = 0$$

$$V_1 = V_+(1 + sCR) \quad \ldots (ii)$$

Put the value of V_1 in equation (i), we get

$$V_+(1 + sCR)(2 + sCR) - V_+ = sCRV_0$$

$$V_+((1 + sCR)(2 + sCR) - 1) = V_0 sCR$$

$$V_+ = \frac{V_0(sCR)}{[(1 + sCR)(2 + sCR)] - 1}$$

$$\frac{V_0}{V_+} = \frac{[(1 + sCR)(2 + sCR)] - 1}{sCR}$$

$$\frac{V_0}{V_+} = \frac{(1 + sCR)(2 + sCR) - 1}{sCR}$$

$$\frac{V_0}{V_+} = \frac{2 + sCR + 2sCR + s^2C^2R^2 - 1}{sCR}$$

$$\frac{V_0}{V_+} = \frac{2 + 3sCR + s^2C^2R^2 - 1}{sCR}$$

$$\frac{V_0}{V_+} = \frac{2}{j\omega CR} + 3 + j\omega CR - \frac{1}{j\omega CR}$$

$$\frac{V_0}{V_+} = \frac{1}{j\omega CR} + 3 + j\omega CR$$

$$\frac{V_0}{V_+} = \frac{1 + 3j\omega CR + j^2\omega^2C^2R^2}{j\omega CR}$$

$$\frac{V_0}{V_+} = \frac{1 + 3j\omega CR - \omega^2C^2R^2}{j\omega CR}$$

$$\frac{V_0}{V_+} = \left[\frac{1 - \omega^2C^2R^2}{j\omega CR} + 3\right]$$

The given circuit is non-inverting amplifier, so the voltage gain is,

$$\frac{V_0}{V_+} = \left(1 + \frac{R_1}{R}\right)$$

$$\frac{V_0}{V_+} = \left(\frac{1 - \omega^2C^2R^2}{j\omega CR} + 3\right) = \left(1 + \frac{R_1}{R}\right) \quad \ldots (iii)$$

For sustained oscillations the overall phase shift should be 360° so imaginary part must be zero (OR) in other words the R.H.S quantity is real so to make L.H.S real , we have to make imaginary part zero.

$$1 - \omega^2C^2R^2 = 0$$

$$\omega = \frac{1}{RC}$$

$$f = \frac{1}{2\pi RC}$$

From equation (iii). For sustained oscillation.

$$0 + 3 = 1 + \frac{R_1}{R}$$

$$\left(1 + \frac{R_1}{R}\right) = 3$$

$$\frac{R_1}{R} = 2$$

$$R_1 = 2R$$

(B) $2R$ and $\frac{1}{2\pi RC}$

Key Points

- **Barkhausen Criterion for Oscillation:** For sustained oscillations, the loop gain magnitude must be unity ($|A\beta| = 1$), and the total phase shift around the loop must be 0° or 360°.
- **Frequency of Oscillation:** The imaginary part of the closed-loop transfer function equation is equated to zero to ensure zero phase shift, yielding $\omega = \frac{1}{RC}$.
- **Gain Condition:** Equating the real parts determines the required non-inverting amplifier gain to overcome network attenuation, which resolves to $A \geq 3$ for this specific topology, enforcing $R_1 \geq 2R$.

Q3.23.1 MCQ | 2023 | 2M | Ans: A | ● ○ ○

Given circuit can be drawn as shown below,

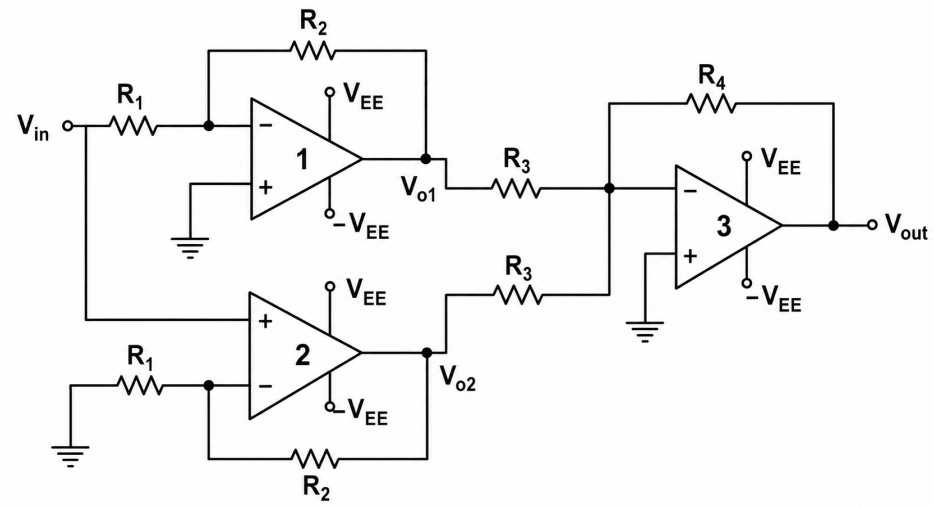

Fig. Solution Q3.23.1

From the above figure, the Op-Amp 1 is an inverting amplifier and Op-Amp 2 is a non inverting amplifier,

$$V_{o1} = -\frac{R_2}{R_1} \times V_{in} \quad \ldots (i)$$

$$V_{o2} = \left(1 + \frac{R_2}{R_1}\right) V_{in} \quad \ldots (ii)$$

Applying KCL at V_- of Op-Amp 3,($V_- = 0$ because of virtual short)

$$\frac{0 - V_{o1}}{R_3} + \frac{0 - V_{o2}}{R_3} + \frac{0 - V_{out}}{R_4} = 0$$

$$\frac{-V_{out}}{R_4} = \frac{V_{o1}}{R_3} + \frac{V_{o2}}{R_3}$$

$$V_{out} = -\frac{R_4}{R_3} V_{o1} - \frac{R_4}{R_3} V_{o2}$$

$$V_{out} = \frac{-R_4}{R_3} [V_{o1} + V_{o2}] \quad \ldots (iii)$$

Substituting equations (i) and (ii) in equation (iii), we get

$$V_{out} = \frac{-R_4}{R_3} \left[\frac{-R_2}{R_1} V_{in} + V_{in} + \frac{R_2}{R_1} V_{in} \right]$$

$$V_{out} = \frac{-R_4}{R_3} V_{in}$$

$$\frac{V_{out}}{V_{in}} = -\frac{R_4}{R_3}$$

$$\boxed{\text{(A)} \quad -\frac{R_4}{R_3}}$$

Key Points

- **Inverting Summer:** The rightmost op-amp acts as an inverting summing amplifier, combining the two intermediate signals according to the formula $V_{out} = -\left(\frac{R_4}{R_3} V_{o1} + \frac{R_4}{R_3} V_{o2}\right)$.

Q3.22.1 MSQ | 2022 | 1M | Ans: B,C | ● ○ ○

Given circuit is shown below,

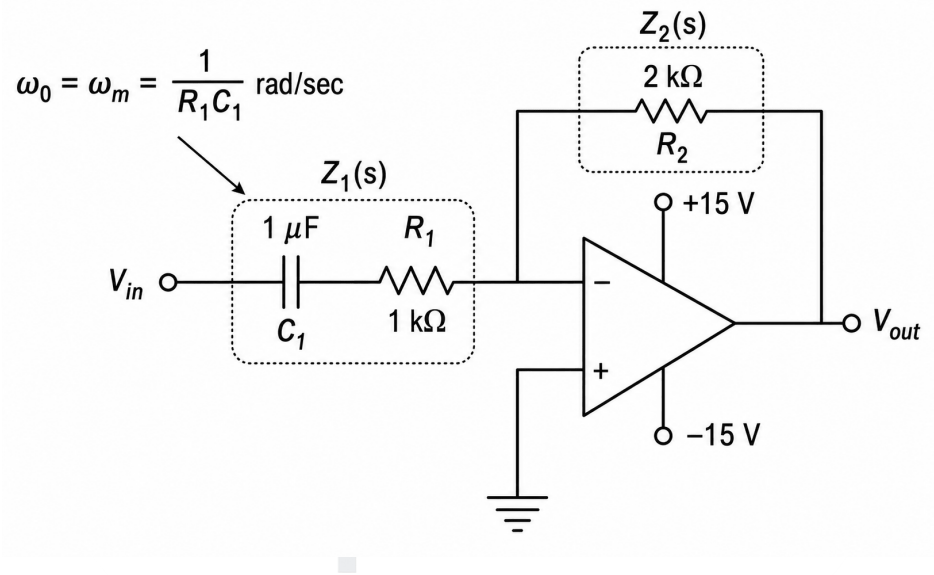

Fig. Solution Q3.22.1

The above circuit is a practical differentiator. The transfer function is given by,

$$\frac{V_{out}(s)}{V_{in}(s)} = \frac{-Z_2(s)}{Z_1(s)} = \frac{-R_2}{R_1 + \frac{1}{C_1 s}} = \frac{-sC_1R_2}{1 + sC_1R_1}$$

$$T(s) = \left|\frac{V_{out}(s)}{V_{in}(s)}\right| = \frac{Z_2(s)}{Z_1(s)} = \frac{sC_1R_2}{1 + sC_1R_1}$$

Put $s = j\omega$,

$$T(j\omega) = \frac{j\omega C_1R_2}{1 + j\omega C_1R_1}$$

$$\text{At } \omega = 0, T(0) = \frac{0}{1+0} = 0$$

$$\text{At } \omega = \infty, T(\infty) = \frac{jC_1R_2}{\frac{1}{\infty} + jC_1R_1} = \frac{R_2}{R_1}$$

So, circuit behave as HPF, because it passes only high frequency and block low frequency.
The 3 dB frequency of practical differentiator circuit is,

$$\omega_{3dB} = \frac{1}{R_1 C_1} = \frac{1}{1000 \times 1\,\mu F} = 1000\,\text{rad/sec}$$

(B) The circuit is a high pass filter.
(C) The 3 dB frequency is 1000 rad/s.

Key Points

- **Practical Differentiator:** The addition of R_1 in series with the input capacitor C_1 limits the high-frequency gain, effectively transforming the ideal differentiator into a high-pass filter.

Q3.22.2 MCQ 2022 2M Ans: C ●●○

The given circuit is an astable multivibrator/free running oscillator.

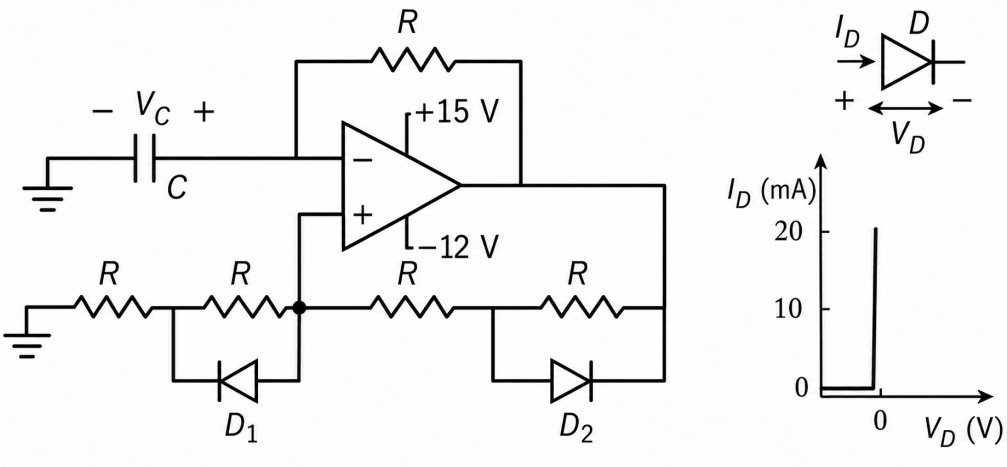

Fig. Solution Q3.22.2

Case 1 : When $V_0 = 15$ V, D_1 is ON and D_2 is OFF then above circuit becomes as,

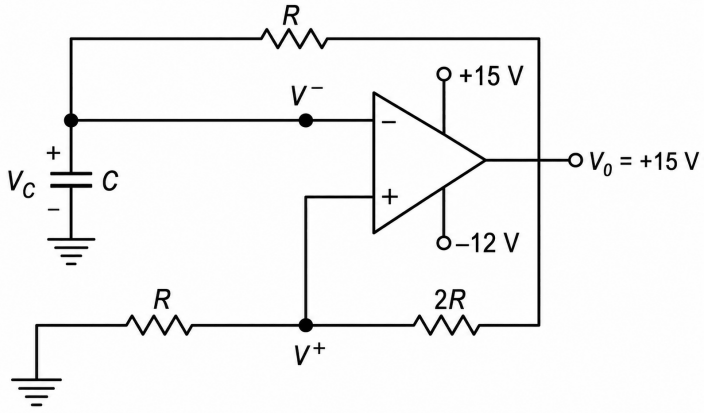

Fig. Solution Q3.22.2

Apply voltage division rule to calculate V^+,

$$V^+ = \frac{R \times V_0}{2R + R} = \frac{15 \times R}{3R} = \frac{15}{3} = 5\,\text{V}$$

Maximum voltage across capacitor is (i.e. capacitor charged upto $V_{UTP} = V^+$),

$$[V_c(t)]_{\max} = 5\,\text{V} = V_{UTP}$$

Case 2 : When $V_0 = -12\,\text{V}$, D_1 is OFF and D_2 is ON, then above circuit becomes as,

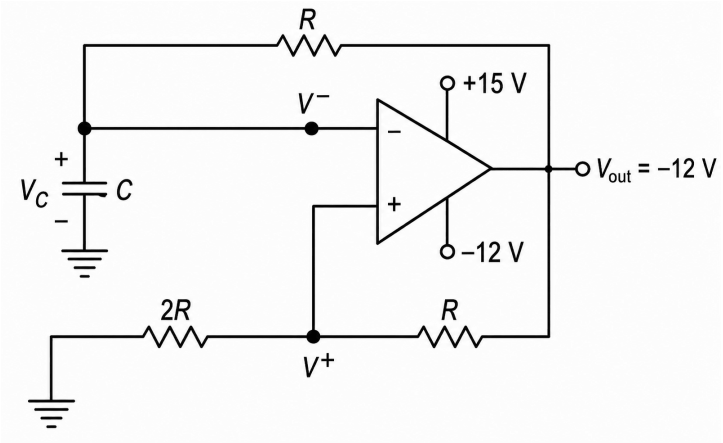

Fig. Solution Q3.22.2

Applying voltage division rule to calculate V^+, So,

$$V^+ = V_{out} \times \frac{2R}{2R + R} = \frac{-12}{3} \times 2 = -8\,\text{V}$$

Minimum capacitor voltage is (i.e. capacitor charged upto $V_{LTP} = V^+$),

$$[V_c(t)]_{\min} = -8\,\text{V} = V_{LTP}$$

Thus,

$$V_{c\,\max} - V_{c\,\min} = 5 - (-8) = 13\,\text{V}$$

(C) 13 V

Key Points

- **Astable Multivibrator:** A circuit using an operational amplifier with positive feedback that continuously toggles between its positive and negative saturation states without external triggering.
- **Threshold Voltages:** The upper threshold point (V_{UTP}) and lower threshold point (V_{LTP}) establish the boundaries for the capacitor's voltage limits.

Q3.22.3 MCQ 2022 2M Ans: A ● ○ ○

The given Op-Amp circuit is shown below,

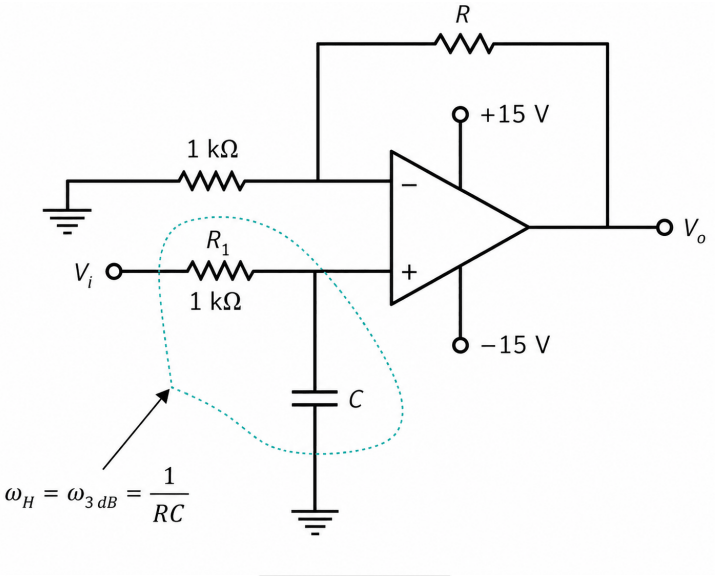

Fig. Solution Q3.22.3

The given Op-Amp circuit is first order active low pass filter, so low frequency ($\omega = 0$) gain of given low pass filter is,

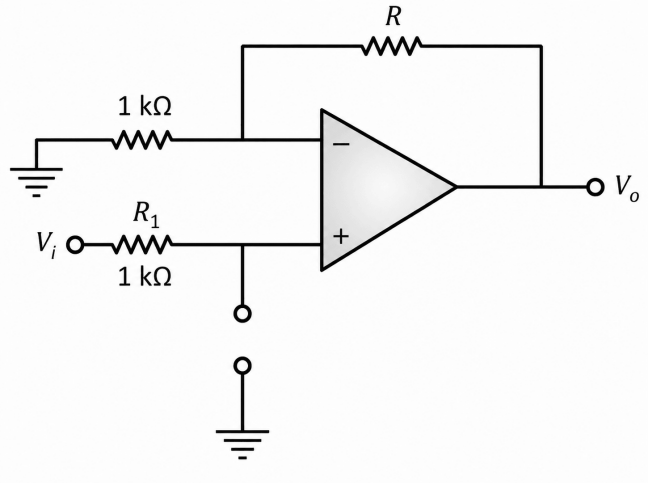

Fig. Solution Q3.22.3

The above Op-Amp circuit becomes as non-inverting closed loop Op-Amp, so gain (A_V) is,

$$A_V = \left[1 + \frac{R}{1}\right] = 1 + R \quad \ldots (i)$$

The 3-dB cut-off frequency of given low pass filter circuit is,

$$\omega_{3dB} = \frac{1}{R_1 C} \text{ rad/sec} \quad \ldots (ii)$$

According to given bode magnitude plot,

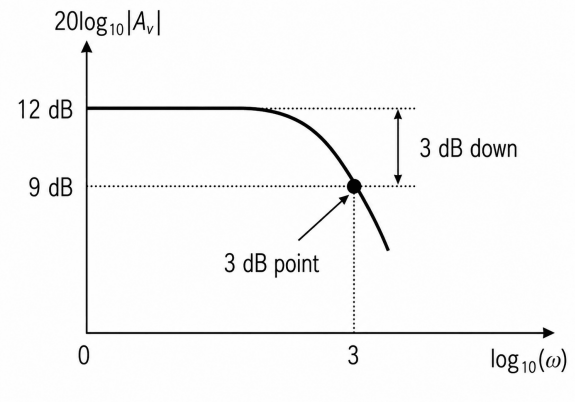

Fig. Solution Q3.22.3

Low frequency ($\omega = 0$) gain from bode magnitude plot is,

$$12\,\text{dB} = 20\log_{10}(A_V)$$

$$A_V = 10^{\left[\frac{12}{20}\right]}$$

$$1 + R = 4$$

$$R = 3\,\text{k}\Omega$$

The 3-dB frequency from bode magnitude plot,

$$\log_{10}\omega_{3dB} = 3$$

$$\omega_{3dB} = 10^3$$

$$\omega_{3dB} = 1000\,\text{rad/sec}$$

Thus,

$$\omega_{3dB} = \frac{1}{R_1 C} = 1000$$

$$C = \frac{1}{1000R_1} = \frac{1}{1000 \times 1000} = 1\,\mu\text{F}$$

$$\boxed{\text{(A)} \quad R = 3\,\text{k}\Omega,\ C = 1\,\mu\text{F}}$$

Key Points

- **Active Low-Pass Filter:** The circuit consists of a passive RC low-pass network at the input of a non-inverting operational amplifier, isolating the filter from the load and providing voltage gain.
- **DC Gain Evaluation:** At low frequencies ($\omega \to 0$), the capacitor acts as an open circuit. The gain is solely determined by the non-inverting amplifier configuration: $A_V = 1 + \frac{R_f}{R_i}$.
- **Bode Plot Analysis:** The flat low-frequency magnitude gives the DC gain in decibels ($20\log_{10}|A_V|$), while the corner frequency (where the gain falls by 3 dB) identifies the pole of the RC network ($\omega_{3dB} = \frac{1}{RC}$).

Q3.21.1 MCQ | 2021 | 2M | Ans: MTA | ● ○ ○

For this inverting connection with a fixed reference, let the non-inverting input voltage be V_x and $r = R_F/R_{IN} > 0$. In linear operation,

$$V_{OUT} = (1 + r)V_x - rV_{IN}.$$

Subtracting the two given operating points eliminates V_x:

$$6 - 1 = -r(1 - 0.1), \qquad r = -50/9.$$

No positive resistor ratio can satisfy the stated positive slope. The ideal op-amp load does not reverse that conclusion. The official 2021Q26 key is MTA (marks to all), confirmed in the archived paper/key; the question carries 2 marks. **Conclusion:** No listed option satisfies the ideal inverting-amplifier equation with positive resistor values. The official key awards marks to all; discarding the minus sign does not yield a valid resistor ratio.

Q3.21.2 MCQ | 2021 | 1M | Ans: A | ● ○ ○

Given circuit is shown below,

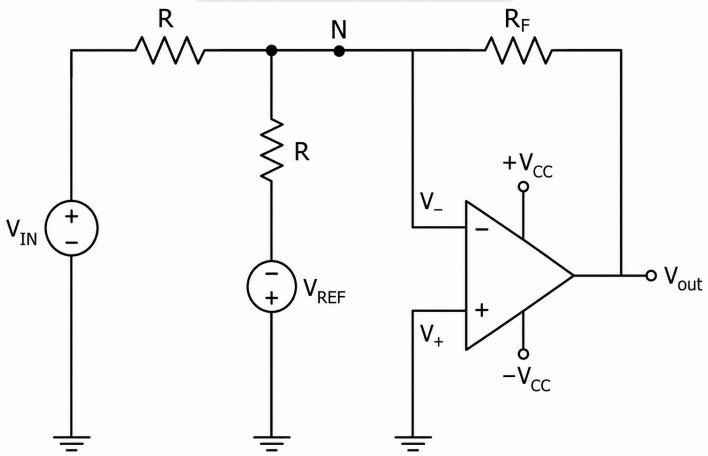

Fig. Solution Q3.21.2

Here, Op-Amp is ideal, negative feedback is present and according to question $|V_{IN}| \ll |V_{CC}|$ and $|V_{REF}| \ll |V_{CC}|$, so Op-Amp will not saturate and hence virtual ground concept is applicable. So,

$$V_+ = V_- = 0$$

Thus applying KCL at node N,

$$\frac{V_- - V_{IN}}{R} + \frac{V_- + V_{REF}}{R} + \frac{V_- - V_{out}}{R_F} = 0$$

$$\frac{0 - V_{IN}}{R} + \frac{0 + V_{REF}}{R} + \frac{0 - V_{out}}{R_F} = 0$$

$$\frac{V_{out}}{R_F} = \frac{V_{REF} - V_{IN}}{R}$$

$$V_{out} = \frac{R_F}{R}[V_{REF} - V_{IN}]$$

According to question, $V_{out} = 0$

$$0 = \frac{R_F}{R}[V_{REF} - V_{IN}]$$

$$V_{REF} - V_{IN} = 0$$

$$V_{REF} = V_{IN}$$

(A) $V_{IN} = V_{REF}$

Q3.21.3 NAT | 2021 | 2M | Ans: -12 | ● ● ○ | CLICK HERE FOR VIDEO SOLUTION

Since both capacitors in the given circuit are in parallel,the circuit can be redrawn as

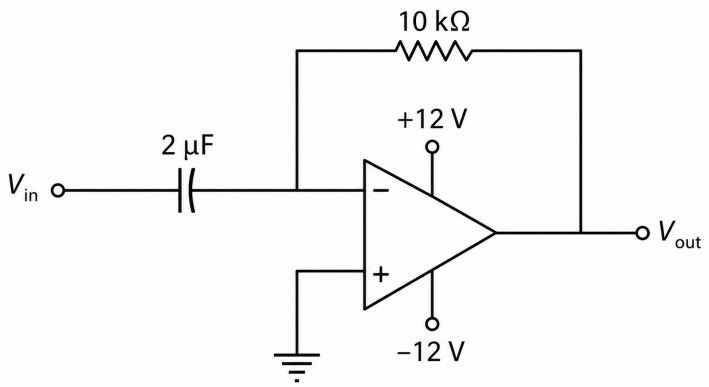

Fig. Solution Q3.21.3

As capacitors are initially uncharged i.e.,

$$V_C(0^-) = 0\,\text{V}$$

Therefore, at $t = 0^+$, capacitor will be short circuit as $V_C(0^-) = V_C(0^+)$. So, above circuit can be redrawn as,

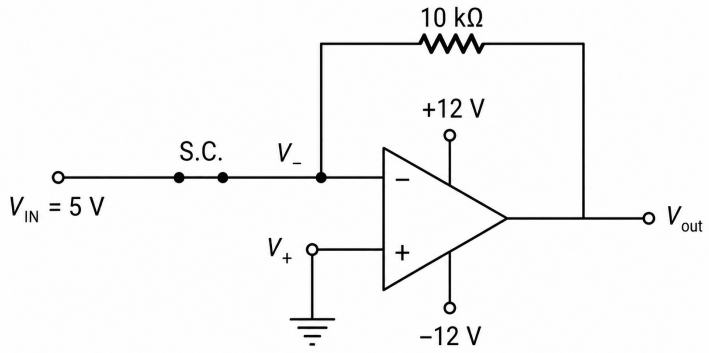

Fig. Solution Q3.21.3

From above circuit it is clear that,

$$V_- = V_{IN} = +5\,\text{V}$$

$$V_+ = 0\,\text{V}$$

Here, $V_- \neq V_+$ It means, voltages at both terminals of op-amp are unequal and fixed so virtual ground concept is not valid here and op-amp will work as comparator.

Thus, $V_- > V_+$. So, $V_{out} = -V_{sat} = -12\,\text{V}$
Hence, the output voltage V_{out} of this circuit at $t = 0^+$ is $\mathbf{-12}\,\text{V}$.

$$\boxed{V_{out}(t = 0^+) = -12\ \text{V}}$$

For t>0:
Since the output is saturated, virtual short concept is no longer valid. The circuit is given by,

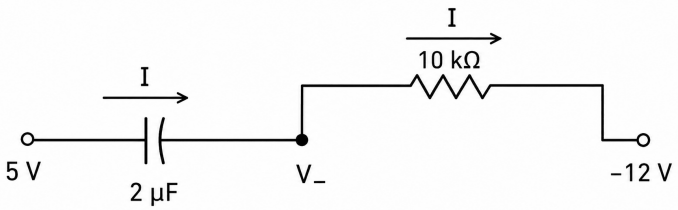

Fig. Solution Q3.21.3

Now V_- is no longer fixed at 0 and can vary. The circuit is a simple RC circuit and

$$V_-(0^+) = 5\ \text{V}$$

and at $t \to \infty$, the capacitor is fully charged and is open circuited, so

$$V_-(\infty) = -12\ \text{V}$$

So, as the capacitor is charging the V_- voltage falls and when it crosses 0 V(say it became -0.1 V), V_+ becomes grater than V_- and the output immediately switches to +12 V.
Now as $V_{out} = +12$ V, the new target of V_- becomes +12 V i.e, $V_-(\infty) = 12$ V. So V_- again starts increasing from 0 V towards +12 V. But when it again crosses 0 V(say it became +0.1 V) the output again switches to -12 V and this cycle repeats and ultimately V_- is stuck at 0 V which means all 5 V is dropped across capacitor so it is open circuited and no current flows through the circuit. So $V_{out} = 0$ V.

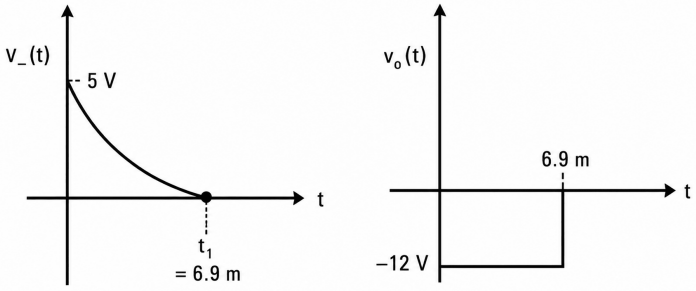

Fig. Solution Q3.21.3

So, at $t = t_1$, $V_-(t) = 0$ and $V_{out} = 0$ we can find t_1 as follows.

$$V_-(t) = V_-(\infty) + (V_-(0) - V_-(\infty))e^{\frac{-t}{RC}}$$

$$0 = -12 + (5 + 12)e^{\frac{-t_1}{20m}} \qquad \because V_-(t_1) = 0$$

$$t_1 = 6.9\text{ms}$$

Now, the same analysis can be applied for $t > 20$ ms as the input goes to 0 V.

Key Points

- When an op-amp is driven with a large differential input that causes it to saturate, it no longer operates in its linear region, and the virtual ground assumption ($V_+ = V_-$) becomes invalid.
- Without virtual ground, the op-amp acts as a comparator. If the inverting terminal voltage (V_-) is greater than the non-inverting terminal voltage (V_+), the output swings to the negative saturation voltage ($-V_{sat}$).

Q3.20.1 MCQ | 2020 | 1M | Ans: D | ● ○ ○

Given circuit is a non-inverting Schmitt trigger

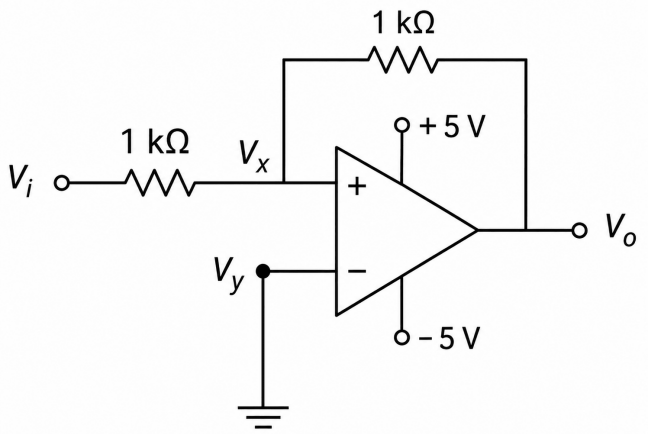

Fig. Solution Q3.20.1

From given figure, $V_y = 0$

$$V_x = \frac{V_i \times 1 + V_o \times 1}{1 + 1}$$

$$V_x = \frac{V_i + V_o}{2}$$

If $V_x > V_y$ then $V_o = +V_{sat}$ and If $V_x < V_y$ then $V_o = -V_{sat}$

Assume $V_o = +5$ V, then

$$\frac{V_i + V_o}{2} > 0$$

$$\frac{V_i + 5}{2} > 0$$

$$V_i > -5$$

so, if $V_i > -5$ then V_0 switches from -5 V to +5 V.

Assume $V_0 = -5$ V, then

$$\frac{V_i + V_o}{2} < 0$$

$$\frac{V_i - 5}{2} < 0$$

$$V_i < 5$$

So, if $V_i < 5$ then V_0 will switch from +5 V to -5 V.

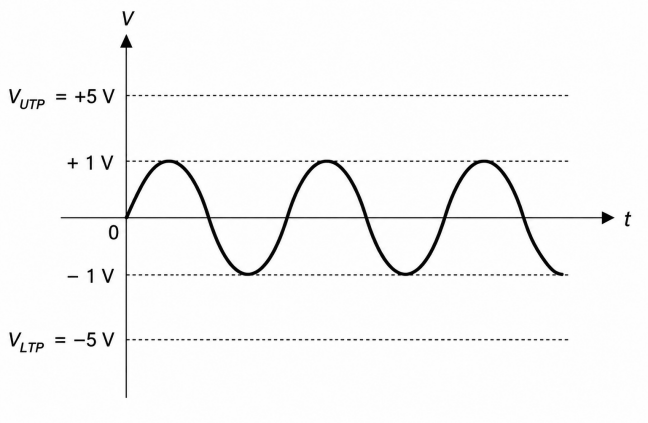

Fig. Solution Q3.20.1

Here, input voltage V_i is not crossing either lower and upper threshold voltage so output will not change and remain in its initial state of either +5 V or −5 V depending on how input arrived initially. Therefore, output of given Schmitt trigger is constant of either +5 V or −5 V

(D) constant of either + 5 V or − 5 V

Key Points

- In a schmitt trigger, the transition of output states requires the input voltage to exceed the upper threshold point (V_{UTP}) or drop below the lower threshold point (V_{LTP}).
- If the input signal's peak amplitude (1 V) is smaller than the magnitude of the threshold voltages (±5 V), the op-amp will not toggle and remains saturated at its initial state (either +5 V or −5 V).

Q3.20.2 NAT 2020 1M Ans: 6

Given Op-Amp circuit is shown in below figure,

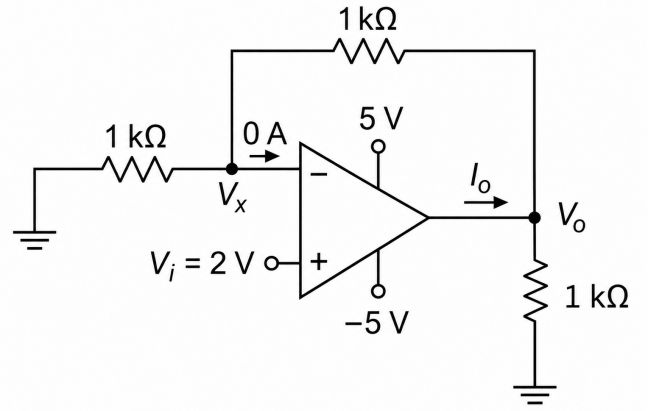

Fig. Solution Q3.20.2

Since this is a -ve feedback Op-Amp circuit. So we can apply virtual grounded concept. So, $V_x = V_i = 2$ V.
Applying KCL at node V_x,

$$\frac{V_x - 0}{1\text{ k}} + \frac{V_x - V_o}{1\text{ k}} + 0 = 0$$

$$\frac{2 - 0}{1\text{ k}} + \frac{2 - V_o}{1\text{ k}} = 0$$

$$V_o = 4\,\text{V}$$

Applying KCL at node V_o,

$$\frac{V_o - V_x}{1\,\text{k}} + \frac{V_o - 0}{1\,\text{k}} - I_o = 0$$

$$\frac{4-2}{1\,\text{k}} + \frac{4}{1\,\text{k}} = I_o$$

$$I_o = 6\,\text{mA}$$

$$\boxed{I_0 = 6\,\text{mA}}$$

Q3.20.3 MCQ 2020 2M Ans: D ●○○

Given circuit is shown in below figure,

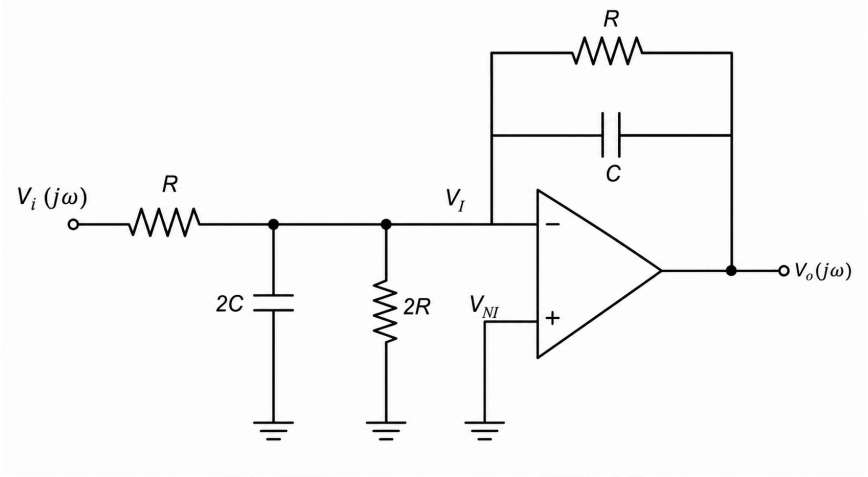

Fig. Solution Q3.20.3

This is an inverting integrator circuit. From virtual ground concept,

$$V_I = V_{N_I} = 0$$

Applying KCL at V_I,

$$\frac{V_I - 0}{2R} + \frac{V_I - 0}{\frac{1}{2sC}} + \frac{V_I - V_i}{R} + \frac{V_I - V_0}{\frac{1}{sC}} + \frac{V_I - V_0}{R} = 0$$

Putting $V_I = 0$,

$$\frac{0 - V_i}{R} + \frac{0 - V_0}{\frac{1}{sC}} + \frac{0 - V_0}{R} = 0$$

$$\frac{-V_i}{R} = \frac{V_0(1 + sRC)}{R}$$

$$V_0 = \frac{-V_i}{(1 + sRC)}$$

$$\frac{V_0}{V_i} = H(s) = \frac{-1}{1 + sRC}$$

Pole at $s = \frac{-1}{RC}$, so

$$\omega_H = \frac{1}{RC}$$

$$f_H = \frac{1}{2\pi RC} = \frac{1}{2\pi \times 2 \times 10^3 \times 10^{-6}}$$

$$f_H = 79.58\,\text{Hz}$$

(D) 79.58

Key Points

- **Cut-off Frequency:** For a single-pole system with the transfer function $H(s) = \frac{A}{1+sRC}$, the −3 dB cut-off frequency is determined by the pole location, which gives $f_H = \frac{1}{2\pi RC}$.

Q3.18.1 MCQ | 2018 | 1M | Ans: C

Given : Op-Amp and zener are ideal and $V_Z = 2.5\,\text{V}, V_{C_1}(0^-) = V_{C_2}(0^-) = 0$

(i) For $0 < t < 2.5$ ms :

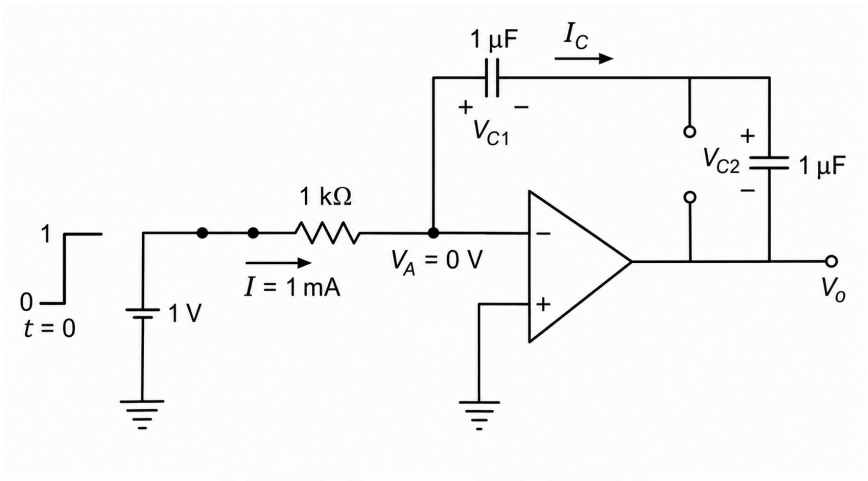

Fig. Solution Q3.18.1

At $t = 0$, the initial voltages across capacitors are zero. Hence the zener diode is open circuited. Now when the voltage source is connected, current I starts flowing in the circuit and same current flows through the capacitors and charges them. When the voltage across C_2 is grater than 2.5 V, then the zener diode goes into breakdown region.

Applying KCL at node A,

$$I_C = I = \frac{1-0}{1\,\text{k}\Omega} = 1\,\text{mA}$$

From figure current through both the capacitors is constant and hence the voltage across capacitors is linear. Hence voltage across C_1 and C_2 for $t > 0$ are given as,

$$V_{C_1}(t) = \frac{1}{C_1}\int_0^t I_C\,dt = 1000t \qquad 0 < t < 1$$

$$V_{C_2}(t) = \frac{1}{C_2}\int_0^t I_C\,dt = 1000t \qquad 0 < t < 1$$

Applying KVL equation starting from point A and ending at output V_0 as,

$$-V_A + V_{C_1} + V_{C_2} + V_0 = 0 \quad [\text{where } V_A = 0]$$

$$V_0 = -(V_{C_1} + V_{C_2}) \quad \ldots(\text{i})$$

At $t = 2.5$ msec

$$V_{C_1}(t) = V_{C_2}(t) = 1000 \times 2.5 \times 10^{-3} = 2.5\,\text{V}$$

(ii) For $t > 2.5$ msec :

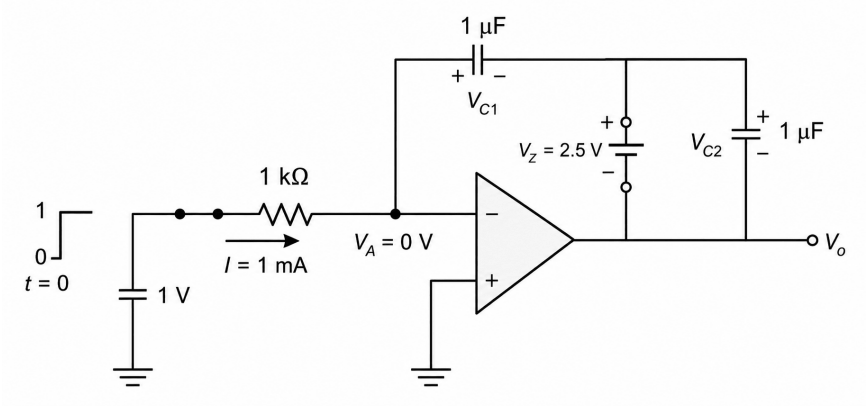

Fig. Solution Q3.18.1

At $t = 2.5$ msec, breakdown of zener occurs and $V_{C_2}(t)$ is now fixed at 2.5 volts.
Voltage across C_1 will further increases linearly with time from 2.5 volts.

$$V_0 = -(V_{C_1} + V_{C_2})$$

$$V_0 = -(V_{C_1} + V_Z)$$

$$[\text{As after } t = 2.5\,\text{msec}, V_{C_2} = V_Z]$$

$$-10 = -(V_{C_1} + 2.5)$$

$$V_{C_1} = 7.5\,\text{V}$$

Hence, voltage across C_1 will be 7.5 volts when output crosses -10 V.
As C_1 is charging linearly with a rate of 1000 V/sec, hence time taken by it to get charged upto 7.5 V is $t = 7.5$ msec

(C) 7.5

Key Points

- The voltage across a linearly charging capacitor is given by $V_C(t) = \frac{1}{C}\int I_C dt$, producing a linear ramp $V_C(t) = \left(\frac{I}{C}\right)t$.

Q3.18.2 NAT 2018 2M Ans: 0.5 ●●○

Given, Op-amp is ideal and saturation value of output voltage is ±15 V.

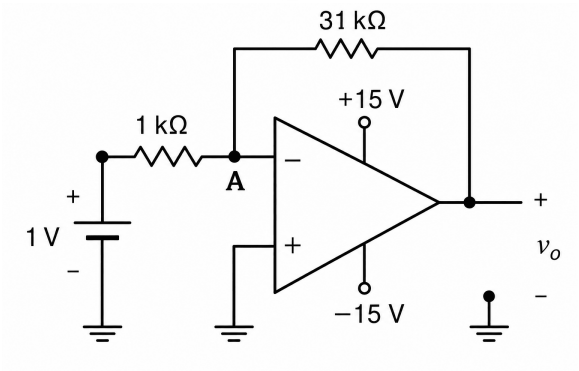

Fig. Solution Q3.18.2

We have to find potential at node A, which seems to be 0 V according to virtual ground. But the virtual ground concept is valid only if output is not saturated.
The above circuit is an inverting amplifier and its output is,

$$V_0 = -\frac{R_f}{R} \times V_{in}$$

$$V_0 = -\frac{31}{1} \times 1 = -31\text{ V}$$

From figure maximum value of output

$$V_0 \text{ max} = \pm V_{sat} = \pm 15\text{ V}$$

As $|V_0| > |V_{sat}|$ Therefore, virtual ground concept is not valid and $V_A \neq 0$ and output will not rise more than -15 V and gets fixed there.

$$V_0 = -15\text{ V}$$

Applying KCL at node A,

$$\frac{V_A - 1}{1} + \frac{V_A - V_0}{31} = 0$$

$$31(V_A - 1) + \{V_A - (-15)\} = 0$$

$$32V_A = 31 - 15$$

$$V_A = \frac{16}{32} = 0.5\text{ V}$$

$$\boxed{V_A = 0.5\text{ V}}$$

Key Points

- **Virtual Ground Validity:** The virtual ground assumption ($V_+ = V_-$) is only valid when the op-amp operates within its linear region. If the theoretical output voltage exceeds the power supply limits, the op-amp saturates.
- **Saturation Behavior:** Once saturated, the op-amp output is clamped to the saturation limit ($\pm V_{sat}$), and the negative feedback loop can no longer maintain the virtual short between input terminals.

Mistakes to be avoided

- Do not directly assume virtual ground and make $V_A = 0$.

Q3.17.1 MCQ | 2017 | 1M | Ans: B | ● ○ ○

Given, $V_{sat} = +15\text{ V}$, $-V_{sat} = -15\text{ V}$ and given circuit is shown below,

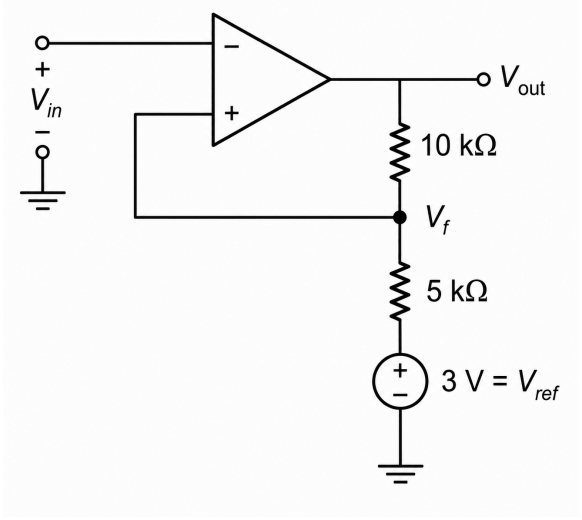

Fig. Solution Q3.17.1

The given circuit is a schmitt trigger circuit.

(i) Calculation of upper threshold voltage (V_{UT}) :

If $V_{out} = +V_{sat} = 15\,\text{V}$ then $V_f = V_{UT}$

Applying KCL at V_f,

$$\frac{V_f - V_0}{10} + \frac{V_f - V_{ref}}{5} = 0$$

$$\frac{V_f - 15}{10} + \frac{V_f - 3}{5} = 0$$

$$V_f - 15 + 2V_f - 6 = 0 \implies 3V_f = 21$$

$$V_f = V_{UT} = 7\,\text{V}$$

(ii) Calculation of lower threshold voltage (V_{LT}) :

If $V_{out} = -V_{sat} = -15\,\text{V}$ then $V_f = V_{LT}$

Applying KCL at V_f,

$$\frac{V_f - V_0}{10} + \frac{V_f - V_{ref}}{5} = 0$$

$$\frac{V_f + 15}{10} + \frac{V_f - 3}{5} = 0$$

$$V_f + 15 + 2V_f - 6 = 0 \implies 3V_f = -9$$

$$V_f = V_{LT} = -3\,\text{V}$$

(B) +7 V and -3 V

Q3.17.2 NAT 2017 2M Ans: 44.37 ● ○ ○

Given, $A_0 = 10^5$ V/V, $f_c = BW = 8$ Hz and given circuit is shown below,

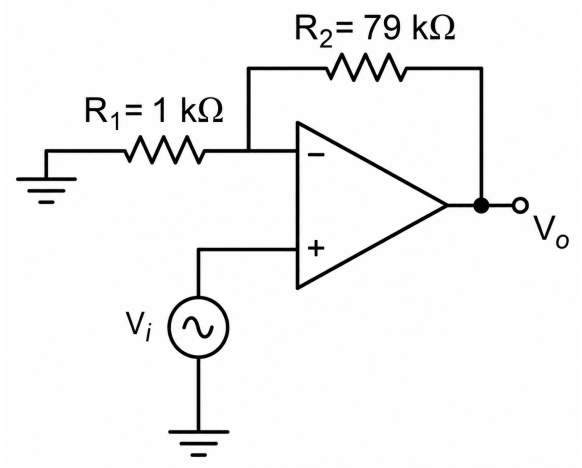

Fig. Solution Q3.17.2

For practical non-inverting Op-Amp, voltage gain is given by,

$$\frac{V_0}{V_i} = A_f = \frac{A_{OL}}{1 + A_{OL}\beta}$$

where $A_{OL} = 10^5$ is the open loop gain and β is the feedback factor.
Due to high open loop gain, we can assume ideal Op-Amp. Hence, for ideal non-inverting Op-Amp voltage gain is given by,

$$\frac{V_0}{V_i} = A_f = 1 + \frac{R_2}{R_1}$$

$$A_f = 1 + \frac{79}{1} = 80$$

Concept : Gain bandwidth product of closed loop system = Gain bandwidth product of open loop system

$$A_f \times BW' = A_{OL} \times BW$$

$$80 \times BW' = 10^5 \times 8$$

$$BW' = 10\,\text{kHz}$$

Now, a compensated op-amp is an op-amp which has been intentionally modified to make it stable when negative feedback is applied. This is usually done by adding a dominant compensation capacitor inside the op-amp. So the op-amp now behaves like a first order low pass system whose gain is,

$$A'_f = \frac{A_f}{\sqrt{1 + \left(\frac{f}{BW'}\right)^2}}$$

Hence, gain of amplifier at 15 kHz is,

$$A'_f = \frac{80}{\sqrt{1 + \left(\frac{15}{10}\right)^2}} = 44.37$$

Hence, the voltage gain of the amplifier at 15 kHz, in V/V is 44.37.

Voltage gain = 44.37

Key Points

- For a compensated operational amplifier, the product of the open-loop gain (A_{OL}) and the open-loop cut-off frequency (BW) is constant and equals the product of the closed-loop midband gain (A_f) and the closed-loop cut-off frequency (BW'). $A_{OL} \times BW = A_f \times BW'$.
- The op-amp acts as a first-order low-pass filter. Its gain magnitude at any frequency f is given by $A(f) = \frac{A_{vo}}{\sqrt{1+(\frac{f}{BW'})^2}}$, where A_{v_o} is the DC closed-loop gain.

Q3.16.1 MCQ 2016 1M Ans: A ●○○

This is a Wien-bridge oscillator. The frequency-selective positive-feedback network selects the oscillation frequency; the diode/resistor network controls amplitude through the negative-feedback gain. At small amplitudes both antiparallel diodes are OFF and the non-inverting gain is approximately $1 + 22.1/10 = 3.21$, allowing startup. As the voltage across the diode branch grows, one diode conducts on each polarity. This reduces the effective feedback resistance and the amplifier gain toward the value 3 needed for sustained sinusoidal oscillation. The purpose is to limit the amplitude before hard rail saturation, not to alternate between the supply rails to make a square wave.

(A) Amplitude stabilization

This is nonlinear amplitude regulation; exact waveform purity depends on the real diode and op-amp characteristics.

Q3.16.2 NAT | 2016 | 2M | Ans: 15 | ● ○ ○

Given,

1. $+V_{CC} = +15\,\text{V},\ -V_{CC} = -15\,\text{V}$
2. $V_1, V_3, V_5 \ldots V_{N-1} = 1, \frac{1}{3}, \frac{1}{5} \ldots$
3. $V_2, V_4, V_6 \ldots V_N = \frac{-1}{2}, \frac{-1}{4}, \frac{-1}{6} \ldots$

Given circuit is shown below,

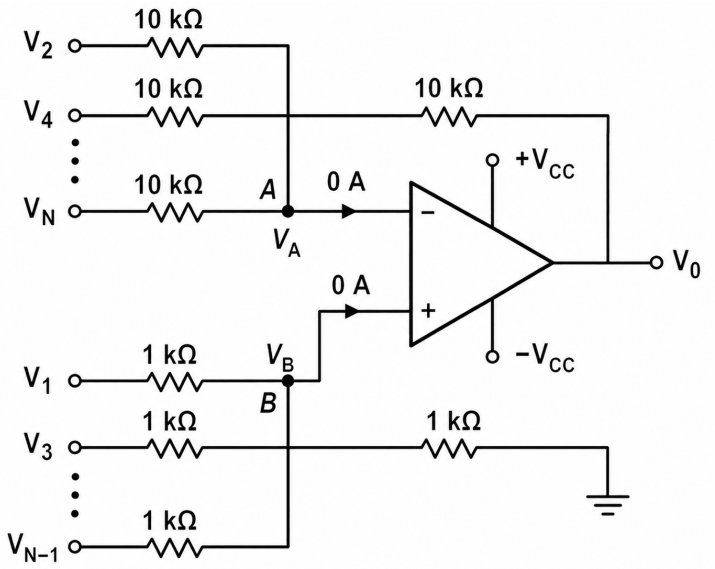

Fig. Solution Q3.16.2

Applying KCL at node A,

$$\frac{V_A - V_2}{10} + \frac{V_A - V_4}{10} + \cdots + \frac{V_A - V_N}{10} + \frac{V_A - V_0}{10} = 0$$

$$\left(V_A \times \frac{N}{2}\right) - (V_2 + V_4 + \cdots + V_N) + V_A - V_0 = 0$$

$$V_A \times \left(\frac{N}{2} + 1\right) - (V_2 + V_4 + \cdots + V_N) = V_0 \quad \ldots (i)$$

Applying KCL at node B,

$$\frac{V_B - V_1}{1} + \frac{V_B - V_3}{1} + \cdots + \frac{V_B - V_{N-1}}{1} + \frac{V_B}{1} = 0$$

$$\left(V_B \times \frac{N}{2}\right) - (V_1 + V_3 + \cdots + V_{N-1}) + V_B = 0$$

$$V_B \times \left(\frac{N}{2} + 1\right) = V_1 + V_3 + \cdots + V_{N-1} \quad \ldots (ii)$$

From virtual ground concept, V_+ (i.e. V_B) = V_- (i.e. V_A) So,

$$V_B \times \left(\frac{N}{2} + 1\right) = V_A \times \left(\frac{N}{2} + 1\right) \quad \ldots (iii)$$

From equation (i), (ii) and (iii),

$$(V_1 + V_3 + \cdots + V_{N-1}) + (-V_2 - V_4 - \cdots - V_N) = V_0$$

$$\left(1+\frac{1}{3}+\frac{1}{5}+\cdots+\frac{1}{N-1}\right)+\left(\frac{1}{2}+\frac{1}{4}+\cdots+\frac{1}{N}\right)=V_0$$

$$V_0=1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\frac{1}{5}+\cdots+\frac{1}{N}$$

Now, as $N \to \infty$, number of term in the expression of V_0 in above equation will become infinite. So, $V_0 = \infty$
Now because V_0 becomes more than V_{sat} i.e. V_{CC}, hence Op-Amp will enter into saturation and hence,

$$V_0 = V_{sat} = +V_{CC} = +15\,\text{V}$$

$$\boxed{V_0 = 15\,\text{V}}$$

Key Points

- **Harmonic Series Divergence:** The derived output voltage takes the form of a harmonic series $(1+\frac{1}{2}+\frac{1}{3}+\cdots+\frac{1}{N})$. A fundamental mathematical property of the harmonic series is that it diverges to infinity as $N \to \infty$.

Q3.16.3 NAT | 2016 | 2M | Ans: 0.798 | ● ● ○

Given circuit is shown below,

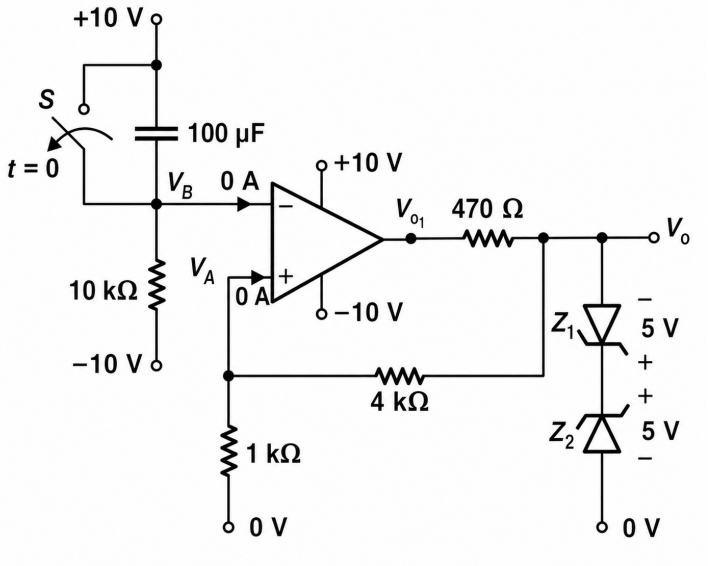

Fig. Solution Q3.16.3

Given figure represent astable multivibrator (positive feedback circuit).
Op-Amp output is given by,

$$V_{o1} = \pm V_{sat} = \pm 10\,\text{V}$$

Circuit output, $V_0 = \pm 5\,\text{V}$ [Due to back to back connected Zener diodes]

V_A and V_B	V_{o1}	Z_1	Z_2	V_0
$V_B > V_A$ ($V_- > V_+$)	$-V_{sat} = -10\,\text{V}$	Break down (Reverse bias)	ON (Forward bias)	$-5\,\text{V}$
$V_B < V_A$ ($V_- < V_+$)	$+V_{sat} = +10\,\text{V}$	ON (Forward bias)	Break down (Reverse bias)	$+5\,\text{V}$

Initially switch is closed i.e. initial voltage across capacitor is 0 V (uncharged capacitor).

At $t < 0$ / $t = 0^-$:

$$V_C(0^-) = 0\,\text{V}$$

$$V_B(0^-) = 10 - V_C(0^-) = 10\,\text{V}$$

If $V_0 = 5$ V, i.e. Z_1 is ON, and Z_2 is in breakdown region. The upper trigger point voltage is given by,

$$V_{UTP} = V_A = \frac{1}{1+4} \times 5 = 1\,\text{V} \quad \text{[By VDR]}$$

If $V_0 = -5$ V, i.e. Z_1 is in breakdown region, and Z_2 is ON. The lower trigger point voltage is given by,

$$V_{LTP} = V_A = \frac{1}{1+4} \times (-5) = -1\,\text{V} \quad \text{[By VDR]}$$

At $t = 0$, S is opened and at $t = \infty$, capacitor becomes in steady state, hence it will act as open circuit.

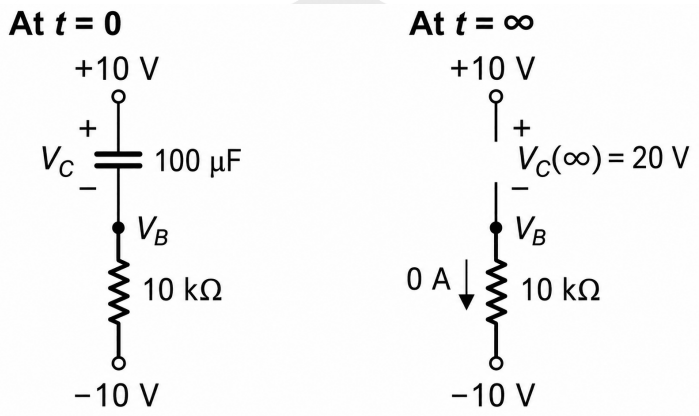

Fig. Solution Q3.16.3

Capacitor voltage is given by,

$$V_C(t) = V_C(\infty) + \left[V_C(0^+) - V_C(\infty)\right] e^{-t/\tau}$$

(i) $V_C(0^-) = V_C(0^+) = 0\,\text{V}$

(ii) $\tau = RC = 10 \times 10^3 \times 100 \times 10^{-6} = 1\,\text{sec}$

(iii) $V_C(\infty) = 20\,\text{V}$

$$V_C(t) = 20(1 - e^{-t}) \quad \ldots\text{(i)}$$

From figure,

$$V_B(t) = 10 - V_C(t) = 10 - (20 - 20e^{-t})$$

$$V_B(t) = -10 + 20e^{-t} \quad \ldots\text{(ii)}$$

At $t = 0$, $V_B = 10$ V, V_A may be -1 V or $+1$ V.

Therefore, $V_B > V_A$ and hence

$$V_{o1} = -V_{sat} = -10\,\text{V}$$

Then, $V_0 = -5$ V,

$$V_A = V_{LTP} = -1\,\text{V}$$

$$V_{o1} = +V_{sat}$$

Hence, $V_0 = +5$ V

So, V_0 changes from -5 V to $+5$ V at the time instant t_1 when $V_B(t)$ is reduced to -1 V.

Method 1

$$-10 + 20e^{-t_1} = -1 \quad \text{[From equation (ii)]}$$

$$e^{-t_1} = \frac{9}{20}$$

$$e^{t_1} = \frac{20}{9}$$

$$t_1 = \ln\left(\frac{20}{9}\right) = 0.798 \text{ sec}$$

Method 2

Calculation of t_1 from capacitor voltage : From figure,

$$10 = V_C(t) + V_B(t)$$

$$V_B(t) = 10 - V_C(t)$$

V_0 changes from -5 V to 5 V at $t = t_1$.
When $V_B(t) = -1$ V then $V_C(t) = 11$

$$20 - 20e^{-t_1} = 11 \quad \text{[From equation (ii)]}$$

$$20e^{-t_1} = 9$$

$$e^{-t_1} = \frac{9}{20}$$

$$e^{t_1} = \frac{20}{9}$$

$$t_1 = \ln\left(\frac{20}{9}\right) = 0.798 \text{ sec}$$

The time at which V_0 changes state is 0.798V

Key Points

- **Astable Multivibrator:** An op-amp circuit with both positive and negative feedback that continuously toggles between two unstable states, generating a square or rectangular wave output.
- **Zener Diode Clamping:** Back-to-back Zener diodes limit (clamp) the output voltage to a specific range (e.g., ±5 V) by undergoing Zener breakdown in one direction and forward-biasing in the other.

Q3.16.4 NAT 2016 2M Ans: 400 to 425 ●○○

The drawn offset source adds to the 25 mV input, so the effective non-inverting voltage is 30 mV. The resistive divider gives $V_- = V_o/16$. With finite open-loop gain 100,

$$V_o = 100(30 - V_o/16),$$

where all voltages in this equation are mV. Hence

$$V_o = \frac{48000}{116} = 413.7931 \text{ mV}.$$

$$V_o \approx 413.8 \text{ mV}$$

Q3.16.5 NAT | 2016 | 2M | Ans: -1 | ● ○ ○

Given :

(i) $R_1 = R_2 = R_3 = 1\,\Omega$

(ii) $L = 1\,\mu\text{H},\ C = 1\,\mu\text{F}$

(iii) Input voltage, $V_{in} = \cos(10^6 t)$

Transforming the domain by replacing inductors and capacitors with their impedance, the circuit can be redrawn as,

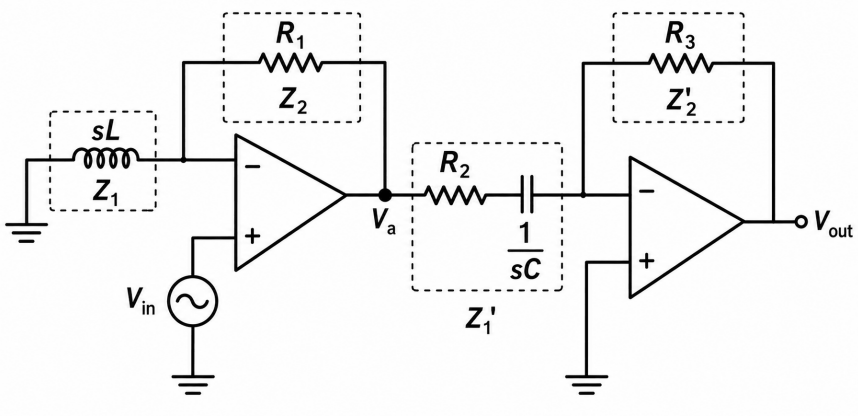

Fig. Solution Q3.16.5

Let gain of first stage is $A_1(s)$ and gain of second stage is $A_2(s)$.

Transfer function of non-inverting Op-Amp is given by,

$$A_1(s) = \frac{V_a}{V_{in}} = 1 + \frac{Z_2}{Z_1} = 1 + \frac{R_1}{Ls}$$

$$A_1(s) = 1 + \frac{1}{10^{-6}s}$$

Transfer function of inverting Op-Amp is given by,

$$A_2(s) = \frac{V_{out}}{V_a} = \frac{-Z_2'}{Z_1'} = \frac{-R_3}{R_2 + \frac{1}{Cs}}$$

$$A_2(s) = \frac{-R_3 Cs}{1 + sR_2C}$$

$$A_2(s) = \frac{-10^{-6}s}{1 + 10^{-6}s}$$

Overall gain is given by,

$$\frac{V_{out}}{V_{in}} = \frac{V_{out}}{V_a} \times \frac{V_a}{V_{in}}$$

$$A(s) = A_1(s) \times A_2(s)$$

$$A(s) = \left(1 + \frac{1}{s \times 10^{-6}}\right)\left(\frac{-s \times 10^{-6}}{1 + s \times 10^{-6}}\right) = -1$$

Hence, the overall voltage gain (V_{out}/V_{in}) of the circuit is -1.

$$\boxed{\frac{V_{out}}{V_{in}} = -1}$$

Key Points

- The overall transfer function of cascaded non-loading op-amp stages is the product of their individual transfer functions: $A(s) = A_1(s) \times A_2(s)$.
- **S-Domain Analysis:** For circuits with energy storage elements (L, C), replace them with their s-domain impedances $(Z_L = Ls, Z_C = \frac{1}{Cs})$ to simplify the calculation of transfer functions algebraically.

Q3.15.1 NAT 2015 2M Ans: 250 ● ○ ○

Given circuit is shown below,

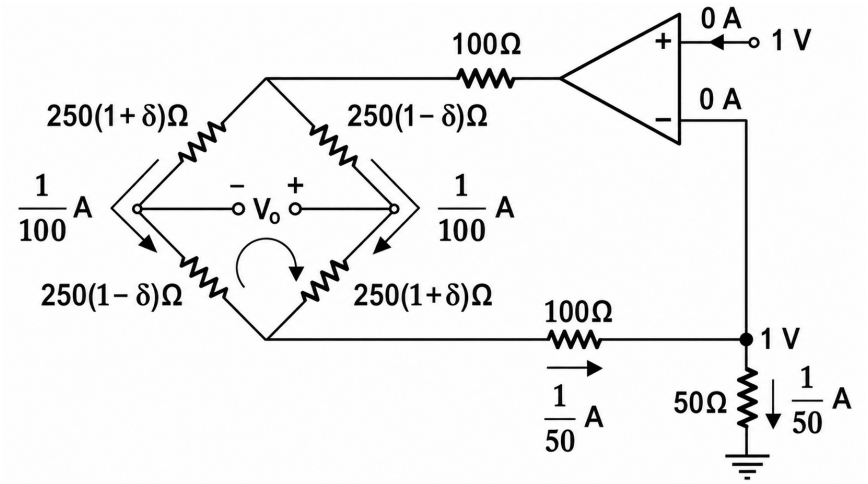

Fig. Solution Q3.15.1

From virtual ground concept,

$$V_+ = V_- = 1\,\text{V}$$

Since op-amp terminals draw no current, same current flows through 100 Ω and 50 Ω resistors.

$$I_{50\Omega} = I_{100\Omega} = \frac{1}{50}\,\text{A}$$

Applying KVL in the loop shown in figure,

$$-V_0 + \frac{1}{100}[250(1+\delta) - 250(1-\delta)] = 0$$

$$V_0 = \frac{1}{100}[250(1+\delta) - 250(1-\delta)]$$

$$V_0 = \frac{1}{100} \times 500\delta$$

$$V_0 = 5\delta = 5 \times 0.05 = 0.25\,\text{V}$$

$$V_0 = 250\,\text{mV}$$

The bridge output voltage V_0 is 250 mV

.

Key Points

- **Bridge Circuit Symmetry:** Due to the symmetry of the bridge resistances $(250(1+\delta)+250(1-\delta)=500\,\Omega$ on both branches), the total current entering the bottom node divides equally between the two branches.

Q3.15.2 MCQ 2015 2M Ans: D ● ○ ○

Given circuit is,

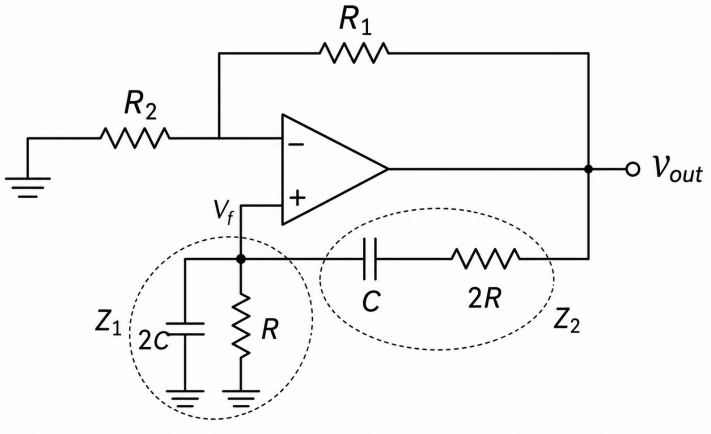

Fig. Solution Q3.15.2

The circuit is a Wein bridge oscillator. It consists of amplifier part and a frequency selective feedback network. The amplifier is a non-inverting amplifier whose gain is,

$$A=\frac{V_{out}}{V_f}=1+\frac{R_1}{R_2}$$

The feedback factor is given by,

$$\beta(s)=\frac{V_f}{V_{out}}=\frac{Z_1}{Z_1+Z_2}$$

where,

$$Z_1=R\parallel\frac{1}{2sC}=\frac{R\cdot\frac{1}{2sC}}{R+\frac{1}{2sC}}=\frac{R}{1+2sRC}$$

$$Z_2=2R+\frac{1}{sC}=\frac{1+2sRC}{sC}$$

$$\beta(s)=\frac{\frac{R}{1+2sRC}}{\frac{R}{1+2sRC}+\frac{1+2sRC}{sC}}$$

$$\beta(s)=\frac{RsC}{RsC+(1+2sRC)^2}$$

$$\beta(s)=\frac{sRC}{sRC+(1+2sRC)^2}$$

$$\beta(s)=\frac{sRC}{sRC+1+4sRC+4s^2R^2C^2}$$

$$\beta(s)=\frac{sRC}{1+5sRC+4s^2R^2C^2}$$

Put $s=j\omega$,

$$\beta(j\omega)=\frac{j\omega CR}{1+5j\omega RC+4(j\omega)^2R^2C^2}$$

$$\beta(j\omega) = \frac{j\omega CR}{1 - 4\omega^2R^2C^2 + j5\omega RC}$$

From Barkhausen criteria, the total phase shift should be zero, for which β should have zero phase. Hence

$$1 - 4\omega^2R^2C^2 = 0$$

$$4\omega^2R^2C^2 = 1$$

$$\omega^2 = \frac{1}{4R^2C^2}$$

$$\omega_0 = \frac{1}{2RC}$$

From Bark Hausen Criterion, loop gain should be 1. So, $A\beta = 1$, we get

$$A\left(\frac{1}{5}\right) = 1 \qquad \because \text{ at } \omega = \omega_0, \beta(\omega_0) = 1/5$$

$$1 + \frac{R_1}{R_2} = 5 \implies R_1 = 4R_2$$

(D) $\omega_0 = \dfrac{1}{2RC}, \qquad R_1 = 4R_2$

Q3.15.3 NAT | 2015 | 1M | Ans: 1.5 | ● ○ ○

When switch is in position A :

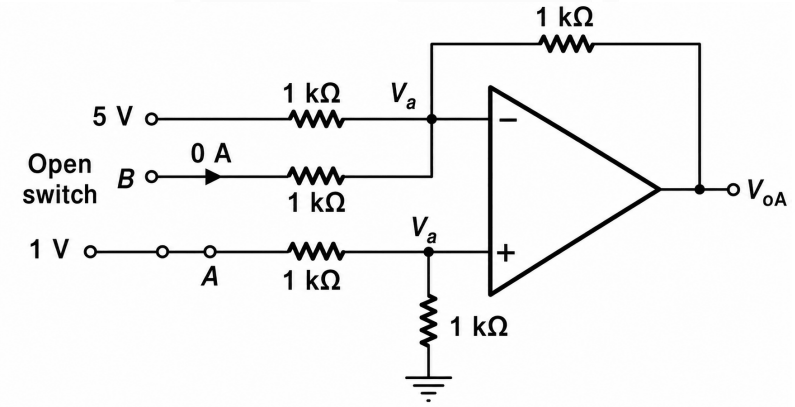

Fig. Solution Q3.15.3

From virtual ground concept,

$$V_+ = V_- = V_a$$

Applying KCL at inverting terminal,

$$\frac{V_a - 5}{1} + \frac{V_a - V_{0A}}{1} = 0$$

$$V_{0A} = 2V_a - 5$$

Apply voltage division rule at non-inverting terminal,

$$V_a = \frac{1}{1+1} \times 1 = 0.5\,\text{V}$$

$$\therefore V_{0A} = 2 \times 0.5 - 5 = -4\,\text{V}$$

When switch is at position B :

Non-inverting input, $V_a = 0\,\text{V}$ and due to virtual short inverting terminal voltage also becomes 0.

Applying KCL at inverting terminal,

$$\frac{0-1}{1} + \frac{0-5}{1} + \frac{0-V_{0B}}{1} = 0$$

$$-1 - 5 - V_{0B} = 0$$

$$V_{0B} = -6\,\text{V}$$

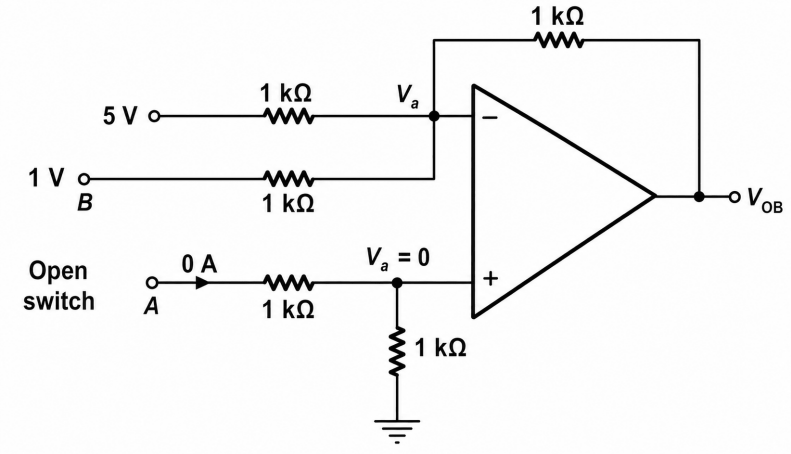

Fig. Solution Q3.15.3

Therefore,

$$\frac{V_{0B}}{V_{0A}} = \frac{-6}{-4} = 1.5$$

$$\boxed{\frac{V_{0B}}{V_{0A}} = 1.5}$$

Q3.15.4 NAT | 2015 | 1M | Ans: 1 | ● ○ ○

Given,

(i) Saturation level = $\pm V_{sat} = \pm 5\,\text{V}$

(ii) Hysteresis width, $V_H = 500\,\text{mV}$

Given circuit is,

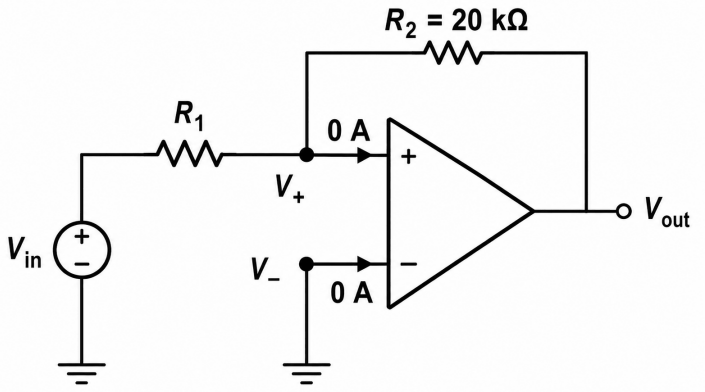

Fig. Solution Q3.15.4

The circuit is a non inverting Schmitt trigger circuit and, op-amp with positive feedback does not follow virtual ground concept.

By using superposition theorem, voltage V_+ is given by,

$$V_+ = \frac{V_{in} \times R_2}{R_1 + R_2}\bigg|_{V_{out}=0} + \frac{V_{out} \times R_1}{R_1 + R_2}\bigg|_{V_{in}=0}$$

Case 1 : When $V_{out} = +V_{sat}$, Then

$$V_+ = \frac{V_{in}R_2}{R_1 + R_2} + \frac{V_{sat}R_1}{R_1 + R_2}$$

$V_{out} = +V_{sat}$ if $V_+ > 0$ i.e.

$$\frac{V_{in}R_2}{R_1 + R_2} + \frac{V_{sat}R_1}{R_1 + R_2} > 0$$

$$V_{in} > -V_{sat}\frac{R_1}{R_2} \quad \ldots (i)$$

Case 2 : When $V_{out} = -V_{sat}$ Then

$$V_+ = \frac{V_{in}R_2}{R_1 + R_2} - \frac{V_{sat}R_1}{R_1 + R_2}$$

$V_{out} = -V_{sat}$ if $V_+ < 0$

$$\frac{V_{in}R_2}{R_1 + R_2} - \frac{V_{sat}R_1}{R_1 + R_2} < 0$$

$$V_{in} < \frac{V_{sat}R_1}{R_2} \quad \ldots (ii)$$

Transfer characteristics is shown below,

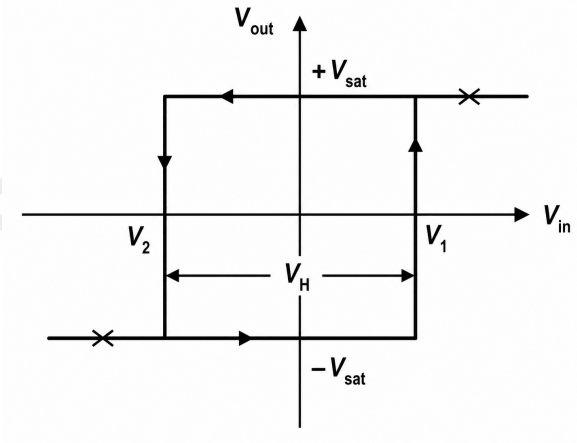

Fig. Solution Q3.15.4

where, $V_1 = \frac{V_{sat}R_1}{R_2}$ and $V_2 = \frac{-V_{sat}R_1}{R_2}$

Hysteresis voltage or width is given by,

$$V_H = V_1 - V_2 = \frac{V_{sat}R_1}{R_2} - \left(\frac{-V_{sat}R_1}{R_2}\right)$$

$$V_H = 2V_{sat}\frac{R_1}{R_2}$$

$$0.5 = 2 \times 5 \times \frac{R_1}{20}$$

$$\boxed{R_1 = 1\,\text{k}\Omega}$$

Key Points

- **Hysteresis Width (V_H):** The voltage difference between the upper and lower threshold levels (V_{UTP} and V_{LTP}). For this specific configuration, it is determined by the ratio of the feedback resistors: $V_H = 2V_{sat}\left(\frac{R_1}{R_2}\right)$.

Q3.15.5 NAT 2015 2M Ans: 12 ● ○ ○

Given circuit is shown below,

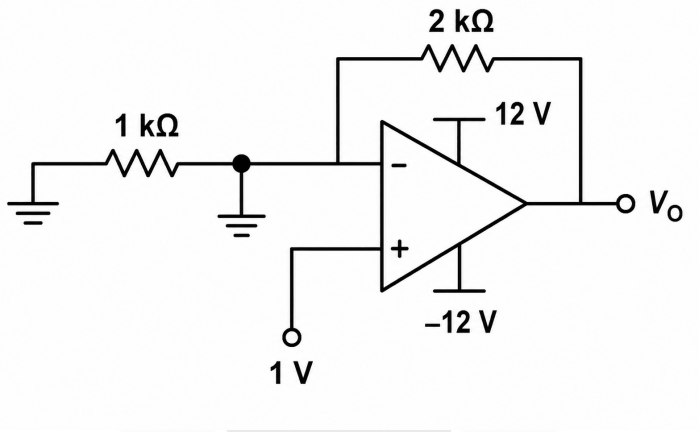

Fig. Solution Q3.15.5

From the above circuit,

$$V_+ = 1\,\text{V},\ V_- = 0\,\text{V}$$

Virtual ground concept is not applicable when both input terminals are known.
So, ideal Op-Amp act as a comparator circuit.
From figure,

$$V_+ > V_-$$

So,

$$V_0 = +V_{sat} = +12\,\text{V}$$

Hence, the output voltage V_0 is **12 V**.

$$\boxed{V_0 = 12\,\text{V}}$$

Key Points

- When negative feedback is disabled (in this case, by hardwiring the inverting terminal to ground), the op-amp operates in open-loop mode and acts as a comparator.

Q3.15.6 NAT 2015 1M Ans: 159.15 ● ○ ○

Method 1

Circuit can be redrawn as,

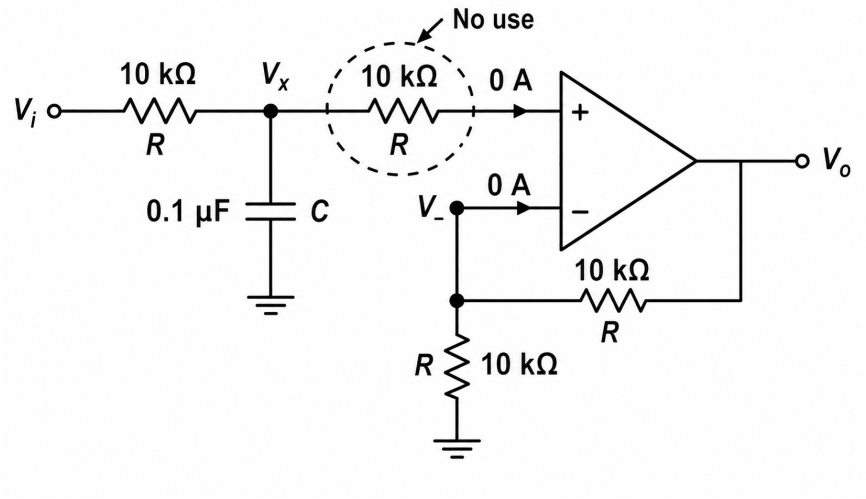

Fig. Solution Q3.15.6

The circuit has negative feedback so using virtual short concept $V_- = V_x$
From non-inverting terminal : Change the circuit in transform domain

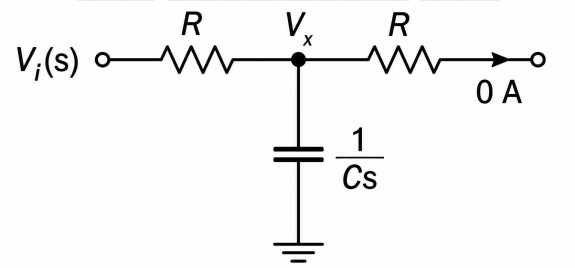

Fig. Solution Q3.15.6

Voltage V_x is given by,

$$V_x = \frac{\frac{1}{Cs}}{R + \frac{1}{Cs}} \times V_i(s) = \frac{V_i(s)}{1 + RCs}$$

Using KCL at inverting terminal,

$$\frac{V_x}{R} + \frac{V_x - V_0(s)}{R} = 0$$

$$V_0(s) = 2V_x = \frac{2V_i(s)}{1 + RCs}$$

Voltage gain is given by,

$$A(s) = \frac{V_0(s)}{V_i(s)} = \frac{2}{1 + RCs}$$

Put $s = j\omega$ in above equation,

$$A(j\omega) = \frac{2}{1 + j\omega RC}$$

$$|A(j\omega)| = \frac{2}{\sqrt{1 + (\omega RC)^2}}$$

At $\omega = 0$, $|A(j0)| = A_0 = 2$ At 3-dB cut-off frequency,

$$|A(j\omega_c)| = \frac{A_0}{\sqrt{2}}$$

$$\frac{2}{\sqrt{1+(\omega_c RC)^2}} = \frac{A_0}{\sqrt{2}} = \frac{2}{\sqrt{2}}$$

$$\frac{4}{1+(\omega_c RC)^2} = 2$$

$$2 = 1 + (\omega_c RC)^2$$

$$1 = (\omega_c RC)^2$$

$$\omega_c RC = \pm 1$$

Taking the positive value for frequency,

$$\omega_c = \frac{1}{RC}$$

$$f_c = \frac{1}{2\pi RC}$$

$$f_c = \frac{1}{2\pi \times 10^4 \times 10^{-7}} = 159.15\,\text{Hz}$$

Method 2

The circuit in Method-1 can be redrawn in the form of a standard first order low pass filter as,

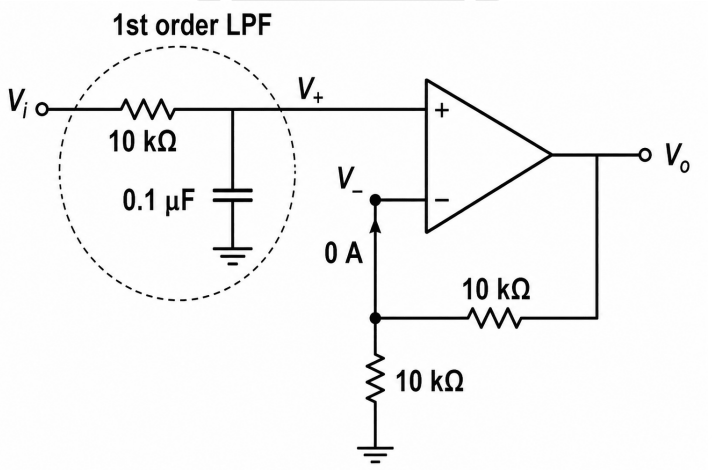

Fig. Solution Q3.15.6

For first order LPF, 3-dB cut-off frequency is given by,

$$f_H = \frac{1}{2\pi RC} = \frac{1}{2\pi \times 10\,\text{k}\Omega \times 0.1\,\mu\text{F}}$$

$$f_H = \frac{1}{2\pi \times 10^4 \times 0.1 \times 10^{-6}} = 159.15\,\text{Hz}$$

$$\boxed{f_H = 159.15\ \text{Hz}}$$

Key Points

- **First-Order Low-Pass Filter (LPF):** The input RC network forms a simple low-pass filter. The 3-dB cutoff frequency (the frequency at which the power drops to half its midband value, or voltage gain drops by $1/\sqrt{2}$) is fundamentally governed by $f_c = \frac{1}{2\pi RC}$.

Q3.15.7 NAT | 2015 | 2M | Ans: 1 | ● ○ ○

Given, $\frac{V_0}{V_i} = -12$

Given circuit is shown below,

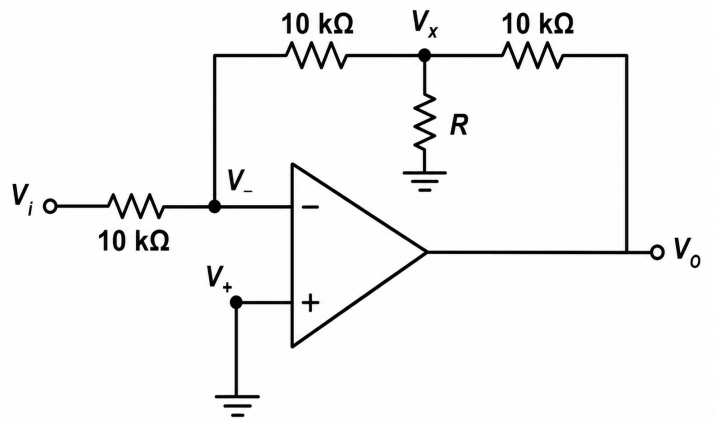

Fig. Solution Q3.15.7

From virtual ground concept,

$$V_+ = V_- = 0\,\text{V}$$

Applying KCL at inverting terminal,

$$\frac{0 - V_i}{10} + \frac{0 - V_x}{10} = 0$$

$$V_x = -V_i \quad \ldots(\text{i})$$

Applying KCL at V_x,

$$\frac{V_x}{10} + \frac{V_x}{R} + \frac{V_x - V_0}{10} = 0$$

$$-\frac{V_i}{10} - \frac{V_i}{R} - \frac{V_i}{10} = \frac{V_0}{10} \quad [\text{From equation (i)}]$$

$$-1 - \frac{10}{R} - 1 = \left(\frac{V_0}{V_i}\right)$$

$$-\frac{10}{R} - 2 = -12 \quad \left[\text{Given} : \frac{V_0}{V_i} = -12\right]$$

$$\frac{10}{R} = 10 \implies R = 1\,\text{k}\Omega$$

Hence, the value of R is **1 kΩ**.

$$\boxed{R = 1\text{k}\Omega}$$

Key Points

- Using a T-network in the feedback loop allows for the realization of high voltage gains without requiring extremely large feedback resistor values.

Q3.14.1 NAT | 2014 | 1M | Ans: 3.18 | ● ○ ○

Given,

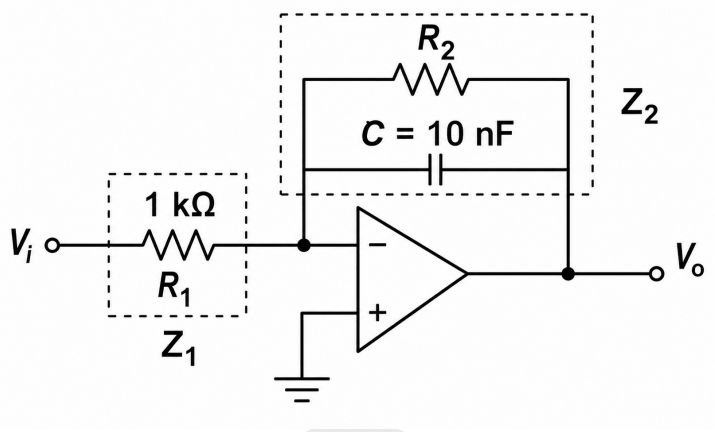

Fig. Solution Q3.14.1

$$Z_1 = R_1$$

$$Z_2 = R_2||C = \frac{R_2\left(\frac{1}{j\omega C}\right)}{R_2 + \frac{1}{j\omega C}} = \frac{R_2}{1 + j\omega R_2 C}$$

For the inverting amplifier:

$$\frac{V_o}{V_i} = -\frac{Z_2}{Z_1}$$

$$\frac{V_o}{V_i} = \frac{-\frac{R_2}{1+j\omega R_2 C}}{R_1} = -\frac{(R_2/R_1)}{1 + j\omega R_2 C}$$

The magnitude of the voltage gain will be:

$$\left|\frac{V_o}{V_i}\right| = \frac{(R_2/R_1)}{\sqrt{1 + (\omega R_2 C)^2}}$$

At $\omega = 0$:

$$A_{max} = \left|\frac{V_o}{V_i}\right| = \frac{R_2}{R_1}$$

$$\therefore \frac{V_o}{V_i} = \frac{A_{max}}{1 + j\omega R_2 C}$$

$$\left|\frac{V_o}{V_i}\right| = \frac{A_{max}}{\sqrt{1 + (\omega R_2 C)^2}}$$

At $\omega = \omega_c$ (Cut-off frequency)

$$\left|\frac{V_o}{V_i}\right| = \frac{A_{max}}{\sqrt{2}}$$

$$\frac{A_{max}}{\sqrt{2}} = \frac{A_{max}}{\sqrt{1 + (\omega_c R_2 C)^2}}$$

$$2 = 1 + (\omega_c R_2 C)^2$$

$$1 = \omega_c R_2 C$$

$$\omega_c = \frac{1}{R_2 C}$$

$$f_c = \frac{1}{2\pi R_2 C}$$

$$R_2 = \frac{1}{2\pi f_c C} = \frac{1}{2\pi \times 5 \times 10^3 \times 10 \times 10^{-9}}$$

$$\boxed{R_2 = 3.18\,\text{k}\Omega}$$

Key Points

- **Cut-off Frequency (f_c):** The frequency at which the gain magnitude drops to $1/\sqrt{2}$ of its maximum (DC) value. It is determined exclusively by the feedback network components: $f_c = \frac{1}{2\pi R_2 C}$.

Q3.14.2 MCQ 2014 2M Ans: C ● ○ ○

Given,

(i) R_i is finite i.e. internal input current exists.

(ii) A_{OL} is infinite i.e. virtual ground concept is valid.

(iii) $V_{ios} = 0$ i.e. output voltage due to offset voltage does not exist.

Given circuit is shown below,

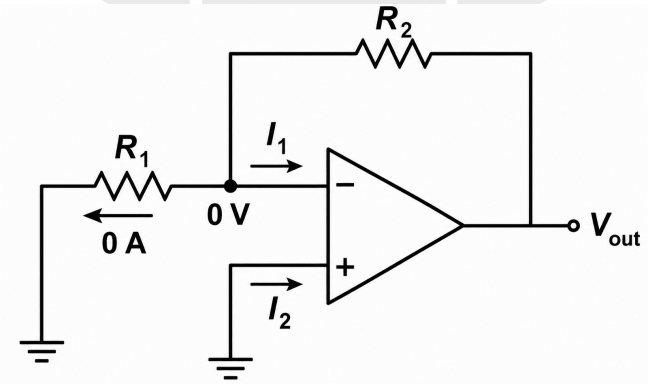

Fig. Solution Q3.14.2

From virtual ground concept, $V_+ = V_- = 0$

Applying KCL at inverting terminal,

$$\frac{0}{R_1} + \frac{0 - V_{out}}{R_2} + I_1 = 0$$

$$V_{out} = I_1 R_2$$

$$\boxed{\text{(C)} \quad I_1 R_2}$$

Key Points

- Unlike a completely ideal op-amp, an op-amp with finite input impedance will have non-zero currents (I_1, I_2) flowing into its input terminals.

Q3.14.3 MCQ 2014 2M Ans: D ● ○ ○

Given circuit is shown below,

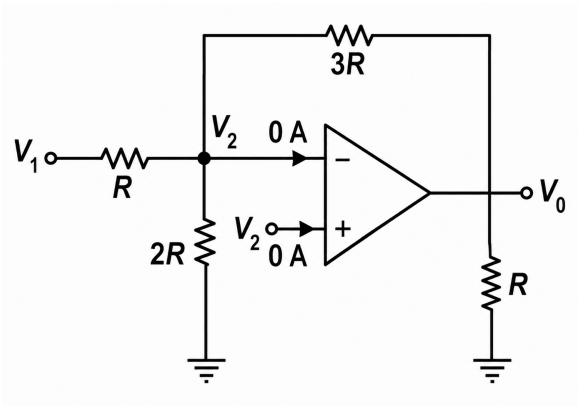

Fig. Solution Q3.14.3

From virtual ground concept, $V_+ = V_- = V_2$.
Applying KCL at inverting terminal,

$$\frac{V_2 - V_1}{R} + \frac{V_2}{2R} + \frac{V_2 - V_0}{3R} = 0$$

$$V_0 = 3V_2 - 3V_1 + \frac{3}{2}V_2 + V_2 = -3V_1 + \frac{11}{2}V_2$$

$$\boxed{\text{(D)} \quad -3V_1 + \frac{11}{2}V_2}$$

Q3.14.4 MCQ 2014 1M Ans: D ● ○ ○

Given,

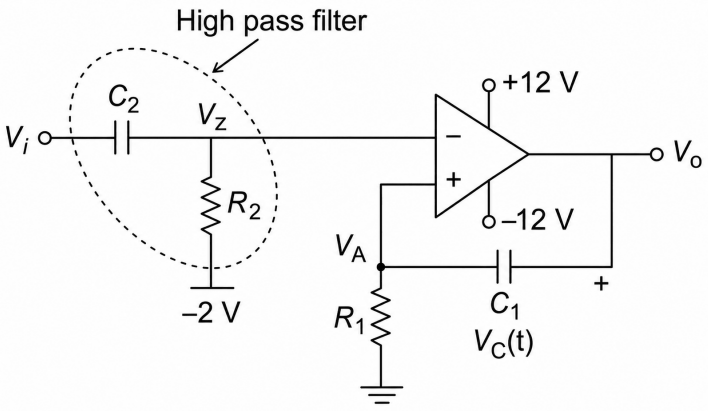

Fig. Solution Q3.14.4

Let, $V^+ = V_A$ and $V^- = V_Z$
Here V_i acts as a trigger to change the output state and when it is applied, the capacitor C_1 charges until the output state returns to the original state.
Initially, the capacitor is uncharged. Due to positive feedback, let $V_0 = +V_{sat}$
Now, the voltage across capacitor is given by:

$$V_c(t) = V_f + (V_i - V_f)e^{-t/\tau}$$

With $V_i =$ initial voltage $= 0\,\text{V}$ and $V_f =$ final voltage $= +V_{sat}$, we can write:

$$V_c(t) = V_{sat} + (0 - V_{sat})e^{-t/R_1C_1}$$

$$V_c(t) = V_{sat}\left(1 - e^{-t/R_1C_1}\right)$$

$$V_c(t) = V_{sat} - V_{sat}e^{-t/R_1C_1}$$

$$V_{sat} - V_c(t) = V_{sat}e^{-t/R_1C_1} \quad \ldots(1)$$

From the given figure, applying KVL,

$$V_0 - V_c(t) - V_A = 0 \quad \text{and} \quad V_0 = V_{sat}$$

$$V_{sat} - V_c(t) - V_A = 0$$

$$V_{sat} - V_c(t) = V_A \quad \ldots(2)$$

From (1) & (2), we get:

$$V_A = V_{sat}e^{-t/R_1C_1}$$

Let the trigger input applied, be in the triangular, then the voltage at V_Z will be a pulse as the circled part in the figure is a differentiator circuit.

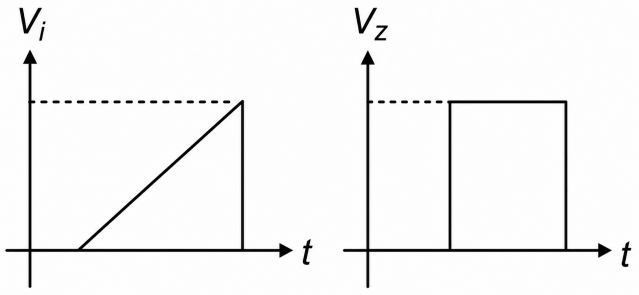

Fig. Solution Q3.14.4

From the below diagram, the voltage at V_+ is a exponentially decaying waveform and as long as it stays above the voltage at V_- output is $+V_{sat}$ but when voltage at V_+ goes below V_-, the output is flipped to $-V_{sat}$ and when the pulse goes down at V_-, output switches back to $+V_{sat}$.
Hence by applying a trigger at input, we will get a short pulse at the output. Hence it is a monostable multivibrator circuit.

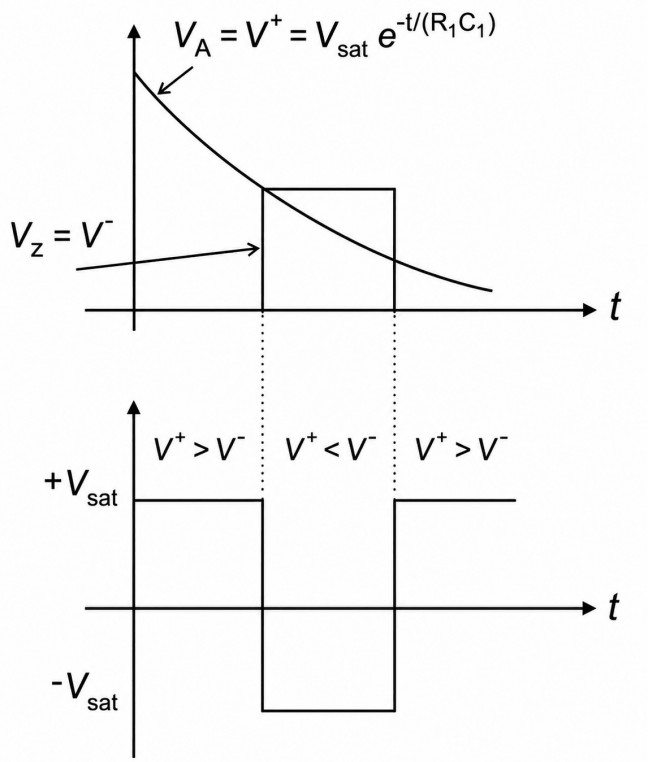

Fig. Solution Q3.14.4

(D) a monostable multivibrator

Key Points

- **Monostable Multivibrator:** A circuit with exactly one stable state. When triggered by an external pulse, it temporarily switches to a quasi-stable state for a predetermined duration before returning to its stable state.

Q3.13.1 MCQ | 2013 | 2M | Ans: C | ● ○ ○

Given circuit is shown below,

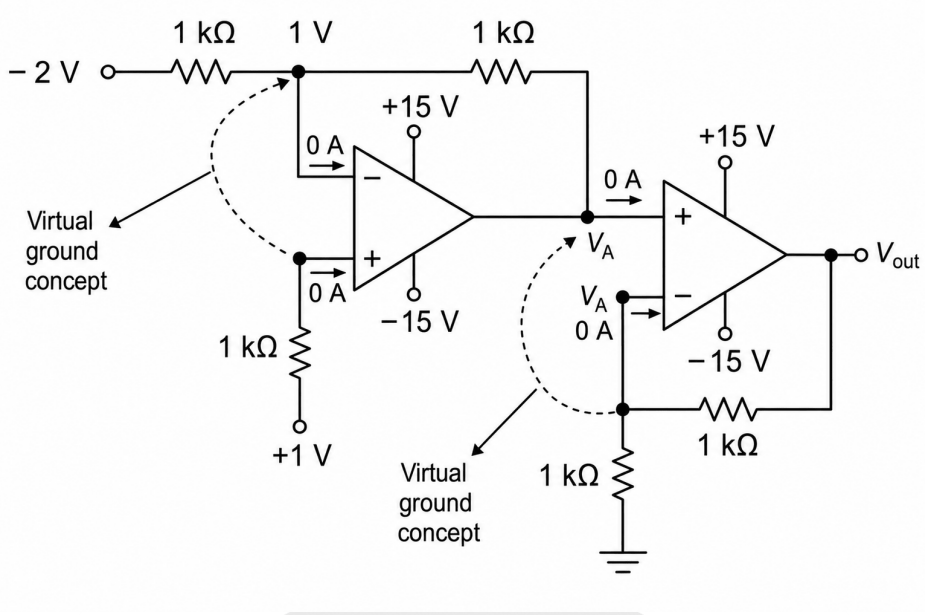

Fig. Solution Q3.13.1

Applying KCL at inverting terminal of first Op-Amp,

$$\frac{1+2}{1}+\frac{1-V_A}{1}=0$$

$$V_A = 4\,\text{V}$$

Apply voltage divider rule at inverting terminal of second op-amp,

$$V_A = \frac{V_{\text{out}} \times 1}{1+1}$$

So, output of second Op-Amp is,

$$V_{\text{out}} = \left[1+\frac{1}{1}\right] V_A$$

$$V_{\text{out}} = 2V_A = 2 \times 4 = 8\,\text{V}$$

(C) 8

Q3.12.1 MCQ 2012 2M Ans: B ● ○ ○

Given circuit is shown below,

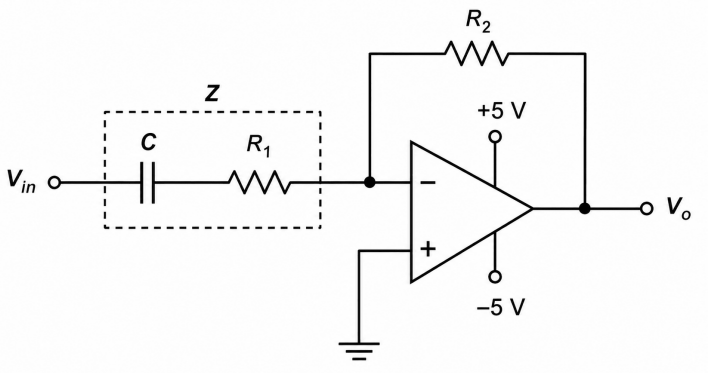

Fig. Solution Q3.12.1

$$Z = R_1 + X_C = R_1 + \frac{1}{sC} = \left[\frac{R_1Cs + 1}{sC}\right]$$

Voltage gain for inverting Op-Amp is given by,

$$\frac{V_0}{V_{in}} = \frac{-R_2}{Z}$$

So,

$$\frac{V_0}{V_{in}} = \frac{-R_2}{\left[\frac{R_1Cs+1}{sC}\right]} = \frac{-R_2Cs}{(1 + R_1Cs)}$$

$$\left|\frac{V_0}{V_{in}}\right| = \frac{R_2Cs}{1 + R_1Cs}$$

$$T(s) = \left|\frac{V_0}{V_{in}}\right| = \frac{\frac{R_2}{R_1}}{1 + \frac{1}{R_1Cs}}$$

$$T(j\omega) = \frac{\frac{R_2}{R_1}}{1 + \frac{1}{j\omega R_1C}}$$

$$T(j\omega) = \frac{\frac{R_2}{R_1}}{1 - \frac{j}{\omega R_1C}} \quad \ldots(\text{i})$$

Frequency response of first order high pass filter is given by,

$$T(j\omega) = \frac{V_0}{V_{in}} = \frac{A_0}{1 - \frac{j\omega_L}{\omega}} \quad \ldots(\text{ii})$$

From equation (i) and (ii), 3-dB frequency is,

$$\omega_{3\text{dB}} = \omega_L = \frac{1}{R_1C} \text{ rad/sec}$$

(B) high pass filter with $f_{3\text{dB}} = \frac{1}{R_1C}$ rad/s

Key Points

- **Active High-Pass Filter:** An inverting operational amplifier configuration with a capacitor in series with the input resistor acts as a first-order high-pass filter.

Q3.11.1 MCQ 2011 1M Ans: D ● ○ ○

Case 1 : At low frequency ($\omega \to 0$) :

Inductive reactance,

$$X_L = \omega L \to 0 \quad [\text{Short Circuit}]$$

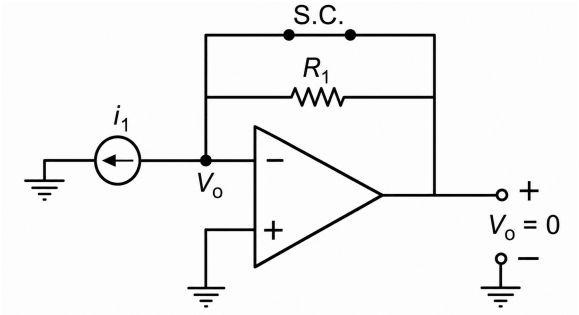

Fig. Solution Q3.11.1

From virtual ground concept,

$$V_+ = V_- = 0\,\text{V}$$

So, $V_0 = 0\,\text{V}$

Case 2 : At high frequency ($\omega \to \infty$) :

Inductive impedance,

$$X_L = \omega L \to \infty \quad [\text{O.C.}]$$

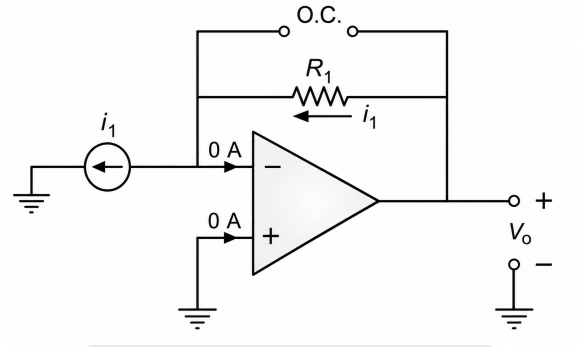

Fig. Solution Q3.11.1

From figure,

$$i_1 = \frac{V_0 - 0}{R_1} \implies V_0 = i_1 R_1$$

From case 1 and case 2, output voltage can be written as given below,

$$V_0 = \begin{cases} 0 & ; \quad \text{At low frequency} \\ i_1 R_1 & ; \quad \text{At high frequency} \end{cases}$$

This circuit passes high frequency current signal and stop low frequency current signal. So, the given Op-Amp acts like a high pass filter.

(D) high pass filter

Key Points

- This circuit is referred as current to voltage converter.
- At low frequencies ($\omega \to 0$), an inductor acts as a short circuit ($X_L \to 0$). At high frequencies ($\omega \to \infty$), it acts as an open circuit ($X_L \to \infty$).
- By evaluating the circuit's response at extreme frequencies (0 and ∞), one can quickly deduce the type of filter without deriving the full transfer function. Since it blocks DC and passes high frequencies, it is a high-pass filter.

Q3.10.1 MCQ 2010 1M Ans: A ●○○

Given circuit can be redrawn as shown below,

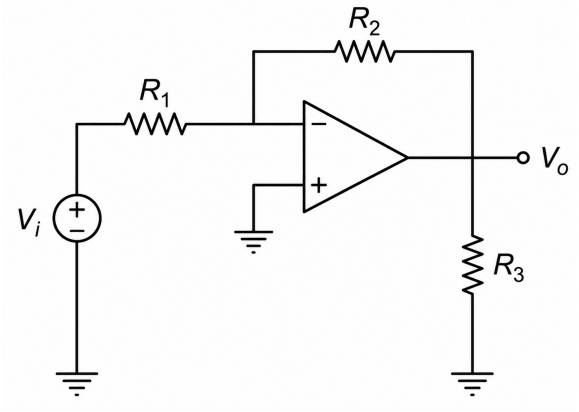

Fig. Solution Q3.10.1

From virtual ground concept,

$$V_+ = V_- = 0$$

Applying KCL at inverting terminal,

$$\frac{0 - V_i}{R_1} + \frac{0 - V_0}{R_2} = 0$$

$$\frac{V_0}{V_i} = -\frac{R_2}{R_1}$$

Note : There is no use of R_3 for calculating gain.

(A) $\dfrac{-R_2}{R_1}$

Q3.10.2 MCQ 2010 2M Ans: B

Given

(i) Op-Amp is ideal and has negative feedback. So virtual ground concept is applicable.

(ii)The diodes are practical so assume a cut-in voltage of $V_\gamma = 0.7$ V.

Method 1

Case 1 : For $V_i > 0$,

Since all the voltages connected to the inverting terminal are positive, inverting terminal voltage (V_-) > Non-inverting terminal voltage (V_+)

Hence, V_0' goes towards $-V_{sat}$ and when $V_0' = -0.7$ V

Diode D_2 is forward biased i.e. ON ($V_\gamma = 0.7$ V).

Diode D_1 is reverse biased i.e. OFF (open circuit).

Hence the modified circuit is shown below,

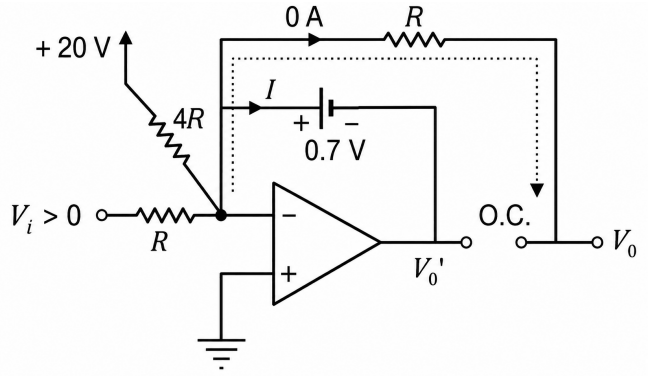

Fig. Solution Q3.10.2

Applying KVL in above shown loop,

$$0 + 0 \times R + V_0 = 0$$

$$V_0 = 0\text{ V}$$

Case 2 : For $V_i < 0$,

Inverting terminal voltage (V_-) < Non-inverting terminal voltage (V_+)

So, V_0' goes towards $+V_{sat}$ and when $V_0' = 0.7$ V, then

Diode D_1 is forward biased i.e. ON ($V_\gamma = 0.7$ V)

Diode D_2 is reverse biased i.e. OFF (Open circuit).

Hence the modified circuit is shown below,

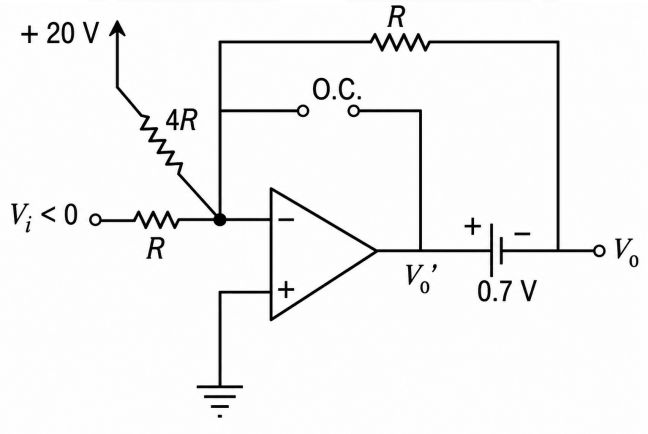

Fig. Solution Q3.10.2

Due to virtual short $V_- = V_+ = 0$. Applying KCL at inverting terminal,

$$\frac{0 - V_i}{R} + \frac{0 - 20}{4R} + \frac{0 - V_0}{R} = 0$$

$$V_0 = -V_i - 5 \quad \ldots (i)$$

(i) If $V_i = -5$ V, then $V_0 = 0$ V

(ii) If $V_i = -10$ V, then $V_0 = +5$ V

(iii) If $V_i = -1$ V, then $V_0 = -4$ V

Above figure represents inverting Op-Amp. Hence, output should be inverted form of input. Therefore, equation (i) is only valid for $V_i < -5$ V.

From case 1 and case 2, output voltage V_0 can be written as given below,

$$V_0 = \begin{cases} 0 & ; \quad V_i \geq -5\text{ V} \\ -V_i - 5 & ; \quad V_i < -5\text{ V} \end{cases}$$

Hence option(B) is correct.

Method 2

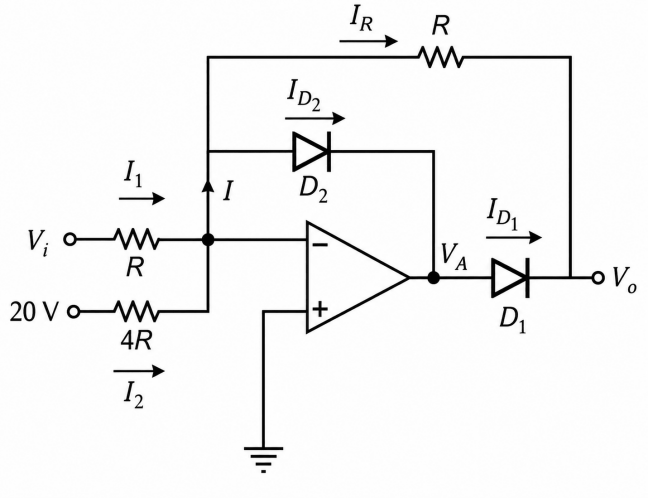

Fig. Solution Q3.10.2

In case of closed loop system i.e. for valid virtual ground concept,

$$V_- = V_+ = 0 \text{ volt}$$

$$I_1 = \frac{V_i - 0}{R} \quad \text{and} \quad I_2 = \frac{20}{4R}$$

From figure, $I = I_1 + I_2 = \frac{V_i}{R} + \frac{5}{R}$

Condition for D_2 to be ON,

$$I_{D_2} > 0$$

Since, $I_{D_2} = I - I_R$, $I > 0$

$$\frac{V_i}{R} + \frac{5}{R} > 0$$

$$\mathbf{V_{in} > -5 \text{ volt}}$$

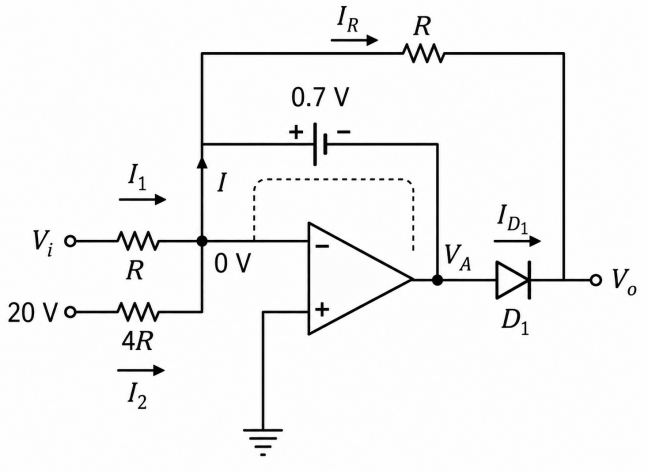

Fig. Solution Q3.10.2

Applying KVL in feedback path,

$$0 + 0.7 + V_A = 0$$

$$V_A = -0.7 \text{ volt}$$

Hence, diode D_1 becomes reverse biased. i.e. OFF (open circuit). Therefore, $I_R = 0$
Hence, the modified circuit is shown below,

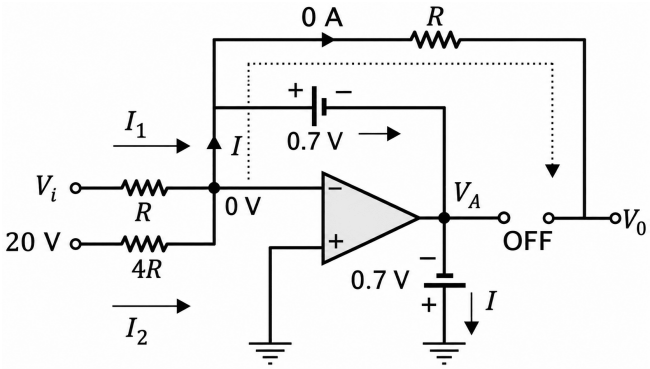

Fig. Solution Q3.10.2

Applying KVL in loop shown in figure,

$$0 + 0 \times R + V_0 = 0$$

$$V_0 = 0 \text{ volt}$$

When $V_i \leq -5$ Volt,
Condition of diodes will becomes reverse and hence the modified circuit is shown below,

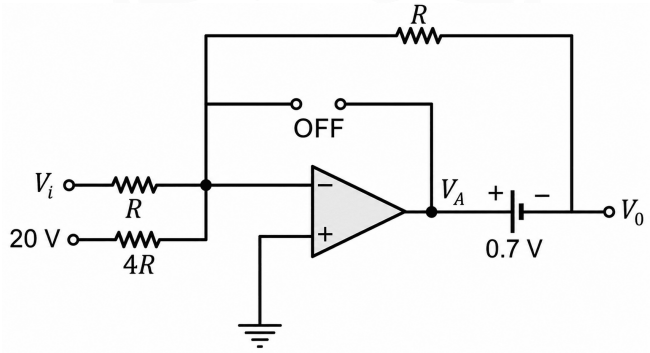

Fig. Solution Q3.10.2

This is a standard inverting adder. So,

$$V_0 = -\left[\frac{R}{R}V_i + \frac{R}{4R} \times 20\right]$$

$$V_0 = -[V_i + 5]$$

$$V_0 = -V_i - 5$$

Output voltage is given by,

$$V_0 = \begin{cases} 0 & ; \quad V_i > -5 \\ -V_i - 5 & ; \quad V_i < -5 \end{cases}$$

Hence, the correct option is (B).

Key Points

- **Precision Rectifier:** A circuit utilizing an operational amplifier to rectify signals with amplitudes smaller than the forward voltage drop of a diode, effectively acting as an ideal diode.

Q3.09.1 MCQ 2009 2M Ans: A ● ○ ○

Given circuit is,

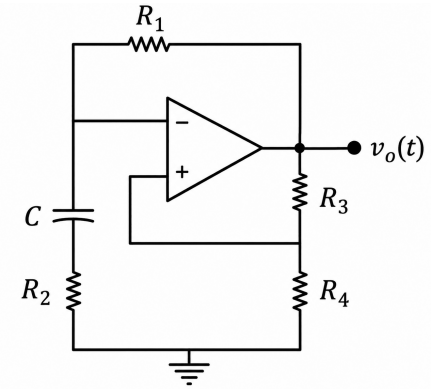

Fig. Solution Q3.09.1

Amplitude of $v_o(t)$:
Since the op-amp has a positive feedback the output voltage $v_o(t)$ switches strictly between the saturation levels $+V_{sat}$ and $-V_{sat}$. The amplitude of this output is determined entirely by the op-amp's internal properties and power supply limits. It is independent of the passive components in the circuit.
Therefore, the amplitude does not depend on R_2.
Frequency of $v_o(t)$:
The frequency of oscillation depends on the rate at which the capacitor C charges and discharges. By analyzing the negative feedback loop, the current $i_c(t)$ flowing through the capacitor must pass through both R_1 and R_2. The equivalent resistance seen by the capacitor during the charging and discharging phases is the series combination of R_1 and R_2. The time constant of the circuit is given by,

$$\tau = (R_1 + R_2)C$$

Since the time constant τ depends on the value of R_2, the charging time and consequently the frequency of oscillation of $v_o(t)$ depend on R_2.
Hence, only the frequency of $v_o(t)$ depends on R_2.

(A) only the frequency

Q3.08.1 MCQ 2008 2M Ans: B ● ○ ○

Given
(i) $I - V$ characteristics of diode is given by,

$$I = I_0\left(e^{\frac{V}{V_T}} - 1\right) \quad \text{where } V_T = 25\,\text{mV}$$

(ii) Reverse saturation current,

$$I_0 = 1\,\mu\text{A} = 0.001\,\text{mA}$$

Given circuit is shown below,

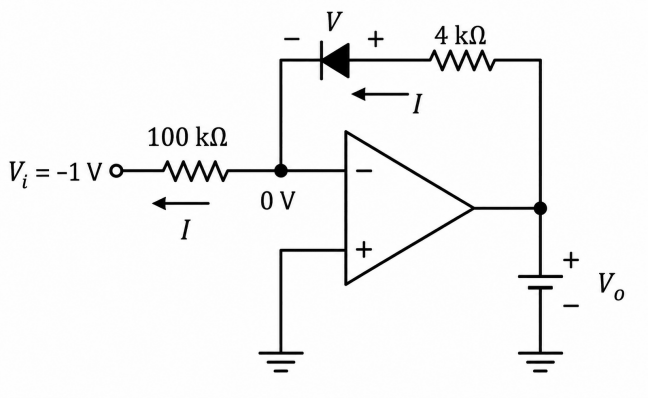

Fig. Solution Q3.08.1

From virtual ground concept,

$$V_+ = V_- = 0$$

Applying KCL at inverting terminal,

$$I = \frac{0-(-1)}{100} = 0.01\,\text{mA}$$

In case of forward bias diode current flow from p to n is given by,

$$I = I_0\left[e^{\frac{V}{V_T}} - 1\right] = 0.01\,\text{mA}$$

$$0.001(e^{V/V_T} - 1) = 0.01$$

$$e^{V/V_T} = 10 + 1$$

$$V = V_T \ln(11) = 25 \times \ln(11) = 0.0599\,\text{V}$$

Output voltage is given by,

$$V_0 = V + 4I + 0$$

$$V_0 = 0.0599 + 4 \times 0.01 = 0.1\,\text{Volt}$$

(B) 0.1 V

Q3.08.2 MCQ 2008 2M Ans: B ● ○ ○

Method 1

Assume $R_1 = 1\,\Omega$, $R_2 = 1\,\Omega$, $L = 1\,\text{H}$ and $C = 1\,\text{F}$

Transform domain :

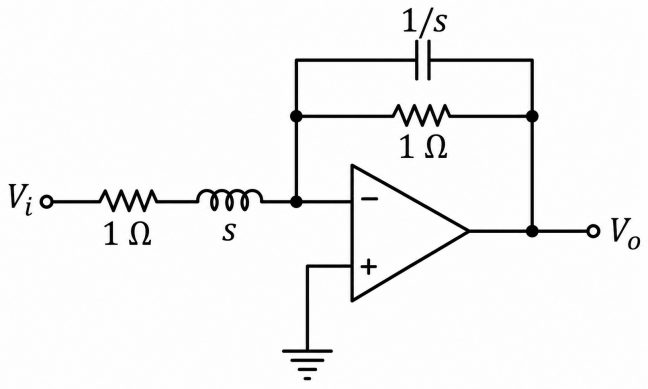

Fig. Solution Q3.08.2

From virtual ground concept,

$$V_+ = V_- = 0$$

Applying KCL at inverting terminal,

$$\frac{0 - V_i}{1 + s} + \frac{0 - V_0}{1} + \frac{0 - V_0}{1/s} = 0$$

$$V_0(1 + s) = \frac{-V_i}{1 + s}$$

$$T(s) = \frac{V_0}{V_i} = \frac{-1}{(s + 1)^2}$$

$$|T(s = 0)| = |-1| = 1$$

$$|T(s = \infty)| = 0$$

This circuit passes low frequency signal and stops high frequency signal. So, the given Op-Amp circuit acts like a low pass filter. Hence, the correct option is (B).

Method 2

Case 1 : At low frequency ($\omega \to 0$) :

Capacitive impedance is,

$$X_C = \frac{1}{\omega C} \to \infty \quad \text{[O.C.]}$$

Inductive impedance is,

$$X_L = \omega L \to 0 \quad \text{[S.C.]}$$

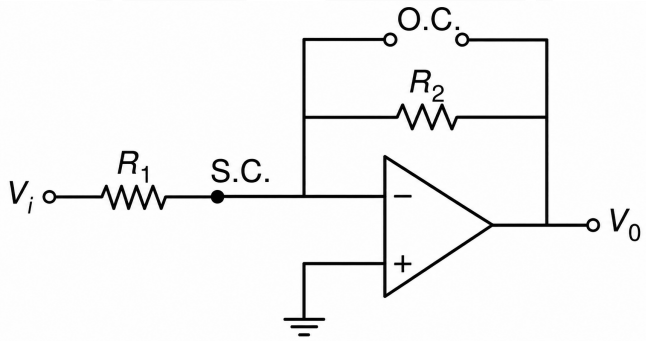

Fig. Solution Q3.08.2

Output of inverting Op-Amp is given by,

$$V_0 = \frac{-R_2}{R_1} V_i$$

Case 2 : At high frequency ($\omega \to \infty$) :

Capacitive impedance is,

$$X_C = \frac{1}{\omega C} \to 0 \quad \text{[S.C.]}$$

Inductive impedance is,

$$X_L = \omega L \to \infty \quad \text{[O.C.]}$$

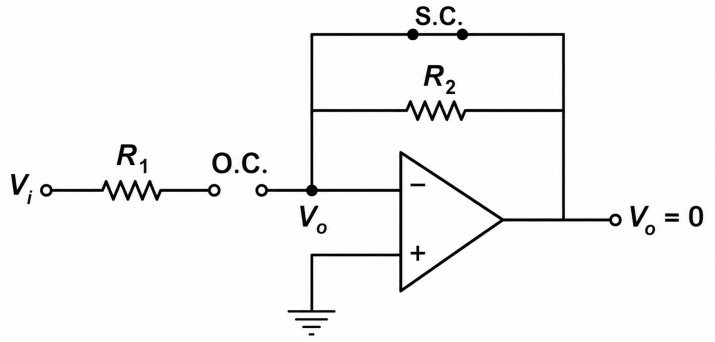

Fig. Solution Q3.08.2

From above circuit, $V_0 = 0$ [Virtual ground]
From case 1 and case 2 output voltage V_0 can be written as given below,

$$V_0 = \begin{cases} \frac{-R_2}{R_1} \times V_i & ; \quad \text{At low frequency} \\ 0 & ; \quad \text{At high frequency} \end{cases}$$

Hence, the given circuit represents low pass filter.

(B) low pass filter

Q3.08.3 MCQ 2008 2M Ans: C ● ○ ○

Given :
(i) $V_i = -12$ V to $+12$ V triangular wave
(ii) $V_0 = -15$ V to $+15$ V
Schmitt trigger circuit :

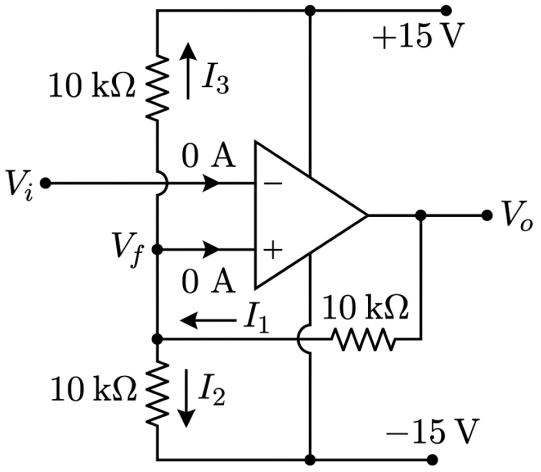

Fig. Solution Q3.08.3

Modified figure is given by,

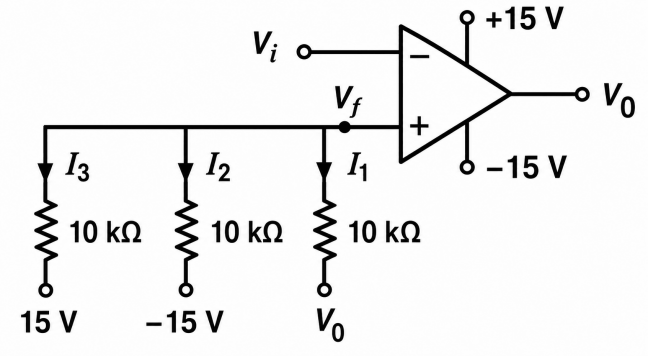

Fig. Solution Q3.08.3

Applying KCL at the non-inverting terminal of above figure,

$$\frac{V_f - V_0}{10} + \frac{V_f - (-15)}{10} + \frac{V_f - 15}{10} = 0$$

$$3V_f - V_0 + 15 - 15 = 0$$

$$3V_f = V_0$$

$$V_f = \frac{V_0}{3}$$

Since, $V_0 = -15$ V to $+15$ V So, $V_f = -5$ V to $+5$ V

(C) -5V and +5V

Q3.07.1 MCQ 2007 2M Ans: C ● ○ ○

Given circuit is,

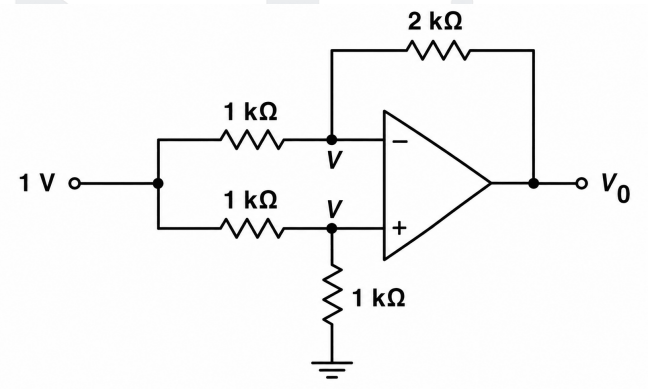

Fig. Solution Q3.07.1

From virtual ground concept,

$$V_+ = V_- = V$$

Applying KCL at non inverting terminal,

$$\frac{V}{1} + \frac{V - 1}{1} = 0$$

$$V = \frac{1}{2} \quad \ldots (i)$$

Applying KCL at inverting terminal,

$$\frac{V-1}{1}+\frac{V-V_0}{2}=0$$

$$3V-V_0=2 \quad \ldots(\text{ii})$$

From equation (i) and equation (ii),

$$V_0=-0.5\,\text{V}$$

(C) -0.5V

Q3.07.2 MCQ 2007 2M Ans: D ● ○ ○

Given

(i) Diode current equation,

$$I=I_s\exp(V/V_T)$$

(ii) When $V_i=2\,\text{V}\Rightarrow V_0=V_{o1}$ and When $V_i=4\,\text{V}\Rightarrow V_0=V_{o2}$

Given circuit is shown below,

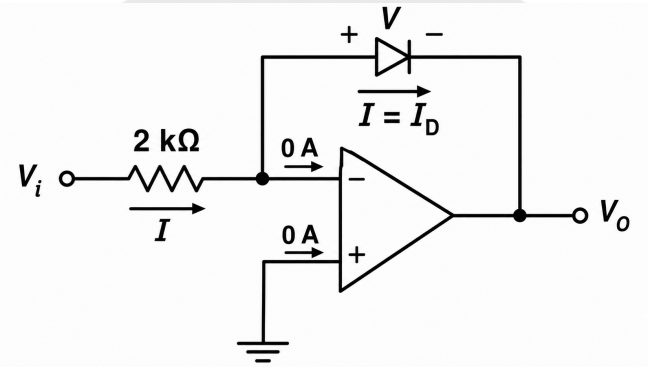

Fig. Solution Q3.07.2

PrepFusion

Method 1

Output of logarithmic Op-Amp is given by,

$$V_0=-\eta V_T\ln\left[\frac{V_i}{RI_s}\right]$$

where, η(Ideality factor) = 1

I_s = reverse saturation current

$V_s=V_i$ = Input voltage

$$V_0=-V_T\ln\left[\frac{V_i}{RI_s}\right] \quad \ldots(\text{i})$$

$$V_{o1}=-V_T\ln\left[\frac{2}{RI_s}\right] \quad \ldots(\text{ii})$$

$$V_{o2}=-V_T\ln\left[\frac{4}{RI_s}\right] \quad \ldots(\text{iii})$$

$$V_{o1}-V_{o2}=-V_T\ln\left[\frac{2}{RI_s}\right]+V_T\ln\left[\frac{4}{RI_s}\right]$$

$$V_{o1}-V_{o2}=V_T\ln\left[\frac{4}{RI_s}\times\frac{RI_s}{2}\right]$$

$$V_{o1} - V_{o2} = V_T \ln[2]$$

Method 2

From virtual ground concept,

$$V_+ = V_- = 0$$

Applying KVL in feedback loop,

$$0 + V + V_0 = 0$$

$$V = -V_0 \quad \ldots(\text{i})$$

Diode current is given by,

$$I_D = I = I_s(e^{V/V_T} - 1)$$

Because for forward bias diode,

$$e^{V/V_T} \gg 1$$

So, $I_D = I = I_s e^{V/V_T}$

From figure,

$$I = \frac{V_i - 0}{R} = I_s e^{V/V_T} \quad \ldots(\text{ii})$$

From equation (i) and (ii),

$$\frac{V_i}{R} = I_s e^{-V_0/V_T} \quad \ldots(\text{iii})$$

For $V_i = 2\,\text{V}$,

$$\frac{V_i}{R} = \frac{2}{2} = 1\,\text{mA} = I_s e^{-V_{o1}/V_T} \quad \ldots(\text{iv})$$

For $V_i = 4\,\text{V}$,

$$\frac{V_i}{R} = \frac{4}{2} = 2\,\text{mA} = I_s e^{-V_{o2}/V_T} \quad \ldots(\text{v})$$

From equation (iv) and (v),

$$\frac{2}{1} = \frac{e^{-V_{o2}/V_T}}{e^{-V_{o1}/V_T}} = e^{\left[\frac{V_{o1}-V_{o2}}{V_T}\right]}$$

$$2 = e^{\left[\frac{V_{o1}-V_{o2}}{V_T}\right]}$$

$$\ln 2 = \frac{V_{o1} - V_{o2}}{V_T}$$

$$V_{o1} - V_{o2} = V_T \ln 2$$

$$\boxed{\text{(D)} \quad V_{o1} - V_{o2} = V_T \ln 2}$$

Key Points

- **Logarithmic Amplifier:** When a diode is placed in the feedback loop of an operational amplifier, the output voltage is proportional to the natural logarithm of the input voltage.

Q3.07.3 MCQ | 2007 | 2M | Ans: A | ● ○ ○

Given circuit in s-domain is,

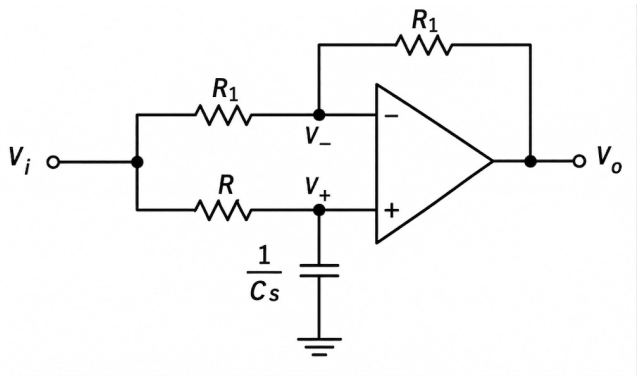

Fig. Solution Q3.07.3

From virtual ground concept,

$$V_+ = V_- = V$$

Applying KCL at non-inverting terminal,

$$\frac{V - V_i}{R} + \frac{V}{1/Cs} = 0$$

$$V\left(Cs + \frac{1}{R}\right) = \frac{V_i}{R}$$

$$V = \frac{V_i}{1 + sRC} \quad \ldots (i)$$

Applying KCL at inverting terminal,

$$\frac{V - V_i}{R_1} + \frac{V - V_0}{R_1} = 0$$

$$2V - V_i = V_0$$

From equation (i),

$$\frac{2V_i}{1 + sRC} - V_i = V_0$$

$$T(s) = \frac{V_0(s)}{V_i(s)} = \frac{1 - sRC}{1 + sRC}$$

$$\boxed{\text{(A)} \quad \frac{1 - sRC}{1 + sRC}}$$

Key Points

- **All-Pass Filter Transfer Function:** The derived transfer function $T(s) = \frac{1-sRC}{1+sRC}$ is the standard form for a first-order all-pass filter. It provides a unity voltage gain ($|T(j\omega)| = 1$) across all frequencies.
- **Phase Shift:** While the amplitude is unattenuated, an all-pass filter introduces a frequency-dependent phase shift, varying from 0 to $-180°$ (or 0 to $-\pi$ radians) for this specific configuration.

Q3.07.4 MCQ 2007 2M Ans: C ● ○ ○

Given, $V_i = V_1 \sin(\omega t)$,

$$V_0 = V_2 \sin(\omega t + \phi)$$

From first part (above question),

$$T(s) = \frac{V_0(s)}{V_i(s)} = \frac{1 - sRC}{1 + sRC}$$

$$T(j\omega) = \frac{1 - j\omega RC}{1 + j\omega RC}$$

$$\angle T(j\omega) = -\tan^{-1}(\omega RC) - \tan^{-1}(\omega RC)$$

$$\phi = \angle T(j\omega) = -2\tan^{-1}(\omega RC)$$

ω	ϕ	**Remark**
0	0	max
∞	$-\pi$	min

(C) $-\pi$ and 0

Q3.05.1 MCQ 2005 1M Ans: B ● ○ ○

The ideal op-amp input pin draws zero current, but the source sees the external 10 kΩ resistor leading to a virtual-ground summing node. Those are different input resistances.

$$I_i = \frac{V_i - 0}{10\,\text{k}\Omega}, \qquad R_i = 10\,\text{k}\Omega.$$

The current through the 30 kΩ feedback resistor balances this source current; their signed sum is zero because no current enters the op-amp pin. That sum is not the source current itself.

(B) 10 kΩ

Q3.05.2 MCQ 2005 2M Ans: C ● ○ ○

Given circuit is shown below,

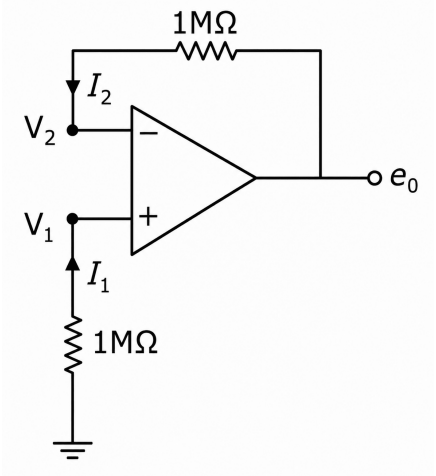

Fig. Solution Q3.05.2

$$V_1 = -I_1 \times (1\,\text{M}\Omega)$$

From virtual ground concept,

$$V_+(\text{i.e. } V_1) = V_-(\text{i.e. } V_2)$$

$$V_2 = V_1 = -I_1 \times (1\,\text{M}\Omega)$$

From figure,

$$e_0 = V_2 + (I_2 \times 1\,\text{M}\Omega)$$

$$e_0 = -I_1 \times (1\,\text{M}\Omega) + I_2 \times (1\,\text{M}\Omega)$$

$$e_0 = (I_2 - I_1) \times 1\,\text{M}\Omega$$

where, $(I_2 - I_1)$ is input offset current. Since e_0 is directly proportional to input offset current, option (C) is correct.

(C) Input offset current only

Key Points

- **Input Bias Current:** The DC currents (I_1 and I_2) flowing into the non-inverting and inverting terminals of a practical operational amplifier required to bias the internal transistors.
- **Input Offset Current:** The difference between the two input bias currents, defined as $I_{io} = |I_1 - I_2|$. It represents the inherent mismatch between the input stage transistors.

Q3.05.3 MCQ 2005 2M Ans: A ●○○

Given circuit in s-domain is given by,

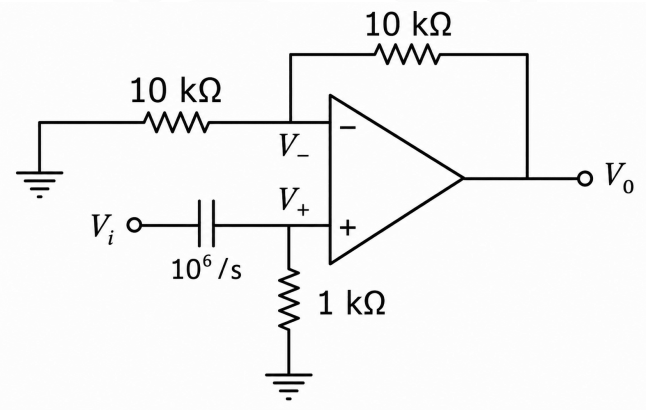

Fig. Solution Q3.05.3

From virtual ground concept,

$$V_+ = V_- = V$$

Applying KCL at non-inverting terminal,

$$\frac{V}{1000} + \frac{V - V_i}{10^6/s} = 0$$

$$V\left[1 + \frac{s}{10^3}\right] = \frac{sV_i}{10^3} \quad \ldots(\text{i})$$

Applying KCL at inverting terminal,

$$\frac{V - V_0}{10^4} + \frac{V - 0}{10^4} = 0$$

$$V = \frac{V_0}{2} \quad \ldots (ii)$$

From equation (i) and (ii),

$$\frac{V_0}{2}\left[\frac{s+10^3}{10^3}\right] = \frac{sV_i}{10^3}$$

$$V_0 = \frac{2sV_i}{s+10^3}$$

$$\frac{V_0}{V_i} = \frac{2s}{s+10^3}$$

Put $s = j\omega$ in above equation,

$$\frac{V_0}{V_i} = \frac{j2\omega}{j\omega + 10^3}$$

For $\omega = 0$, $\quad V_0/V_i = 0$
For $\omega = \infty$, $\quad V_0/V_i = 2$
So, the given circuit represent a high pass filter.

$$\frac{V_0}{V_i} = \frac{2s}{s+10^3} = \frac{2}{1+10^3/s}$$

$$\frac{V_0(j\omega)}{V_i(j\omega)} = T(j\omega) = \frac{2}{1-j10^3/\omega} \quad \ldots (iii)$$

Frequency response of 1st order high pass filter is given by,

$$T(j\omega) = \frac{K}{1-j(\omega_L/\omega)} \quad \ldots (iv)$$

Comparing equation (iii) and (iv), Lower cut-off frequency = 10^3 rad/sec .

(A) high pass, 1000 rad/sec.

Q3.05.4 MCQ 2005 2M Ans: B ● ○ ○

Given circuit is,

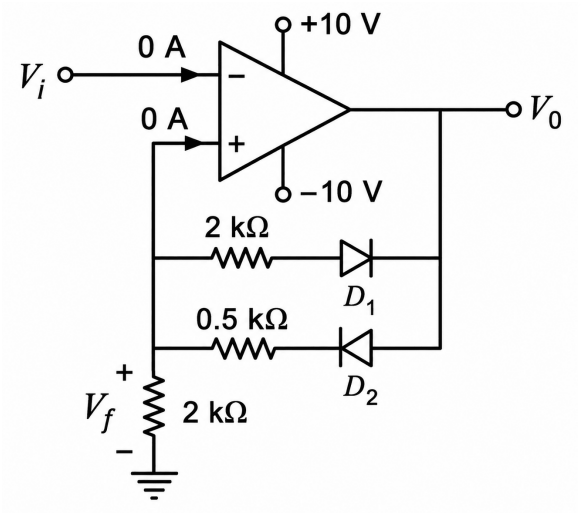

Fig. Solution Q3.05.4

The given Op-Amp circuit has a positive feedback. It is referred as Schmitt trigger circuit.
V_0 can be either $+V_{sat}$ $(+10\text{ V})$ or $-V_{sat}$ (-10 V)

1. When $V_0 = +10\text{ V}$, D_2 is ON (i.e. short circuit) and D_1 is OFF (i.e. open circuit)

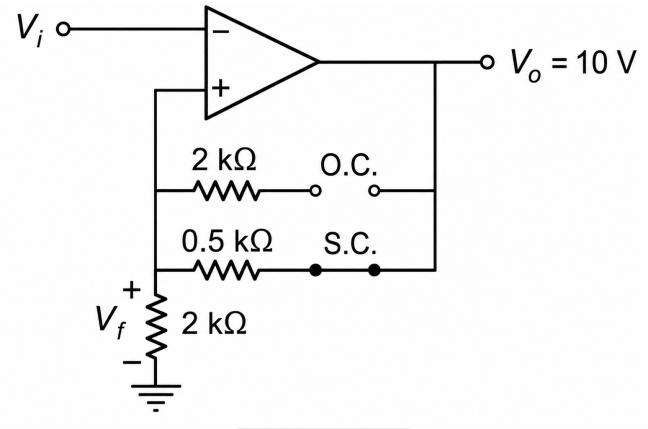

Fig. Solution Q3.05.4

$$V_f = \frac{2}{2+0.5} \times V_0 = \frac{10 \times 2}{2+0.5} = 8\text{ V}$$

$$V_f = V_{UTP} = 8\text{ V}$$

where, V_{UTP} = Upper trigger (threshold) point voltage

Concept of V_{UTP} :

(i) If $V_i < V_{UTP}$, then inverting terminal will be at lower potential than non-inverting terminal. Hence, $V_0 = +V_{sat}$

(ii) If $V_i > V_{UTP}$, then non-inverting terminal will be at lower potential than inverting terminal. Hence, $V_0 = -V_{sat}$

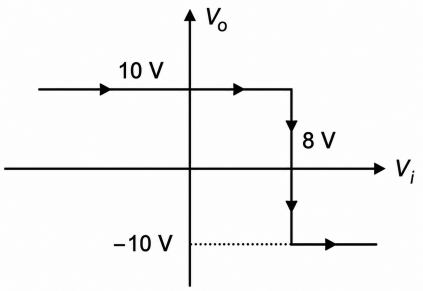

Fig. Solution Q3.05.4

2. When $V_0 = -10\text{ V}$, D_1 is ON (i.e. short circuit) and D_2 is OFF (i.e. open circuit)

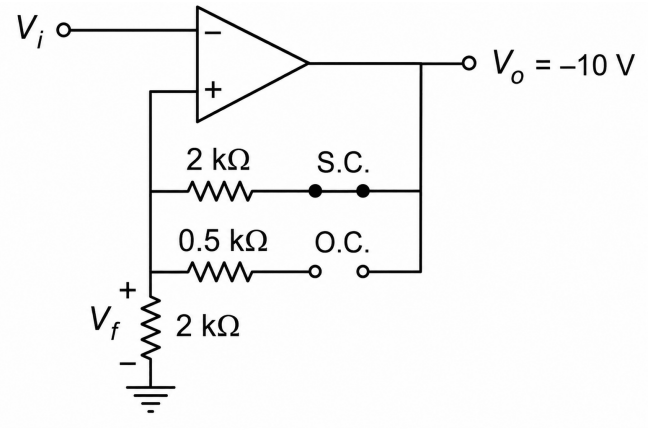

Fig. Solution Q3.05.4

$$V_f = \frac{2}{2+2} \times V_0 = \frac{-10 \times 2}{2+2} = -5\,\text{V}$$

$$V_f = V_{LTP} = -5\,\text{V}$$

where, V_{LTP} = Lower trigger (threshold) point voltage

Concept of V_{LTP} :

(i) If $V_i < V_{LTP}$, then inverting terminal will be at lower potential than non-inverting terminal. Hence, $V_0 = +V_{sat}$

(ii) If $V_i > V_{LTP}$, then non-inverting terminal will be at lower potential than inverting terminal. Hence, $V_0 = -V_{sat}$

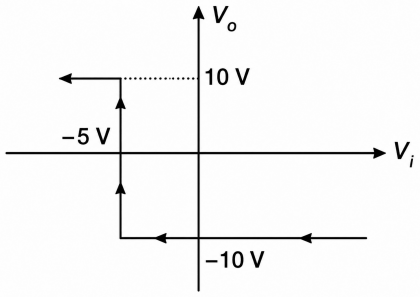

Fig. Solution Q3.05.4

Hence, the correct option is (B).

Q3.04.1 MCQ | 2004 | 1M | Ans: B | ● ○ ○

Consider non-inverting Op-Amp circuit shown below,

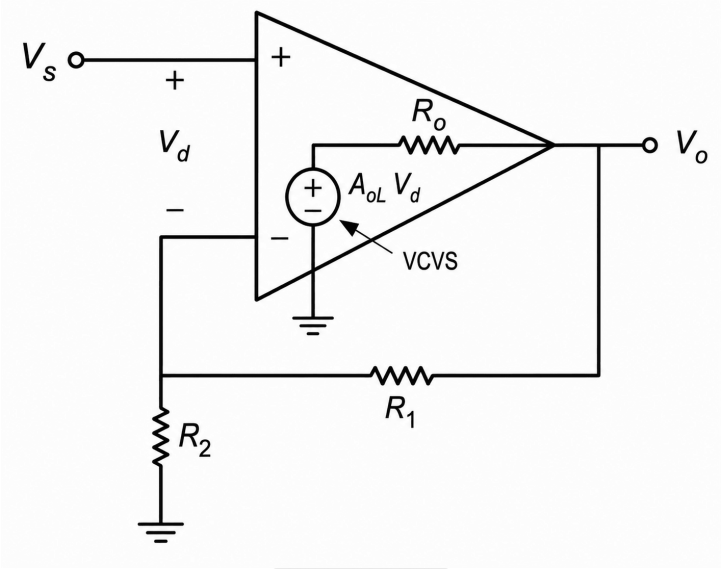

Fig. Solution Q3.04.1

Here the op-amp output voltage is dependent on the difference in input terminal voltages. So, ideal Op-Amp is an ideal voltage controlled voltage source.

(B) voltage controlled voltage source

Key Points

- **Voltage Controlled Voltage Source (VCVS):** The output voltage is generated by a dependent voltage source whose value is directly proportional to the differential input voltage ($V_d = V_+ - V_-$). Therefore, it acts as a VCVS governed by the equation $V_0 = A_{OL}V_d$, where A_{OL} is the open-loop gain.

Q3.04.2 MCQ 2004 1M Ans: A ●○○

Method 1

Case 1 : At low frequency ($\omega \to 0$) : Capacitive impedance,

$$X_C = \frac{1}{\omega C} = \infty \quad \text{[O.C.]}$$

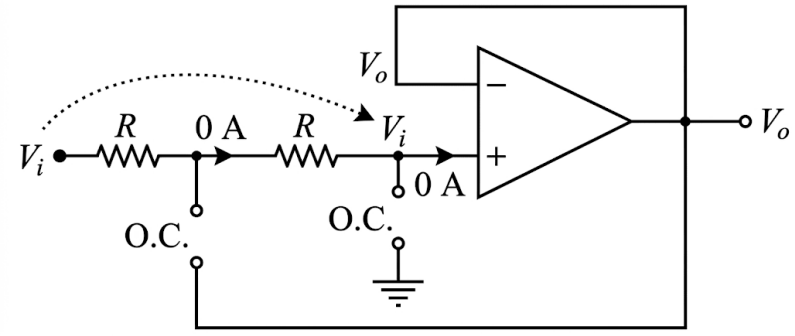

Fig. Solution Q3.04.2

From virtual ground concept,

$$V_+ = V_-$$

Since, $V_+ = V_i, V_- = V_0, V_0 = V_i$

Case 2 : At high frequency ($\omega \to \infty$) : Capacitive impedance,

$$X_C = \frac{1}{\omega C} = 0 \quad \text{[S.C.]}$$

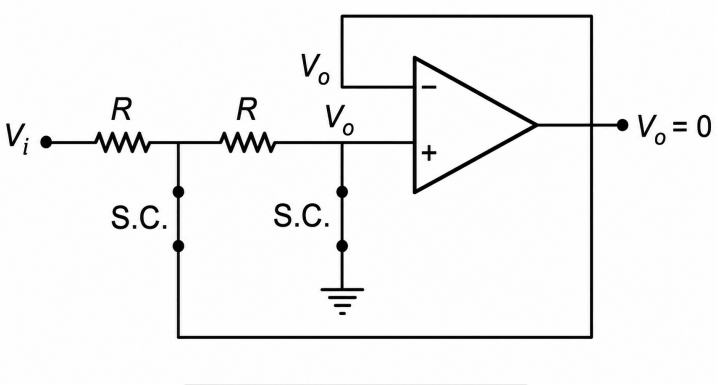

Fig. Solution Q3.04.2

From virtual ground concept,

$$V_+ = V_-$$

Since, $V_+ = 0, V_- = V_0, V_+ = 0, \quad V_- = V_0$

From case 1 and case 2, output voltage is given by,

$$V_0 = \begin{cases} V_i & ; \quad \text{At low frequency} \\ 0 & ; \quad \text{At high frequency} \end{cases}$$

So, the given Op-Amp acts like a low pass filter. Hence, the correct option is (A).

Method 2

Let us assume, $R = 1\,\Omega$ and $C = 1\,\text{F}$

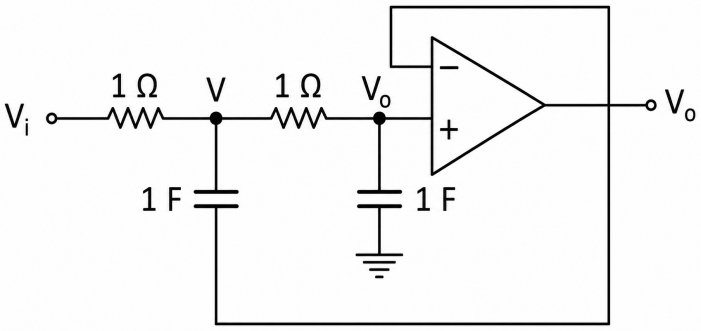

Fig. Solution Q3.04.2

From virtual ground concept,

$$V_+ = V_- = V_0$$

Applying KCL at node V,

$$\frac{V - V_i}{1} + \frac{V - V_0}{1/s} + \frac{V - V_0}{1} = 0$$

$$(s+2)V - V_i = V_0(s+1)$$

$$V = \frac{(s+1)V_0 + V_i}{s+2} \quad \ldots(\text{i})$$

Applying KCL at node V_0,

$$\frac{V_0 - V}{1} + \frac{V_0 - 0}{1/s} = 0$$

$$(s+1)V_0 = V \quad \ldots(\text{ii})$$

From equation (i) and (ii),

$$\frac{(s+1)V_0 + V_i}{s+2} = (s+1)V_0$$

$$V_0\left[\frac{s+1}{s+2} - (s+1)\right] = \frac{-V_i}{s+2}$$

$$\frac{V_0}{V_i} = \frac{-1}{(s+1) - (s+1)(s+2)}$$

$$\frac{V_0}{V_i} = \frac{1}{s^2 + 2s + 1}$$

$$\left.\frac{V_0}{V_i}\right|_{s=0} = 1, \quad \left.\frac{V_0}{V_i}\right|_{s=\infty} = 0$$

So, the given circuit acts as a low pass filter.

(A) low-pass filter

Key Points

- **Sallen-Key Topology:** The given circuit is a classic Sallen-Key active second order low-pass filter configuration.

Q3.04.3 MCQ | 2004 | 2M | Ans: A | ● ○ ○

Given circuit is shown below,

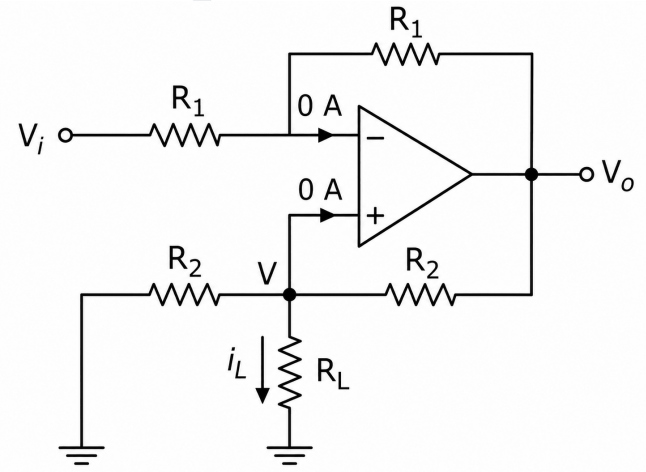

Fig. Solution Q3.04.3

From virtual ground concept,

$$V_+ = V_- = V$$

Applying KCL at inverting terminal,

$$\frac{V - V_i}{R_1} + \frac{V - V_0}{R_1} = 0$$

$$V = \frac{V_0 + V_i}{2} \quad \ldots (i)$$

Applying KCL at non-inverting terminal,

$$\frac{V}{R_2} + i_L + \frac{V - V_0}{R_2} = 0$$

$$R_2 i_L = V_0 - 2V \quad \ldots (ii)$$

From equation (i) and (ii),

$$R_2 i_L = V_0 - V_0 - V_i$$

$$i_L = \frac{-V_i}{R_2}$$

(A) $-\frac{V_i}{R_2}$

Key Points

- **Voltage-to-Current Converter:** The given circuit acts as an ideal current source for the load R_L. The load current i_L is entirely independent of the load resistor R_L and is determined solely by the input voltage and the scaling resistor R_2.

Q3.03.1 MCQ | 2003 | 1M | Ans: B | ● ○ ○

Given, Peak to peak voltage $(V_{p-p}) = 8\,\text{V}$

$$2 \times \text{peak voltage } (V_p) = 8\,\text{V}$$

$$\text{Peak voltage } (V_p) = 4\,\text{V}$$

So, $V_i(t) = 4 \sin \omega t$

Given circuit is shown below,

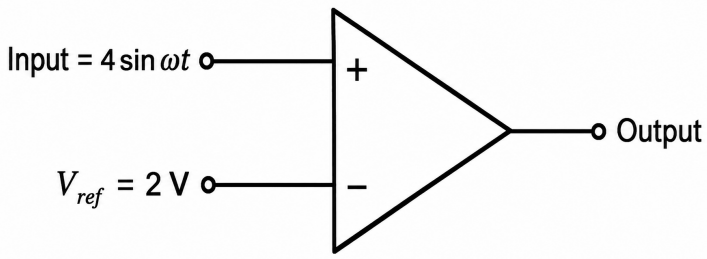

Fig. Solution Q3.03.1

$$V_i = 4 \sin \omega t$$

$$V_i(\omega t = \pi/6) = 4 \sin 30^\circ = 2\,\text{V}$$

$$V_i(\omega t = 5\pi/6) = 4 \sin 150^\circ = 2\,\text{V}$$

The given circuit is a comparator circuit and hence its output is given by,

$$V_0 = \begin{cases} +V_{sat} & ; \quad V_+ > V_- \\ -V_{sat} & ; \quad V_+ < V_- \end{cases}$$

Hence, the output waveform of comparator circuit is shown below,

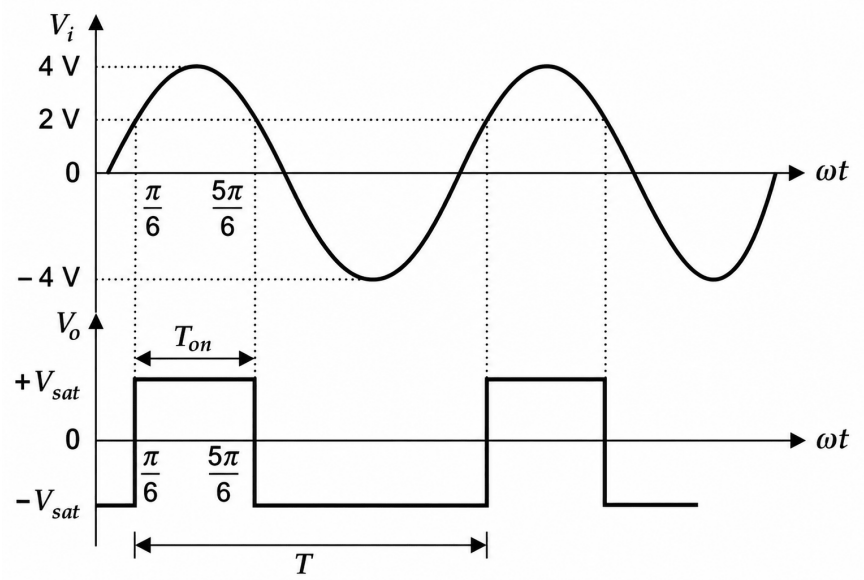

Fig. Solution Q3.03.1

$$\text{Duty cycle} = \frac{T_{ON}}{T} = \frac{\frac{5\pi}{6} - \frac{\pi}{6}}{2\pi} = \frac{\frac{4\pi}{6}}{2\pi} = \frac{1}{3}$$

$$\boxed{\text{(B)} \quad \frac{1}{3}}$$

Key Points

- For a periodic pulse wave, the duty cycle is defined as the fraction of one period in which a signal is active (high). It is calculated as Duty Cycle = $\frac{T_{ON}}{T}$, where T_{ON} is the high time and T is the total period.

Q3.03.2 MCQ 2003 1M Ans: C ●○○

Given, $A_d = 48\,\text{dB}$, $A_c = 2\,\text{dB}$

Common mode rejection ratio is given by,

$$\text{CMRR} = \left|\frac{A_d}{A_c}\right|$$

$$\text{CMRR (dB)} = 20\log|A_d| - 20\log|A_c|$$

$$\text{CMRR (dB)} = A_d(\text{dB}) - A_c(\text{dB})$$

$$\text{CMRR (dB)} = 48\,\text{dB} - 2\,\text{dB} = 46\,\text{dB}$$

$$\boxed{\text{(C)} \quad 46\,\text{dB}}$$

Key Points

- **Common Mode Rejection Ratio (CMRR):** A critical figure of merit for differential amplifiers, defining its ability to reject noise or interference signals that are common to both input terminals. It is mathematically defined as the magnitude of the ratio of differential voltage gain to common-mode voltage gain.

Q3.03.3 MCQ 2003 2M Ans: A ● ○ ○

Given circuit is,

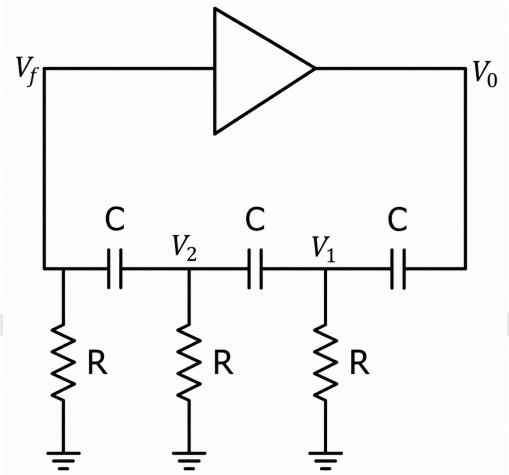

Fig. Solution Q3.03.3

Let the feedback network voltages be V_0, V_1, V_2, and V_f, where V_0 is the amplifier output and V_f is the feedback voltage applied to the amplifier input.
Let $x = sRC$, using nodal analysis at node V_f,

$$(V_f - V_2)sC + \frac{V_f}{R} = 0$$

$$(1 + x)V_f = xV_2$$

$$V_f = \frac{x}{1 + x}V_2$$

Applying KCL at V_2,

$$(V_2 - V_1)sC + (V_2 - V_f)sC + \frac{V_2}{R} = 0$$

Substituting V_f,

$$(1 + 3x + x^2)V_2 = x(1 + x)V_1$$

$$V_2 = \frac{x(1 + x)}{1 + 3x + x^2}V_1$$

Applying KCL at V_1,

$$(V_1 - V_0)sC + (V_1 - V_2)sC + \frac{V_1}{R} = 0$$

Substituting V_2 and simplifying,

$$\frac{V_f}{V_0} = \frac{x^3}{x^3 + 6x^2 + 5x + 1}$$

Hence the feedback transfer function becomes,

$$\beta(s) = \frac{(sRC)^3}{(sRC)^3 + 6(sRC)^2 + 5(sRC) + 1}$$

For oscillation, the total phase shift around the loop must be 360°. Since the amplifier is ideal and inverting, it contributes 180°. Therefore, the RC network must contribute an additional 180°.
Substituting $s = j\omega$ and defining $v = \omega RC$, gives

$$\beta(j\omega) = \frac{-jv^3}{(1-6v^2) + j(5v - v^3)}$$

For the RC network to provide a phase shift of 180°, the denominator must be purely imaginary. Hence,

$$1 - 6v^2 = 0$$

Therefore,

$$v^2 = \frac{1}{6}$$

$$\omega_0 = \frac{1}{RC\sqrt{6}}$$

Hence, the frequency of oscillation is

$$f_0 = \frac{\omega_0}{2\pi} = \frac{1}{2\pi RC\sqrt{6}}$$

$$\boxed{\text{(A)} \quad \frac{1}{2\pi\sqrt{6}\,RC}}$$

Key Points

- For three identical RC sections, $\omega_0 = \dfrac{1}{RC\sqrt{6}}$.

Q3.03.4 MCQ | 2003 | 2M | Ans: B | ● ○ ○

Method 1

Apply superposition theorem,

(i) Consider only $V_1 = 2\,V$:

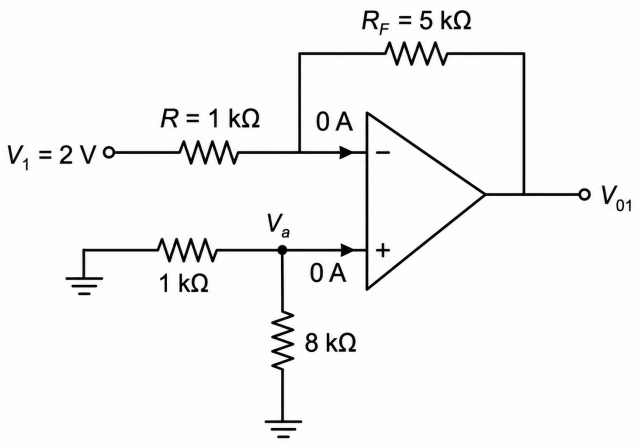

Fig. Solution Q3.03.4

From figure, $V_a = 0\,V$
Output of inverting Op-Amp is given by,

$$V_{0_1} = \left(\frac{-R_F}{R}\right) V_1 = \left(\frac{-5}{1}\right) \times 2$$

$$V_{0_1} = -10 \text{ Volt}$$

(ii) Consider only $V_2 = 3$ V :

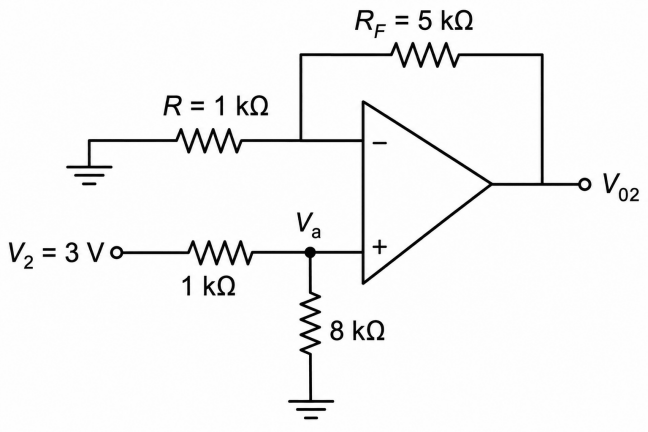

Fig. Solution Q3.03.4

By voltage division rule,

$$V_a = \left[\frac{8}{8+1}\right] \times 3 = \frac{8}{9} \times 3 = \frac{8}{3} \text{ Volt}$$

Output of non-inverting Op-Amp is given by,

$$V_{0_2} = \left[1 + \frac{R_F}{R}\right] V_a$$

$$V_{0_2} = \left[1 + \frac{5}{1}\right] \times \frac{8}{3} = 6 \times \frac{8}{3} = 16 \text{ Volt}$$

According to superposition theorem, total output voltage V_0 is given by,

$$V_0 = V_{0_1} + V_{0_2} \quad \text{[Superposition theorem]}$$

$$V_0 = (16 - 10) = 6 \text{ Volt}$$

Method 2

From virtual ground concept,

$$V_- = V_+ = V_a$$

From given figure,

$$V_a = \frac{3 \times 8}{1 + 8} = \frac{8}{3} \text{ V} \quad \text{[By VDR]}$$

Applying KCL at inverting terminal,

$$\frac{V_a - V_1}{1} + \frac{V_a - V_{out}}{5} = 0$$

$$\frac{\frac{8}{3} - 2}{1} + \frac{\frac{8}{3} - V_{out}}{5} = 0 \quad [V_1 = 2 \text{ V}]$$

$$\frac{40}{3} - 10 + \frac{8}{3} - V_{out} = 0$$

$$V_{out} = \frac{48}{3} - 10 = 16 - 10 = 6 \text{ V}$$

(B) 6V

Q3.02.1 MCQ 2002 1M Ans: B ● ○ ○

Given,
(i) Gain × (3 dB Bandwidth) = 1 MHz
(ii) $(\text{gain})_{\text{dB}} = 20\,\text{dB}$

$$20\log_{10}(\text{gain}) = 20$$

$$\text{Gain} = 10^1 = 10$$

So, $10 \times (3\,\text{dB Bandwidth}) = 1\,\text{MHz}$

$$(3\,\text{dB Bandwidth}) = 10^5\,\text{Hz} = 100\,\text{kHz}$$

(B) 100 KHz

Q3.02.2 MCQ 2002 2M Ans: C ● ○ ○

The slew rate is defined as:

$$S.R = \left(\frac{dV_0}{dt}\right)_{\max}$$

When the input is a sinusoid given as:

$$V_i(t) = V_m \sin(2\pi f t)$$

And the voltage gain of the op-amp is A_v, the output voltage will be:

$$V_0 = A_v V_i$$

$$V_0 = A_v V_m \sin(2\pi f t)$$

Now, the rate of change in output will be:

$$\frac{dV_0}{dt} = A_v V_m 2\pi f \cos(2\pi f t)$$

The **maximum rate of change (Slew rate)** for undistorted output will be:

$$\left(\frac{dV_0}{dt}\right)_{\max} = A_v V_m 2\pi f_{\max} \quad \ldots(1)$$

Calculation:
Gain = 40 dB, i.e.

$$20\log_{10} A_v = 40$$

$$A_v = 100$$

Putting on the respective values in Equation (1), we can write:

$$\frac{1}{10^{-6}} = 100 \times V_m \times 2\pi \times 20 \times 10^3$$

$$V_m = \frac{1}{2\pi \times 2} = 79.5\,\text{mV}$$

$\therefore$ The input level must not exceed above 79.5 mV otherwise output will be distorted.

(C) 79.5 mV

Key Points

- **Slew Rate (SR):** The maximum rate of change of the output voltage of an operational amplifier. It sets a limit on the maximum frequency and amplitude of a signal that can be passed without distortion.
- **Slew Rate Equation:** For a sinusoidal output $V_0(t) = V_p \sin(2\pi ft)$, the maximum rate of change is $V_p 2\pi f$. To avoid slew-rate distortion, this must be less than or equal to the op-amp's SR: SR $\geq 2\pi f_{\max} V_{p,\text{out}}$.

Q3.02.3 MCQ 2002 2M Ans: A ●○○

Given circuit is,

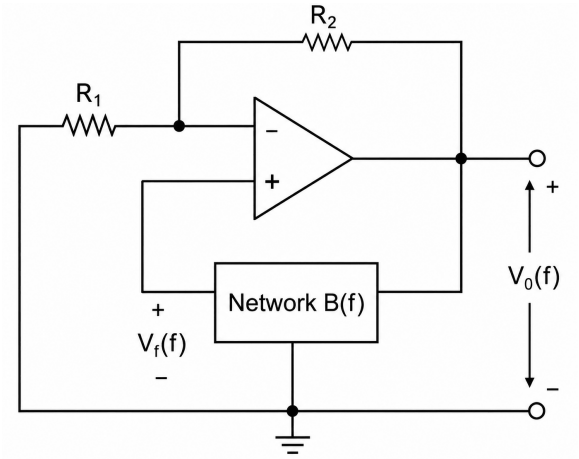

Fig. Solution Q3.02.3

The circuit employs positive feedback through the network $B(f)$. At the oscillation frequency,

$$B(f_0) = \frac{V_f(f)}{V_o(f)} = \frac{1}{6}\angle 0^\circ$$

To sustain oscillations, the Barkhausen criterion must be satisfied:

$$A\beta = 1\angle 0^\circ$$

Since $\beta = B(f_0) = \frac{1}{6}$,

$$A = \frac{1}{\beta} = 6$$

The above op-amp with R_1 and R_2 is a non-inverting amplifier whose gain is

$$A = 1 + \frac{R_2}{R_1}$$

$$1 + \frac{R_2}{R_1} = 6 \implies \frac{R_2}{R_1} = 5$$

$$R_2 = 5R_1$$

$$\boxed{\text{(A)} \quad R_2 = 5R_1}$$

Mistakes to be avoided

- Do not use $A = \frac{R_2}{R_1}$. Check properly where the feedback input is connected.

Q3.01.1 MCQ 2001 1M Ans: A ● ○ ○

Input Resistance (R_i)
An ideal OP-AMP possesses an **infinite input resistance**. This means:

- The resistance looking into the amplifier's input terminals (both the non-inverting '+' and inverting '−' terminals) is infinitely large.
- As a result, no current flows into either input terminal. For an ideal OP-AMP, the input currents I_+ and I_- are both equal to zero ($I_+ = I_- = 0$).
- This infinite input impedance prevents the OP-AMP from drawing any current from the signal source. It ensures that the input signal voltage is accurately applied to the amplifier without being affected by the amplifier's own input circuitry (i.e., no loading effect on the source).

Open-Loop Voltage Gain (A)
An ideal OP-AMP is characterized by an **infinite open-loop voltage gain**. This signifies:

- The open-loop gain (A) is the amplification provided by the OP-AMP before any external feedback is applied. For an ideal device, $A = \infty$.
- An infinite gain implies that the amplifier can produce an extremely large output voltage change for even a minuscule difference between the input voltages ($V_+ - V_-$).
- In practical amplifier circuits that utilize negative feedback, this very high gain forces the voltage difference between the two input terminals to be essentially zero ($V_+ - V_- \approx 0$). This principle is often referred to as the "virtual short" condition, which is fundamental for analyzing OP-AMP circuits with negative feedback.

Output Resistance (R_0)
An ideal OP-AMP has a **zero output resistance**. This means:

- The output resistance (R_0) is the internal resistance that appears when looking back into the output terminal of the amplifier.
- For an ideal OP-AMP, $R_0 = 0$. This ensures that the amplifier can deliver its full output voltage to any connected load, regardless of the current drawn by that load.
- A zero output resistance guarantees that the output voltage remains constant and is not affected by the load impedance connected to the output. There is no voltage drop across any internal resistance.

(a) $R_i = \infty,\ A = \infty,\ R_o = 0$

Q3.01.2 MCQ 2001 2M Ans: B ● ● ○

Given circuit is shown below,

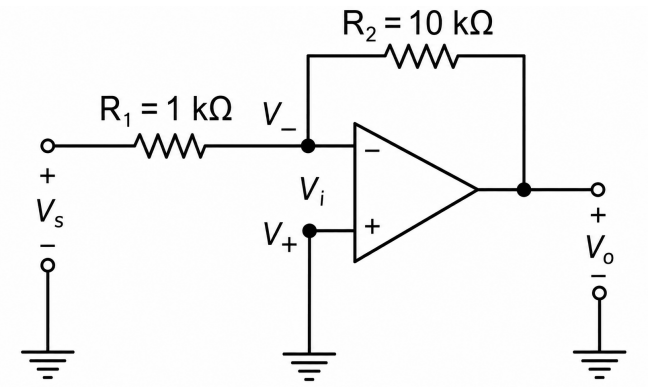

Fig. Solution Q3.01.2

This is a practical inverting Op-amp due to finite open loop gain 100 i.e. $A_{OL} = 100$
From figure,

$$V_d = V_i = V_+ - V_-$$

$$A_{OL} = \frac{V_0}{V_d} = \frac{V_0}{V_i}$$

$$V_0 = A_{OL} \times V_d$$

$$V_0 = A_{OL}(V_+ - V_-)$$

$$V_+ = 0\,\text{V} \quad \text{[From figure]}$$

$$V_0 = 100(0 - V_-)$$

$$V_0 = -100V_- \quad \ldots(\text{i})$$

Calculation of V_- :
Applying KCL at inverting terminal,

$$\frac{V_- - V_s}{1} + \frac{V_- - V_0}{10} = 0$$

$$11V_- - 10V_s = V_0$$

$$V_- = \frac{10V_s + V_0}{11} \quad \ldots(\text{ii})$$

From equation (i) and (ii),

$$V_0 = -100\left(\frac{10V_s + V_0}{11}\right)$$

$$V_0 + \frac{100V_0}{11} = \frac{-1000V_s}{11}$$

$$\frac{111V_0}{11} = \frac{-1000V_s}{11}$$

$$\frac{V_0}{V_s} = \frac{-1000}{111}$$

$$\frac{V_0}{V_s} = -9.009$$

(B) -9

Key Points

- **Finite Open-Loop Gain:** When an op-amp has a finite open-loop gain (A_{OL}), the virtual ground concept ($V_- = V_+$) is no longer strictly true. You must use the fundamental relationship $V_0 = A_{OL}(V_+ - V_-)$.

Mistakes to be avoided

$$\frac{V_0}{V_S} = \frac{A_{OL}}{1 + A_{OL}\beta}$$

This result is valid for practical non-inverting Op-Amp. Do not use this for inverting configurations.

Q3.01.3 MCQ 2001 2M Ans: A ● ○ ○

Given circuit in s-domain is given below.

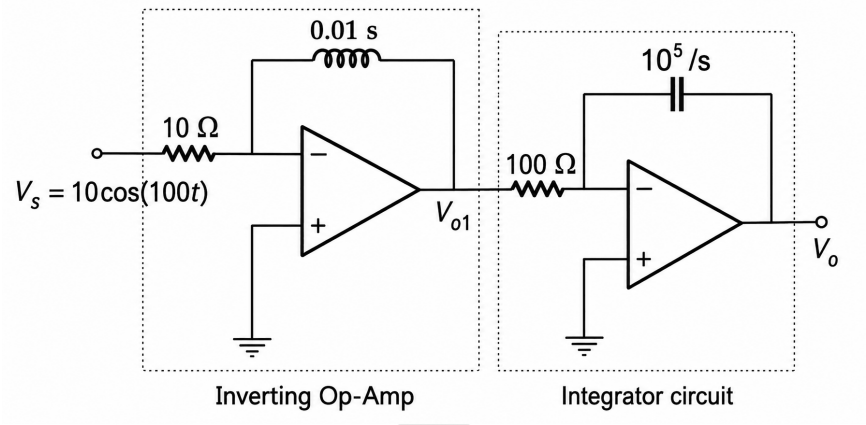

Fig. Solution Q3.01.3

From virtual ground concept,

$$V_+ = V_- = 0$$

Gain of inverting Op-Amp is given by,

$$\frac{V_{0_1}}{V_s} = \frac{-0.01s}{10} = -10^{-3}s$$

$$V_{0_1} = -10^{-3}sV_s \quad \ldots (i)$$

Applying KCL at inverting terminal of second Op-Amp,

$$\frac{0 - V_{0_1}}{100} + \frac{0 - V_0}{10^5/s} = 0$$

$$V_0 = \frac{-10^3}{s}V_{0_1}$$

From equation (i),

$$V_0 = \frac{-10^3}{s} \times (-10^{-3}sV_s)$$

$$V_0 = V_s = 10\cos(100t)$$

$$\boxed{\text{(A)} \quad 10\cos(100t)}$$

Q3.00.1 MCQ 2000 1M Ans: D ● ○ ○

Given circuit is shown below,

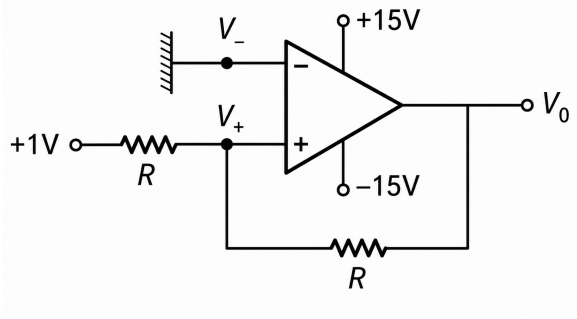

Fig. Solution Q3.00.1

Due to positive feedback, virtual ground concept is not valid (i.e. $V_+ \neq V_-$).
From figure, $V_- = 0\,\text{V}$
V_+ will be positive due to applied positive voltage. So,

$$V_+ > V_-$$

Hence, $V_0 = +V_{sat} = +15\,\text{V}$

(D) +15 V

Mistakes to be avoided

- Applying the virtual ground concept ($V_+ = V_-$) to a circuit with positive feedback. This is a very common error. Virtual ground is strictly valid only when negative feedback dominates.

Q3.00.2 MCQ 2000 1M Ans: C ● ○ ○

Given circuit in s-domain is given as

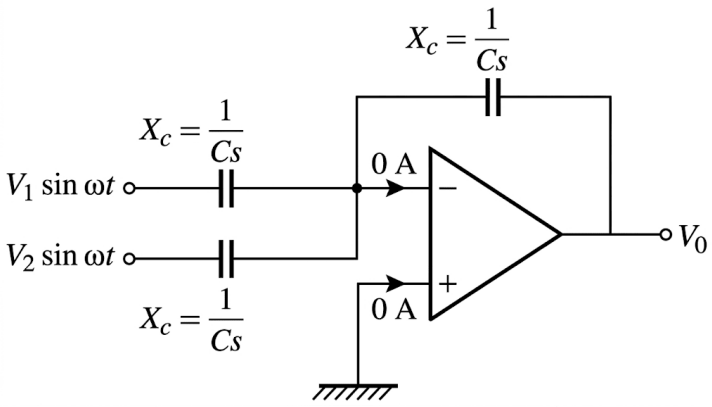

Fig. Solution Q3.00.2

From virtual ground concept,

$$V_+ = V_- = 0\,\text{V}$$

Applying KCL at inverting terminal,

$$\frac{0 - V_1 \sin \omega t}{1/Cs} + \frac{0 - V_2 \sin \omega t}{1/Cs} + \frac{0 - V_0}{1/Cs} = 0$$

$$V_0 = -(V_1 + V_2) \sin \omega t$$

(C) $-(V_1 + V_2) \sin \omega t$

Q3.00.3 MCQ 2000 1M Ans: A ● ○ ○

Given circuit is shown below,

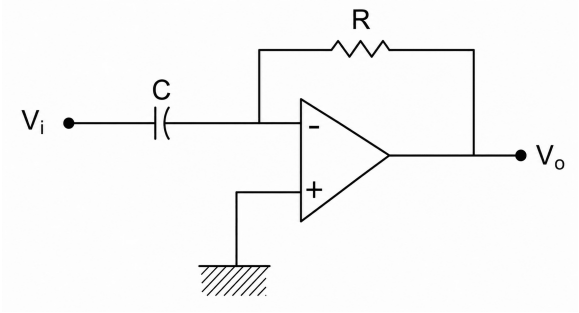

Fig. Solution Q3.00.3

It is an ideal differentiator circuit.
Waveform of input is shown below,

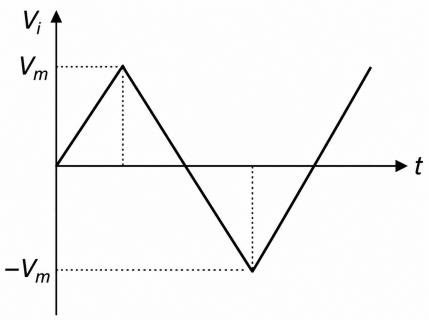

Fig. Solution Q3.00.3

Output of differential circuit is given by,

$$V_0 = -RC\frac{d}{dt}V_i$$

Hence, output waveform is shown below,

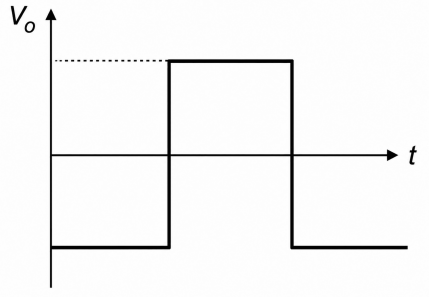

Fig. Solution Q3.00.3

So, $\frac{d}{dt}$ (Triangular wave) = Square wave

(A) square wave

Q3.00.4 MCQ | 2000 | 2M | Ans: C

Given, Input offset voltage, $V_{ios} = 5\,\text{mV}$
Open loop voltage gain, $A_{OL} = 10000$
Given circuit is shown below,

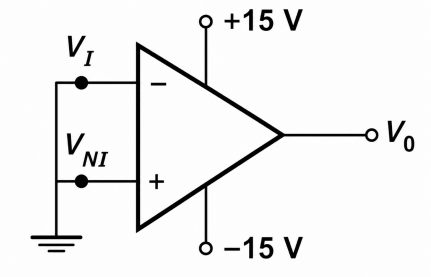

Fig. Solution Q3.00.4

Method 1

Open loop gain, $A_{OL} = \frac{V_0}{V_d}$

$$A_{OL} \neq \infty \quad \text{i.e.} \quad V_d \neq 0$$

$$V_d = V_{NI} - V_I$$

Case 1 : If $V_{NI} = V_{ios}$, then $V_I = 0$

$$V_d = V_{ios} - V_I = 5\,\text{mV} - 0$$

$$V_d = 5\,\text{mV} \quad \ldots (\text{i})$$

Case 2 : If $V_I = V_{ios}$, then $V_{NI} = 0$

$$V_d = V_{NI} - V_{ios} = 0 - 5\,\text{mV}$$

$$V_d = -5\,\text{mV} \quad \ldots (\text{ii})$$

From case 1 and case 2, differential input voltage V_d can be written as given below,

$$V_d = \pm 5\,\text{mV}$$

Output voltage is given by,

$$V_0 = A_{OL} V_d$$

$$V_0 = 10000 \times (\pm 5\,\text{mV}) = \pm 50\,\text{V}$$

But output voltage cannot be greater than saturation voltage. Therefore, $V_0 = \pm 15\,\text{V}$

Method 2

Output voltage $V_0 = A_{OL} \times V_{ios}$

$$V_0 = 10^4 \times 5 \times 10^{-3}$$

$$V_0 = 50\,\text{V}$$

Since, in Open loop Op-Amp, output can not be greater than $\pm V_{sat}$. Therefore output voltage will be $+V_{sat} (= 15\,\text{V})$ or $-V_{sat} (= -15\,\text{V})$.

(C) + 15 V or −15 V

Key Points

- **Input Offset Voltage (V_{ios}):** The small internal mismatch in the differential input stage, modeled as a small DC voltage source in series with one of the input terminals. Its exact polarity is usually unknown, hence the ± consideration.

Q3.99.1 NAT | 1999 | 5M | Ans: * | ● ○ ○

Given circuit is,

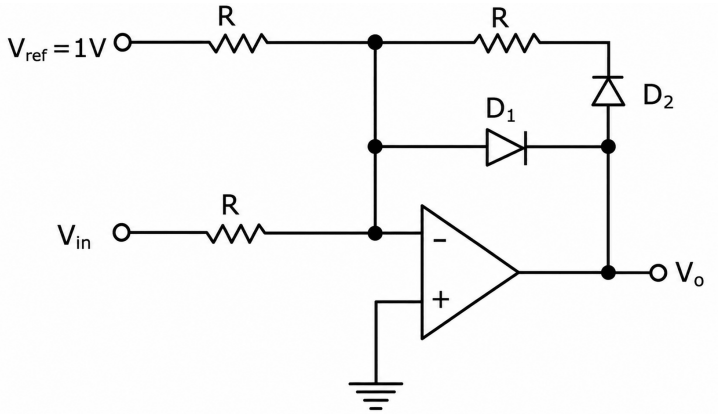

Fig. Solution Q3.99.1

The op-amp maintains negative feedback as long as at least one of the diodes is ON.

Case 1: Initially assume both diodes are OFF

When both diodes are OFF, there is no feedback connection, so the op-amp is in open loop mode(comparator). Since $V_+ = 0$ and assuming $V_{in} = +2$V, V_- is a positive voltage and as $V_- > V_+$ output will go to towards $-V_{sat}$. So D1 becomes ON and D2 becomes OFF.

Case 2: When D1 is ON and D2 is OFF

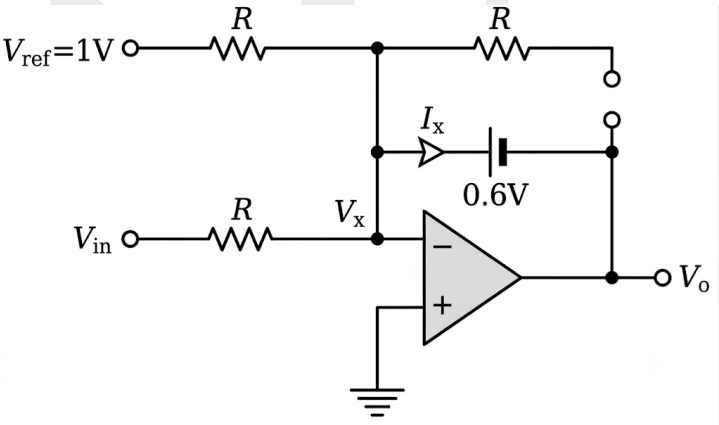

Fig. Solution Q3.99.1

Due to the virtual ground concept, $V_x = V_+ = 0$ V.

Apply KCL at node V_x:

$$\frac{V_{in} - 0}{R} + \frac{V_{ref} - 0}{R} = I_x$$

$$I_x = \frac{V_{in}}{R} + \frac{1}{R} = \frac{V_{in} + 1}{R} \quad (\text{since } V_{ref} = 1\text{ V})$$

Diode D1 is ON as long as the current $I_x > 0$

$$\frac{V_{in}+1}{R} > 0 \implies V_{in} > -1\,\text{V}$$

When D1 is ON, the forward voltage drop is 0.6 V.

$$V_x - V_o = 0.6\,\text{V} \implies 0 - V_o = 0.6\,\text{V} \implies V_o = -0.6\,\text{V}$$

$$\therefore V_0 = -0.6\,\text{V} \quad \text{for } -1 < V_{in} < 2\,\text{V}$$

Case 3: When D2 is ON and D1 is OFF

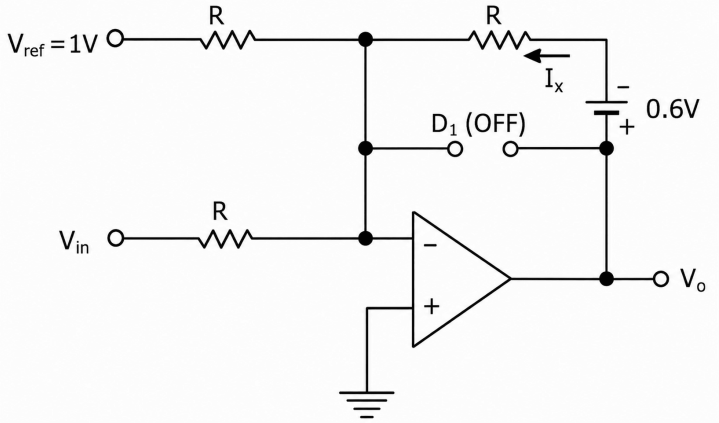

Fig. Solution Q3.99.1

Applying KCL at V_x,

$$\frac{V_{ref}-0}{R} + \frac{V_{in}-0}{R} + I_x = 0$$

$$\frac{V_{in}+1}{R} + I_x = 0 \implies I_x = -\left(\frac{V_{in}+1}{R}\right)$$

Diode D2 is ON as long as current $I_x > 0$

$$\frac{V_{in}+1}{R} < 0 \implies V_{in} < -1\,\text{V}$$

Now apply KCL at V_x again,

$$\frac{V_{in}-0}{R} + \frac{1-0}{R} + \frac{V_0-0.6-0}{R} = 0$$

$$V_{in} + V_0 + 0.4 = 0 \implies V_0 = -V_{in} - 0.4$$

$$\therefore V_0 = -V_{in} - 0.4\,\text{V} \quad \text{for } -2 < V_{in} < -1\,\text{V}$$

Transfer Characteristic Summary:

$$V_o = \begin{cases} -V_{in} - 0.4\,\text{V} & \text{for } -2 < V_{in} < -1\,\text{V} \\ -0.6\,\text{V} & \text{for } -1 < V_{in} < 2\,\text{V} \end{cases}$$

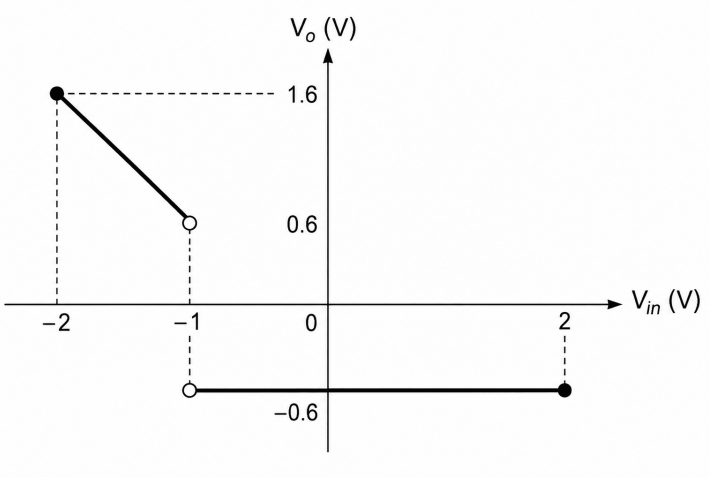

Fig. Solution Q3.99.1

Q3.98.1 MCQ 1998 1M Ans: D

The comparator of the op-amp does not contain any feedback and it will generate a rectangular wave at the output with saturation voltage levels when the input is a sine wave.

When $V_+ > V_-$, the output is $+V_{sat}$

When $V_+ < V_-$, the output is $-V_{sat}$

Since $V_- = 0$, the output will be a square wave as sine wave has one half cycle of positive values and one half cycle of negative values in one period.

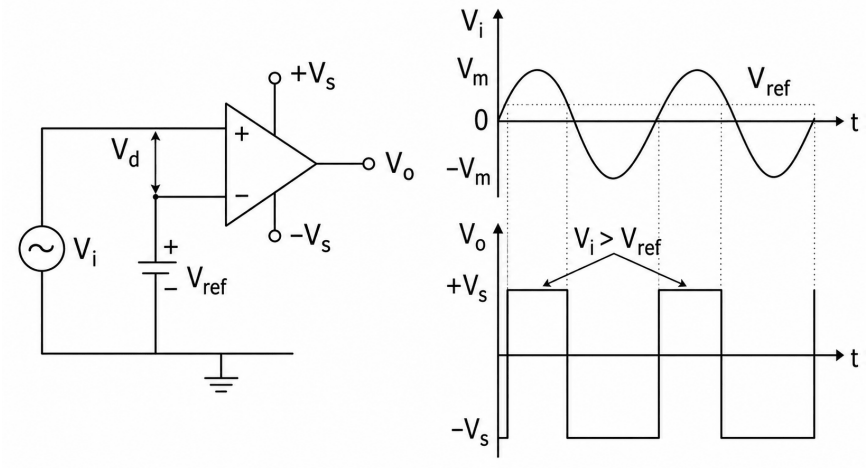

Fig. Solution Q3.98.1

(D) a square wave

Q3.98.2 NAT | 1998 | 5M | Ans: * | ● ○ ○

Given circuit is,

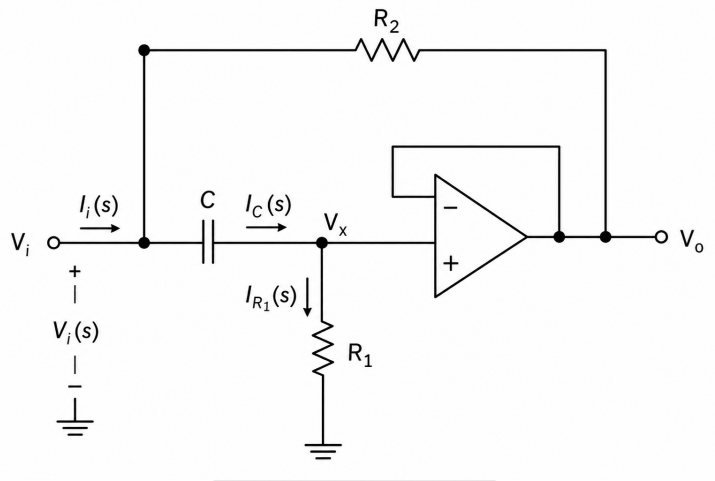

Fig. Solution Q3.98.2

The op-amp is configured as a voltage follower. Therefore,

$$V_o = V_+ = V_x$$

Applying KCL at node V_x,

$$\frac{V_x}{R_1} + sC(V_x - V_i) = 0$$

$$V_x\left(\frac{1}{R_1} + sC\right) = sCV_i$$

$$V_x = \frac{sCR_1}{1 + sCR_1} V_i$$

The input current is

$$I_i = I_C + I_{R_2}$$

where

$$I_C = sC(V_i - V_x) = \frac{sCV_i}{1 + sCR_1}$$

$$I_{R_2} = \frac{V_i - V_o}{R_2} = \frac{V_i - V_x}{R_2} = \frac{V_i}{R_2(1 + sCR_1)}$$

Hence,

$$I_i = \frac{V_i}{1 + sCR_1}\left(sC + \frac{1}{R_2}\right)$$

$$\therefore Z_{in} = \frac{V_{in}}{I_{in}} = \frac{1 + sCR_1}{sC + \frac{1}{R_2}}$$

$$\boxed{Z_{in} = R_2\frac{1 + sCR_1}{1 + sCR_2}}$$

Substituting $s = j\omega$,

$$Z_{in} = R_2\frac{1 + j\omega CR_1}{1 + j\omega CR_2}$$

Multiplying by the complex conjugate of the denominator,

$$Z_{in} = R_2 \frac{1 + \omega^2 C^2 R_1 R_2 + j\omega C(R_1 - R_2)}{1 + (\omega C R_2)^2}$$

Here the imaginary part is,

$$Im\{Z_{in}\} = \frac{\omega C R_2 (R_1 - R_2)}{1 + (\omega C R_2)^2}$$

The impedance is inductive if the imaginary part is positive. Therefore, the circuit behaves inductively when

$$R_1 > R_2$$

Q3.97.1 MCQ | 1997 | 1M | Ans: D | ● ○ ○

Given circuit is,

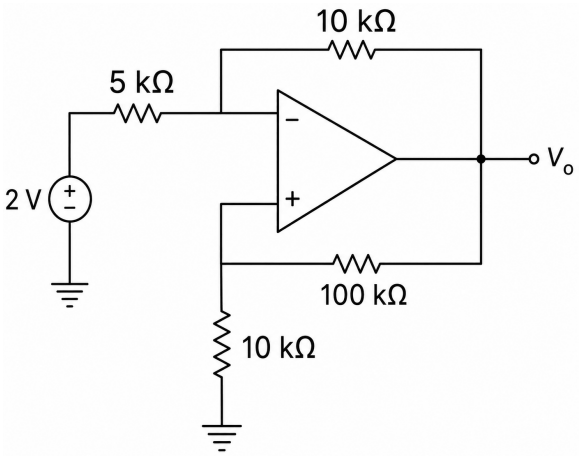

Fig. Solution Q3.97.1

Here the op-amp has both positive and negative feedback. To find out which feedback is dominating, find out input terminal voltages by shorting DC sources.

$$V_- = V_0 \left(\frac{5}{5+10}\right) = \frac{V_0}{3}$$

$$V_+ = V_0 \left(\frac{10}{10+100}\right) = \frac{V_0}{11}$$

Since $V_- > V_+$, negative feedback is dominating.
From virtual ground concept, $V_+ = V_- = (V_0/11)$
Applying KCL at inverting terminal,

$$\frac{V_- - 2}{5} + \frac{V_- - V_0}{10} = 0$$

$$-4 + 2V_- + V_- - V_0 = 0$$

$$3V_- - V_0 = 4$$

$$3\left(\frac{V_0}{11}\right) - V_0 = 4$$

$$\frac{3V_0}{11} - V_0 = 4$$

$$\frac{-8V_0}{11} = 4 \implies V_0 = \frac{-44}{8} = -5.5\,\text{V}$$

(D) -5.5 V

Key Points

- When op-amp has both positive and negative feedback, find out which feedback is dominating by calculating input terminal voltages by deactivating DC sources.

Q3.97.2 NAT | 1997 | 3M | Ans: * | ● ○ ○

Given circuit is,

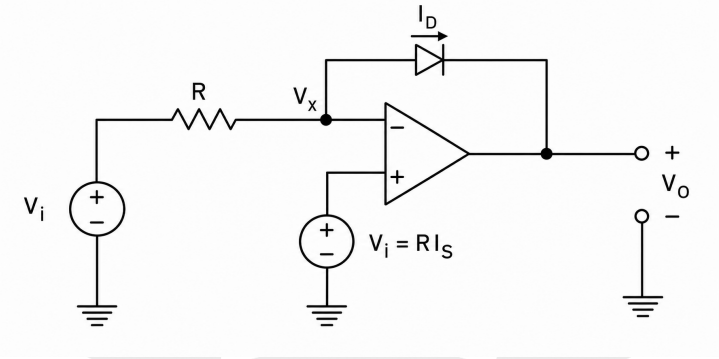

Fig. Solution Q3.97.2

For an ideal operational amplifier operating with negative feedback,

$$V_- = V_+ = RI_s$$

Let the voltage at the inverting terminal be V_x.

$$V_x = RI_s$$

Applying KCL at the inverting node,

$$I_D = \frac{V_i - V_x}{R}$$

$$I_D = \frac{V_i - RI_s}{R}$$

The diode current is given by

$$I_D = I_s\left(e^{\frac{V_D}{kT/q}} - 1\right)$$

Since the diode is connected between the op-amp output and the inverting node,

$$V_D = V_x - V_0 = RI_s - V_o$$

Therefore,

$$\frac{V_i - RI_s}{R} = I_s\left(e^{\frac{RI_s - V_O}{kT/q}} - 1\right)$$

$$e^{\frac{RI_s - V_O}{kT/q}} = 1 + \frac{V_i - RI_s}{RI_s} = \frac{V_i}{RI_s}$$

Taking natural logarithm,

$$\boxed{V_o = RI_s - \frac{kT}{q}\ln\left(\frac{V_i}{RI_s}\right)}$$

Q3.97.3 NAT | 1997 | 2M | Ans: 5.45 | ● ○ ○

From above question,

$$V_o = RI_s - \frac{kT}{q}\ln\left(\frac{V_i}{RI_s}\right)$$

Given,

$$R = 100\,\text{k}\Omega, \qquad I_s = 1\,\mu\text{A}, \qquad \frac{kT}{q} = 25\,\text{mV}$$

Setting $V_o = 0$,

$$0 = (100\times10^3)(1\times10^{-6}) - 0.025\ln\left(\frac{V_i}{0.1}\right)$$

$$\ln\left(\frac{V_i}{0.1}\right) = \frac{0.1}{0.025} = 4$$

$$\frac{V_i}{0.1} = e^4$$

$$V_i = 0.1\,e^4 = 5.45\,\text{V}$$

$$\boxed{V_i = 5.45\,\text{V}}$$

Q3.97.4 NAT | 1997 | 1M | Ans: * | ● ○ ○

Given circuit is,

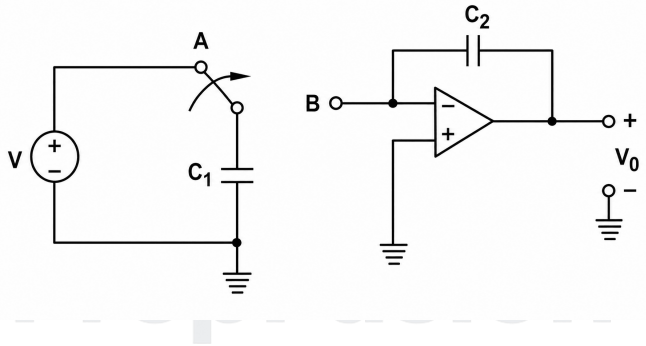

Fig. Solution Q3.97.4

when the switch is connected to point A, capacitor C_1 is charged to voltage V.
Hence, the charge stored on C_1 is

$$Q = C_1V$$

When the switch is moved to point B, the inverting input of the ideal op-amp is at virtual ground. Therefore, the entire charge stored on C_1 is transferred to node B during each switching cycle.
Thus, charge transferred per switching operation is

$$\Delta Q = C_1V$$

If the switching process is repeated f times per second, then the total charge transferred per second is

$$Q_{\text{sec}} = f\,\Delta Q$$

$$Q_{\text{sec}} = f\,C_1V$$

Therefore, the charge transferred per second from node A to node B is

$$\boxed{Q_{\text{sec}} = f\,C_1V}$$

Q3.97.5 NAT 1997 2M Ans: *

For each switching cycle, the charge transferred from node A to node B is

$$\Delta Q = C_1 V$$

If the switching frequency is f cycles per second, the average current injected into node B is

$$I_{avg} = \frac{\Delta Q}{T} = f\,\Delta Q$$

$$I_{avg} = f\,C_1 V$$

Since the op-amp is ideal, the inverting input draws no current. Therefore, the entire current flows through the feedback capacitor C_2 which is given by

$$i_{C_2} = C_2 \frac{dv}{dt}$$

where v = Voltage across $C_2 = 0 - V_0 = -V_0$. Hence,

$$I_{avg} = C_2 \frac{d(-V_o)}{dt}$$

$$I_{avg} = -C_2 \frac{dV_o}{dt}$$

$$f\,C_1 V = -C_2 \frac{dV_o}{dt}$$

$$\frac{dV_o}{dt} = -\frac{f\,C_1}{C_2} V$$

Hence, the average rate of change of the output voltage is

$$\boxed{\frac{dV_o}{dt} = -\frac{f\,C_1}{C_2} V}$$

Q3.97.6 NAT 1997 1M Ans: *

From above question, the average current flowing into B node is,

$$I_{avg} = f\,C_1 V$$

Suppose the capacitor C_1 and the switch are removed and replaced by a resistor R connected between nodes A and B, the circuit becomes,

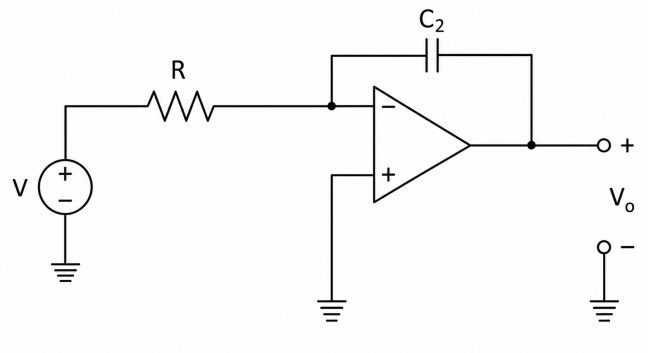

Fig. Solution Q3.97.6

The current through resistor is,

$$I_R = \frac{V - 0}{R}$$

$$I_R = \frac{V}{R}$$

To obtain the same average rate of change of the output voltage, the current through the resistor must equal the average current produced by the switched-capacitor network:

$$\frac{V}{R} = f\,C_1 V$$

$$\frac{1}{R} = f\,C_1$$

$$R = \frac{1}{f\,C_1}$$

Hence, the value of the resistor required is

$$\boxed{R = \frac{1}{f\,C_1}}$$

Q3.97.7 NAT | 1997 | 1M | Ans: -1000 | ● ○ ○

From above solution Q3.97.5, the average rate of change of the output voltage is

$$\frac{dV_o}{dt} = -\frac{f\,C_1}{C_2} V$$

Given

$$f = 10^4\ \text{s}^{-1}$$

$$C_1 = 100\,\text{pF} = 100 \times 10^{-12}\,\text{F}$$

$$C_2 = 10\,\text{pF} = 10 \times 10^{-12}\,\text{F}$$

$$V = 10\,\text{mV} = 10 \times 10^{-3}\,\text{V}$$

Substituting these values,

$$\frac{dV_o}{dt} = -\frac{\left(10^4\right)\left(100 \times 10^{-12}\right)}{10 \times 10^{-12}}\left(10 \times 10^{-3}\right)$$

$$\frac{dV_o}{dt} = -\left(10^4\right)(10)\left(10 \times 10^{-3}\right)$$

$$\frac{dV_o}{dt} = -10^3\ \text{V/s}$$

Therefore, the average rate of change of the output voltage is

$$\boxed{\frac{dV_o}{dt} = -1000\ \text{V/s}}$$

Q3.96.1 MCQ | 1996 | 2M | Ans: A | ● ○ ○

Given circuit is,

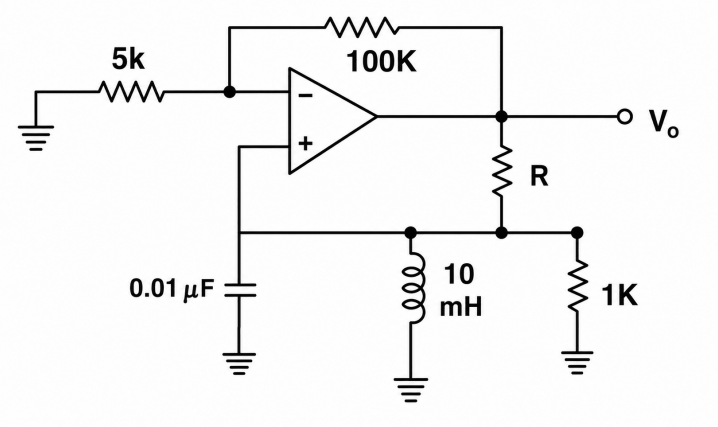

Fig. Solution Q3.96.1

The op-amp is configured as a non-inverting amplifier. The closed-loop gain is

$$A = \frac{V_0}{V_-} = 1 + \frac{100\,\text{k}\Omega}{5\,\text{k}\Omega}$$

$$A = 21$$

The positive-feedback network connected to the non-inverting terminal consists of a parallel RLC circuit:

$$L = 10\,\text{mH} = 10^{-2}\,\text{H}$$

$$C = 0.01\,\mu\text{F} = 10^{-8}\,\text{F}$$

At resonance, the impedance of the parallel RLC network is maximum and the phase shift is zero. Therefore, the Barkhausen criterion requires

$$A\beta = 1$$

where

$$\beta = \frac{V_+}{V_o}$$

At resonance, the parallel RL branch offers infinite impedance and becomes open-circuited, so the feedback factor is determined by the voltage divider formed by R and 1 kΩ:

$$\beta = \frac{1000}{R + 1000}$$

Applying the Barkhausen criterion,

$$21\left(\frac{1000}{R + 1000}\right) = 1$$

$$R + 1000 = 21000$$

$$R = 20000\,\Omega$$

$$\boxed{R = 2 \times 10^4\,\Omega}$$

The oscillation frequency is the resonant frequency of the LC network:

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{(10^{-2})(10^{-8})}} = \frac{1}{10^{-5}}$$

$$\boxed{\omega_0 = 10^5\,\text{rad/s}}$$

$$\boxed{\text{(A)} \quad 10^5\,\text{rad/sec}, 2 \times 10^4\,\Omega}$$

Q3.95.1 NAT 1995 1M Ans: 250 ● ○ ○

Given: $V_{pp} = 12\,\text{V}$ and Slew Rate (SR) $= 12\,\text{V}/\mu\text{s}$

For a zero-crossing detector, the output is a square wave switching between the positive and negative saturation levels. During each transition, the output must swing from $-12\,\text{V}$ to $+12\,\text{V}$ (or vice versa), i.e., a total voltage change of

$$\Delta V = 24\,\text{V}$$

The minimum time required for one transition is

$$t_r = \frac{\Delta V}{SR}$$

$$t_r = \frac{24}{12\,\text{V}/\mu\text{s}}$$

$$t_r = 2\,\mu\text{s}$$

There are two such transitions in one complete cycle of the output waveform. Hence, the minimum period of oscillation is

$$T_{\min} = 2t_r$$

$$T_{\min} = 4\,\mu\text{s}$$

Therefore, the maximum frequency without slew-rate limitation is

$$f_{\max} = \frac{1}{T_{\min}}$$

$$f_{\max} = \frac{1}{4 \times 10^{-6}}$$

$$f_{\max} = 0.25 \times 10^6\,\text{Hz}$$

Thus,

$$\boxed{f_{\max} = 250\,\text{kHz}}$$

Key Points

- Transition time is given by
$$t_r = \frac{\Delta V}{SR}.$$
- Two transitions occur during each cycle of the output waveform.

Q3.95.2 NAT 1995 1M Ans: 33 ● ○ ○

Given circuit is,

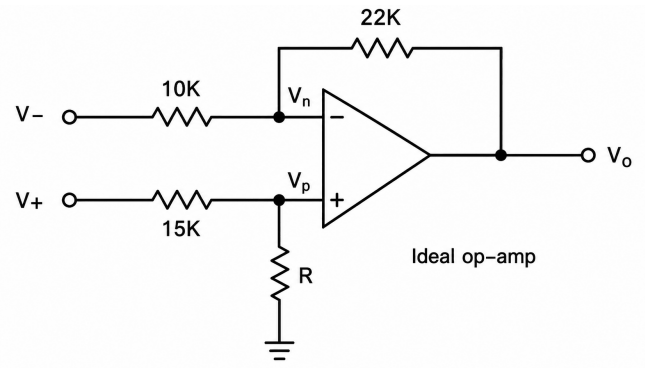

Fig. Solution Q3.95.2

Since the op-amp is ideal and has negative feedback, $V_p = V_n$. Using voltage division at the non-inverting terminal,

$$V_p = \frac{R}{R + 15\,\text{k}\Omega} V_+$$

Applying KCL at the inverting terminal,

$$\frac{V_n - V_-}{10\,\text{k}\Omega} + \frac{V_n - V_o}{22\,\text{k}\Omega} = 0$$

Substituting $V_n = V_p$,

$$\frac{V_p - V_-}{10\,\text{k}\Omega} + \frac{V_p - V_0}{22\,\text{k}\Omega} = 0$$

$$\therefore V_o = \left(1 + \frac{22}{10}\right) V_p - \frac{22}{10} V_-$$

$$V_o = 3.2\,V_p - 2.2\,V_-$$

$$V_o = 3.2\left(\frac{R}{R + 15\,\text{k}\Omega}\right) V_+ - 2.2 V_-$$

The gains of V_+ and V_- must be equal in magnitude, hence

$$3.2\left(\frac{R}{R + 15\,\text{k}\Omega}\right) = 2.2$$

$$\frac{R}{R + 15\,\text{k}\Omega} = \frac{22}{32} = \frac{11}{16}$$

$$16R = 11R + 165\,\text{k}\Omega$$

$$5R = 165\,\text{k}\Omega$$

$$\boxed{R = 33\,\text{k}\Omega}$$

Q3.93.1 NAT | 1993 | 1M | Ans: 0.02 | ● ○ ○

Given circuit is,

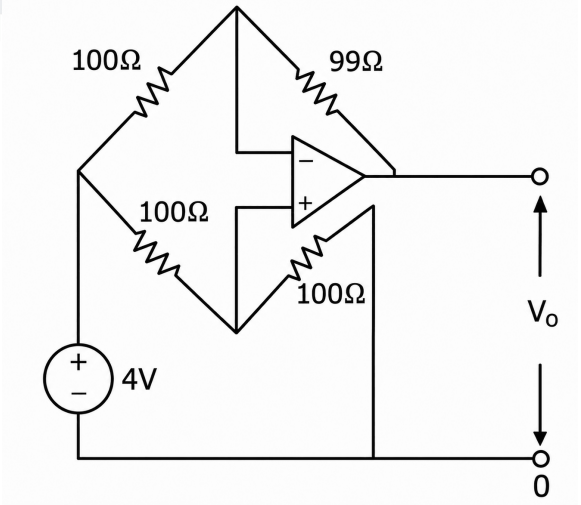

Fig. Solution Q3.93.1

The op-amp has negative feedback. So by virtual short concept, $V_- = V_+$
Applying voltage division rule at non inverting terminal,

$$V_+ = 4 \times \frac{100}{100 + 100} = 4 \times \frac{1}{2} = 2$$

Applying KCL at inverting terminal,

$$\frac{V_- - 4}{100} + \frac{V_- - V_0}{99} = 0$$

$$0.99V_- - 4 \times 0.99 + V_- - V_0 = 0$$

$$\therefore V_0 = 1.99V_- - 3.96$$

Since $V_- = V_+ = 2$ V,

$$V_0 = 1.99 \times 2 - 3.96 = 0.02$$

$$\boxed{V_0 = 0.02 \text{ V}}$$

Q3.93.2 NAT | 1993 | 2M | Ans: 6 | ● ○ ○

Given circuit is,

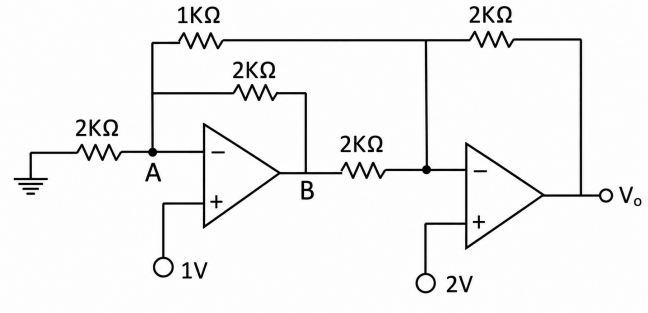

Fig. Solution Q3.93.2

Given that both op-amps are ideal and operate with negative feedback, $V_A = 1$ V for the first op-amp and $V_- = V_+ = 2$ V for the second op-amp.

Applying KCL at node A,

$$\frac{1}{2\,\text{k}\Omega} + \frac{1 - V_B}{2\,\text{k}\Omega} + \frac{1 - 2}{1\,\text{k}\Omega} = 0$$

$$1 + (1 - V_B) - 2 = 0$$

$$V_B = 0\,\text{V}$$

Now apply KCL at the inverting terminal of the second op-amp,

$$\frac{2 - V_B}{2\,\text{k}\Omega} + \frac{2 - 1}{1\,\text{k}\Omega} + \frac{2 - V_o}{2\,\text{k}\Omega} = 0.$$

$$\frac{2}{2\,\text{k}\Omega} + \frac{1}{1\,\text{k}\Omega} + \frac{2 - V_o}{2\,\text{k}\Omega} = 0.$$

$$2 + 2 + (2 - V_o) = 0$$

$$6 - V_o = 0$$

$$\boxed{V_0 = 6\,\text{V}}$$

Q3.92.1 MCQ 1992 2M Ans: C ● ○ ○

Given $V_{ios} = 1$ mV and given circuit is,

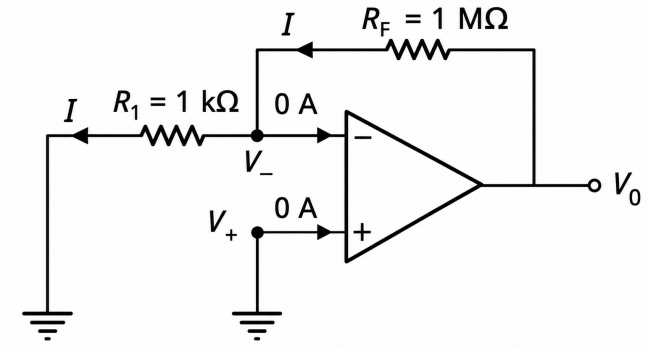

Fig. Solution Q3.92.1

Input offset voltage is given by,

$$V_{ios} = V_+ - V_-$$

Output voltage due to input offset voltage V_{ios} is given by,

$$V_0 = \left[1 + \frac{R_f}{R}\right] V_{ios} \quad \ldots (i)$$

Case 1 : If $V_+ = 1$ mV and $V_- = 0$ V

$$V_{ios} = 1 \text{ mV}$$

From equation (i),

$$V_0 = \left[1 + \frac{10^6}{10^3}\right] \times 10^{-3} \approx 1 \text{ V}$$

Case 2 : If $V_- = 1$ mV and $V_+ = 0$ V

$$V_{ios} = -1 \text{ mV}$$

From equation (i),

$$V_0 = \left[1 + \frac{10^6}{10^3}\right] \times (-10^{-3}) \approx -1 \text{ V}$$

From case 1 and case 2, output voltage V_0 can be written as given below,

$$V_0 = \pm 1 \text{ V}$$

(C) ± 1V

Key Points

- **Input Offset Voltage (V_{ios}):** Modeled as a small DC voltage source in series with one of the input terminals. Its exact polarity is typically unknown, so both positive and negative extremes ($\pm V_{ios}$) must be considered.

Q3.92.2 MCQ 1992 2M Ans: C ● ○ ○

Given circuit is shown below,

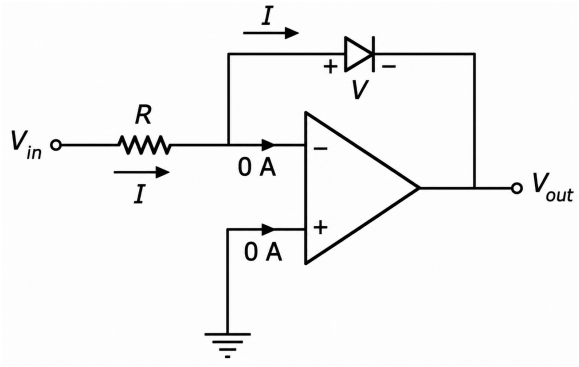

Fig. Solution Q3.92.2

From virtual ground concept,

$$V_- = V_+ = 0\,\text{V}$$

From figure,

$$I = \frac{V_{in} - 0}{R} = \frac{V_{in}}{R} \quad \ldots(\text{i})$$

Since, the same current I is flowing through the diode, hence

$$I = I_0\left(e^{\frac{V}{\eta V_T}} - 1\right)$$

$$I = I_0\left(e^{\frac{V}{V_T}} - 1\right) \quad [\text{Assume } \eta = 1] \ldots(\text{ii})$$

From equation (i) and (ii),

$$\frac{V_{in}}{R} = I_0\left[e^{\frac{V}{V_T}} - 1\right]$$

So, $\quad V = V_T \ln\left(1 + \frac{V_{in}}{I_0 R}\right) \quad \ldots(\text{iii})$

Applying KVL in feedback loop,

$$0 + V + V_0 = 0$$

$$V = -V_0$$

From equation (iii),

$$V_0 = -V_T \ln\left(1 + \frac{V_{in}}{I_0 R}\right)$$

$$V_0 \propto \ln(V_{in})$$

So, given figure represent the logarithmic amplifier.

(C) a logarithmic amplifier

Key Points

- **Logarithmic Amplifier (Log Amp):** An op-amp circuit that utilizes the non-linear exponential current-voltage relationship of a PN junction (diode or BJT) in its feedback path to produce an output voltage proportional to the natural logarithm of the input voltage.

Q3.92.3 NAT | 1992 | 2M | Ans: -4 | ● ○ ○

Given circuit is shown below,

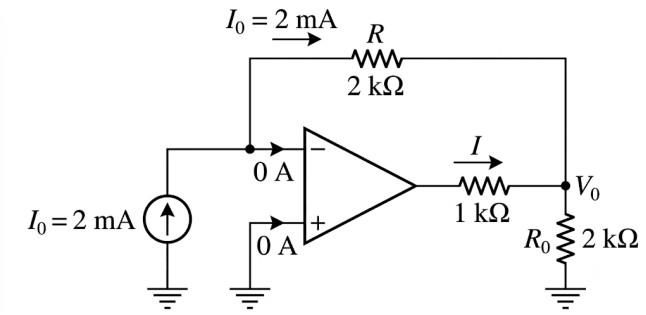

Fig. Solution Q3.92.3

Method 1

From virtual ground concept,

$$V_+ = V_- = 0\,\text{V}$$

Applying KCL at inverting terminal,

$$\frac{0 - V_0}{R} - I_0 = 0$$

$$\frac{0 - V_0}{2} - 2 = 0$$

$$V_0 = -4\,\text{V}$$

Applying KCL at output node,

$$-I_0 - I + \frac{V_0}{R_0} = 0$$

$$-2 - I + \frac{-4}{2} = 0$$

$$I = -4\,\text{mA}$$

Hence, the current I through 1 kΩ resistor is $-\mathbf{4}$ mA.

Method 2

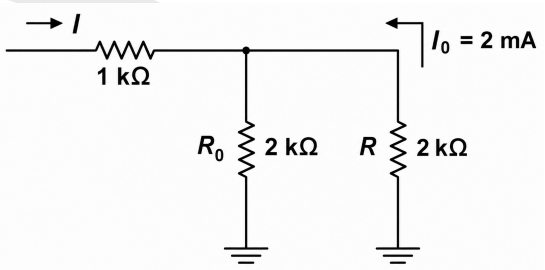

Fig. Solution Q3.92.3

By using current division rule,

$$-I_0 = \frac{I \times R_0}{R_0 + R}$$

$$-2 = \frac{I \times 2}{2 + 2}$$

$$I = -4\,\text{mA}$$

$$\boxed{I = -4\,\text{mA}}$$

Q3.91.1 MCQ | 1991 | 1M | Ans: C | ● ○ ○

Given two non-inverting identical op-amps have unity gain and a gain of 20 respectively.
For identical operational amplifier gain bandwidth product should be constant.

$$\text{Gain} \times \text{Bandwidth} = \text{Constant}$$

i.e.

$$A_1 \times \text{BW}_1 = A_2 \times \text{BW}_2$$

$$\text{BW}_2 = \frac{A_1 \times \text{BW}_1}{A_2} = \frac{1 \times \text{BW}_1}{20} = \frac{\text{BW}_1}{20}$$

Hence, as compared to the unity gain amplifier, the amplifier with gain twenty has less bandwidth.

(C) less bandwidth

Key Points

- **Gain-Bandwidth Product (GBP):** For a standard internally compensated voltage-feedback operational amplifier, the product of the closed-loop voltage gain and its corresponding −3 dB bandwidth is a constant value ($A \times BW = \text{Constant}$).

Q3.91.2 NAT 1991 1M Ans: *

The given circuit is an integrator not a differentiator. A differentiator is a high pass filter whose circuit is given by interchanging the capacitor and resistor in the integrator circuit.

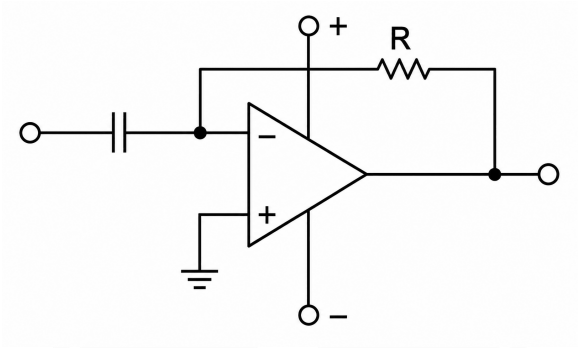

Fig. Solution Q3.91.2

Q3.91.3 NAT 1991 5M Ans: *

Given op-amp saturation voltages are ±15 V. Given circuit is,

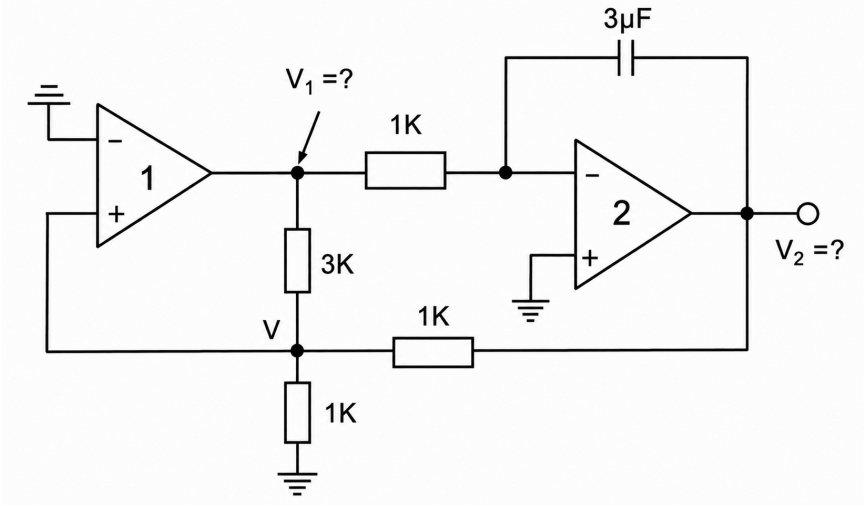

Fig. Solution Q3.91.3

Here op-amp 1 has positive feedback and $V_- = 0, V_+ = V$. Hence the possible output voltages are

$$V_1 = \begin{cases} +15, & V > 0 \\ -15, & V < 0 \end{cases}$$

Applying KCL at V,

$$\frac{V-0}{1}+\frac{V-V_1}{3}+\frac{V-V_2}{1}=0$$

$$7V_+ = V_1 + 3V_2$$

$$V=\frac{V_1+3V_2}{7}$$

The op-amp 2 is configured as an integrator circuit whose input is V_1 and output is V_2

$$V_2=\frac{-1}{RC}\int V_1(t)\,\mathrm{dt}$$

$$V_2=\frac{-1}{1\times10^3\times3\times10^{-6}}\int V_1(t)\,\mathrm{dt}$$

$$\therefore V_2=\frac{-1000}{3}\int V_1(t)\,\mathrm{dt}$$

Case-1: Initially assume $V_1 = +15\text{V}$ and $V_2 = 0$. Then

$$V_2=\frac{-1000}{3}\int_0^t 15\,\mathrm{dt}$$

$$V_2=-5000\int_0^t \mathrm{dt}=-5000\,\mathrm{t}$$

Now the voltage V_1 will switch to -15 V when V just crosses 0 V.

$$V=0 \implies \frac{V_1+3V_2}{7}=0$$

$$V_1+3V_2=0$$

$$15-15000t=0 \implies t=1\text{ ms}$$

Hence at $t = 1$ ms, V_1 switches to -15 V and $V_2(t = 1\text{ms}) = -5$ V.

Case 3: Now $V_1 = -15$ V and V_2 is already at -5 V. So

$$V_2(t>1)=-5-\frac{1000}{3}\int -15\,\mathrm{dt}$$

$$V_2(t)=-5+5000(t-1)$$

Now, the voltage V_1 will again switch to -15 V when V just crosses zero again.

$$V_1+3V_2=0$$

$$-15-15+15000(t-1)=0 \implies t=3\text{ ms}$$

Hence at t=3 ms, V_1 switches from -15 to +15 V and $V_2(t = 3) = 5$ V.

At steady state the voltage V_2 varies between -5 and +5V linearly (triangular wave) whereas V_1 varies between -15 and +15 V (square wave).

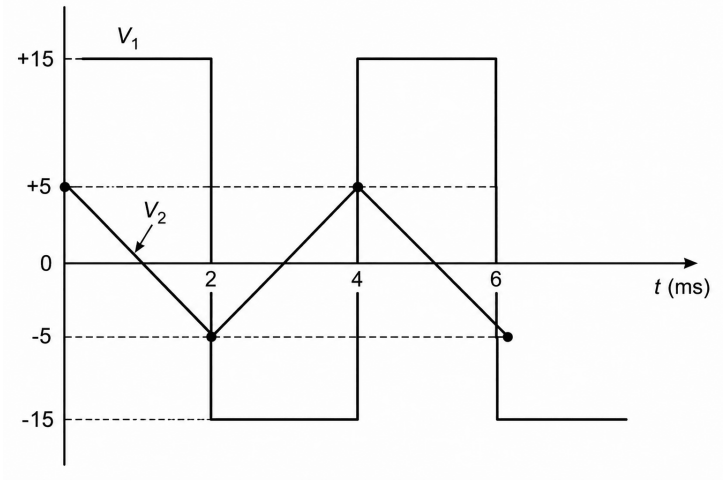

Fig. Solution Q3.91.3

Q3.90.1 MCQ | 1990 | 2M | Ans: C

Method 1

CMRR is given by,

$$\text{CMRR} = \left|\frac{\text{Differential mode gain}}{\text{Common mode gain}}\right| = \left|\frac{A_d}{A_c}\right|$$

(i) Calculation of A_d :

$$A_d = \frac{V_0}{V_d}$$

where, $V_d = V_{NI} - V_I = V_+ - V_-$. Assume $V_{NI} = \frac{V_s}{2}, V_I = \frac{-V_s}{2}$

$$V_d = \frac{V_s}{2} - \left(\frac{-V_s}{2}\right) = V_s$$

Hence, $A_d = \frac{V_0}{V_d} = \frac{V_0}{V_s}$

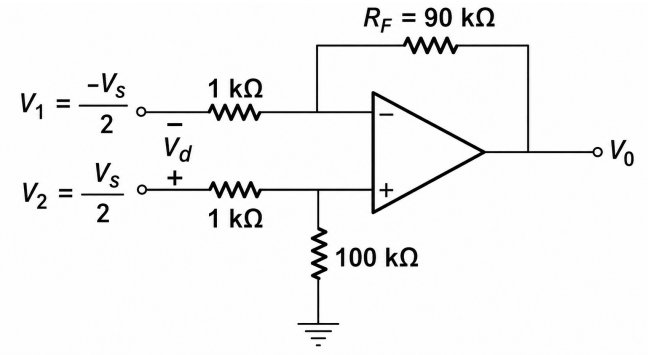

Fig. Solution Q3.90.1

Output of standard subtractor circuit, by using **superposition theorem**, is given by,

$$V_0 = \frac{V_s}{2}\left[\frac{100}{1+100}\right]\left[1+\frac{90}{1}\right] - \frac{V_s}{2}\left[\frac{-90}{1}\right]$$

$$V_0 = \frac{V_s}{2}\left[\frac{100}{101}\right][91] + \left[\frac{90}{2}\right]V_s$$

$$\frac{V_0}{V_s} = \frac{50}{101}[91] + 45$$

$$A_d = \frac{V_0}{V_s} = 90.04$$

(ii) Calculation of A_c :

$$A_c = \frac{V_0}{V_c}$$

where, $V_c = \frac{V_{NI}+V_I}{2} = \frac{V_+ + V_-}{2}$. Assume $V_{NI} = V_s, V_I = V_s$

$$V_c = \frac{V_s + V_s}{2} = V_s$$

Hence, $A_c = \frac{V_0}{V_c} = \frac{V_0}{V_s}$

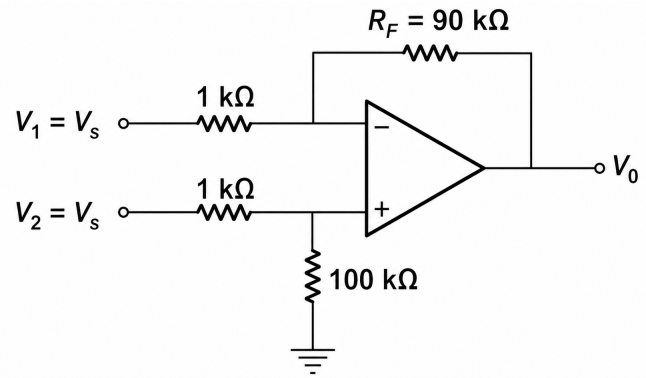

Fig. Solution Q3.90.1

Output of standard subtractor circuit, by using **superposition theorem**, is given by,

$$V_0 = V_s\left[\frac{100}{1+100}\right]\left[1+\frac{90}{1}\right] + V_s\left[\frac{-90}{1}\right]$$

$$V_0 = V_s\left[\frac{100}{101}\right][91] - V_s[90] = 0.099V_s$$

$$A_c = \frac{V_0}{V_s} = 0.099$$

$$\text{CMRR} = \left|\frac{A_d}{A_c}\right| = \left|\frac{90.04}{0.099}\right| = 909.49$$

$$\text{CMRR} \approx 900$$

Hence, the correct option is (C).

Method 2

CMRR is given by,

$$\text{CMRR} = \left|\frac{A_d}{A_c}\right|$$

where, $A_d = \frac{A_1 - A_2}{2}$

$$A_c = A_1 + A_2$$

(i) Calculation of A_1 :

$$A_1 = \left.\frac{V_0}{V_1}\right|_{V_2=0}$$

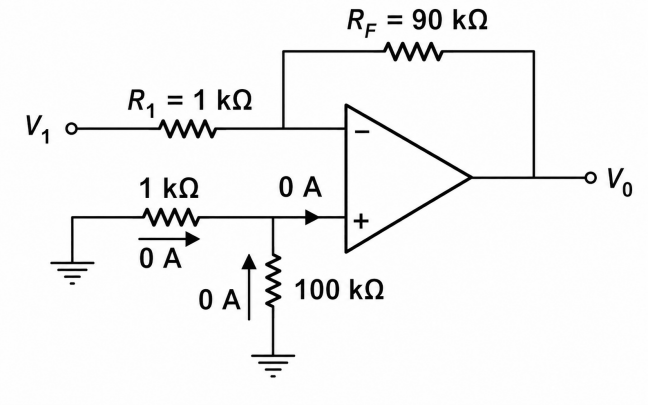

Fig. Solution Q3.90.1

Above figure represents an inverting Op-Amp. Hence, gain of inverting Op-Amp is given by,

$$A_1 = \frac{-R_F}{R_1} = \frac{-90}{1} = -90$$

(ii) Calculation of A_2 :

$$A_2 = \left.\frac{V_0}{V_2}\right|_{V_1=0}$$

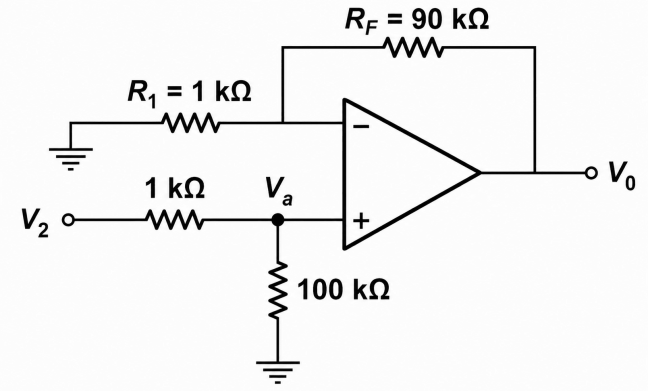

Fig. Solution Q3.90.1

Above figure represents a non-inverting amplifyer.

$$V_a = \frac{100}{100+1}V_2 \quad [\text{By VDR}] \ldots (\text{i})$$

Output of non-inverting amplifier is given by,

$$V_0 = \left(1 + \frac{R_F}{R_1}\right)V_a$$

From equation (i),

$$V_0 = \left(1 + \frac{90}{1}\right)\left(\frac{100}{100+1}V_2\right)$$

$$A_2 = \frac{V_0}{V_2} = \left(1 + \frac{90}{1}\right)\left(\frac{100}{100+1}\right) = 90.1$$

Hence,

$$\text{CMRR} = \left|\frac{A_d}{A_c}\right| = \left|\frac{(A_1 - A_2)}{2(A_1 + A_2)}\right|$$

$$\text{CMRR} = \left|\frac{(-90 - 90.1)}{2(90.1 - 90)}\right| = \left|\frac{-180.1}{2 \times 0.1}\right|$$

$$\text{CMRR} = 900.5 \approx 900$$

(C) 900

Q3.90.2 MCQ 1990 2M Ans: D

Method 1

Closed loop gain of non-inverting Op-Amp is given by,

$$A_F = \frac{A_{OL}}{1 + A_{OL}\beta}$$

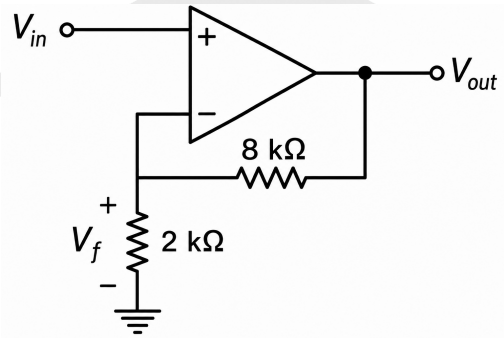

Fig. Solution Q3.90.2

It is a non-inverting Op-Amp. Feedback gain is given by,

$$\beta = \frac{V_f}{V_{out}}$$

$$V_f = \frac{2}{2 + 8} \times V_{out} \quad \text{[By VDR]}$$

$$\frac{V_f}{V_{out}} = \frac{2}{10} = \frac{1}{5} = \beta$$

Given : $A_{OL} = 45$ So,

$$A_F = \frac{45}{1 + 45 \times \frac{1}{5}} = 4.5$$

Hence, the correct option is (D).

Method 2

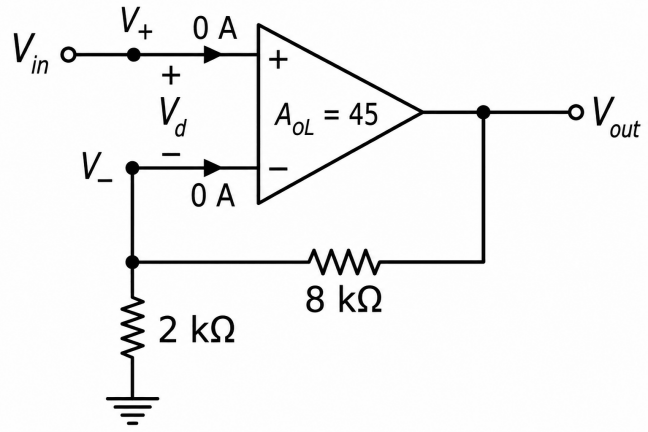

Fig. Solution Q3.90.2

For the above Op-Amp circuit,

$$A_{OL} = 45 = \frac{V_0}{V_d} = \frac{V_0}{V_+ - V_-} = \frac{V_0}{V_{in} - V_-} \quad \ldots (i)$$

From figure,

$$V_- = \frac{V_0 \times 2}{2+8} = \frac{V_0}{5} \quad \text{[By VDR]} \quad \ldots (ii)$$

From equation (i) and (ii),

$$45 = \frac{V_0}{V_{in} - \frac{V_0}{5}} = \frac{5V_0}{5V_{in} - V_0}$$

$$9 = \frac{V_0}{5V_{in} - V_0}$$

$$45V_{in} - 9V_0 = V_0$$

$$10V_0 = 45V_{in}$$

$$\frac{V_0}{V_{in}} = 4.5$$

(D) 4.5

Key Points

- Method 1 is valid for only non-inverting op-amp whereas method 2 is valid for inverting op-amp as well.

Q3.89.1 MSQ | 1989 | 2M | Ans: B,C | ● ○ ○

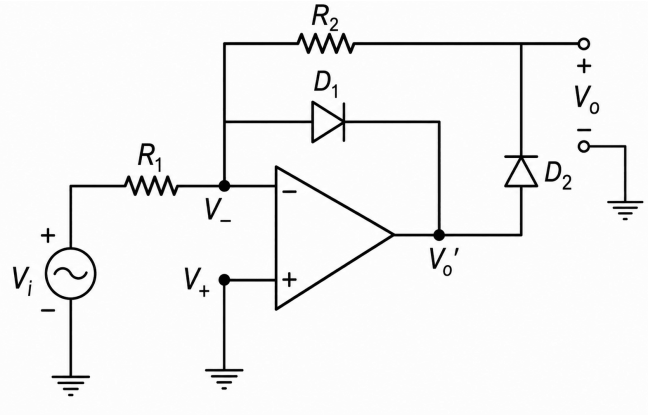

Fig. Solution Q3.89.1

Given circuit consist of two diodes and initially at $t = 0$ the states of diode are unknown so it is indetermine system.

Case 1 : When $V_i > 0$, then

$$\left[\begin{array}{c}\text{Inverting terminal}\\ \text{voltage } (V_-)\end{array}\right] > \left[\begin{array}{c}\text{Non-inverting}\\ \text{terminal voltage } (V_+)\end{array}\right]$$

Then, V_0' goes towards negative saturation. When $V_0' = -0.7\,\text{V}$, then diode D_1 becomes forward biased i.e. ON (forward voltage drop = 0.7 V).

Diode D_2 becomes reverse biased i.e. OFF (Open circuit).

Hence, the modified circuit is shown below,

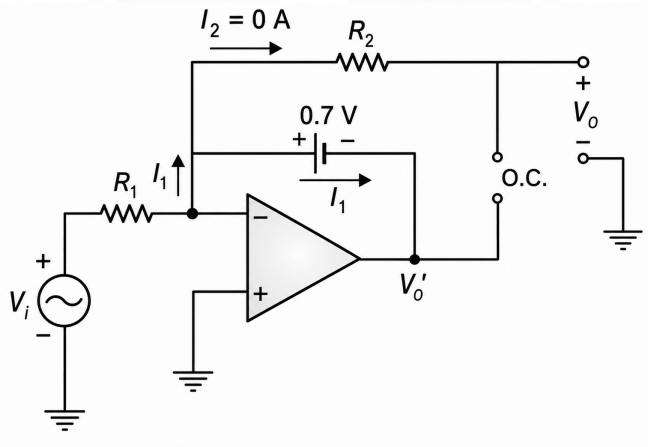

Fig. Solution Q3.89.1

Virtual ground concept is valid for Negative feedback closed system. From virtual ground concept,

$$V_- = V_+ = 0$$

Since, entire current I_1 is flowing through the diode D_1. So, $I_2 = 0\,\text{A}$

From figure, $I_2 = \frac{0 - V_0}{R_2} = 0$. Hence, $V_0 = -0 \times R_2 = 0\,\text{V}$

Case 2 : When $V_i < 0$, then

$$\left[\begin{array}{c}\text{Inverting terminal}\\ \text{voltage } (V_-)\end{array}\right] < \left[\begin{array}{c}\text{Non-inverting}\\ \text{terminal voltage } (V_+)\end{array}\right]$$

Then, V_0' goes towards positive saturation.

When $V_0' = 0.7\,\text{V}$, then diode D_1 becomes reverse biased i.e. OFF (Open circuit).

Diode D_2 becomes forward biased i.e. ON (forward voltage drop = 0.7 V). Hence, the modified circuit is shown below,

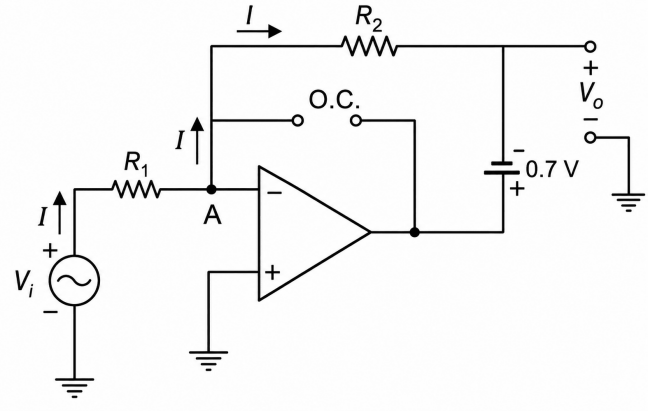

Fig. Solution Q3.89.1

From the virtual ground concept,

$$V_- = V_A = V_+ = 0$$

Modified circuit represents inverting Op-Amp circuit.
Hence, output of inverting Op-Amp circuit is given by,

$$V_0 = \frac{-R_2}{R_1} \times V_i$$

From case 1 and case 2, output voltage V_0 can be written as given below,

$$V_0 = \begin{cases} 0 & ; \quad V_i \geq 0\,\text{V} \\ \frac{-R_2}{R_1} V_i & ; \quad V_i \leq 0\,\text{V} \end{cases}$$

(B) For $V_i > 0$, $V_0 = 0$

(C) For $V_i < 0$, $V_0 = -\frac{R_2}{R_1} V_i$

Q3.88.1 MCQ 1988 2M Ans: C

Given, $R = \sqrt{L/C}$, $\omega = \frac{1}{\sqrt{LC}}$
Given circuit in s-domain is shown below,

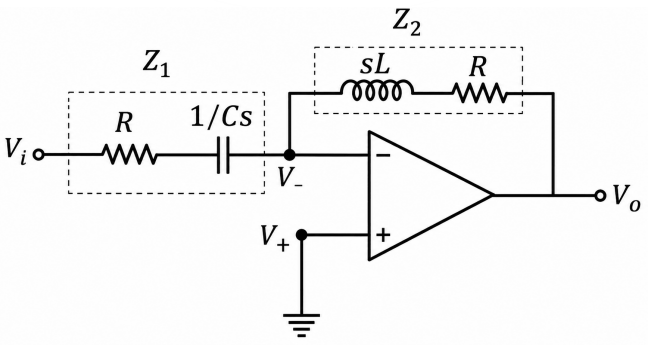

Fig. Solution Q3.88.1

From virtual ground concept,

$$V_+ = V_- = 0$$

Let us assume, $V_i = V_1\angle 0°$. From figure,

$$Z_1 = R + \frac{1}{Cs} = \frac{1 + sRC}{sC} \quad \ldots(\text{i})$$

$$Z_2 = R + sL \quad \ldots(\text{ii})$$

Applying KCL at inverting terminal,

$$\frac{0 - V_i}{Z_1} + \frac{0 - V_0}{Z_2} = 0$$

Hence, output of inverting Op-Amp is given by,

$$V_0 = \frac{-Z_2}{Z_1} V_i \quad \ldots(\text{iii})$$

From equation (i), (ii) and (iii),

$$V_0(s) = \frac{-(R + sL)}{\frac{1+sRC}{sC}} V_i$$

$$V_0(s) = \frac{-(R + sL)}{1 + sRC} \times sC \times V_i$$

$$V_0(j\omega) = \frac{-(R + j\omega L)}{1 + j\omega RC} \times j\omega C \times V_i$$

Given : $R = \sqrt{\frac{L}{C}}, \omega = \frac{1}{\sqrt{LC}}$

$$V_0(j\omega) = \frac{-\left(\sqrt{\frac{L}{C}} + j\frac{1}{\sqrt{LC}} \times L\right)}{1 + j\frac{1}{\sqrt{LC}} \times \sqrt{\frac{L}{C}} \times C} \times j\frac{1}{\sqrt{LC}} \times C \times V_i$$

$$V_0(j\omega) = \frac{-\left(\sqrt{\frac{L}{C}} + j\sqrt{\frac{L}{C}}\right)}{1 + j\sqrt{\frac{L}{C}} \times \sqrt{\frac{C}{L}}} \times j\sqrt{\frac{C}{L}} \times V_i$$

$$V_0(j\omega) = -\sqrt{\frac{L}{C}} \times \left(\frac{1 + j}{1 + j}\right) \times j\sqrt{\frac{C}{L}} \times V_i$$

$$V_0(j\omega) = -jV_i$$

$$V_0(j\omega) = 1\angle -\frac{\pi}{2} \times V_1\angle 0°$$

$$V_0(j\omega) = V_1\angle -\frac{\pi}{2} = V_1\angle \frac{3\pi}{2}$$

Hence, phase difference between V_i and V_0 is $-\frac{\pi}{2}$ or $\frac{3\pi}{2}$.

$$\boxed{\text{(C)} \quad \frac{3\pi}{2}}$$

Q3.87.1 NAT | 1987 | 2M | Ans: 100 | ● ○ ○

Given CMRR of given Op-Amp is 60 dB.
Op-Amp circuit without bridge is shown below,

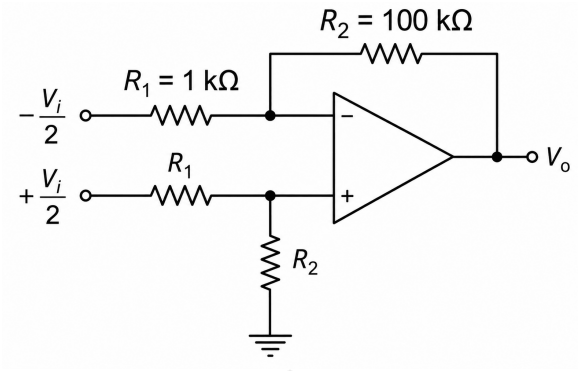

Fig. Solution Q3.87.1

From above circuit, differential gain is given by,

$$|A_d| = \frac{R_2}{R_1} = 100$$

Bridge circuit of given Op-Amp circuit is shown below,

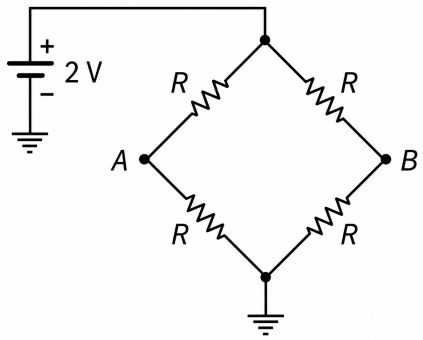

Fig. Solution Q3.87.1

In the above circuit bridge is balance. Hence,

$$V_A = \frac{R}{R+R} \times 2 = 1 \text{ Volt}$$

$$V_B = \frac{R}{R+R} \times 2 = 1 \text{ Volt}$$

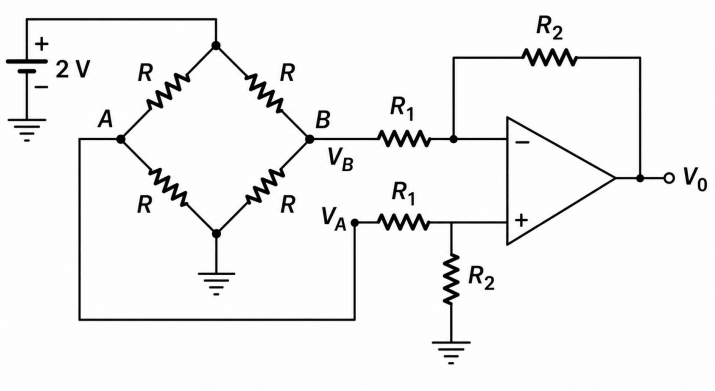

Fig. Solution Q3.87.1

Differential input voltage is given by,

$$V_d = V_A - V_B = 0$$

Output voltage when CMRR $\neq \infty$, is given by,

$$V_0 = A_d V_d + A_c V_c$$

Since, $V_d = 0$. Hence, $V_0 = A_c V_c$ ... (i)
Common mode input voltage is given by,

$$V_c = \frac{V_A + V_B}{2} = 1 \text{ Volt}$$

Since, $\text{CMRR}|_{dB} = 20\log(\text{CMRR})$

$$60 = 20 \log \text{CMRR}$$

$$\text{CMRR} = (10)^{60/20} = 1000$$

$$\text{CMRR} = \left|\frac{A_d}{A_c}\right|$$

Hence, $A_c = \frac{A_d}{\text{CMRR}} = \frac{100}{1000} = 0.1$ From equation (i),

$$V_0 = 0.1 \times 1 = 0.1 \text{ Volt} = 100 \text{ mV}$$

$$\boxed{V_0 = 100 \text{ mV}}$$

Key Points

- **Common Mode Rejection Ratio (CMRR):** It is defined as the ratio of differential mode gain (A_d) to common mode gain (A_c). Expressed in decibels, CMRR (dB) = $20\log_{10}(|A_d/A_c|)$.
- **Non-Ideal Output Voltage:** For an op-amp with a finite CMRR, the output voltage is a superposition of both differential and common-mode amplifications: $V_0 = A_d V_d + A_c V_c$, where $V_d = V_+ - V_-$ is the differential input and $V_c = (V_+ + V_-)/2$ is the common-mode input.
- **Balanced Bridge:** A balanced Wheatstone bridge ($R_1/R_2 = R_3/R_4$) produces zero differential voltage ($V_A = V_B = V_d = 0$), meaning the entire input to the subsequent differential amplifier acts as a pure common-mode signal.

SOLUTIONS

IV

BJTs

Topics

1. BJT Physics and Biasing
2. Current Mirror Circuit Using BJT
3. Small Signal Analysis of BJT Amplifiers
4. Differential Amplifiers and Feedback Topologies Using BJTs

Unit Snapshot*

Stream: EC	**Questions:** 127	**MCQ:** 79	**MSQ:** 3	**NAT:** 42
1M: 49	**2M:** 70	**Descriptive:** 8	**Avg Weightage** ~4M	**Range:** 1M(2016)–6M(2010)

**See preface for how Avg Weightage and Range are calculated.*

Q4.26.1 MCQ 2026 1M Ans: D ● ○ ○

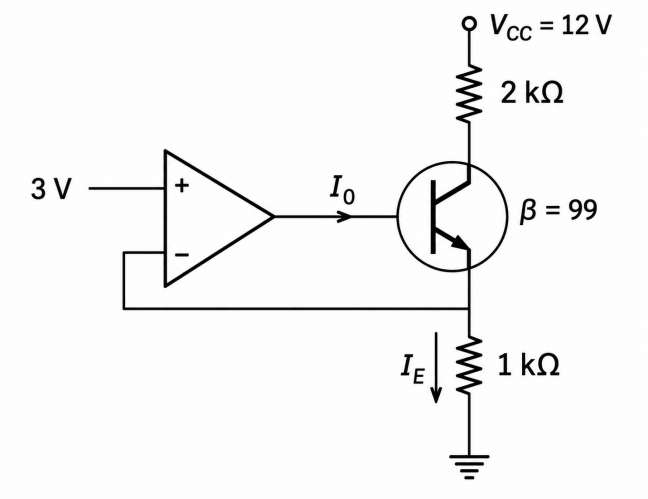

Fig. Solution Q4.26.1

From the virtual short concept of the ideal op-amp,

$$V^- = V^+ = 3\,\text{V}$$

The voltage at the emitter of the BJT is connected to the inverting terminal, so the emitter voltage $V_E = 3\,\text{V}$. The emitter current I_E can be calculated as:

$$I_E = \frac{V_E}{1\,\text{k}\Omega} = \frac{3}{1\,\text{k}\Omega} = 3\,\text{mA}$$

The output current of the op-amp I_0 flows directly into the base of the BJT, meaning $I_0 = I_B$. Using the standard BJT current relationship for the active region:

$$I_0 = I_B = \frac{I_E}{1+\beta}$$

$$I_0 = \frac{3\,\text{mA}}{1+99} = \frac{3}{100}\,\text{mA} = 0.03\,\text{mA}$$

$$I_0 = 30\,\mu\text{A}$$

(D) $30\,\mu\text{A}$

Key Points

- **BJT Current Relationship:** In the active region, the relationship between the emitter current and base current is defined by $I_E = (1+\beta)I_B$. The op-amp only needs to supply the required base current.

Q4.26.2 MCQ 2026 2M Ans: A ● ○ ○

If the actual collector current at its bias point is I_C, differentiating the Early model gives

$$r_o = \frac{V_A + V_{CE}}{I_C}.$$

The common approximation $r_o \approx V_A/I_C$ neglects V_{CE} relative to V_A. From the bias divider, $V_{th} = 2.4$ V and $R_{th} = 20\,\text{k}\Omega$. With total DC emitter resistance 1.5 kΩ,

$$I_B = 10\,\mu\text{A}, \quad I_C = 0.99\,\text{mA}.$$

Thus $g_m = 39.6$ mS, $r_\pi = 2.5$ kΩ, and $V_{CE} \approx 5.55$ V. The two output-resistance estimates are about 106.6 and 101.0 kΩ. At both input frequencies the 10 microfarad bypass has negligible impedance, leaving 500 ohms emitter degeneration. The 100 nF input coupling reactance is 100 ohms at the lower frequency, small relative to the approximately 52.5 kΩ input resistance. The unloaded output coupling capacitor draws no current. Neglecting r_o initially,

$$A_v = -\frac{\beta R_C}{r_\pi + (\beta + 1)R_E} \approx -9.43.$$

Including the finite r_o with collector/emitter nodal equations gives about -9.40 with either convention. Both terms pass with inversion; A, about -9.1 times the input, is the closest listed approximation. No exact equality to the rounded option is claimed.

Q4.25.1 MCQ 2025 1M Ans: A ● ○ ○

Given circuit is,

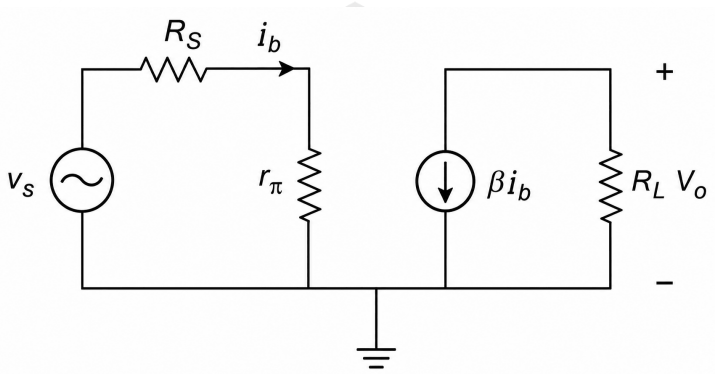

Fig. Solution Q4.25.1

From input loop,

$$I_b = \frac{V_s}{R_s + r_\pi}$$

From output loop,

$$V_0 = -\beta I_b \times R_L$$

$$V_0 = -\beta \times \frac{V_s}{R_s + r_\pi} \times R_L$$

$$\therefore A_v = \frac{V_0}{V_s} = \frac{-\beta R_L}{R_S + r_\pi}$$

$$\boxed{\text{(A)} \quad \frac{-\beta R_L}{R_S + r_\pi}}$$

Q4.25.2 MCQ 2025 1M Ans: C ● ○ ○

Given BJT is biased in active region with $\beta = 100$. $I_B = 10\,\mu$A and $I_E = (1 + \beta)I_B = 101 \times 10\,\mu\text{A} = 1.01$ mA

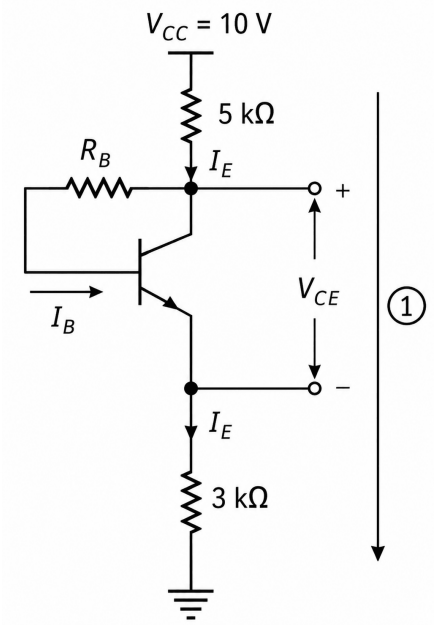

Fig. Solution Q4.25.2

Applying KVL in loop (1),

$$V_{CC} - 5\text{k}I_E - V_{CE} - 3\text{k}I_E = 0$$

$$10 - 5 \times 10^3 I_E - 3 \times 10^3 I_E = V_{CE}$$

$$V_{CE} = 10 - 8 \times 10^3 \times 1.01 \times 10^{-3}$$

$$V_{CE} = 1.92\,\text{V}$$

(C) 1.92

Key Points

- **Collector-to-Base Bias Configuration:** In this circuit, the feedback resistor R_B is connected directly to the collector terminal. Therefore, the current flowing through the upper 5 kΩ resistor is the sum of the collector current (I_C) and the base current (I_B), which is exactly equal to the emitter current (I_E).

Q4.25.3 MCQ 2025 2M Ans: A ● ○ ○

Given Two identical transistor Q_1 and Q_2 are biased in active region.
$\beta = 120$, $V_Z = 5\,\text{V}$, $I_Z = 25\,\text{mA}$, $I_L = 12\,\text{mA}$ and $V_{EB1} = V_{EB2} = 0.7\,\text{V}$.

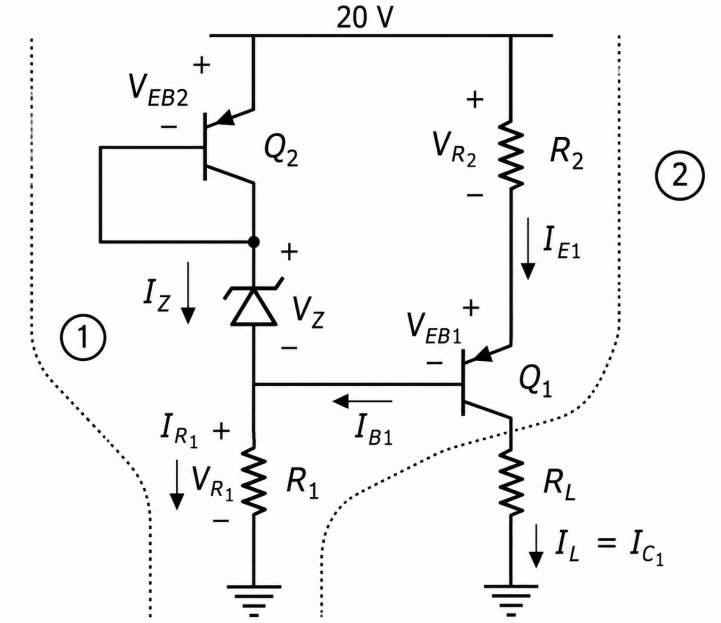

Fig. Solution Q4.25.3

From the given circuit,

$$I_L = I_{C1} = 12\,\text{mA}$$

$$I_{B1} = \frac{I_{C1}}{\beta} = \frac{12}{120} = 0.1\,\text{mA}$$

$$I_{E1} = 0.1 + 12 = 12.1\,\text{mA}$$

From figure,

$$I_{R1} = I_{B1} + I_Z = 0.1 + 25 = 25.1\,\text{mA}$$

Voltage across R_1 : Applying KVL in loop (1),

$$20 - V_{EB2} - V_Z - V_{R_1} = 0$$

$$V_{R1} = 20 - 0.7 - 5 = 14.3\,\text{V}$$

$$R_1 = \frac{V_{R1}}{I_{R1}} = \frac{14.3\,\text{V}}{25.1\,\text{mA}}$$

$$R_1 = 0.569\,\text{k}\Omega \approx 0.6\,\text{k}\Omega$$

Voltage across R_2 : Applying KVL in loop (2),

$$20 - V_{R2} - V_{EB1} - V_{R1} = 0$$

$$V_{R2} = 20 - 0.7 - 14.3$$

$$V_{R2} = 20 - 15 = 5\,\text{V}$$

$$R_2 = \frac{V_{R2}}{I_{E1}} = \frac{5}{12.1\,\text{mA}} = 0.413\,\text{k}\Omega \approx 0.4\,\text{k}\Omega$$

(A) 0.6 and 0.4

Q4.20.1 NAT | 2020 | 2M | Ans: 800 | ● ○ ○

Given circuit is,

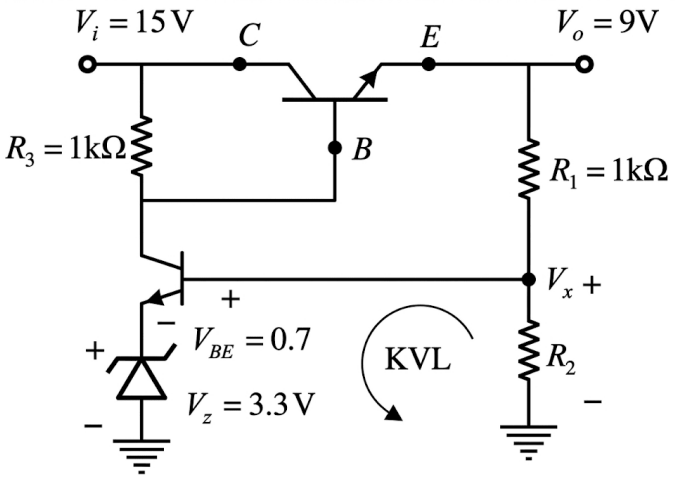

Fig. Solution Q4.20.1

Since the base current of both the transistors is negligible, the current through R_1 will pass straight to R_2. Therefore R_1, R_2 will be in series. Using voltage divider rule, we can write:

$$\frac{V_O \times R_2}{R_1 + R_2} = V_x \quad \ldots (1)$$

Applying KVL from V_x to Zener to ground, we can write:

$$V_x - V_{BE} - 3.3 = 0$$

Given:

$$V_{BE} = 0.7$$

$$V_x = 3.3 + 0.7 = 4\,V$$

Putting this in equation (1), we get:

$$\frac{V_O \times R_2}{R_1 + R_2} = 4$$

$$\frac{9 \times R_2}{1000 + R_2} = 4$$

$$9R_2 = 4(1000 + R_2)$$

$$9R_2 = 4000 + 4R_2$$

$$5 \times R_2 = 4000$$

$$\boxed{R_2 = 800\,\Omega}$$

Key Points

- When the base current is negligible, the feedback resistors (R_1 and R_2) form an unloaded voltage divider.
- By applying Kirchhoff's Voltage Law (KVL) in the bottom loop, we can get the voltage at the feedback node.

Q4.20.2 MCQ | 2020 | 2M | Ans: A

DC Analysis : For DC analysis all the capacitors are open-circuited as shown below.

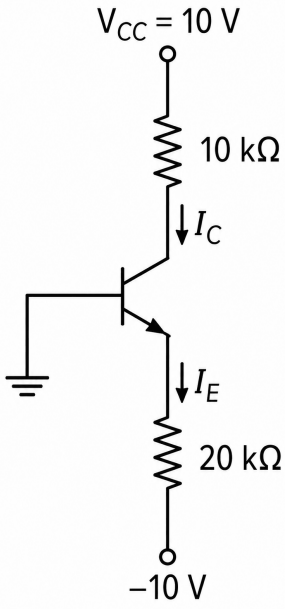

Fig. Solution Q4.20.2

Given that the base current is negligible, so we can write:

$$I_C \approx I_E$$

Applying KVL for the base-emitter loop, we get:

$$0 - V_{BE} - I_C(20\,\text{k}\Omega) + 10 = 0$$

$$I_C = \frac{10 - 0.7}{20\,\text{k}\Omega}$$

$$I_C = 0.465\,\text{mA}$$

$$g_m = \frac{I_C}{V_T} = \frac{0.465\,\text{mA}}{26\,\text{mV}}$$

$$g_m = 0.01788\,\text{A/V}$$

AC analysis: Given that for AC analysis, capacitors are shorted and r_0 is very high. Hence the The small signal model is drawn as:

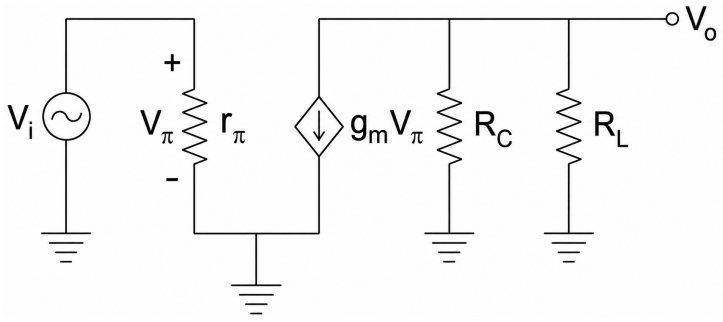

Fig. Solution Q4.20.2

Here $V_i = V_\pi$. Applying KVL in output loop,

$$V_0 = -g_m V_\pi (R_C || R_L)$$

The small-signal gain will be:

$$A_v = -g_m (R_C \parallel R_L)$$

$$A_v = -(0.01788)(10\,\text{k}\Omega \parallel 10\,\text{k}\Omega)$$

$$A_v = -0.01788 \times 5\,\text{k}\Omega$$

$$A_v = -89.42$$

(A) -89.42

Key Points

- **DC Analysis:** During DC analysis, capacitors act as open circuits.
- **Transconductance (g_m):** It is a key small-signal parameter defined as $g_m = \frac{I_C}{V_T}$, where $V_T = \frac{kT}{q}$ is the thermal voltage.
- **AC Analysis:** For small-signal AC analysis, coupling and bypass capacitors act as short circuits, and DC voltage sources are grounded.

Q4.17.1 MCQ 2017 1M Ans: D ●○○

In a Common Emitter (CE) amplifier, a small parasitic capacitance C_{bc} exists between the base and collector terminals. Since the CE amplifier provides a large voltage gain, the voltage across this capacitor is amplified. As a result, the capacitor appears much larger when viewed from the input side. This phenomenon is known as the **Miller Effect**.
The effective capacitance seen at the input is

$$C_M = C_{bc}\,(1 + |A_v|)$$

where A_v is the voltage gain of the amplifier.
Since $|A_v|$ is usually large, the effective input capacitance increases significantly. The upper cutoff frequency of an amplifier is given by

$$f_H = \frac{1}{2\pi RC}$$

From the above relation, the cutoff frequency is inversely proportional to capacitance. Therefore, an increase in the effective capacitance due to the Miller effect causes the upper cutoff frequency to decrease.

$$C \uparrow \implies f_H \downarrow$$

Hence, the Miller effect reduces the high-frequency response and narrows the bandwidth of the amplifier.

(D) Decrease in the high-frequency cutoff frequency

Key Points

- Miller effect primarily affects the high-frequency performance of a CE amplifier.
- The lower cutoff frequency is mainly decided by the coupling and bypass capacitors whereas the upper cutoff frequency is mainly decided by the parasitic capacitances.

Q4.17.2 NAT | 2017 | 2M | Ans: 0.90 | ● ○ ○

At the 2 V high input, the available base current is

$$I_B = \frac{2 - 0.7}{12\,\text{k}\Omega} = 0.108333\,\text{mA}.$$

The collector saturation current is

$$I_{C,sat} = \frac{5 - 0.2}{4.8\,\text{k}\Omega} = 1\,\text{mA}.$$

Therefore $\beta_{\min} = I_{C,sat}/I_B = 120/13$, and

$$\alpha_{\min} = \frac{\beta_{\min}}{1 + \beta_{\min}} = \frac{120}{133}.$$

$$\alpha_{\min} = 0.9022556\ldots$$

0.90 (two decimal places)

At the zero input level the base is not forward biased and the transistor is in cutoff; this is not a claim of a reverse-biased 0.7 V junction.

Q4.17.3 NAT | 2017 | 2M | Ans: 128 | ● ○ ○

DC Analysis :

The DC equivalent circuit is drawn by deactivating AC sources, open circuiting capacitors.

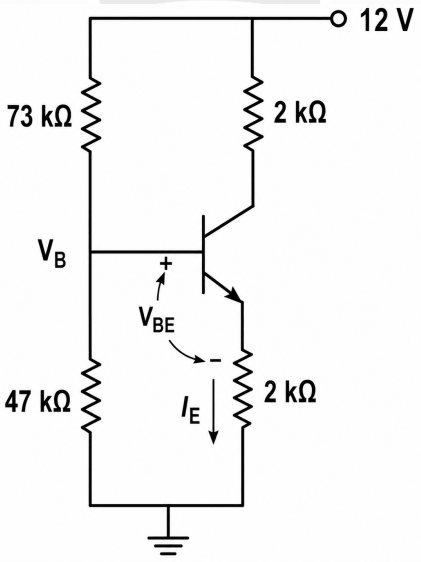

Fig. Solution Q4.17.3

Given that base current is zero so we can apply voltage division rule to get V_B.

$$V_B = 12 \times \left(\frac{47}{47 + 73}\right) = 4.7\,\text{V}$$

Applying KVL in the base emitter loop, we get:

$$V_B - 0.7 - I_E(2) = 0$$

$$\frac{4.7 - 0.7}{2} = I_E = 2\,\text{mA} = I_C$$

$$g_m = \frac{I_c}{V_T} = \frac{2\,\text{mA}}{25\,\text{mV}} = \frac{2}{25}\text{A/V}$$

The AC equivalent circuit is given by shorting capacitors and DC voltage sources. Also given that $r_0 = \infty$.

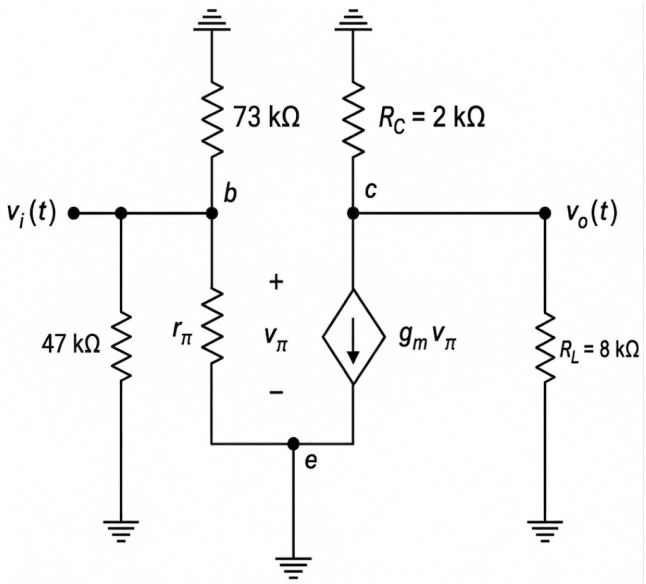

Fig. Solution Q4.17.3

Here $V_i = V_\pi$. Applying KCL at V_0 node, we get

$$V_0 = -g_m V_\pi (R_c || R_L)$$

$$V_0 = -g_m V_i (R_c || R_L)$$

$$|A_v| = \left|\frac{V_0}{V_i}\right| = g_m(R_C \parallel R_L)$$

$$|A_v| = \frac{2}{25}(2\,\text{k}\Omega \parallel 8\,\text{k}\Omega)$$

$$|A_v| = \frac{2}{25} \times \left(\frac{2\,\text{k}\Omega \times 8\,\text{k}\Omega}{2\,\text{k}\Omega + 8\,\text{k}\Omega}\right)$$

$$|A_v| = \frac{2}{25} \times 1.6 \times 10^3 = 128$$

$$\boxed{|A_V| = 128}$$

Key Points

- **DC Analysis (Approximate Method):** When the base current is neglected, the base voltage can be calculated directly using the voltage divider rule: $V_B = V_{CC}\frac{R_2}{R_1+R_2}$.
- **Midband Voltage Gain:** For a common-emitter amplifier with a fully bypassed emitter resistor (by C_E), the voltage gain magnitude is determined by the parallel combination of the collector and load resistors: $|A_v| = g_m(R_C \parallel R_L)$.

Q4.17.4 NAT 2017 1M Ans: 6 ●○○

Given that $\alpha \approx 1 \implies \beta \approx \infty$. Hence $I_C \approx I_E$ and $I_B \approx 0$.
Given circuit is,

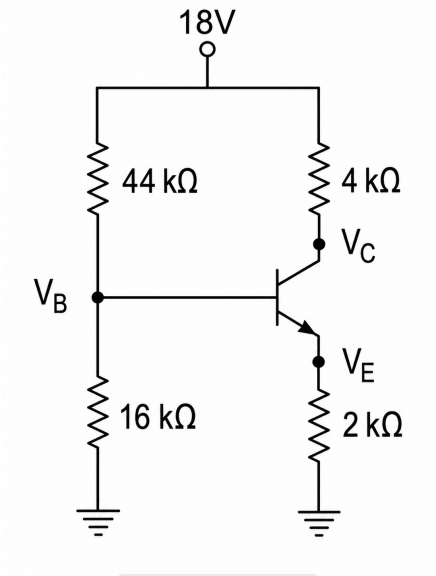

Fig. Solution Q4.17.4

As $I_B \approx 0$,we can apply voltage division rule to get V_B:

$$V_B = \frac{18 \times 16}{16 + 44} = \frac{288}{60} = 4.8\,\text{V}$$

Applying KVL in base-emitter loop,

$$V_B - V_{BE} = V_E$$

$$V_E = 4.8 - 0.8\,\text{V} = 4\,\text{V}$$

$$i_E = \frac{V_E}{2k} = \frac{4}{2k} = 2\,\text{mA} \approx i_C$$

Applying KVL from 18V to V_C, we get

$$18 - 4 \times 2 = V_C$$

$$V_C = 18 - 8 = 10\,\text{V}$$

Now, $V_{CE} = V_C - V_E$

$$V_{CE} = 10 - 4\,\text{V} = 6\,\text{V}$$

$$\boxed{V_{CE} = 6}$$

Key Points

- **Unity Current Gain:** A common-base current gain of $\alpha = 1$ indicates that the collector current is exactly equal to the emitter current ($I_C = I_E$).

Q4.17.5 NAT 2017 2M Ans: 1.175 ●●○

Given that all transistors are identical and for all transistors, $\beta = \infty$

$$I_C = I_s e^{V_{BE}/V_T}$$

where, I_s = reverse saturation current, $V_P = 0.7\,\text{V}$, Thermal voltage, $V_T = 26\,\text{mV}$
Given circuit is shown below,

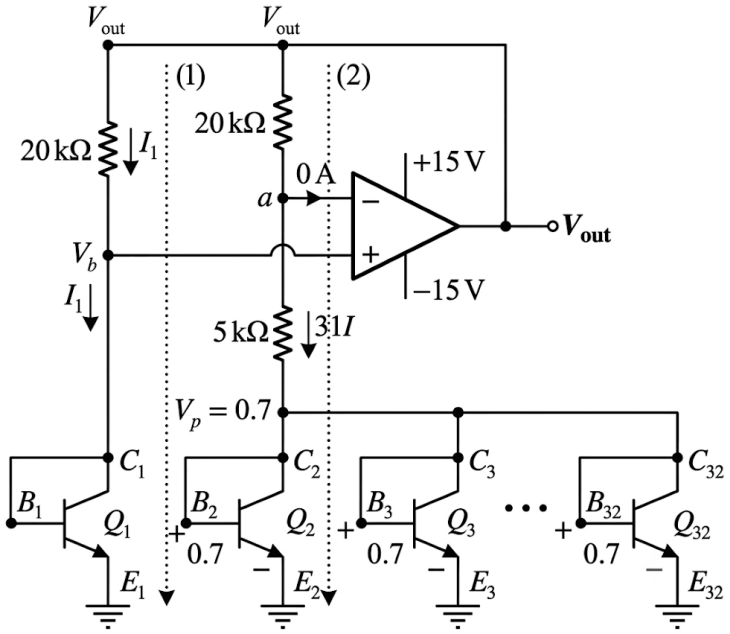

Fig. Solution Q4.17.5

From the above figure,

$$V_P = V_{BE_2} = V_{BE_3} = \cdots = V_{BE_{32}} = 0.7\,\text{V}$$

Above circuit can be redrawn as given below,

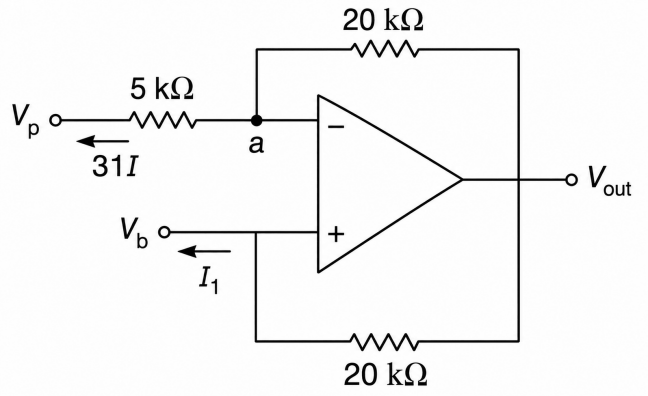

Fig. Solution Q4.17.5

From virtual ground concept,

$$V_a = V_b$$

Applying KCL at node a,

$$\frac{V_a - V_{out}}{20} + \frac{V_a - V_P}{5} = 0$$

$$\frac{V_{out} - V_b}{20} = \frac{V_b - 0.7}{5}$$

$$V_{out} - V_b = 4V_b - 2.8$$

$$V_{out} = 5V_b - 2.8 \quad \ldots (i)$$

Applying KVL in loop (1),

$$-V_{out} + 20I_1 + 0.7 = 0 \quad \ldots (ii)$$

Applying KVL in loop (2),

$$-V_{out} + 20 \times 31I + 5 \times 31I + 0.7 = 0 \quad \ldots \text{(iii)}$$

From equation (ii) and (iii),

$$20I_1 = 25 \times 31I$$

$$20I_S e^{V_b/V_T} = 25 \times 31 I_S e^{V_P/V_T}$$

$$e^{\frac{V_b - V_P}{V_T}} = \frac{775}{20}$$

$$\frac{V_b - V_P}{V_T} = \ln \frac{775}{20}$$

$$V_b - 0.7 = 0.026 \times 3.657$$

$$V_b = 0.795\,\text{V}$$

From equation (i),

$$V_{out} = 5 \times 0.795 - 2.8 = 1.175\,\text{V}$$

Hence, the output voltage V_{out} is **1.175** V.

$$\boxed{V_{out} = 1.175\,\text{V}}$$

Key Points

- **Current Summation:** The current flowing through the 5 kΩ resistor supplies 31 identical transistors biased at the same $V_{BE} = 0.7$ V, meaning it carries $31 \times I_C$ of a single transistor.

Q4.17.6 NAT 2017 2M Ans: 50 ●○○

DC Analysis :

Given that $I_C \approx I_E = 2.6\,\text{mA}$. Since both BJTs are in series, they will have the same collector and emitter currents.

$$g_m = \frac{I_C}{V_T} = \frac{2.6\,\text{mA}}{26\,\text{mV}} = \frac{1}{10}\,\text{A/V} = 0.1\,\text{A/V}$$

AC Analysis :

The AC equivalent circuit is drawn by deactivating DC sources and short-circuiting capacitors.

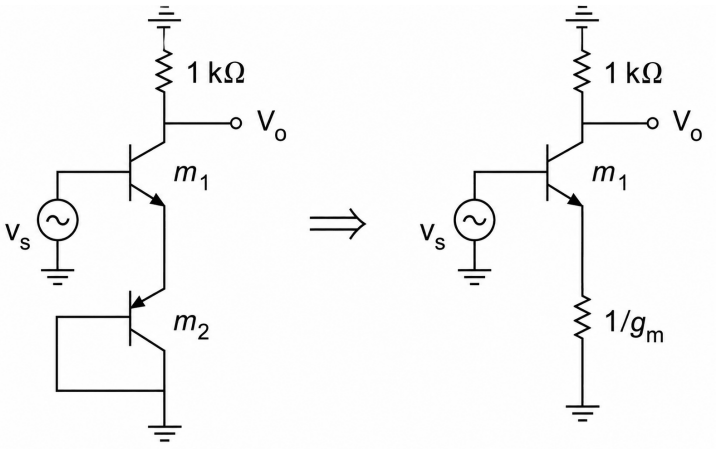

Fig. Solution Q4.17.6

Here the bottom transistor (m_2) is a diode-connected transistor, so the resistance seen through any of its terminal is $r_e = \frac{1}{g_m}$. So Replacing the transistor m_1 with its equivalent small-signal model, we get:

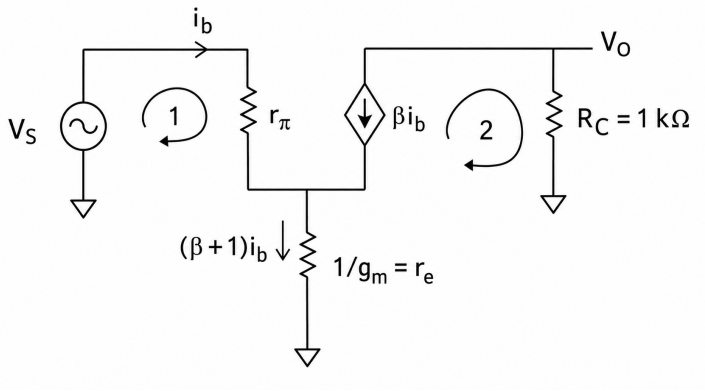

Fig. Solution Q4.17.6

Applying KVL in loop 2, we get output voltage.

$$V_o = -\beta i_b R_C$$

Apply KVL in the input loop (Loop 1):

$$V_s - i_b r_\pi - (\beta + 1) i_b r_e = 0$$

Where $r_\pi = (\beta + 1) r_e \approx \beta r_e$.

$$V_s = i_b r_\pi + (\beta + 1) i_b r_e \approx 2\beta i_b r_e$$

The voltage gain is:

$$\frac{V_o}{V_s} = \frac{-\beta i_b R_C}{2\beta i_b r_e} = \frac{-R_C}{2r_e}$$

Since $r_e = \frac{1}{g_m} = 10\,\Omega$ and $R_C = 1\,\text{k}\Omega$:

$$\frac{V_o}{V_s} = \frac{-1000}{2(10)} = -50$$

The magnitude of the voltage gain is 50.

$$\boxed{\frac{V_0}{V_s} = 50}$$

Key Points

- **Diode-Connected Transistor:** A BJT with its base and collector shorted together acts as a two-terminal device (a diode). Its small-signal AC resistance is roughly $r_e = 1/g_m$.
- **Transconductance:** The small-signal transconductance g_m is determined by the DC collector current: $g_m = I_C/V_T$.
- **Common Emitter with Emitter Degeneration:** The voltage gain of a CE amplifier with an unbypassed emitter resistance R_E is approximately $-R_C/(r_e + R_E)$. In this case, the diode-connected transistor acts as $R_E = r_e$, leading to a gain of $-R_C/(2r_e)$.

Q4.16.1 NAT 2016 1M Ans: 570 to 610 ●○○

The matched transistors have equal saturation currents and negligible base currents. Their base-emitter difference is

$$V_{BE1} - V_{BE2} = V_T \ln(I_1/I_2).$$

That voltage appears across the emitter resistor of the second transistor. Therefore

$$R_2 = \frac{V_T}{I_2} \ln \frac{I_1}{I_2} = 260 \ln 10\,\Omega.$$

$$\boxed{R_2 \approx 598.672\,\Omega}$$

Q4.16.2 MCQ 2016 1M Ans: A ● ○ ○

We know that:

$$|X_C| = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

The parasitic capacitances are in pF (10^{-12} F). So, $|X_C|$ becomes too large ($\approx \infty$) for the mid freq. band so parasitic capacitance act like open circuit.

The bypass and coupling capacitances are in μF (10^{-6} F) so, X_C becomes very small (≈ 0) for the mid-frequency band so, they behave as a short circuit. Hence, the correct option is (A).

(A) The device parasitic capacitances behave like open circuits, whereas coupling and bypass capacitances behave like short circuits.

Key Points

- **Mid-Band Operation:** In the mid-band frequency range of an amplifier, the frequency is sufficiently high such that large external capacitors (coupling and bypass) have negligible reactance and act as short circuits.
- **Parasitic Capacitances:** At these same frequencies, the signal frequency is low enough that the very small internal device capacitances (parasitic capacitances) have extremely high reactance and act as open circuits.
- **Capacitive Reactance:** The behavior is governed by $X_C = \frac{1}{2\pi f C}$, demonstrating the inverse relationship between the capacitance value and its reactance at a given frequency.

Q4.16.3 NAT 2016 1M Ans: 0.5 ● ○ ○

Given $V_{BE_1} = V_{EB_2} = 0.7$ V and $\beta_1 = 100$, $\beta_2 = 50$

Method 1

Given circuit is shown below,

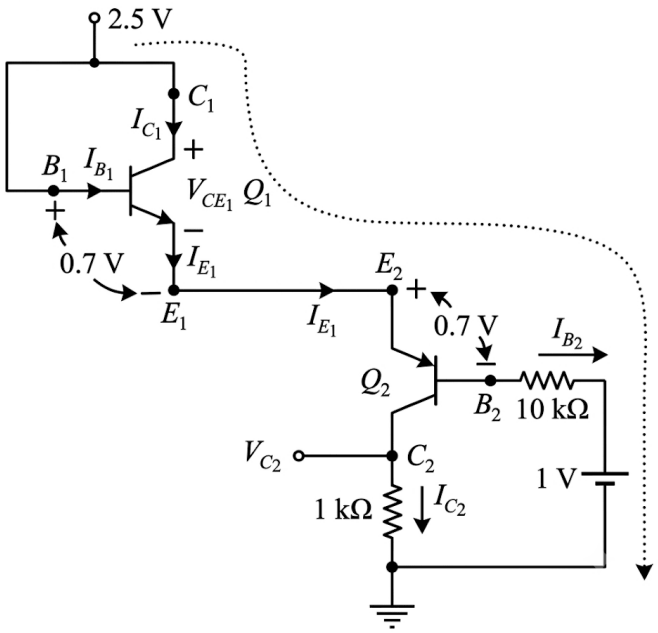

Fig. Solution Q4.16.3

For Q_1 : $V_{B_1} = V_{C_1}$

Hence, $V_{CE_1} = V_{BE_1} = 0.7$ V

Applying KVL in loop shown in figure,

$$-2.5 + V_{CE_1} + 0.7 + I_{B_2} \times 10 + 1 = 0$$

$$2.5 = 0.7 + 0.7 + 10I_{B_2} + 1$$

$$I_{B_2} = \frac{2.5 - 1.4 - 1}{10} = 0.01\,\text{mA}$$

$$I_{C_2} = \beta_2 I_{B_2} = 0.5\,\text{mA}$$

$$V_{C_2} = I_{C_2} \times 1 = 0.5 \times 1 = 0.5\,\text{V}$$

Hence, the dc voltage V_{C_2} is **0.5** V.

Method 2

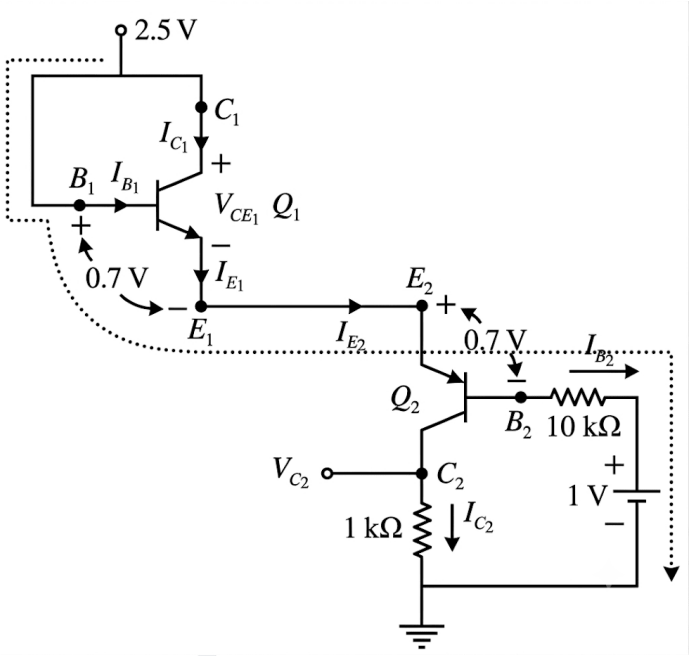

Fig. Solution Q4.16.3

Applying KVL in loop shown in figure,

$$-2.5 + 0.7 + 0.7 + 10 \times 10^3 I_{B_2} + 1 = 0$$

$$I_{B_2} = \frac{2.5 - 2.4}{10 \times 10^3} = \frac{0.1}{10} \times 10^{-3} = 0.01\,\text{mA}$$

$$I_{C_2} = \beta_2 I_{B_2} = 50 \times 0.01 \times 10^{-3} = 0.5\,\text{mA}$$

From the figure,

$$V_{C_2} = I_{C_2} \times 1 \times 10^3$$

$$V_{C_2} = 0.5 \times 10^{-3} \times 1 \times 10^3$$

$$\boxed{V_{C_2} = 0.5}$$

Q4.15.1 NAT | 2015 | 2M | Ans: 83.86 | ● ○ ○

Given $I_1 = 80$ mA, $I_2 = 4$ mA and $V_T = 26$ mV at 27^o.
The given circuit is,

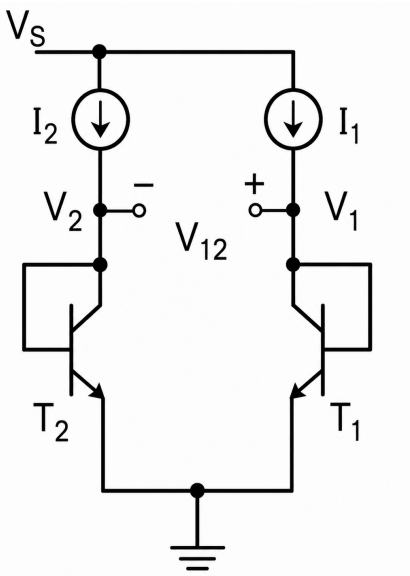

Fig. Solution Q4.15.1

Applying KVL we get,

$$-V_{BE1} + V_{12} + V_{BE2} = 0$$

$$\therefore V_{12} = V_{BE1} - V_{BE2}$$

But $I_1 \propto e^{\frac{V_{BE1}}{V_T}}$ and $I_2 \propto e^{\frac{V_{BE2}}{V_T}}$

$$\frac{I_1}{I_2} = e^{\left(\frac{V_{BE1}-V_{BE2}}{V_T}\right)} = \frac{80}{4} = 20$$

$$\therefore V_{BE1} - V_{BE2} = V_T \ln 20$$

We know $V_T \propto T$(in Kelvin). Therefore

$$V_{T_{50^o}}(27 + 273) = V_{T_{27^o}}(50 + 273) \implies V_{T_{50^o}} = \frac{323 \times 26 \text{ mV}}{300} = 28 \text{ mV}$$

$$\therefore V_{12} = V_{BE1} - V_{BE2} = V_T \ln 20 = 83.88 \text{ mV}$$

$$\boxed{V_{12} = 83.88 \text{ mV}}$$

Key Points

- The relationship between collector current and base-emitter voltage in a BJT is proportional to $e^{\frac{V_{BE}}{V_T}}$.
- Thermal voltage can be calculated using the relationship $\frac{V_{T_1}}{V_{T_2}} = \frac{T_1}{T_2}$ where temperature is in Kelvin.

Q4.15.2 NAT | 2015 | 2M | Ans: 2.8 | ● ○ ○

Given

$$V_{in} = 20 \pm 20\%,\ V_0 = 10\,\text{V},\ I_L = 200\,\text{mA}$$

$$V_{in} = 20 \pm 20\% = [24\,\text{V}, 16\,\text{V}]$$

Given circuit is,

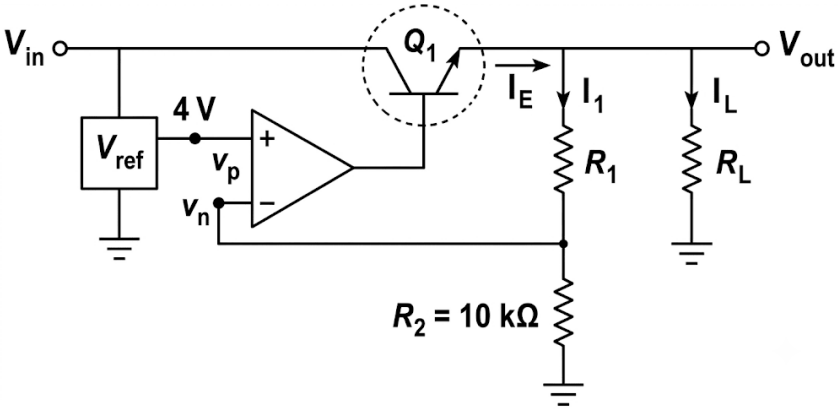

Fig. Solution Q4.15.2

For maximum power dissipation, we consider maximum V_{in} i.e, 24 V
From the circuit, we have:

$$V_{CE} = V_{in} - V_{out} = 24 - 10 = 14\,\text{V}$$

Applying virtual ground concept, we get:

$$V_n = V_p = 4\,\text{V}$$

So:

$$I_{R2} = \frac{4}{10} = 0.4\,\text{mA}$$

No current will flow through the op-amp terminals, i.e.

$$I_1 = I_{R2} = 0.4\,\text{mA} \quad \text{and} \quad I_L = 200\,\text{mA}$$

$$I_E = I_1 + I_L$$

$$I_E = 200.4\,\text{mA}$$

$$\therefore I_C = I_E = 200.4\,\text{mA}$$

Now:

$$P(Q_{1max}) = V_{CE} \times I_C$$

$$= 14 \times 200.4 \times 10^{-3}$$

$$\boxed{P_{Q1} = 2.80\,\text{Watt}}$$

Key Points

- The power dissipated by a series pass transistor is calculated as the product of the collector-emitter voltage (V_{CE}) and the collector current (I_C).
- To find the maximum power dissipation, the upper limit of the input voltage tolerance must be considered.

Q4.15.3 NAT 2015 2M Ans: -1 ● ○ ○

In the given circuit, trans-conductance for both the transistors will be the same and will be equal to g_m, as the transistors are identical.

For the given circuit, the upper transistor acts as a diode-connected load and the resistance seen into any of its terminals is equal to $1/g_m$.

The circuit can be redrawn as,

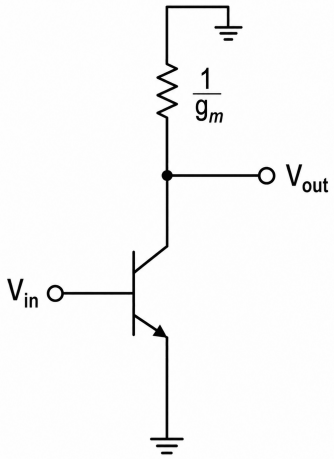

Fig. Solution Q4.15.3

The above circuit is a simple common emitter amplifier whose voltage gain is given by,

$$A_v = -g_m \times (r_0 || R_c) = -g_m \times \left(r_0 || \frac{1}{g_m}\right)$$

Since $r_0 \gg 1/g_m \implies r_0||(1/g_m) \approx 1/g_m$

$$A_v = -g_m \times \frac{1}{g_m} = -1$$

Key Points

- A transistor with its base and collector shorted acts as a diode-connected load, providing a small-signal resistance of approximately $\frac{1}{g_m}$.
- The intrinsic output resistance (r_o) of a transistor is typically much larger than $\frac{1}{g_m}$. Thus, their parallel combination is dominated by the smaller resistance: $r_o \parallel \frac{1}{g_m} \approx \frac{1}{g_m}$.
- The open-circuit small-signal voltage gain (A_v) of a basic common-emitter amplifier with an effective load resistance R_{load} is given by $A_v = -g_m R_{\text{load}}$.

Q4.15.4 NAT 2015 1M Ans: 2 ● ○ ○

Given, Current gain, $\beta = 50$ and $V_{EB} = 600\,\text{mV} = 0.6\,\text{V}$

Given circuit is shown below,

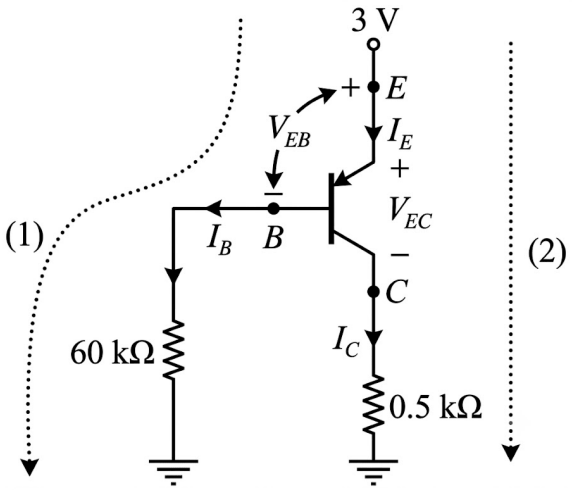

Fig. Solution Q4.15.4

Assume transistor is active region :
Applying KVL in emitter-base loop (1),

$$-3 + V_{EB} + 60\, I_B = 0$$

$$3 = V_{EB} + 60\, I_B$$

Since, $V_{EB} = 0.6\,\text{V}$

$$3 - 0.6 = 60\, I_B$$

$$I_B = \frac{2.4}{60}\,\text{mA} = 0.04\,\text{mA}$$

Applying KVL in emitter collector loop (2),

$$-3 + V_{EC} + 0.5\, I_C = 0$$

$$3 = V_{EC} + 0.5\, I_C$$

$$3 = V_{EC} + 0.5\,\beta I_B \quad [I_C = \beta I_B]$$

$$V_{EC} = 3 - 0.5 \times 50 \times 0.04 = 2\,\text{Volts}$$

$$V_{EC} > V_{EC(sat)}$$

Since, transistor is operate in active region, hence our assumption is correct. Hence, the emitter-collector voltage V_{EC} is 2 V.

$$\boxed{V_{EC} = 2\,\text{V}}$$

Key Points

- When analyzing DC biasing, the standard approach is to assume the transistor is in the active region, calculate the currents using KVL, and then verify the assumption by checking if the junction voltages satisfy the active region conditions (e.g., $V_{EC} > V_{EC(sat)}$).
- Using resistance in kΩ and current in mA within KVL equations directly yields the voltage drop in Volts.

Q4.15.5 NAT 2015 2M Ans: 1475 ● ○ ○

Given $I_S = 10^{-15}$ A, $V_T = 25\,\text{mV}$ and $V_{BE} = 700\,\text{mV}$.
We know,

$$I_C = I_s e^{\frac{V_{BE}}{V_T}} \quad \ldots (1)$$

$$I_C = 10^{-15} \times e^{(700/25)}\,\text{A}$$

$$I_E = \left(1 + \frac{1}{\beta}\right) I_C$$

I_E will be maximum when β is minimum. So when $\beta = 50$,

$$(I_E)_{max} = \left(1 + \frac{1}{50}\right) \times I_C$$

$$= 1475 \times 10^{-6}\,\text{A} = 1475\,\mu\text{A}$$

$$\boxed{(I_E)_{max} = 1475\mu\text{A}}$$

Key Points

- The relationship between collector current and base-emitter voltage is governed by the Ebers-Moll equation: $I_C = I_S e^{\frac{V_{BE}}{V_T}}$.
- The emitter current I_E is the sum of collector current and base current, relating to I_C via $I_E = \frac{\beta+1}{\beta} I_C = \left(1 + \frac{1}{\beta}\right) I_C$.

Q4.14.1 NAT | 2014 | 2M | Ans: 5.7 to 5.9 | ● ○ ○

Unit qualification: The stated saturation current is $I_S = 10^{-13}$ A and asks for transconductance in mA/V. The official accepted range, 5.7–5.9 mA/V, conflicts with the result obtained from these stated units. Using the literal original units,

$$I_C = 10^{-13} e^{0.7/0.025}\,\text{A},$$

$$I_C = 0.1446257\,\text{A},$$

$$g_m = I_C/0.025 = 5.78503\,\text{A/V}.$$

That is 5785.03 mA/V, not 5.78503 mA/V. If the given saturation current were instead 10^{-13} mA, the collector current would be 0.1446257 mA and transconductance 5.78503 mA/V, matching the official interval. Alternatively, changing the requested unit to A/V would match it with the original saturation current. These are alternative problem statements, not consequences of the stated units.

Q4.14.2 NAT | 2014 | 2M | Ans: 1092 to 1094 | ● ○ ○

With the ideal op-amp operating under negative feedback, the divider midpoint equals the 4.7 V zener reference. The 1 kΩ upper divider resistor carries

$$I = \frac{9 - 4.7}{1\,\text{k}\Omega} = 4.3\,\text{mA}.$$

No op-amp input current is drawn, so the same current passes through R:

$$R = \frac{4.7}{0.0043} = 1093.02326\,\Omega.$$

$$\boxed{R \approx 1093.02\,\Omega}$$

Q4.14.3 NAT | 2014 | 2M | Ans: -240 to -230 | ● ● ○

The DC bias divider gives $V_{th} = 3$ V and $R_{th} = 8.25$ kΩ. The DC emitter path includes both 1 kΩ and 10 ohms. Therefore

$$I_B = \frac{3 - 0.7}{8250 + 201(1010)}\,\text{A},$$

$$I_C = 2.1774127\,\text{mA},$$

$$r_\pi = 2296.30435\,\Omega.$$

At midband the 1 kΩ emitter resistor is bypassed, but the 10 ohm resistor remains. With output resistance neglected as in the intended model,

$$A_v = -\frac{\beta R_C}{r_\pi + (\beta+1)(10)},$$

$$\boxed{A_v \approx -232.21768}.$$

Keep the bias and small-signal quantities unrounded until the final calculation; the voltage gain is negative.

Q4.14.4 MCQ 2014 1M Ans: B ● ○ ○

The procedure to find the type of feedback is given below.

Step - 1: Draw AC Model of the circuit by shorting capacitors and deactivating DC sources.

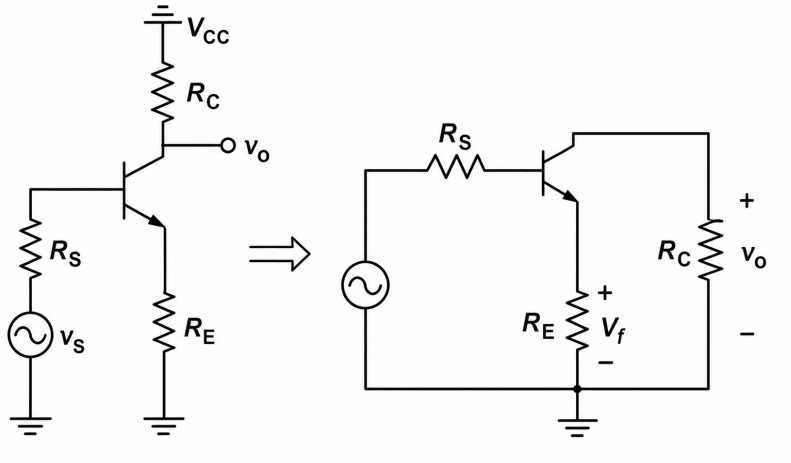

Fig. Solution Q4.14.4

Step-2: Identify the feed back element in circuit. In the above circuit R_E is the feedback element.

Step-3: If feedback element is already connected to output then it indicates voltage sampling otherwise current sampling. R_E is not directly connected to output so it is current sampling.

Step-4: If feedback element is directly connected to input then it indicates shunt mixing otherwise series mixing. Since R_E is not directly connected to input, it is series mixing which means voltage mixing.

So the given circuit is series-series feedback amplifier or current series feedback amplifier.

$$\boxed{\text{(B)} \quad \text{current-series feedback}}$$

Key Points

- **Sampling (Output):** If the feedback network is connected in parallel (shunt) with the output, it samples voltage. If connected in series, it samples current.
- **Mixing (Input):** If the feedback network is connected in series with the input loop, it is series mixing (voltage feedback). If connected in parallel with the input, it is shunt mixing (current feedback).

Q4.14.5 MCQ 2014 1M Ans: B ● ○ ○

Output-definition warning: the source figure labels a differential output between the two matched collectors. Under ideal exact symmetry, common-mode excitation produces equal collector voltages, so that *differential-output* common-mode gain is zero. Its ideal CMRR is infinite, independently of the finite tail resistance. The official 2014Set2Q12 key is B, "CMRR increases". The conventional intended argument instead uses a single-ended output, or finite mismatch producing common-mode leakage. At fixed bias, increasing tail resistance reduces single-ended common-mode gain; approximately

$$|A_{cm,se}| \approx \frac{R_C}{2R_E},$$

so its CMRR increases. That single-ended expression must not be silently applied to the exactly matched differential output shown. Thus the single-ended CMRR trend and the ideal differential-output CMRR describe different output definitions.

Q4.14.6 NAT | 2014 | 1M | Ans: 1.07 | ● ○ ○

Given ,$|V_{BE}| = 0.7\,\text{V}, \beta = 50, R_B = 100\,\text{k}\Omega, V_0 = 5\,\text{V}$.
Transistor circuit is shown below,

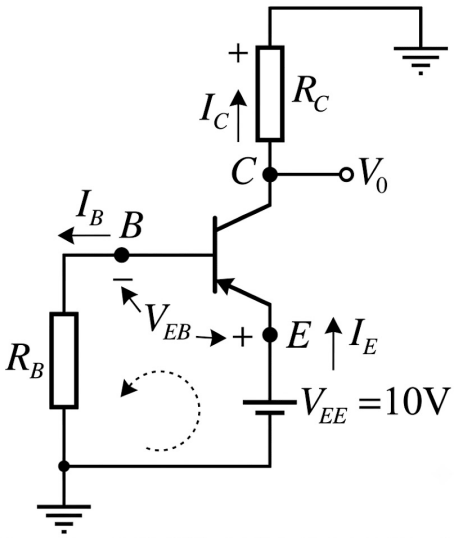

Fig. Solution Q4.14.6

For p-n-p transistor,

$$V_{BE} = -0.7\,\text{V} \implies V_{EB} = +0.7\,\text{V}$$

Assume transistor is operate in active region :
Applying KVL in the input loop,

$$-10 + V_{EB} + I_B R_B = 0$$

$$10 = 0.7 + 100 I_B$$

$$I_B = \frac{10 - 0.7}{100} = \frac{9.3}{100}\,\text{mA}$$

$$I_C = \beta I_B$$

$$I_C = 50 \times \frac{9.3}{100} = 4.65\,\text{mA}$$

Output voltage V_0 is given by,

$$V_0 = I_C R_C$$

$$5 = 4.65 \times R_C$$

$$\boxed{R_C = 1.07\,\text{k}\Omega}$$

From the figure,

$$-V_{EE} + V_{EC} + V_0 = 0$$

$$V_{EC} = 10 - 5 = 5\,\text{V}$$

$$V_{EC} > V_{EC(sat)}$$

Since, transistor is in active region, hence our assumption is correct. Hence, the value of R_C is 1.07 kΩ.

Key Points

- The assumption of active region operation must always be verified by checking if $V_{EC} > V_{EC(sat)}$ after finding the currents and voltages.

Q4.14.7 MCQ 2014 2M Ans: B

Given, DC current gain, $\beta = 50$, $V_{BE} = 0.7\,\text{V}$, this value can be used for both active and saturation region. and $V_{CE(sat)} = 0.2\,\text{V}$

Given circuit is shown below,

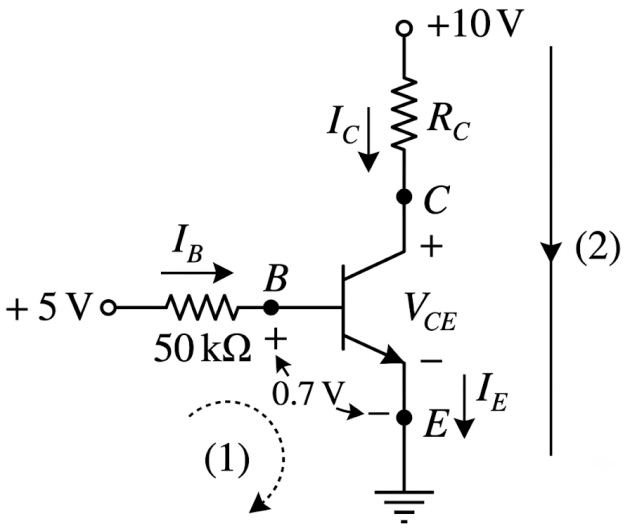

Fig. Solution Q4.14.7

Assume transistor is in saturation region :

Applying KVL in loop (1),

$$-5 + 50\,I_B + 0.7 = 0$$

$$I_B = \frac{5 - 0.7}{50\,\text{k}\Omega} = 0.086\,\text{mA}$$

Applying KVL in loop (2),

$$-10 + I_{C(sat)}R_C + V_{CE(sat)} = 0$$

$$I_{C(sat)} = \frac{10 - V_{CE(sat)}}{R_C} = \frac{10 - 0.2}{R_C}$$

Condition of saturation is given by,

$$I_{B(\text{min})} \le I_B$$

where, $I_{B(\text{min})} = \frac{I_{C(sat)}}{\beta}$. So,

$$\frac{I_{C(sat)}}{\beta} \le I_B$$

$$\frac{10 - 0.2}{50\,R_C} \le I_B$$

$$\frac{9.8}{50\,R_C} \le 0.086$$

$$R_C \ge 2.279\,\text{k}\Omega$$

Above condition of R_C is satisfied only by option (B) i.e. for $R_C = 3\,\text{k}\Omega$, the BJT operates in saturation region.
For $R_C = 1\,\text{k}\Omega$, BJT will operate in active region.
For $R_C = 20\,\text{k}\Omega$, BJT will be still in saturation region (making options C and D incorrect).
Hence, the correct option is B.

(B) For $R_C = 3\,\text{k}\Omega$, the BJT operates in the saturation region

Key Points

- A BJT operates in the saturation region if the base current I_B is greater than or equal to the minimum base current required for saturation: $I_B \geq I_{B(\min)} = \frac{I_{C(sat)}}{\beta}$.
- The saturation collector current $I_{C(sat)}$ is determined by the external circuit components (like R_C and V_{CC}) and the inherent saturation voltage $V_{CE(sat)}$ (usually 0.2 V).
- Assuming an operating region (like saturation) and proving the inequalities (like $I_B \geq I_{B(\min)}$ or $V_{CE} \leq V_{CE(sat)}$) is the standard technique for checking BJT states.

Q4.14.8 MCQ 2014 2M Ans: B

AC analysis: The AC equivalent circuit is drawn by deactivating DC sources.

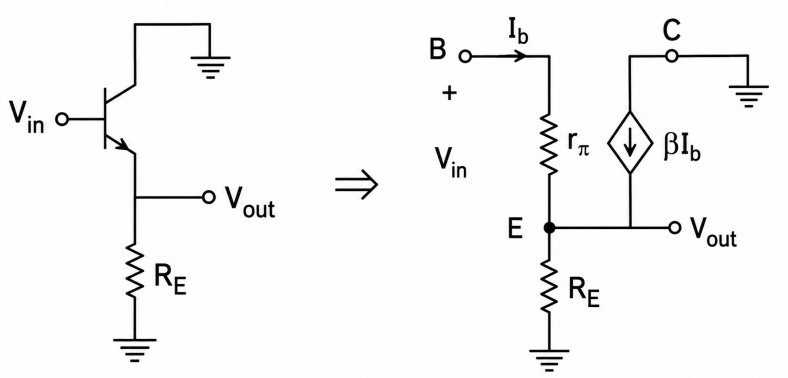

Fig. Solution Q4.14.8

Applying KVL in input and output loops,

$$V_{in} = I_b(r_\pi + (\beta + 1)R_E)$$

$$V_{out} = I_b(\beta + 1)R_E$$

Voltage gain will be:

$$A_V = \frac{V_{out}}{V_{in}} = \frac{(\beta + 1)R_E}{r_\pi + (\beta + 1)R_E}$$

$$A_V = \frac{R_E}{\frac{r_\pi}{\beta+1} + R_E}$$

$$\frac{r_\pi}{\beta + 1} \simeq \frac{1}{g_m}$$

$$A_V = \frac{R_E}{\frac{1}{g_m} + R_E}$$

For a constant small-signal voltage gain, the condition can be written as,

$$\frac{1}{g_m} \ll R_E$$

$$\therefore A_V \simeq \frac{R_E}{R_E} \simeq 1 \quad \text{(constant)}$$

$$\frac{1}{g_m} \ll R_E$$

$$g_m R_E \gg 1$$

$$\frac{I_C}{V_T} R_E \gg 1$$

$$I_C R_E \gg V_T$$

$$\boxed{(B) \quad I_C R_E \gg V_T}$$

Key Points

- For the gain to be nearly constant (approximately 1) and independent of specific transistor parameters, the condition $R_E \gg r_e$ (or $g_m R_E \gg 1$) must be satisfied.

Q4.14.9 NAT 2014 2M Ans: 9.4 ● ○ ○

The high-beta transistor negligibly loads the base divider, so $V_B = 12(10/15) = 8$ V and the emitter quiescent voltage is 7.3 V. The collector is at 12 V. The question explicitly makes the saturation voltage negligible. Therefore the largest emitter voltage before saturation is approximately 12 V, giving positive headroom 4.7 V. Cutoff gives a lower emitter limit 0 V, or negative headroom 7.3 V. The smaller headroom limits an undistorted symmetric signal:

$$V_{o,pp} = 2\min(4.7, 7.3) = 9.4\,\text{V}.$$

$$\boxed{9.4\,\text{V}}$$

Using 0.2 V for saturation would instead give 9.0 V, but that imports a different assumption from the one explicitly stated here. The 9.4 V answer is also consistent with the official 2014Set4Q39 interval 9.39–9.41.

Q4.13.1 MCQ 2013 2M Ans: C ● ○ ○

Given, β is very large. Assume $\beta = \infty$, so

$$I_B = \frac{I_C}{\beta} = \frac{I_C}{\infty} = 0\,\text{A} \quad \text{and} \quad I_C = I_E = 1\,\text{mA}$$

Method 1

The thevenin equivalent circuit of input side self bias circuit is,

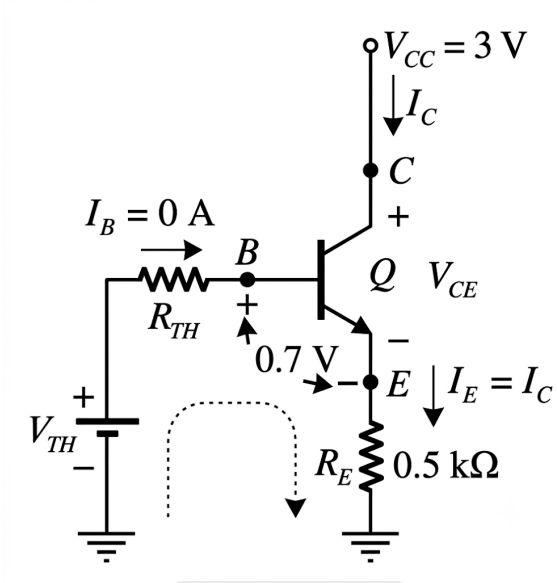

Fig. Solution Q4.13.1

$$V_{TH} = \frac{R_2 \times 3}{R_1 + R_2} = \frac{3R_2}{R_2 + 60} \quad \ldots \text{(i)} \quad \text{[By VDR]}$$

$$R_{TH} = \frac{R_1 \times R_2}{R_1 + R_2} = \frac{60R_2}{R_2 + 60} \quad \ldots \text{(ii)}$$

Applying KVL in the loop,

$$-V_{TH} + 0 \times R_{TH} + V_{BE} + I_E R_E = 0$$

$$V_{TH} = V_{BE} + I_C R_E$$

$$V_{TH} = 0.5 + 0.7 = 1.2\,\text{V}$$

From equation (i),

$$\frac{3R_2}{R_2 + 60} = 1.2$$

$$3R_2 = 1.2R_2 + 72$$

$$R_2 = 40\,\text{k}\Omega$$

Method 2

Given circuit is shown below,
β is very high ($\beta \to \infty$)

$$I_B = \frac{I_C}{\beta} = \frac{I_C}{\infty} = 0\,\text{A}$$

$$I_C = 1\,\text{mA}$$

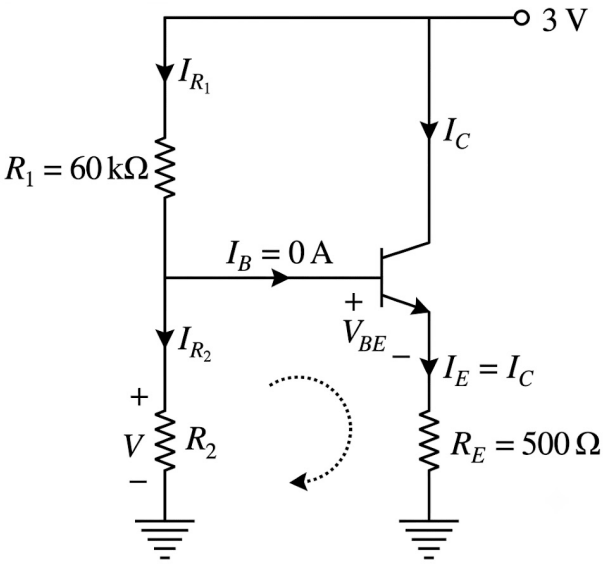

Fig. Solution Q4.13.1

Applying KVL in loop (1),

$$-V + V_{BE} + I_E R_E = 0$$

$$V = 0.7 + 1 \times 10^{-3} \times 500$$

$$V = 0.7 + 0.5 = 1.2\,\text{V}$$

Current flow through resistance R_1 is

$$I_{R_1} = \frac{3 - V}{R_1} = \frac{3 - 1.2}{60 \times 10^3} = \frac{1.8}{60}\,\text{mA}$$

$$I_{R_1} = 0.03\,\text{mA}$$

By ohm's law,

$$V = I_{R_2} R_2$$

Applying KCL at node V,

$$I_{R_1} = I_{R_2} + 0$$

$$I_{R_1} = I_{R_2} = 0.03\,\text{mA}$$

$$R_2 = \frac{V}{I_{R_2}} = \frac{1.2}{0.03 \times 10^{-3}}$$

$$R_2 = 40\,\text{k}\Omega$$

(C) 40

Q4.13.2 MCQ 2013 1M Ans: B ●○○

Given that the op-amp is ideal and is in negative feedback. So we can apply the virtual ground concept.

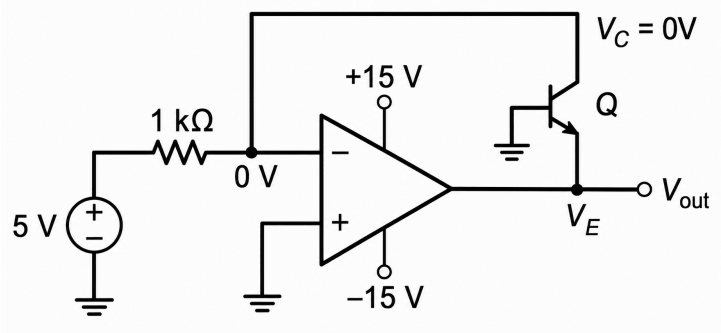

Fig. Solution Q4.13.2

The current through the 1KΩ resistor is

$$I = \frac{5-0}{1\,\text{k}\Omega} = 5\,\text{mA}$$

This current completely flows through the BJT since ideal opamp doesn't take any current.
For BJT $V_C = 0\,\text{V}$ and $V_B = 0\,\text{V}$, $I_C = 5\,\text{mA}$
The base-collector junction is **reverse** biased ($\because$ **zero volts**), therefore the collector current can have a value only if the base-emitter junction is forward biased. Forward bias $\Rightarrow V_{BE} = 0.7\,\text{V}$.

$$V_B - V_E = 0.7$$

$$0 - V_E = 0.7$$

From the given circuit $V_{out} = V_E$ So $V_{out} = -0.7\,\text{V}$

(B) -0.7 V

Mistakes to be avoided

- Do not directly apply KVL in the base-emitter loop without checking the region of operation of transistor.

Q4.12.1 MCQ 2012 1M Ans: C ●○○

Given i_b through the transistor is $1 + 0.1\cos(10000\pi t)$ mA
The DC value of i_b is $i_{b(DC)} = 1$ mA. We know

$$r_\pi = \frac{\beta}{g_m} = \frac{\beta V_T}{I_c(\text{dc})} = \frac{\beta V_T}{\beta i_b(\text{dc})}$$

$$r_\pi = \frac{V_T}{i_b(\text{dc})} = \frac{25\,\text{mV}}{1\,\text{mA}} = 25\,\Omega$$

(C) 25Ω

Q4.12.2 MCQ 2012 2M Ans: D ●○○

Method 1

DC Analysis : The DC equivalent circuit is drawn by opening the capacitors and deactivating AC sources.

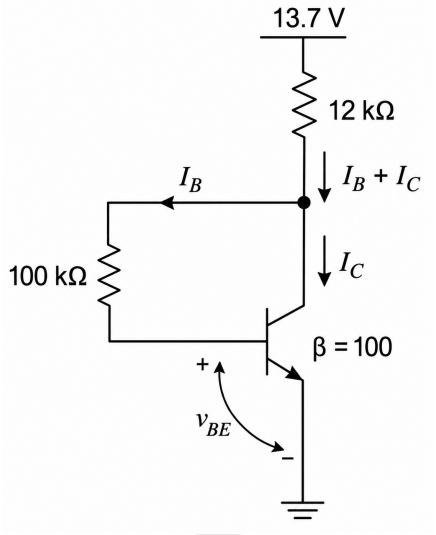

Fig. Solution Q4.12.2

Assume the transistor is in active region. So the relation $I_C = \beta I_B$ is valid.
Applying KVL from power supply to ground,

$$13.7 - 12(I_B + I_C) - 100I_B - V_{BE} = 0$$

$$13.7 - 0.7 = 112I_B + 12I_C$$

$$112\left(\frac{I_C}{\beta}\right) + 12I_C = 13$$

$$\left(\frac{112}{100} + 12\right) I_C = 13 \implies I_C = 0.991 \text{ mA}$$

The transconductance is

$$g_m = \frac{I_C}{V_T} \approx \frac{0.991\,\text{mA}}{25\,\text{mV}} \approx 0.04\,\text{S}.$$

Hence,

$$r_\pi = \frac{\beta}{g_m} = \frac{100}{39.6 \times 10^{-3}} \approx 2.5\,\text{k}\Omega.$$

Small Signal Analysis : The AC equivalent circuit is drawn by shorting capacitors, deactivating DC sources and replacing transistor with its small signal model.
Transforming given input voltage source into equivalent current source.

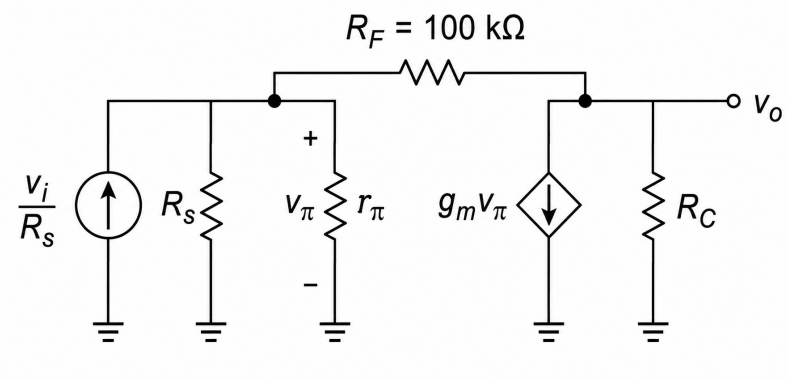

Fig. Solution Q4.12.2

This is a shunt-shunt feedback amplifier. Writing KCL at output node

$$\frac{V_0}{R_C} + g_m v_\pi + \frac{V_0 - V_\pi}{R_F} = 0$$

$$v_0\left[\frac{1}{R_C} + \frac{1}{R_F}\right] + v_\pi\left[g_m - \frac{1}{R_F}\right] = 0$$

Substituting $R_C = 12\,\text{k}\Omega$, $R_F = 100\,\text{k}\Omega$, $g_m = 0.04\,\text{s}$

$$v_0(9.33 \times 10^{-5}) + v_\pi(0.04) = 0$$

$$v_0 = -428.72 V_\pi \quad \ldots(\text{i})$$

Writing KCL at input node

$$\frac{V_i}{R_s} = \frac{V_\pi}{R_s} + \frac{V_\pi}{r_\pi} + \frac{V_\pi - V_0}{R_F}$$

$$\frac{V_i}{R_s} = v_\pi\left[\frac{1}{R_s} + \frac{1}{r_\pi} + \frac{1}{R_F}\right] - \frac{V_0}{R_F}$$

$$\frac{V_i}{R_s} = v_\pi(5.1 \times 10^{-4}) - \frac{V_0}{R_F}$$

Substituting V_π from equation (i)

$$\frac{V_i}{R_s} = \frac{-5.1 \times 10^{-4}}{428.72} v_0 - \frac{V_0}{R_F}$$

$$\frac{V_i}{10 \times 10^3} = -1.16 \times 10^{-6} v_0 - 1 \times 10^{-5} v_0 \qquad R_s = 10\,\text{k}\Omega \text{ (source resistance)}$$

$$\frac{V_i}{10 \times 10^3} = -1.116 \times 10^{-5} v_0$$

$$|A_v| = \left|\frac{V_0}{V_i}\right| = \frac{1}{10 \times 10^3 \times 1.116 \times 10^{-5}} \simeq 10$$

Method 2

Given circuit is,

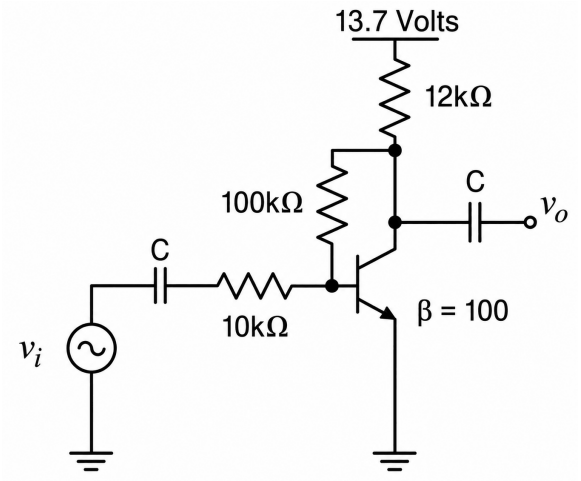

Fig. Solution Q4.12.2

Using Miller effect, the feedback resistor can be split into two resistances, the equivalent AC circuit becomes,

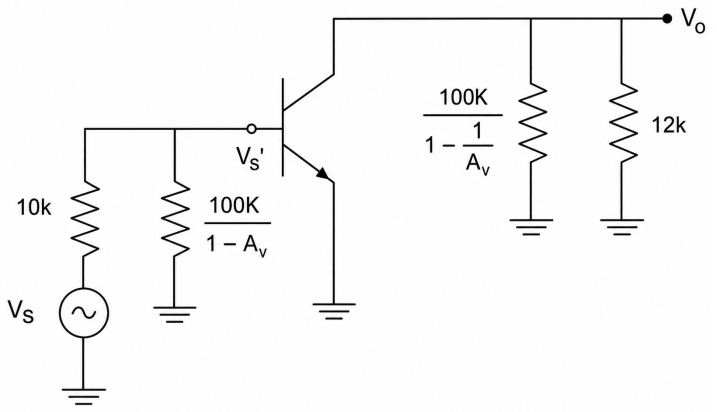

Fig. Solution Q4.12.2

Since the voltage gain will be large, the collector resistance can be approximated to 100K.
It is a simple common emitter amplifier whose voltage gain is

$$A_v = \frac{V_0}{V_S'} = -g_m(R_c \parallel R_L) = -(100\,\text{k} \parallel 12\,\text{k}) \times 0.04 = -428$$

We know the resistance seen into the base of bjt is $Z_i = r_\pi = 2.5\ \text{K}\Omega$.
The impedance seen including the miller resistance is,

$$z_i' = z_i \parallel \left(\frac{100\,\text{k}}{1+428}\right) = 213\,\Omega$$

But

$$A_{vs} = \frac{V_0}{V_s} = \frac{V_0}{V_s'} \times \frac{V_s'}{V_s} = A_v \frac{z_i'}{z_i' + R_s}$$

$$= (-428)\left(\frac{213}{213 + 10\,\text{k}}\right)$$

$$|A_{vs}| \approx 10$$

$$\boxed{\text{(D)} \quad |A_v| \simeq 10}$$

Q4.11.1 MCQ 2011 2M Ans: D ●○○

Given $I_s = 0.6\,\mu\text{A}$, $\alpha = 0.98$ and $I_B = 20\,\mu\text{A}$
Current gain β is given as:

$$\beta = \frac{\alpha}{1-\alpha}$$

$$\beta = \frac{0.98}{1-0.98} = 49$$

Collector current in terms of reverse bias saturation current is given as:

$$I_C = \beta I_B + (1+\beta) I_s$$

$$I_C = 49 \times 20\,\mu\text{A} + (50) \times 0.6\,\mu\text{A}$$

$$I_C = 0.98\,\text{mA} + 0.03\,\text{mA} = \boxed{1.01\,\text{mA}}$$

Key Points

- The relationship between the common-base current gain (α) and the common-emitter current gain (β) is $\beta = \frac{\alpha}{1-\alpha}$.
- The total collector current I_C in active mode accounts for both the amplified base current and the leakage current, given by $I_C = \beta I_B + (1+\beta)I_{CO}$, where I_{CO} (or I_s) is the reverse saturation current of the collector-base junction.

Q4.11.2 MCQ 2011 1M Ans: A ●○○

Given operating frequency is 10Mrad/sec and given circuit is,

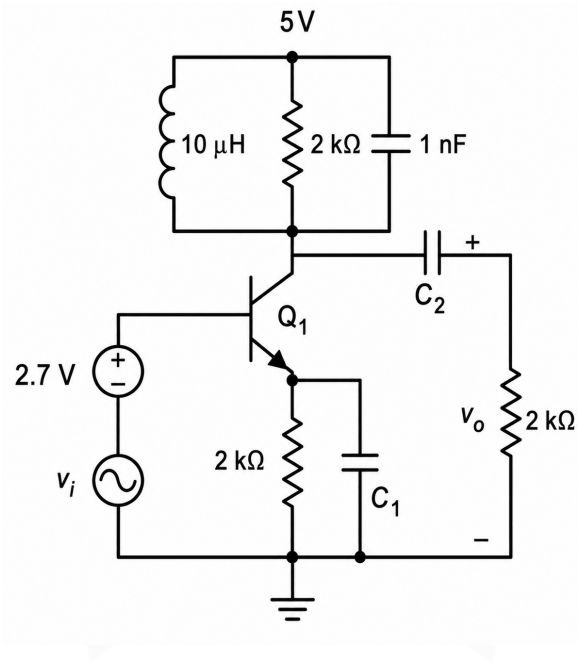

Fig. Solution Q4.11.2

For the parallel RLC circuit, the resonant frequency is

$$\omega_r = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{10\times10^{-6}\times1\times10^{-9}}} = 10\,\text{Mrad/sec}$$

So the given circuit is in resonance and at resonance the impedance of parallel RLC is maximum and purely resistive $= Z_{\max} = R = 2\,\text{k}\Omega$

The given circuit is a common emitter amplifier with emitter bypass capacitor whose gain is

$$A_V = -g_m(Z_C \parallel R_L)$$

Since Z_C is maximum , the gain is also maximum.

(A) maximum

Key Points

- At resonance, the impedance of a parallel RLC tank circuit is purely resistive and reaches its maximum value, which is equal to the parallel resistance R.
- Because the magnitude of the amplifier's voltage gain is directly proportional to the load impedance ($|A_v| \approx g_m|Z_{\text{load}}|$), the gain attains its maximum value at the resonant frequency.

Q4.11.3 MCQ | 2011 | 2M | Ans: C ● ○ ○

Given, β is very large, hence $I_B = \frac{I_C}{\beta} = \frac{I_C}{\infty} = 0\,\text{A}$ and therefore $I_C \approx I_E$, $V_{BE(\text{ON})} = 0.7\,\text{V}$, $V_{CE(\text{sat})} = 0.7\,\text{V}$.
Given circuit is shown below,

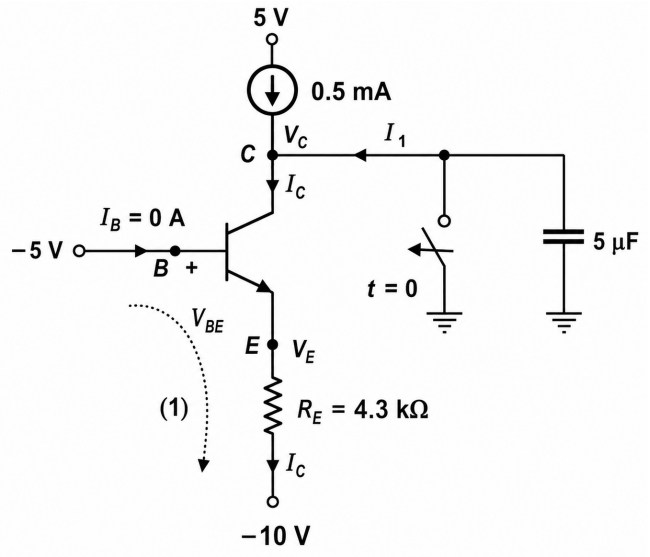

Fig. Solution Q4.11.3

From figure, $-10\,\text{V}$ is more negative as compared to $-5\,\text{V}$. Therefore, input junction is in forward bias.
Applying KVL in input loop,

$$5 + 0.7 + 4.3 \times I_C - 10 = 0$$

$$4.3 I_C = 4.3$$

$$I_C = 1\,\text{mA}$$

At $t = 0^-$, switch is closed.

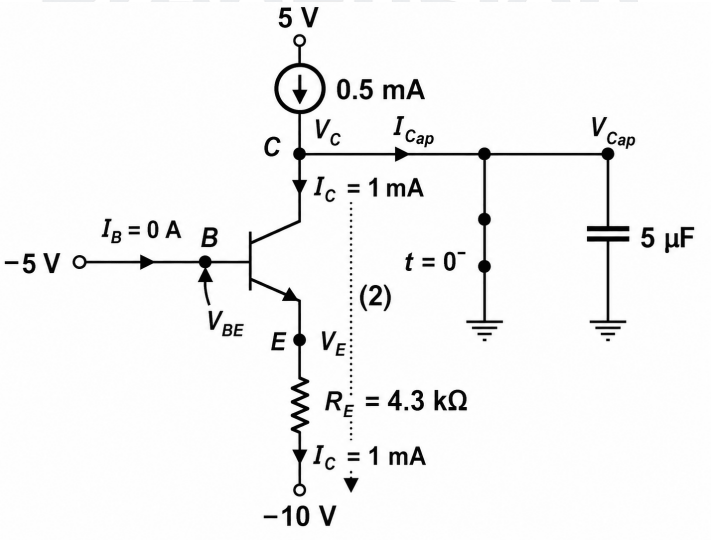

Fig. Solution Q4.11.3

Capacitor voltage $V_C(0^-) = 0\,\text{V}$. Collector voltage $V_C = 0\,\text{V}$
Note : $I_C = 1\,\text{mA}$ is fixed because other elements are fixed, which are responsible for current I_C.

Applying KVL in loop (2),

$$-V_C + V_{CE} + 4.3 \times 1 - 10 = 0$$

$$V_C = V_{CE} + 4.3 \times 1 - 10$$

$$0 = V_{CE} + 4.3 \times 1 - 10$$

$$V_{CE} = 5.7\,\text{V}$$

$$V_{CB} = V_{CE} - V_{BE} = 5.7 - 0.7 = 5\,\text{V}$$

Since the collector is n-type and V_{CB} is positive, collector junction is in reverse bias. Hence at $t = 0^-$ given transistor is in active region.

At $t = 0$, switch is opened.

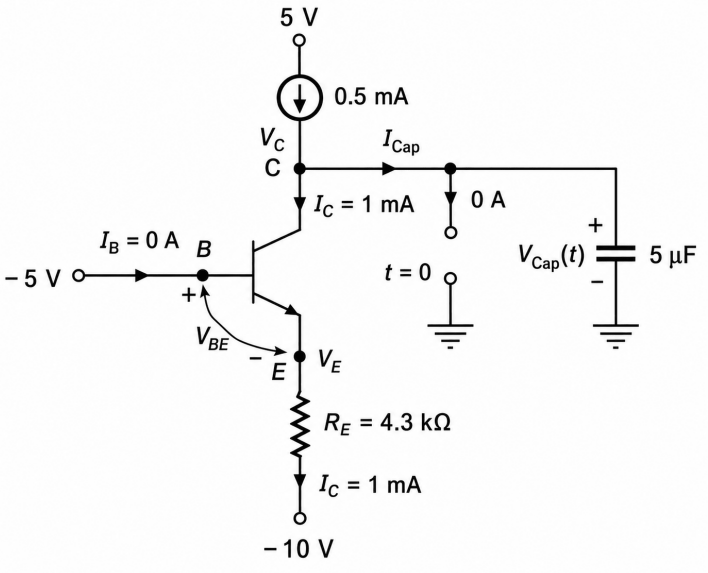

Fig. Solution Q4.11.3

From above figure, $V_C(t) = V_{Cap}(t)$

Applying KCL at collector terminal,

$$-0.5 + I_{Cap} + 1 = 0$$

$$0.5 = I_{Cap} + 1$$

$$I_{Cap} = -0.5\,\text{mA} \quad \text{[Constant value]}$$

Here, capacitor will linearly charge in opposite direction or linearly discharge. Capacitor voltage is given by,

$$V_{Cap}(t) = \frac{1}{C}\int_0^t I_{Cap}(t)dt$$

$$V_{Cap}(t) = \frac{1}{5 \times 10^{-6}}\int_0^t -0.5 \times 10^{-3}dt$$

$$V_{Cap}(t) = -100t \quad \ldots (\text{i})$$

When transistor enters into saturation region,

$$V_{CE(\text{sat})} = 0.7\,\text{V} \quad \text{[Given]}$$

From figure,

$$V_{Cap}(t) = V_{CE} + I_C R_E - 10$$

$$V_{Cap}(t) = 0.7 + 1 \times 4.3 - 10$$

$$V_{Cap}(t) = -5\,\text{V} \quad \ldots \text{(ii)}$$

Equating equation (i) and (ii),

$$-5 = -100 \times t$$

$$t = \frac{5}{100} = 0.05\,\text{sec} = 50\,\text{msec}$$

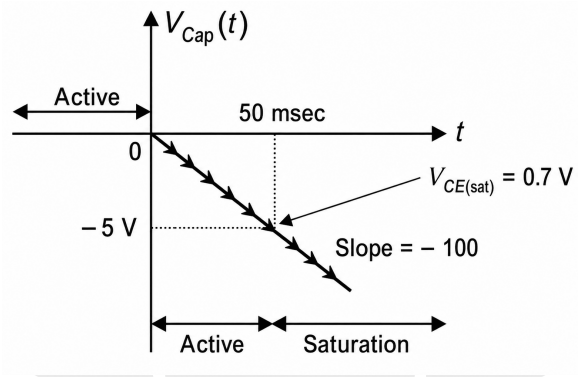

Fig. Solution Q4.11.3

Upto 50 msec transistor is in active region. After 50 msec transistor leaves the active region and goes to saturation region.

(C) 50 ms

Key Points

- If instead of -10 V, R_E was connected to ground, then emitter base junction would be in reverse bias. Proper biasing ensures initial operation in the active region.
- A constant current flowing out of a capacitor node causes the capacitor voltage to decrease linearly with time according to $V(t) = \frac{I}{C}t$.
- The boundary of the active region occurs when the collector-emitter voltage drops to $V_{CE(\text{sat})}$. By substituting $V_{CE} = V_{CE(\text{sat})}$ into the KVL equation, the corresponding collector voltage and the time it takes to reach it can be determined.

Q4.10.1 MCQ 2010 1M Ans: B ● ○ ○

Method 1

Given current mirror circuit is shown below,

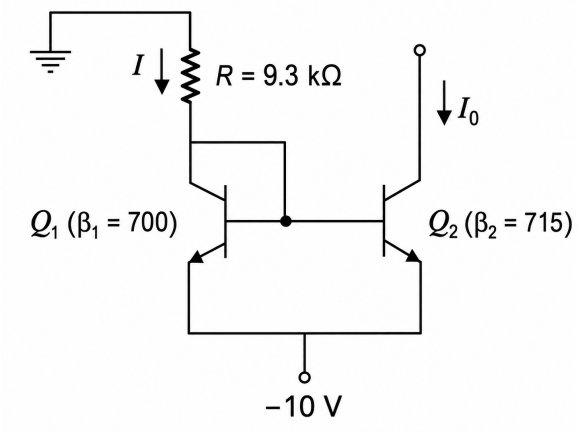

Fig. Solution Q4.10.1

According to current mirror circuit,

$$I = I_0 \quad (\beta \to \infty)$$

For perfectly matched transistor, area of emitter junction, V_{BE} and β of both transistor should be same.

$$I_0 = \left(\frac{\beta}{\beta + 2}\right) I$$

From figure, $\beta_1 = 700$, $\beta_2 = 715$, $\beta_1 \approx \beta_2$
Given : Emitter area of Q_1 is half that of Q_2, Since,

$$I = \frac{V}{R} \qquad \left[R = \frac{\rho l}{A}\right]$$

$$I = \frac{VA}{\rho l}, \qquad I \propto A$$

Hence, current in Q_2 will be twice the current of Q_1. Therefore,

$$I_0 = 2I \quad \ldots (i)$$

Given circuit is shown below,

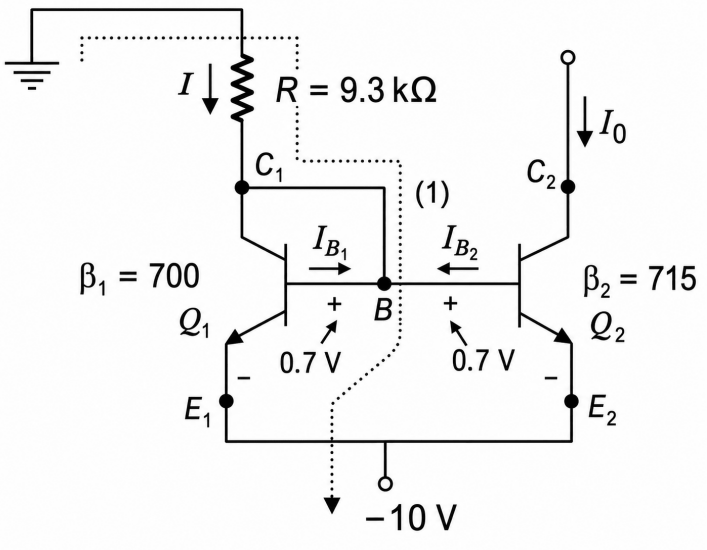

Fig. Solution Q4.10.1

Applying KVL in loop (1) shown in figure,

$$0 + I \times 9.3 + 0.7 - 10 = 0$$

$$I = \frac{9.3}{9.3} = 1\,\text{mA}$$

From equation (i),

$$I_0 = 2 \times 1 = 2\,\text{mA}$$

Hence, the correct option is (B).

Method 2

Given circuit is shown below,

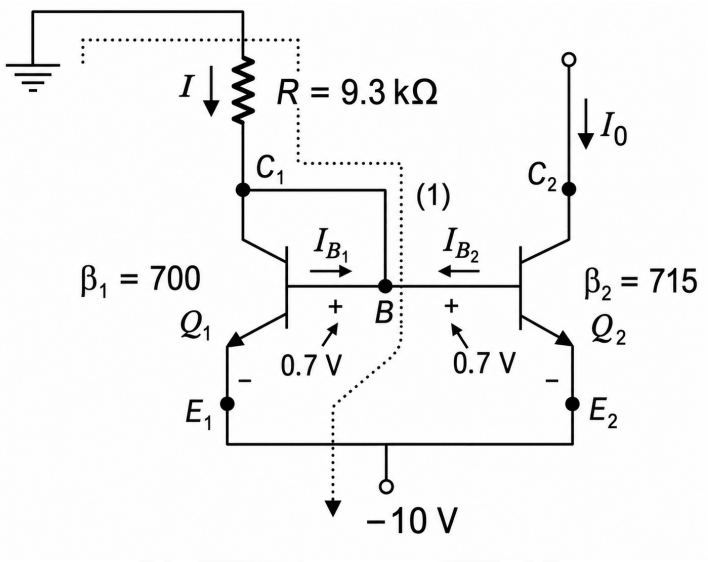

Fig. Solution Q4.10.1

Area $(Q_1) = \frac{1}{2}$Area (Q_2)

$$A_{E_1} = \frac{1}{2}A_{E_2}$$

Applying KVL, in loop (1) shown in figure,

$$0 + I \times 9.3 + 0.7 - 10 = 0$$

$$I = \frac{9.3}{9.3} = 1\,\text{mA}$$

$$I = I_{C_1} = I_{01}e^{V_{BE}/\eta V_T} \quad \ldots(\text{i})$$

$$I_0 = I_{C_2} \cong I_{E_2} = I_{02}e^{V_{BE}/\eta V_T} \quad \ldots(\text{ii})$$

$$I_0 = Aq\left[\frac{D_p}{L_p}p_{n0} + \frac{D_n}{L_n}n_{p0}\right]\left(e^{V_{BE}/\eta V_T} - 1\right)$$

$I_0 \propto A$ (area of emitter base junction $\approx$ area of emitter region)

$$I_{01} \propto A_{E_1}$$

$$I_{02} \propto A_{E_2}$$

$$\frac{I_{01}}{I_{02}} = \frac{KA_1}{KA_2}$$

$$\frac{I_{01}}{I_{02}} = \frac{A_{E_1}}{A_{E_2}} = \frac{A_{E_1}}{2A_{E_1}} = \frac{1}{2}$$

From equation (ii),

$$I_{02} = 2I_{01} \quad \ldots \text{(iii)}$$

$$I_0 = I_{02} e^{V_{BE}/\eta V_T} \quad \ldots \text{(iv)}$$

From equation (iii) and (iv)

$$I_0 = 2I_{01} e^{V_{BE}/\eta V_T}$$

$$I_0 = 2I = 2 \times 1\,\text{mA}$$

$$I_0 = 2\,\text{mA}$$

$$\boxed{\text{(B)} \quad 2\,\text{mA}}$$

Key Points

- Current through a p-n junction is directly proportional to the junction area.
- In a BJT current mirror configuration, if the transistors have different emitter areas but the same base-emitter voltage (V_{BE}), their collector currents scale proportionally to their respective emitter areas: $I_{C2} = I_{C1}\left(\frac{A_{E2}}{A_{E1}}\right)$.

Q4.10.2 MCQ 2010 1M Ans: A ●○○

Case (i): With bypass capacitor, the modified circuit is shown below along with the small-signal model.

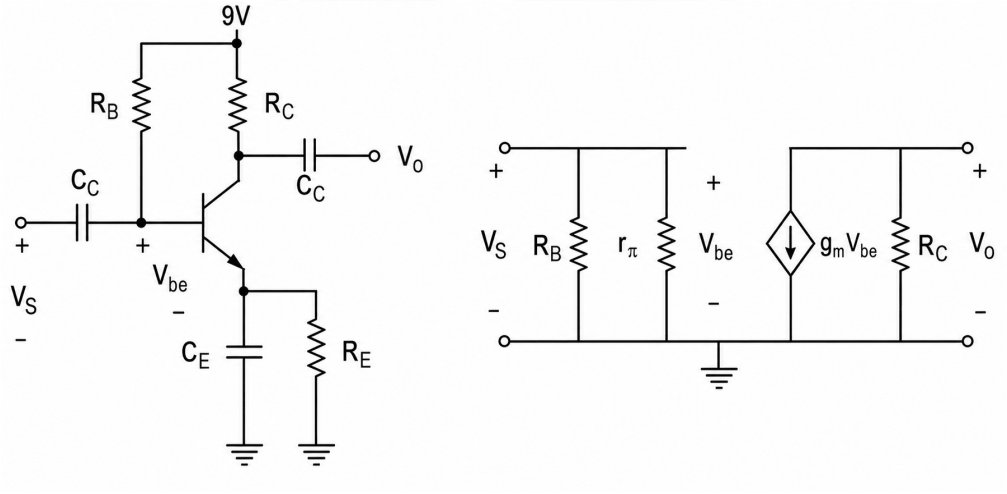

Fig. Solution Q4.10.2

As we know that the capacitors are short-circuited in the AC analysis, the emitter resistor will not be present. The whole input signal applied will come across the BE junction.

$$V_{be} = V_s$$

The output will be

$$V_o = -g_m V_{be} \times R_C$$

$$V_o = -g_m V_s \times R_C$$

$$\text{gain} = \frac{V_o}{V_s} = -g_m \times R_C$$

Input resistance will be

$$R_i = R_B \parallel r_\pi$$

In parallel combination, the resultant resistance will be lesser than the least. So, the input resistance is:

$$R_i \approx r_\pi$$

Case (ii): Without bypass capacitor, The circuit and its small-signal equivalent are shown below:
The voltage across the BE junction is divided due to the presence of the emitter resistor.

$$V_{be} = V_s \times \frac{r_\pi}{R_E + r_\pi}$$

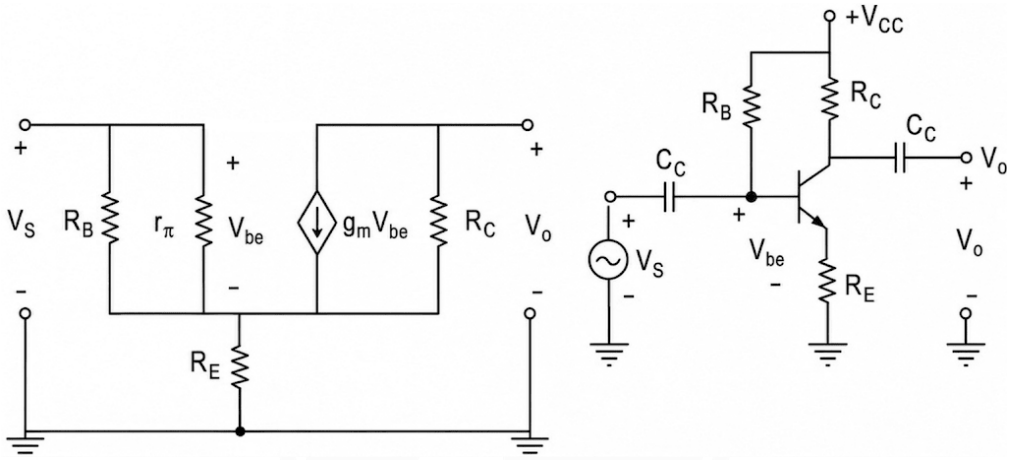

Fig. Solution Q4.10.2

The BE junction voltage is less the above case where the emitter capacitor is considered. So, the gain is reduced here.

$$\text{Gain}_{(\text{without } C_E)} < \text{Gain}_{(\text{with } C_E)}$$

Gain is reduced when the emitter capacitor is not considered.
The input impedance is given by

$$R_i = R_B \parallel (r_\pi + R_E)$$

$$R_i \approx r_\pi + R_E$$

The input resistance is increased when the emitter capacitor is removed.

(A) The input resistance R_i increases and the magnitude of voltage gain A_v decreases

Key Points

- **With C_E (Bypassed):** The circuit acts as a standard Common Emitter (CE) amplifier, providing maximum voltage gain ($A_v \approx -g_m R_C$) and low input resistance ($R_i \approx r_\pi$).
- **Without C_E (Unbypassed):** The emitter resistor R_E introduces negative current-series feedback (emitter degeneration). This sacrifices voltage gain to improve linearity and bandwidth, resulting in a reduced gain ($A_v \approx -\frac{R_C}{R_E}$) and a substantially increased input resistance.

Q4.10.3 MCQ 2010 2M Ans: B ● ○ ○

Given circuit is,

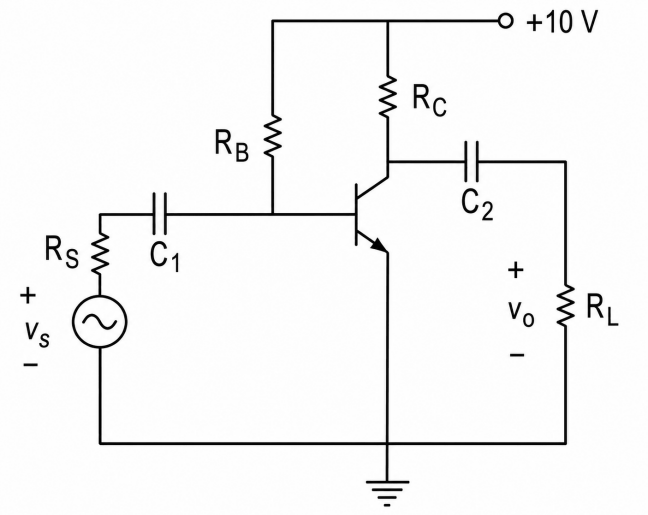

Fig. Solution Q4.10.3

Given parameters are $\beta = 100$, $g_m = 0.3861$ A/V, $r_o = \infty$, $r_\pi = 259\ \Omega$, $R_S = 1$ kΩ, $R_B = 93$ kΩ, $R_C = 250\ \Omega$, $R_L = 1$ kΩ, $C_1 = \infty$ and $C_2 = 4.7\ \mu$F

The small signal model of the given circuit is drawn by shorting capacitors and deactivating DC sources and replacing bjt with its small signal model.

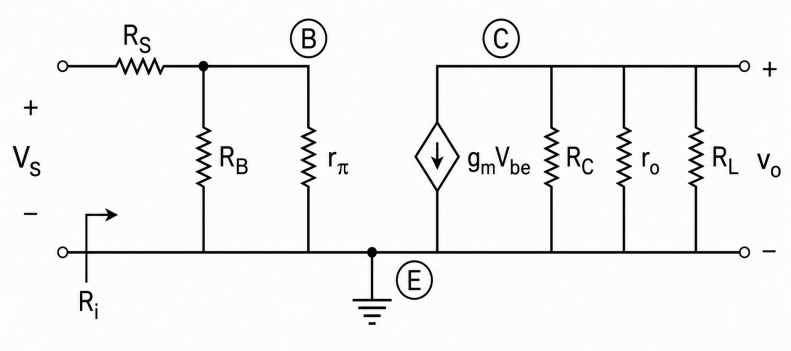

Fig. Solution Q4.10.3

The input resistance is:

$$R_i = R_S + (R_B \parallel r_\pi)$$

$$R_i = 1\,\text{k}\Omega + (93\text{k}\Omega \parallel 0.259\,\text{k}\Omega)$$

$$R_i = 1.258\,\text{k}\Omega = 1258\Omega$$

(B) 1258Ω

Q4.10.4 MCQ | 2010 | 2M | Ans: B | ● ○ ○

The AC model for the given circuit is shown below:

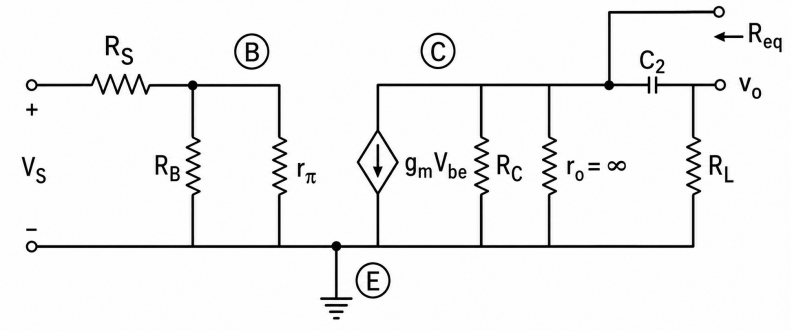

Fig. Solution Q4.10.4

While calculating the time constants the independent sources are de-energized and the equivalent circuit is:

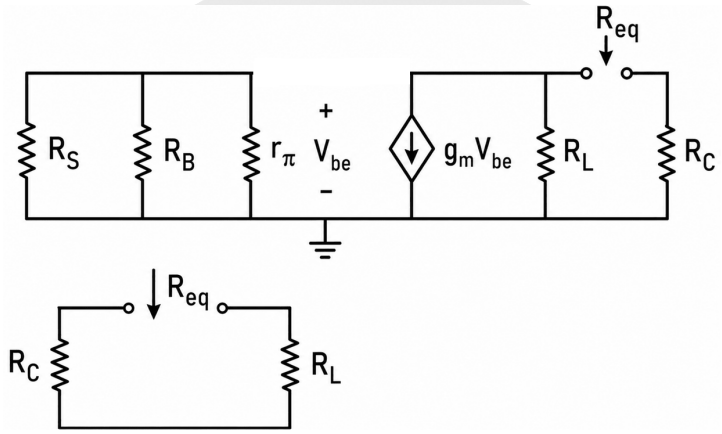

Fig. Solution Q4.10.4

$$R_{eq} = R_C + R_L$$

$$\tau = \text{Time constant} = (R_C + R_L)C_2$$

$$\tau = 1.25 \times 10^3 \times 4.7 \times 10^{-6}$$

The lower cut-off frequency by C_2 is:

$$f_{c2} = \frac{1}{2\pi\tau} = \frac{10^3}{2\pi \times 1.25 \times 4.7}$$

$$f_{c2} = \frac{10^3}{36.895}$$

$$f_{c2} = 27.1\ \text{Hz}$$

(B) 27.1 Hz

Key Points

- To determine the lower cut-off frequency due to a specific coupling capacitor, the open-circuit time constant method is used.
- De-energize all independent AC sources (short voltage sources, open current sources) and find the equivalent Thevenin resistance R_{eq} seen by the capacitor.

- For the output coupling capacitor C_2, the dependent current source $g_m V_{be}$ becomes an open circuit (since $V_s = 0 \implies V_{be} = 0$). The resistance seen by C_2 is simply the series combination $R_C + R_L$.
- The corresponding lower cut-off frequency is then calculated using $f_c = \frac{1}{2\pi\tau} = \frac{1}{2\pi R_{eq} C_2}$.

Q4.09.1 MCQ 2009 2M Ans: D

Given $V_{BE} = 0.6$ V and $\beta = 150$. Given circuit is,

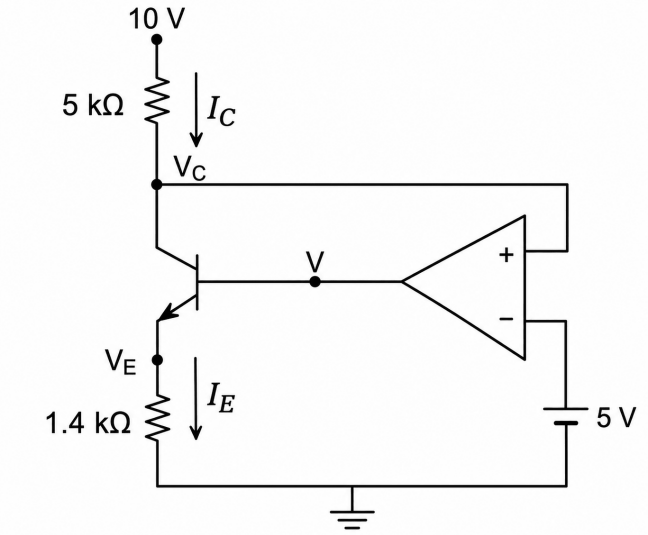

Fig. Solution Q4.09.1

To determine the type of feedback, assume that the op-amp output voltage increases slightly. Then

- Base voltage increases.
- Collector current I_C increases.
- Voltage drop across the 5 kΩ resistor increases.
- Collector voltage V_C decreases.
- Since V_C is connected to the non-inverting input, V_+ decreases.
- The op-amp responds by decreasing its output voltage.

Hence, the circuit opposes the original change, indicating negative feedback. Since the op-amp is ideal,

$$V_+ = V_- = 5\text{ V}.$$

Therefore, $V_C = 5$ V
The collector current is

$$I_C = \frac{10 - 5}{5\,\text{k}\Omega} = 1\,\text{mA}.$$

$$I_B = \frac{I_C}{\beta} = \frac{1\,\text{mA}}{150} = 6.67\,\mu\text{A}.$$

Hence,

$$I_E = I_C + I_B = 1.0067\,\text{mA}.$$

The emitter voltage is

$$V_E = I_E \times 1.4\,\text{k}\Omega = 1.0067\,\text{mA} \times 1.4\,\text{k}\Omega \approx 1.41\,\text{V}.$$

the base voltage (which is also the op-amp output) is

$$V = V_B = V_E + V_{BE} = 1.41 + 0.6 \approx 2.01\,\text{V} \approx 2\,\text{V}.$$

(D) Negative feedback, $V = 2\,\text{V}$

Mistakes to be avoided

- Do not assume positive feedback merely because the collector is connected to the non-inverting input; always check the effect of a small increase in the output.

Q4.09.2 MCQ 2009 2M Ans: D ●○○

Given $h_{ie} = r_\pi = 3\ \text{K}\Omega$, $\beta = 150$ and

$$v_i(t) = A\cos 20t + B\sin 10^6 t$$

Given circuit is,

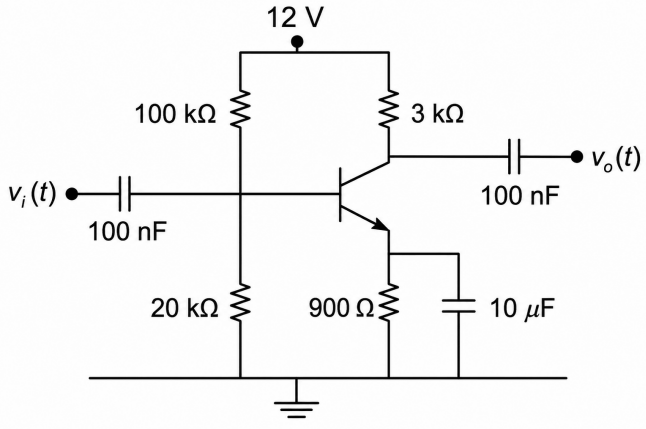

Fig. Solution Q4.09.2

The input signal $v_i(t)$ consists of two frequency components: $A\cos 20t$ (low frequency) and $B\sin 10^6 t$ (high frequency). The common emitter circuit given has a coupling capacitor at input and at output. The presence of 100 nF coupling capacitors and a 10 μF bypass capacitor acts as a high-pass filter, blocking low-frequency signals. Therefore, the amplifier will only pass significant high-frequency components.

At very low frequency, the capacitive reactance is high, and input is blocked, so the $A\cos 20t$ component is heavily attenuated. At high frequency, the capacitive reactance is low and $B\sin 10^6 t$ passes through and gets amplified.

AC Analysis : The AC equivalent circuit is given by deactivating DC sources, shorting capacitors and replacing bjt with its small signal model.

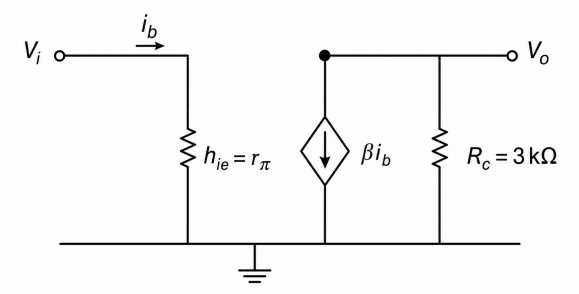

Fig. Solution Q4.09.2

Applying KVL in input and output loops,

$$V_i = h_{ie} i_b$$

$$V_0 = -\beta i_b R_C$$

Amplifier voltage gain:

$$A_v = \frac{V_0}{V_i} = -\frac{\beta \cdot R_C}{h_{ie}}$$

$$A_v = -\frac{150 \times 3000}{3000}$$

$$A_v = -150$$

Only the $B \sin 10^6 t$ term will appear at the output, multiplied by the gain -150:

$$v_o(t) = A_V \times V_i(t) = -150 \times B \sin 10^6 t$$

$$\boxed{\text{(D)} \quad v_o(t) = -150B \sin 10^6 t}$$

Key Points

- Coupling and bypass capacitors in AC transistor amplifiers block DC and act as high-pass filters, attenuating frequencies below their calculated cut-off frequency.

Q4.08.1 MCQ 2008 2M Ans: A

DC Analysis: The equivalent circuit is formed by opening the capacitors and removing AC sources.

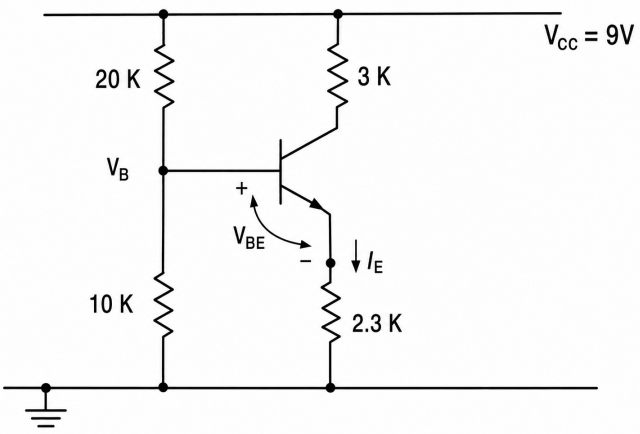

Fig. Solution Q4.08.1

The base is biased by the voltage divider. Since β is large, the base current is negligible.
Using the voltage division rule,

$$V_B = V_{CC}\left(\frac{10\,\text{k}\Omega}{20\,\text{k}\Omega + 10\,\text{k}\Omega}\right) = 9\left(\frac{10}{30}\right) = 3\,\text{V}$$

$$V_E = V_B - V_{BE} = 3 - 0.7 = 2.3\,\text{V}$$

Hence,

$$I_E = \frac{V_E}{R_E} = \frac{2.3\,\text{V}}{2.3\,\text{k}\Omega} = 1\,\text{mA}$$

Therefore,

$$\boxed{\text{(A)} \quad 1\,\text{mA}}$$

Q4.08.2 MCQ 2008 2M Ans: D ● ○ ○

The mid-band gain expression for a CE amplifier with a fully bypassed emitter resistor is,

$$A_v = \frac{V_0}{V_i} = -g_m\,(R_C \parallel R_L)$$

where

$$g_m = \frac{I_C}{V_T} = \frac{I_E}{V_T} = \frac{1\,\text{mA}}{25\,\text{mV}} = 0.04\,\text{S}$$

Hence,

$$A_v = -0.04 \times (3\text{K} \parallel 3\text{K}) = -60$$

$$\boxed{\text{(D)} \quad -60}$$

Key Points

- For a bypassed emitter resistor,

$$A_v \approx -\frac{R_C \parallel R_L}{r_e} = -g_m(R_C \parallel R_L).$$

Q4.07.1 MCQ 2007 2M Ans: B ● ○ ○

Given, $V_{BE} = 0.7\,\text{V}$ and β is large. Assume $\beta = \infty$, so $I_B = \frac{I_C}{\beta} = \frac{I_C}{\infty} = 0\,\text{A}$ and $I_C = I_E$.

Method 1

Given circuit is shown below,

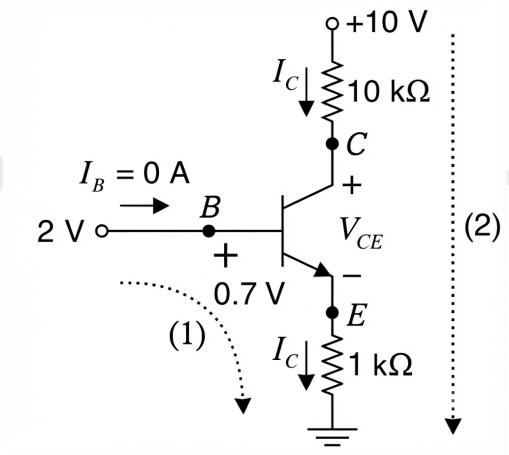

Fig. Solution Q4.07.1

In the given $n - p - n$ transistor, $V_{BE} = 0.7\,\text{V}$ and $V_B = 2\,\text{V}$, so the emitter-base junction is forward biased ($V_B > V_{BE}$).
Applying KVL in loop (1),

$$-2 + 0.7 + 1 \times I_C = 0$$

$$I_C = 1.3\,\text{mA}$$

Applying KVL in loop (2),

$$-10 + 10 \times I_C + V_{CE} + 1 \times I_C = 0$$

$$10 = 10 \times 1.3 + V_{CE} + 1 \times 1.3$$

$$V_{CE} = -4.3\,\text{V}$$

Applying KVL, in the junction of transistor,

$$V_{CB} + V_{BE} = V_{CE}$$
$$V_{CB} + 0.7 = -4.3$$
$$V_{CB} = -5\,\text{V}$$

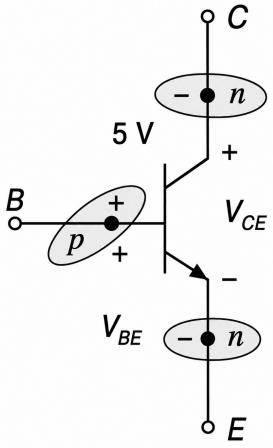

Fig. Solution Q4.07.1

From above figure, collector to base junction is in forward bias. Therefore, the transistor is in saturation region. Hence, the correct option is (B).

Method 2

In the given circuit above, applying KVL in loop (1),

$$-2 + 0.7 + 1 \times I_C = 0$$
$$I_C = 1.3\,\text{mA}$$

Applying KVL in loop (2)

$$-10 + 10 \times I_C + V_{CE} + 1 \times I_C = 0$$
$$-10 + 10 \times 1.3 + V_{CE} + 1.3 = 0$$
$$V_{CE} = -4.3\,\text{V}$$

Since $V_{CE} < V_{CE(\text{sat})} (\approx 0.2V)$, transistor is in saturation region.
Hence, the correct option is (B).

Method 3

Applying KVL in loop (1)

$$-2 + 0.7 + 1 \times I_C = 0$$
$$I_C = 1.3\,\text{mA}$$

Assume transistor is saturation region,

$$V_{CE} = 0.2\,\text{V}$$

Applying KVL, in the output loop,

$$-10 + 10I_{C(\text{sat})} + 0.2 + 1 \times I_{C(\text{sat})} = 0$$
$$I_{C(\text{sat})} = \frac{9.8}{11} = 0.890\,\text{mA}$$

Since $I_{CQ} > I_{C(\text{sat})}$, transistor is remains saturation regions.

(B) saturation

Key Points

- A practical test for saturation is calculating the active-mode collector-emitter voltage (V_{CE}). If the resulting V_{CE} is less than the typical saturation voltage ($V_{CE(\text{sat})} \approx 0.2\,\text{V}$), the transistor is in saturation.
- Alternatively, calculate the saturation collector current $I_{C(\text{sat})}$ by forcing $V_{CE} = V_{CE(\text{sat})}$. If the available active-mode collector current (I_{CQ}) is greater than $I_{C(\text{sat})}$, the transistor is saturated.

Q4.06.1 MCQ | 2006 | 2M | Ans: C | ● ○ ○

The circuit under DC condition is shown in fig below

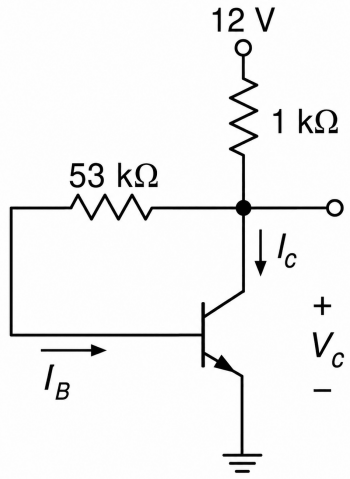

Fig. Solution Q4.06.1

Applying KVL in base emitter loop, we have

$$12 - 1\text{K}(I_C + I_B) - 53\text{K}(I_B) - 0.7 = 0$$

$$12 - 1\text{K}(\beta I_B + I_B) - 53\text{K}(I_B) - 0.7 = 0$$

$$(\beta(1\text{K}) + 54\text{K})I_B = 11.3 \implies I_B = 0.1\ \text{mA}$$

Apply KVL in the collector loop,

$$12 - 1\text{K}(I_B + I_C) - V_C = 0$$

$$V_C = 12 - (\beta + 1)I_B$$

$$V_C = 12 - 6.1 = 5.9\text{V}$$

(C) 6.0 Volts

Q4.06.2 MCQ | 2006 | 2M | Ans: B | ● ○ ○

Given $\beta_{DC} = 60$. It is to be increased by 10%. New β is,

$$\beta' = \frac{110}{100} \times 60 = 66$$

Substituting $\beta' = 66$ with other values in previous solutions

$$I'_B = \frac{11.3}{66k + 54k} = 0.094\ \text{mA}$$

$$V_C' = 12 - 67 \times 0.094 = 5.702$$

Thus the change in V_C is

$$= \frac{5.702 - 5.9}{5.9} \times 100 = -3.35\%$$

So the collector to emitter voltage drop decreases by 3.35%.

(B) decreases by less than or equal to 10%

Q4.06.3 MCQ 2006 2M Ans: A ● ○ ○

The AC equivalent circuit is,

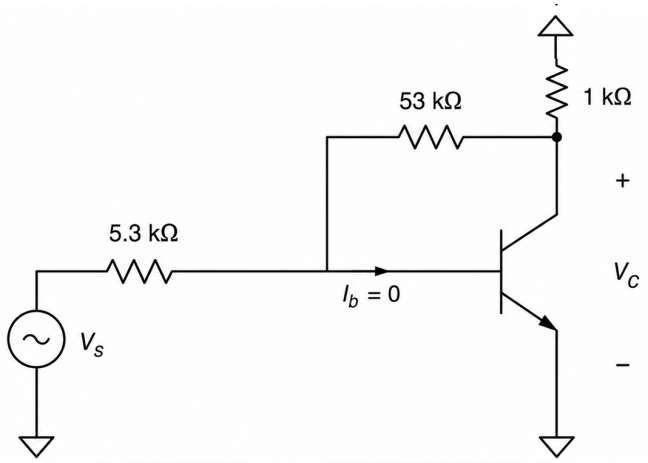

Fig. Solution Q4.06.3

Given $h_{fe} = \beta_{AC} = \infty$, so $i_b = 0$ and there exists negative feedback between collector and emitter through feedback resistor 53KΩ. Let $R_F = 53K\Omega$ and $R_1 = 5.3K\Omega$. Due to negative feedback the gain can be approximated as,

$$A_V = \frac{v_c}{v_s} \approx \frac{-R_F}{R_1} = \frac{-53}{5.3} = -10$$

(A) -10

Q4.06.4 MCQ 2006 2M Ans: C ● ○ ○

Given circuit is,

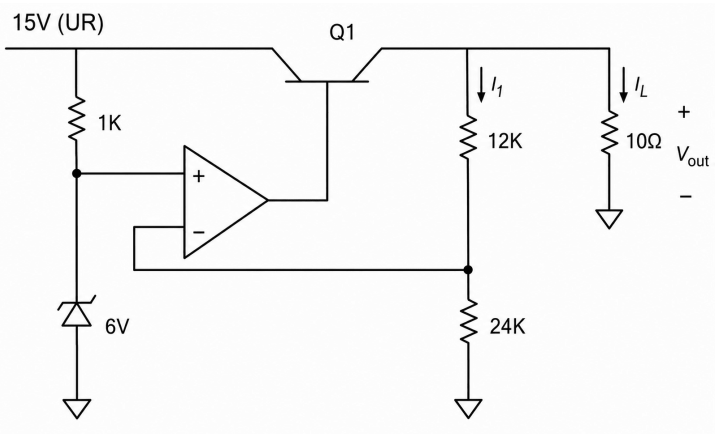

Fig. Solution Q4.06.4

The Zener diode is in breakdown region, thus

$$V_+ = V_Z = 6\,\text{V}$$

Because of negative feedback, virtual short concept is valid, so $V_- = V_+ = 6\text{V}$.

$$I_1 = \frac{V_-}{24\text{K}} = \frac{6}{24\text{K}} = 0.25\,\text{mA}$$

$$\therefore V_0 = I_1(12\text{K} + 24\text{K}) = 0.25 \times 36 = 9\,\text{V}$$

$$I_L = \frac{V_o}{10} = \frac{9}{10} = 0.9\,\text{A}$$

$$I_C \approx I_E = I_1 + I_L \approx I_L = 0.9\,\text{A}$$

$$V_{CE} = V_C - V_E = 15 - 9 = 6\,\text{V}$$

The power dissipated in transistor is

$$P = V_{CE} \times I_C = 6 \times 0.9 = 5.4\,\text{W}$$

(C) 5.4 Watts

Q4.06.5 MCQ 2006 2M Ans: B ● ○ ○

If the unregulated voltage increase by 20%, then the unregulated voltage is $1.2 \times 15 = 18\,\text{V}$, but the $V_Z = V_{in} = 6\,\text{V}$ remain same and hence V_{out} and I_C remain same. There will be change in V_{CE}. Thus,

$$V_{CE} = 18 - 9 = 9\,\text{V}$$

$$I_C = 0.9\,\text{A}$$

Power dissipation

$$P = V_{CE} I_C = 9 \times 0.9 = 8.1\,\text{W}$$

Thus % increase in power is

$$\frac{8.1 - 5.4}{5.4} \times 100 = 50\%$$

(B) increases by 50%

Q4.05.1 MCQ 2005 1M Ans: B ● ○ ○

- CE – CB → Cascode connection
- CE – CC → Cascade connection
- CC – CC → Darlington pair

Where, CE = Common Emitter, CC = Common Collector, CB = Common Base.

(B) CE-CB

Key Points

- A **Cascode** amplifier consists of a Common-Emitter (CE) stage driving a Common-Base (CB) stage. It is widely used to achieve high voltage gain, high input impedance, and excellent high-frequency response by mitigating the Miller effect in the CE stage.
- A **Cascade** connection is a general term for connecting the output of one amplifier stage to the input of the next (e.g., CE-CE or CE-CC) to multiply their individual gains or provide impedance matching.

Q4.05.2 MCQ 2005 2M Ans: C ● ○ ○

Given, $V_{BE} = 0.7$ V, $I_S = 10^{-13}$ A, $V_T \approx 26$ mV
The collector current is

$$I_C = I_S\left(e^{V_{BE}/V_T} - 1\right) \approx I_S e^{V_{BE}/V_T}$$

$$I_C = 10^{-13} e^{0.7/0.026} = 10^{-13} e^{26.92}$$

$$I_C = 10^{-13} \times 4.93 \times 10^{11} = 4.93 \times 10^{-2}\text{ A} = 49.3\text{ mA}$$

Since no information about β is given assume it is high.

$$I_E \approx I_C = 49.3\text{ mA}$$

(C) 49 mA

Q4.05.3 MCQ 2005 2M Ans: B ● ○ ○

Given, $\beta = 50$, $V_{BE} = 0.7$ V
DC equivalent circuit is formed by open circuiting capacitors and deactivating AC sources.

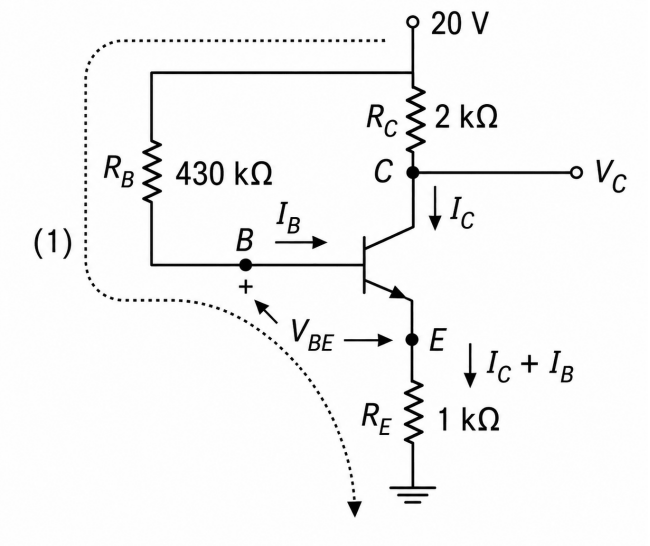

Fig. Solution Q4.05.3

Applying KVL in loop (1),

$$-20 + (I_B \times R_B) + V_{BE} + [R_E \times (I_C + I_B)] = 0$$

$$20 = (I_B \times 430) + 0.7 + [1 \times (\beta I_B + I_B)]$$

$$I_B = \frac{20 - 0.7}{430 + (\beta + 1) \times 1}$$

$$I_B = \frac{19.3}{430 + 51} = 0.04\text{ mA} = 40\,\mu\text{A}$$

$$I_C = \beta I_B = 50 \times 40\,\mu\text{A} = 2000\,\mu\text{A} = 2\text{ mA}$$

From the output circuit,

$$V_C = 20 - (I_C \times R_C) = 20 - (2 \times 2) = 16\text{ V}$$

(B) 40 μA and 16 Volts

Q4.04.1 MCQ 2004 1M Ans: A ● ○ ○

Given, $V_{CE(\text{sat})} = 0.2\,\text{V}$, $\beta = 50$. Transistor circuit is shown below,

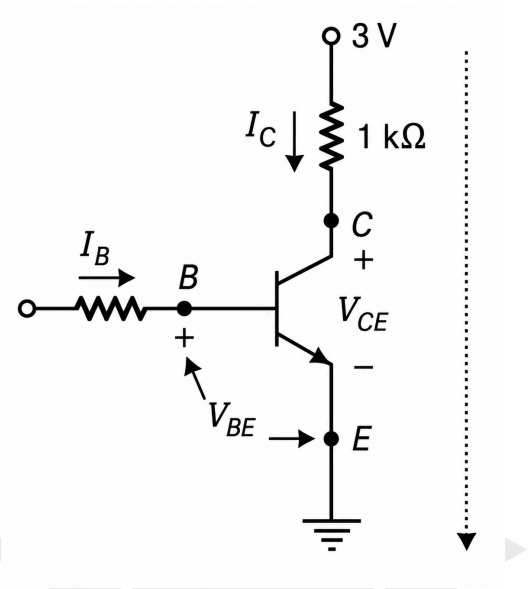

Fig. Solution Q4.04.1

Applying KVL in loop shown,

$$-V_{CC} + I_C R_C + V_{CE} = 0 \quad \ldots(\text{i})$$

For driving the transistor in saturation,

$$V_{CE(\text{sat})} = 0.2\,\text{V}$$

From equation (i),

$$V_{CE(\text{sat})} = V_{CC} - I_{C(\text{sat})} R_C$$

$$0.2 = 3 - I_{C(\text{sat})} \times 1$$

$$I_{C(\text{sat})} = 2.8\,\text{mA}$$

So,

$$I_{B(\text{min})} = \frac{I_{C(\text{sat})}}{\beta} = \frac{2.8}{50} = 56\,\mu\text{A}$$

(A) $56\,\mu\text{A}$

Key Points

- To find the minimum base current required to drive a BJT into saturation, first determine the saturation collector current $I_{C(\text{sat})}$.

Q4.04.2 MCQ 2004 2M Ans: C ● ○ ○

Given $V_{BE} = 0.7\,\text{V}$ and β is large. Assume $\beta = \infty$, so

$$I_B = \frac{I_C}{\beta} = \frac{I_C}{\infty} = 0\,\text{A} \quad \text{and} \quad I_C = I_E.$$

The Thevenin equivalent circuit of the given self bias circuit is,

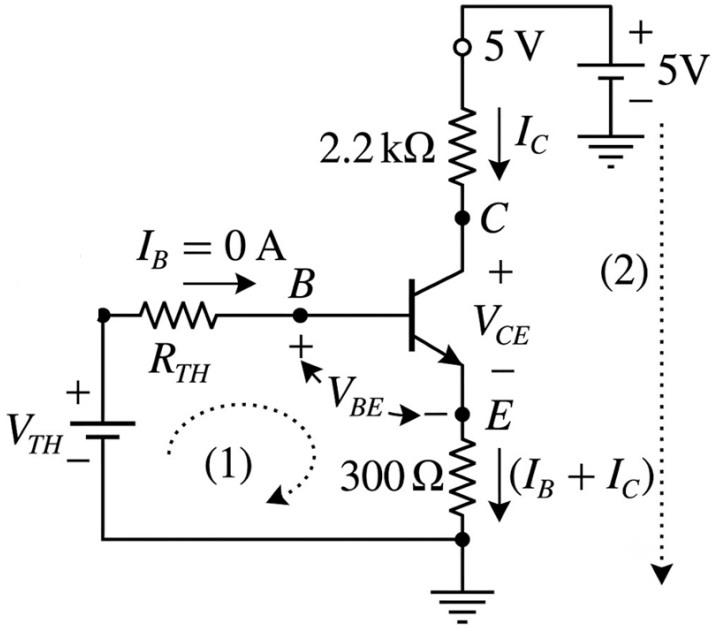

Fig. Solution Q4.04.2

$$V_{TH} = \frac{5 \times R_2}{R_1 + R_2} = \frac{5 \times 1}{4 + 1} = 1\,\text{V} \quad \text{[By VDR]}$$

$$R_{TH} = \frac{R_1 \times R_2}{R_1 + R_2} = \frac{4 \times 1}{4 + 1} = 0.8\,\text{k}\Omega$$

Applying KVL in loop (1),

$$-V_{TH} + (0 \times R_{TH}) + V_{BE} + [R_E \times (0 + I_C)] = 0$$

$$V_{TH} = V_{BE} + (R_E \times I_C)$$

$$I_C = \frac{V_{TH} - V_{BE}}{R_E} = \frac{1 - 0.7}{0.3} = 1\,\text{mA}$$

Applying KVL in loop (2),

$$-5 + (I_C \times 2.2) + V_{CE} + [0.3 \times (0 + I_C)] = 0$$

$$5 = (I_C \times 2.2) + V_{CE} + [0.3 \times I_C]$$

$$V_{CE} = 5 - (2.2 + 0.3) = 2.5\,\text{V}$$

(C) $I_C = 1$ mA, $V_{CE} = 2.5$ V

Q4.04.3 MCQ 2004 2M Ans: D ●○○

Given, $I_C = 1\,\text{mA}, V_T = 25\,\text{mV}$ and $\beta = 100$

For a BJT operating in the active region,

$$g_m = \frac{I_C}{V_T}$$

$$g_m = \frac{1\,\text{mA}}{25\,\text{mV}} = 40\,\text{mA/V}$$

The transconductance and input resistance are related by

$$g_m = \frac{\beta}{r_\pi} \implies r_\pi = \frac{\beta}{g_m}$$

$$r_\pi = \frac{100}{40 \times 10^{-3}} = 2500\,\Omega = 2.5\,\text{k}\Omega$$

(D) $g_m = 40mA/V$ and $r_\pi = 2.5k\Omega$

Q4.03.1 MCQ 2003 2M Ans: A ● ○ ○

Given that for $\beta = 150$,

$$V_{CE} = 3\,\text{V}, I_C = 1.5\,\text{mA}$$

and for $\beta = 200$,

$$V_{CE} = ?, I_C = ?$$

DC equivalent circuit : The DC equivalent circuit is formed by open circuiting capacitors and short circuiting AC voltage sources.

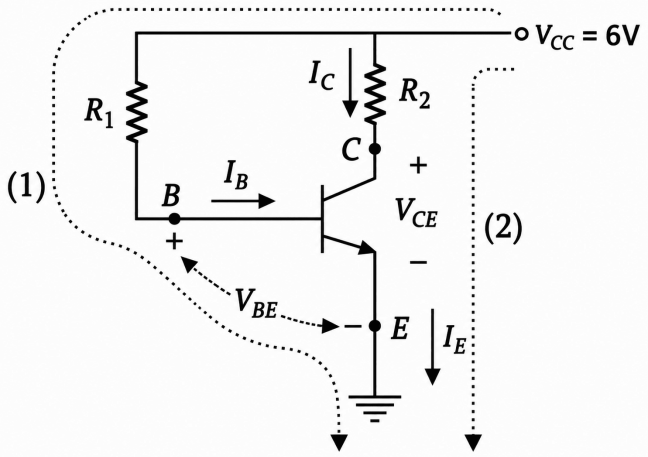

Fig. Solution Q4.03.1

Case 1 : When $\beta = 150$
Applying KVL in loop (2),

$$-V_{CC} + I_C R_2 + V_{CE} = 0$$

$$R_2 = \frac{V_{CC} - V_{CE}}{I_C} = \frac{6-3}{1.5} = 2\,\text{k}\Omega$$

Applying KVL in loop (1),

$$-V_{CC} + I_B R_1 + V_{BE} = 0$$

$$V_{CC} = \left(\frac{I_C}{\beta} \times R_1\right) + V_{BE}$$

$$R_1 = \beta \times \left(\frac{V_{CC} - V_{BE}}{I_C}\right) = 150 \times \left(\frac{6-0.7}{1.5}\right)$$

$$R_1 = 530\,\text{k}\Omega$$

Case 2 : When $\beta = 200$
Applying KVL in loop (1),

$$-V_{CC} + R_1 I_B + V_{BE} = 0$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_1} = \frac{6-0.7}{530} = 0.01\,\text{mA}$$

$$I_C = \beta I_B = 200 \times 0.01 = 2\,\text{mA}$$

Applying KVL in loop (2),

$$-V_{CE} + V_{CC} - I_C R_2 = 0$$

$$V_{CE} = 6 - (2 \times 2) = 2\,\text{V}$$

(A) (2 V, 2 mA)

Q4.03.2 MCQ 2003 2M Ans: C

Given circuit is,

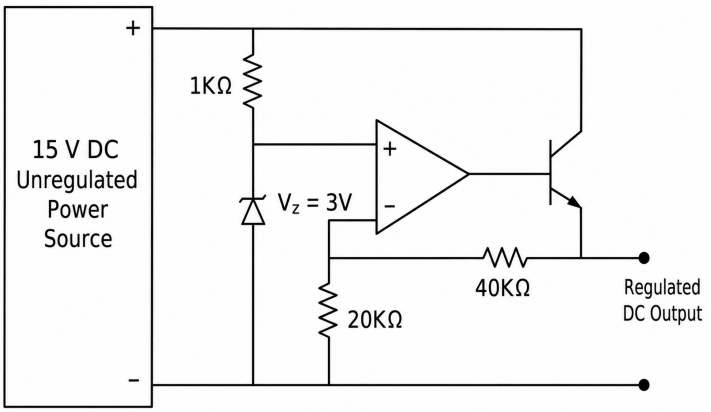

Fig. Solution Q4.03.2

Here the op-amp has negative feedback so virtual short concept is valid. Hence,

$$V_- = V_+ = V_Z = 3\,\text{V}$$

Applying voltage division rule at output,

$$V_- = V_0 \times \left(\frac{20}{20+40}\right)$$

$$V_- = \frac{V_0}{3}$$

$$V_0 = V_- \times 3 = 3 \times 3 = 9\,\text{V}$$

(C) 9 V

Q4.02.1 MCQ 2002 1M Ans: A

Given frequency response is,

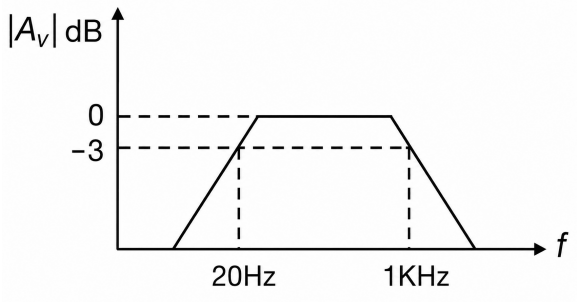

Fig. Solution Q4.02.1

For n identical cascaded first-order stages,

$$f_{L,\text{overall}} = f_L\sqrt{\frac{1}{2^{1/n}-1}},$$

$$f_{H,\text{overall}} = f_H\sqrt{2^{1/n}-1}$$

For $n = 3$,

$$f_{L,\text{overall}} = 20\sqrt{\frac{1}{2^{1/3}-1}} \approx 39.2\,\text{Hz},$$

$$f_{H,\text{overall}} = 1000\sqrt{2^{1/3}-1} \approx 509\,\text{Hz}$$

which rounds to

$$40\,\text{Hz} \quad \text{and} \quad 0.5\,\text{kHz}$$

Hence Option(A) is correct.

Q4.02.2 MCQ 2002 1M Ans: D ●○○

The relation

$$I_C = \beta_{dc} I_B$$

is valid only when the transistor operates in the active region.
When the transistor enters the saturation region, both the base-emitter and base-collector junctions are forward biased. The collector current is then limited by the external collector circuit rather than by the transistor current gain. Therefore,

$$I_C < \beta_{dc} I_B.$$

At the boundary between the active and saturation regions,

$$I_C = \beta_{dc} I_B.$$

Hence, for a transistor in saturation,

$$I_C \leq \beta_{dc} I_B.$$

(D) I_C is less than or equal to $\beta_{dc} I_B$

Q4.02.3 NAT 2002 1M Ans: * ●○○

Given the op-amp is ideal and operates with negative feedback.
Given circuit is,

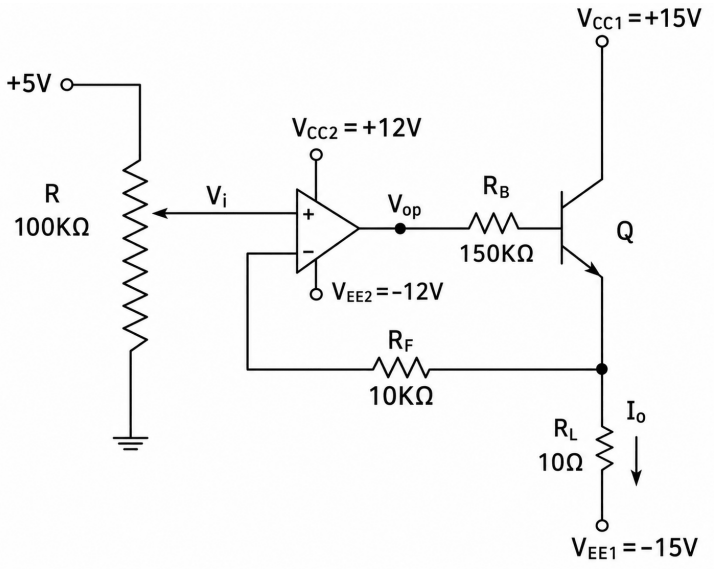

Fig. Solution Q4.02.3

Since the input current to an ideal op-amp is zero, no current flows through the feedback resistor R_F. Therefore, there is no voltage drop across R_F, giving

$$V_- = V_E.$$

Under negative feedback, the op-amp forces

$$V_+ = V_-.$$

Since

$$V_+ = V_i \implies V_E = V_i.$$

The op-amp drives the transistor base such that

$$V_B = V_E + 0.7\,\text{V} = V_i + 0.7\,\text{V},$$

which forward biases the base-emitter junction. The collector is directly connected to

$$V_C = +15\,\text{V}.$$

$$\therefore V_{CE} = V_C - V_E = 15 - V_i.$$

Since the potentiometer varies the input voltage within

$$0 \le V_i \le 5\,\text{V},$$

the collector-emitter voltage satisfies

$$10\,\text{V} \le V_{CE} \le 15\,\text{V}.$$

$$\therefore V_{CE} \gg 0.2\,\text{V},$$

which means the collector-base junction remains reverse biased while the base-emitter junction is forward biased.

The BJT operates in the forward active region.

Q4.02.4 NAT 2002 2M Ans: * ●○○

Given the op-amp is ideal and the transistor operates in the forward active region.
From above question, we know that

$$V_E = V_i$$

The current I_0 is given by,

$$I_o = \frac{V_E - (-15)}{R_L}$$

$$I_o = \frac{V_i + 15}{10}$$

$$I_o = 0.1V_i + 1.5\,\text{A}$$

Q4.02.5 NAT 2002 2M Ans: * ●○○

The original shared figure has a 10 ohm load to -15 V, a 150 kΩ base resistor, and an op-amp limited to plus/minus 12 V. An ideal op-amp has zero input current, so the 10 kΩ feedback resistor has no drop. If linear tracking held, $V_E = V_i = 2$ V would demand 1.7 A and a base drive of 2552.7 V. This is impossible. Once the op-amp saturates, the virtual short no longer applies: keeping 1.7 A while clipping only its output is inconsistent. At positive saturation $V_{op} = 12$ V, write

$$I_E = (\beta + 1)I_B = 100I_B,$$

$$V_E = -15 + 10I_E, \qquad V_B = V_E + 0.7,$$

$$12 = 150000 I_B + V_B.$$

Hence

$$I_B = \frac{26.3}{151000} = 174.172\,\mu\text{A},$$

$$\boxed{I_o = I_E = 17.4172\,\text{mA}},$$

$$V_E = -14.8258\,\text{V}, \qquad \boxed{V_{op} = 12\,\text{V}}.$$

The positive op-amp error is consistent with that rail, and the collector at 15 V keeps the BJT forward active in this constant-V_{BE} model.

Q4.02.6 NAT 2002 1M Ans: *

Given circuit is,

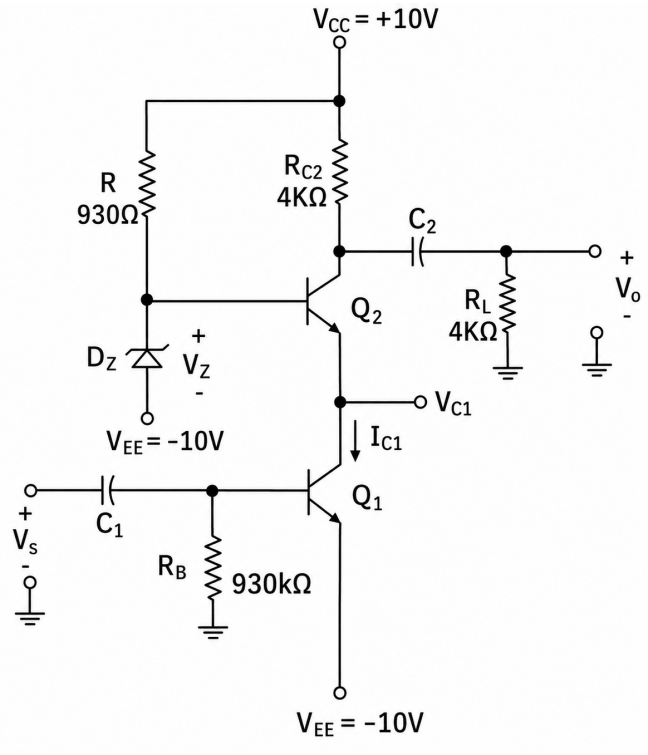

Fig. Solution Q4.02.6

The emitter of transistor Q_1 is directly connected to the -10 V supply, which acts as AC ground. The input signal is applied to the base through the coupling capacitor C_1, while the output is taken from the collector (node V_{C1}).
Hence, the first stage is a **Common-Emitter (CE)** amplifier.
For transistor Q_2, the input is applied to the emitter through node V_{C1} (collector of Q_1), while the base is held at a fixed DC voltage by the Zener diode D_z and is therefore at AC ground. The output is taken from the collector through the coupling capacitor C_2.
Hence, the second stage is a **Common-Base (CB)** amplifier.

Q4.02.7 NAT 2002 2M Ans: *

The DC equivalent circuit is,

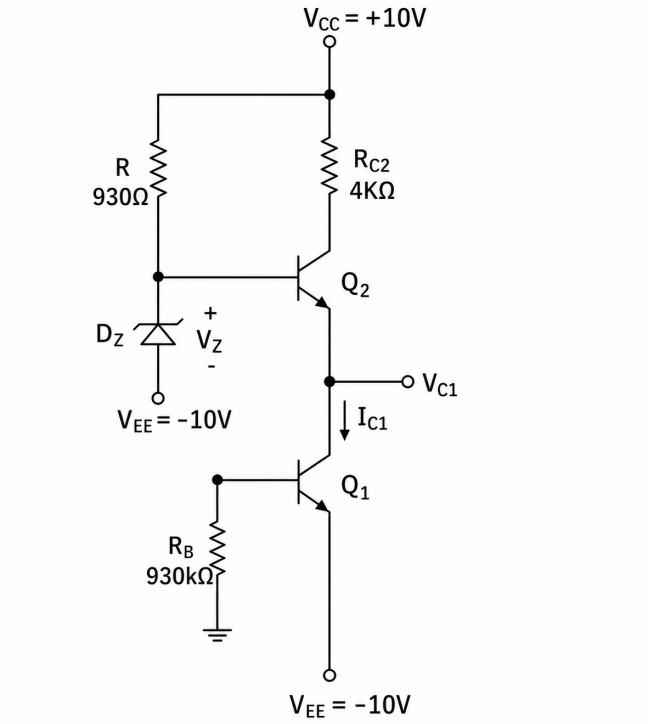

Fig. Solution Q4.02.7

Given $\beta = 100, V_{BE} = 0.7\,\text{V}$ and $V_Z = 10.7\,\text{V}$
Since the Zener diode is in breakdown and has zero dynamic resistance,

$$V_{B2} = V_{EE} + V_Z = -10 + 10.7 = 0.7\,\text{V}$$

Hence, the emitter voltage of transistor Q_2 is

$$V_{E2} = V_{B2} - V_{BE} = 0.7 - 0.7 = 0\,\text{V}$$

The emitter of Q_2 is directly connected to the collector of Q_1.

$$\therefore V_{C1} = V_{E2} = 0\,\text{V}$$

The base voltage of Q_1 is,

$$V_{B1} = V_{E1} + V_{BE1}$$

$$V_{B1} = V_{EE} + 0.7 = -10 + 0.7 = -9.3\,\text{V}.$$

The current through the base resistor is

$$I_{RB} = I_{B1} = \frac{0 - (-9.3)}{930\,\text{k}\Omega} = \frac{9.3}{930 \times 10^3} = 10\,\mu\text{A}.$$

$$I_{C1} = \beta I_{B1} = 100 \times 10\,\mu\text{A} = 1\,\text{mA}.$$

$$\boxed{I_{C1} = 1\,\text{mA} \quad V_{C1} = 0\,\text{V}}.$$

Q4.02.8 NAT | 2002 | 2M | Ans: -80 | ● ○ ○

Given $\beta = 100, r_o = \infty$ and $V_T = 25\,\text{mV}$.
From above question,

$$I_{C1} = 1\,\text{mA}$$

Since the transistors are identical and connected in series,

$$I_{C2} = I_{C1} = 1\,\text{mA}$$

Hence, the transconductance of each transistor is

$$g_m = \frac{I_C}{V_T} = \frac{1\,\text{mA}}{25\,\text{mV}} = 0.04\,\text{S}$$

$$r_\pi = \frac{\beta}{g_m} = \frac{100}{0.04} = 2.5\,\text{k}\Omega$$

Voltage gain of the first stage (CE):
The resistance seen looking into the emitter of Q_2 is $1/g_m$ which acts as collector resistance for Q_1. So the AC equivalent of the common emitter stage is redrawn as,

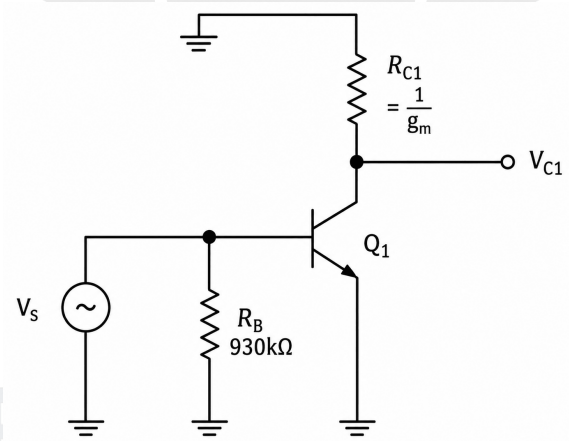

Fig. Solution Q4.02.8

$$R_{C1} \approx \frac{1}{g_m} = \frac{1}{0.04} = 25\,\Omega$$

The voltage gain of common emitter stage without emitter resistor is,

$$A_{v1} = \frac{V_{C_1}}{V_S} = -g_m R_{C1} = -0.04 \times 25 = -1$$

Voltage gain of the second stage (CB):
The AC equivalent circuit of common base stage is given as,

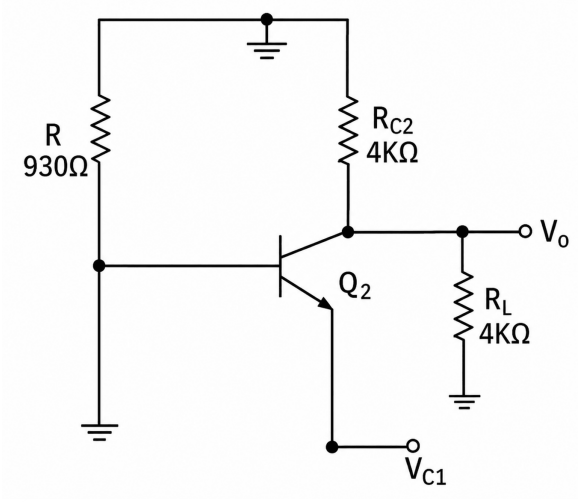

Fig. Solution Q4.02.8

The gain of a common-base stage is

$$A_{v2} = \frac{V_0}{V_{C_1}} = g_m(R_{C2} \parallel R_L) = 0.04 \times 2000 = 80$$

Hence, the overall voltage gain is

$$A_v = A_{v1} \times A_{v2} = \frac{V_{C_1}}{V_S} \times \frac{V_0}{V_{C_1}} = \frac{V_0}{V_S} = (-1)(80) = -80.$$

$$\boxed{A_v = \frac{V_o}{V_s} = -80}$$

Q4.01.1 MCQ | 2001 | 1M | Ans: C | ● ○ ○

The hybrid pi model of BJT is given as

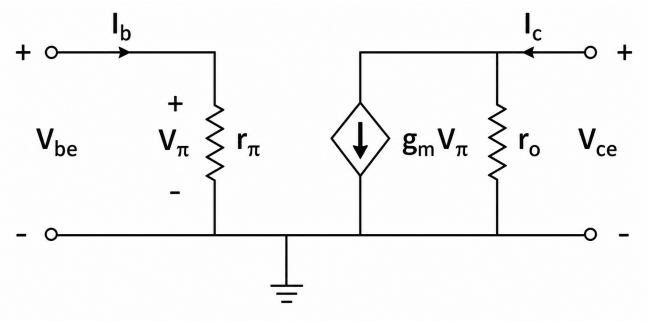

Fig. Solution Q4.01.1

The small-signal collector current is

$$i_c = g_m v_\pi,$$

where

$$v_\pi = i_b r_\pi.$$

$$\therefore i_c = g_m(i_b r_\pi) = g_m r_\pi i_b.$$

Hence, the small-signal current gain is

$$\beta = \frac{i_c}{i_b} = g_m r_\pi.$$

Hence the correct option is (C).

(C) $g_m r_\pi$

Q4.01.2 MCQ 2001 2M Ans: B ●○○

Given, $g_m = 38\,\text{mA/V} = 0.038\,\text{S}$, $C_\mu = 10^{-14}\,\text{F}$, $C_\pi = 4 \times 10^{-13}\,\text{F}$ and $\beta_0 = 90$.
The transition frequency is given by

$$f_T = \frac{g_m}{2\pi(C_\pi + C_\mu)}.$$

$$f_T = \frac{0.038}{2\pi(4 \times 10^{-13} + 10^{-14})} = 1.47 \times 10^{10}\,\text{Hz}.$$

The beta cut-off frequency is related to the transition frequency by

$$f_\beta = \frac{f_T}{\beta_0}.$$

Therefore,

$$f_\beta = \frac{1.47 \times 10^{10}}{90} = 1.64 \times 10^8\,\text{Hz}.$$

Hence, the correct option is (B).

(B) $f_T = 1.47 \times 10^{10} Hz$ and $f_\beta = 1.64 \times 10^8 Hz$

Q4.01.3 MCQ 2001 2M Ans: C ●○○

Given,
(i) Regulated output voltage, $V_0 = 10\,\text{V}$.
(ii) $V_{in(\min)} = 20\,\text{V}$, $V_{in(\max)} = 30\,\text{V}$
(iii) $V_Z = 9.5\,\text{V}$, $V_{BE} = 0.3\,\text{V}$, $\beta = 99$
(iv) Neglect the current through R_B i.e. $I_{R_B} = 0\,\text{A}$
The transistor shunt regulator is shown below,

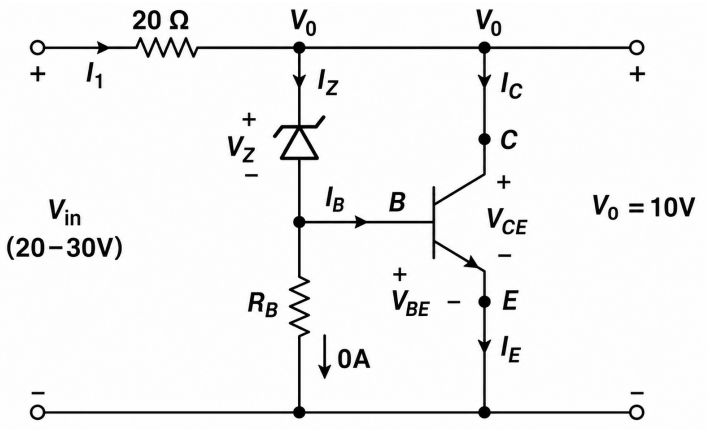

Fig. Solution Q4.01.3

Input current I_1 is given by,

$$I_1 = \frac{V_{in} - V_0}{20}$$

$$I_{1(\min)} = \frac{V_{in(\min)} - V_0}{20} = \frac{20 - 10}{20} = 0.5\,\text{A}$$

$$I_{1(\max)} = \frac{V_{in(\max)} - V_0}{20} = \frac{30 - 10}{20} = 1\,\text{A}$$

From figure, $I_B = I_Z$. Also,

$$I_C = \beta I_B = \beta I_Z$$

From figure,

$$I_1 = I_C + I_Z = \beta I_Z + I_Z = (\beta + 1) I_Z$$

Therefore,

$$I_1 = 100 I_Z \implies I_Z = \frac{I_1}{100}$$

$$I_{Z(\max)} = \frac{I_{1(\max)}}{100} = \frac{1}{100} = 0.01\,\text{A} = 10\,\text{mA}$$

Maximum power dissipation in Zener diode is given by,

$$P_{Z(\max)} = V_Z \times I_{Z(\max)}$$

$$P_{Z(\max)} = 9.5 \times 10\,\text{mA} = 95\,\text{mW}$$

Since, $I_C = \beta I_B = \beta I_Z$

$$I_{C(\max)} = \beta I_{Z(\max)} = 99 \times 10 = 990\,\text{mA}$$

Maximum power dissipation in transistor is given by,

$$P_{T(\max)} = V_{CE} I_{C(\max)}$$

From figure, $V_{CE} = V_0 = 10\,\text{V}$

$$P_{T(\max)} = 10 \times 990\,\text{mA} = 9.9\,\text{W}$$

(C) $P_Z = 95\text{mW}, P_T = 9.9\text{W}$

Q4.01.4 MCQ 2001 2M Ans: MTA

The tank uses a capacitive divider, so the oscillator is Colpitts. The two 2 pF capacitors are in series for the tank resonance:

$$C_{eq} = \frac{C_1 C_2}{C_1 + C_2} = 1\,\text{pF}.$$

The displayed tank inductor is 10 microhenry, giving

$$f_0 = \frac{1}{2\pi\sqrt{LC_{eq}}} = 50.33\,\text{MHz}.$$

Figure/options warning: none of the listed frequencies is 50.33 MHz. Option (B) gives 79.6 MHz, which would require about 4 microhenry with this capacitance. No such effective inductance follows from the shown ideal tank.

Marks to all: none of the printed options is correct.

Q4.01.5 NAT 2001 1M Ans: *

For small-signal analysis, the DC voltage sources V_{CC} and $-V_{EE}$ are replaced by AC ground. The BJT is replaced by its hybrid-π model consisting of the base-emitter resistance r_π and the dependent current source $g_m V_\pi$. The resistor R_L remains connected from the emitter to ground, while the collector is at AC ground.
Hence, the required small-signal equivalent circuit is shown below.

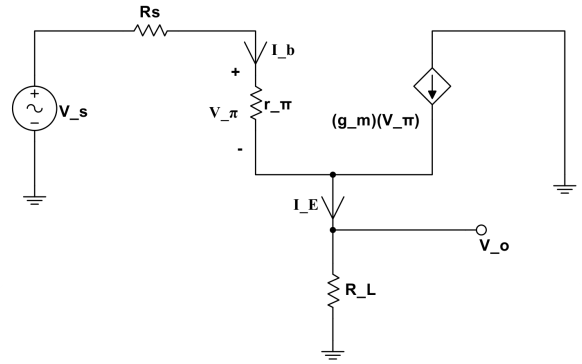

Fig. Solution Q4.01.5

Q4.01.6 NAT 2001 1M Ans: *

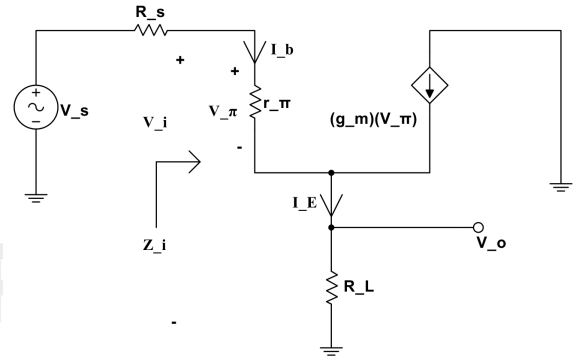

Fig. Solution Q4.01.6

Using the small-signal model, let the input current be I_b.
The emitter current is

$$I_E = (\beta + 1) I_b.$$

Hence, the emitter voltage is

$$V_o = I_E R_L = (\beta + 1) I_b R_L.$$

The base-emitter voltage is

$$V_\pi = I_b r_\pi.$$

Therefore, the input voltage is

$$V_i = V_\pi + V_o = I_b r_\pi + (\beta + 1) I_b R_L.$$

Hence, the input impedance looking into the base is

$$Z_i = \frac{V_i}{I_b} = \frac{I_b r_\pi + (\beta + 1) I_b R_L}{I_b} = r_\pi + (\beta + 1) R_L.$$

$$\boxed{Z_i = r_\pi + (\beta + 1)R_L}$$

Q4.01.7 NAT | 2001 | 1M | Ans: *

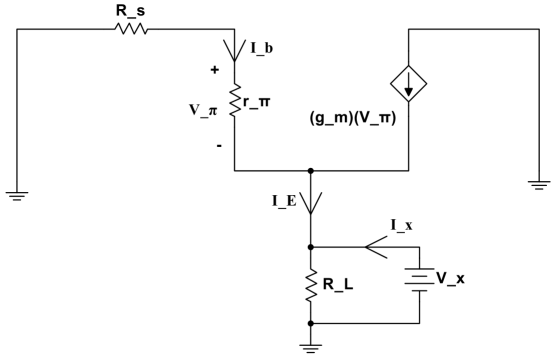

Fig. Solution Q4.01.7

To find Z_o, the input source is set to zero and a test voltage V_x is applied at the emitter. Let the resulting current drawn from the source be I_x.

Applying KVL in the base-emitter loop,

$$I_b R_s + I_b r_\pi + V_x = 0.$$

Hence,

$$I_b = -\frac{V_x}{R_s + r_\pi}.$$

Applying KCL at the emitter node,

$$I_E + I_x = \frac{V_x}{R_L}$$

$$(\beta + 1)I_b + I_x = \frac{V_x}{R_L}$$

Therefore,

$$I_x = \frac{V_x}{R_L} - (\beta + 1)I_b.$$

Substituting the value of I_b,

$$I_x = \frac{V_x}{R_L} + \frac{(\beta + 1)V_x}{R_s + r_\pi}.$$

Hence,

$$\frac{I_x}{V_x} = \frac{1}{R_L} + \frac{\beta + 1}{R_s + r_\pi} = \frac{1}{R_L} + \frac{1}{\frac{R_s + r_\pi}{\beta + 1}}.$$

Therefore,

$$Z_o = \frac{V_x}{I_x} = R_L \parallel \frac{R_s + r_\pi}{\beta + 1}.$$

$$\boxed{Z_o = R_L \parallel \frac{R_s + r_\pi}{\beta + 1}}$$

Q4.01.8 NAT 2001 2M Ans: * ● ● ○

With a capacitor C_L connected across R_L, the effective emitter impedance becomes

$$Z_E = R_L \parallel \frac{1}{j\omega C_L}$$

From Q4.01.6,

$$Z_i = r_\pi + (\beta + 1)Z_E$$

Hence,

$$Z_i = r_\pi + (\beta + 1)\left(R_L \parallel \frac{1}{j\omega C_L}\right)$$

From Q4.01.7,

$$Z_o = Z_E \parallel \frac{R_s + r_\pi}{\beta + 1}$$

Therefore,

$$Z_o = \left(R_L \parallel \frac{1}{j\omega C_L}\right) \parallel \frac{R_s + r_\pi}{\beta + 1}$$

$$\boxed{Z_i = r_\pi + (\beta + 1)\left(R_L \parallel \frac{1}{j\omega C_L}\right)}$$

$$\boxed{Z_o = \left(R_L \parallel \frac{1}{j\omega C_L}\right) \parallel \frac{R_s + r_\pi}{\beta + 1}}$$

Q4.01.9 NAT 2001 1M Ans: * ● ● ○

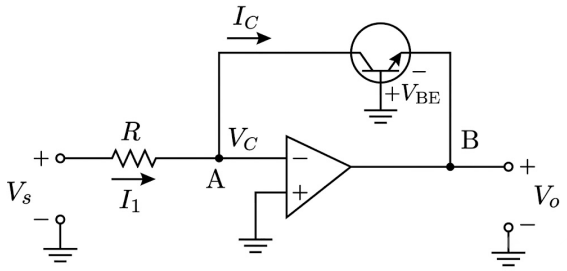

Fig. Solution Q4.01.9

Since the OP-AMP is ideal, $V_A = 0$
Hence the current through the resistor is,

$$I_1 = \frac{V_s}{R}$$

The input current of the OP-AMP is zero
Applying KCL at node A

$$I_C = I_1 = \frac{V_s}{R}$$

For the transistor,

$$I_C = I_s e^{V_{BE}/V_T}$$

Therefore

$$V_{BE} = V_T \ln\left(\frac{I_C}{I_s}\right)$$

Since the base is grounded,

$$V_{BE} = V_B(\text{base}) - V_B(\text{emitter}) = 0 - V_o = -V_o$$

Hence

$$-V_o = V_T \ln\left(\frac{V_s}{RI_s}\right)$$

$$\boxed{V_o = -V_T \ln\left(\frac{V_s}{RI_s}\right)}$$

Q4.01.10 NAT 2001 2M Ans: * ● ● ○

From Q4.01.9,

$$V_o = -V_T \ln\left(\frac{V_s}{RI_s}\right)$$

Substituting $R = 1\,\Omega$, $I_s = 1\,\text{pA}$, $V_T = 25\,\text{mV}$, $V_s = 1\,\text{V}$

$$V_o = -25 \times 10^{-3} \ln\left(\frac{1}{1 \times 10^{-12}}\right)$$

$$V_o \approx -0.69\,\text{V}$$

$$\boxed{-0.69\,\text{V}}$$

Q4.01.11 NAT 2001 2M Ans: * ● ● ○

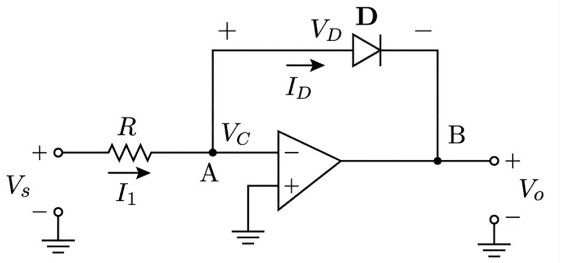

Fig. Solution Q4.01.11

Using the ideal OP-AMP property,

$$V_C = 0$$

Hence the current through the resistor is,

$$I_1 = \frac{V_s}{R}$$

Since the input current of the OP-AMP is zero,

$$I_D = I_1 = \frac{V_s}{R}$$

The diode current is,

$$I_D = I_s e^{V_D/V_T}$$

$$V_D = V_T \ln\left(\frac{I_D}{I_s}\right)$$

From the polarity shown in the figure

$$V_D = V_C - V_o = 0 - V_o = -V_o$$

Hence

$$-V_o = V_T \ln\left(\frac{V_s}{RI_s}\right)$$

$$V_o = -V_T \ln\left(\frac{V_s}{RI_s}\right)$$

$$\boxed{V_o = -V_T \ln\left(\frac{V_s}{RI_s}\right)}$$

Q4.00.1 NAT 2000 Ans: -100 ● ○ ○

At midband, the coupling capacitors C_b and C_c act as short circuits and the emitter bypass capacitor C_e shorts the emitter to AC ground

Hence the small-signal equivalent circuit is shown in the figure

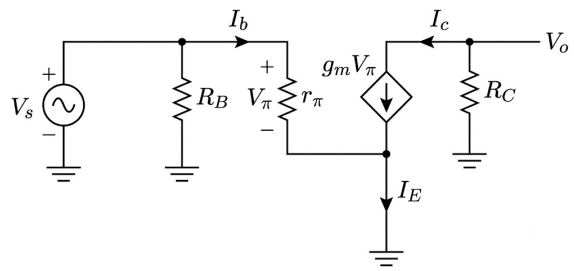

Fig. Solution Q4.00.1

Since the emitter is at AC ground

$$V_\pi = V_s$$

The collector current is

$$I_c = g_m V_\pi$$

The output voltage is

$$V_o = -I_c R_C$$

Substituting $I_c = g_m V_\pi$

$$V_o = -g_m V_\pi R_C$$

Since $V_\pi = V_s$

$$V_o = -g_m R_C V_s$$

Therefore

$$A_v = \frac{V_o}{V_s} = -g_m R_C$$

Now

$$g_m = \frac{I_C}{V_T} = \frac{1.3 \times 10^{-3}}{26 \times 10^{-3}} = 0.05\ \text{S}$$

Hence

$$A_v = -0.05 \times 2000 = -100$$

$$\boxed{A_v = -100}$$

Q4.00.2 NAT 2000 Ans: -3.81 ● ● ○

With C_e removed, the emitter is no longer at AC ground

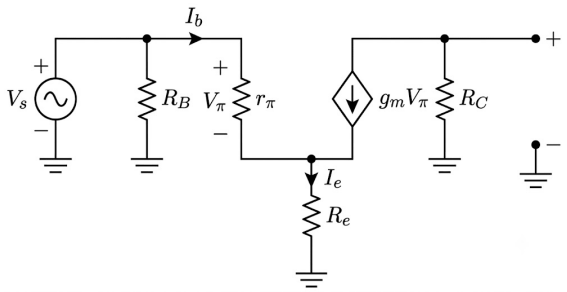

Fig. Solution Q4.00.2

Applying KVL in the input loop

$$V_s = V_\pi + I_e R_E$$

Since

$$I_e = I_c + I_b = \left(1 + \frac{1}{\beta}\right) I_c = \left(1 + \frac{1}{\beta}\right) g_m V_\pi$$

Therefore

$$V_s = V_\pi \left[1 + \left(1 + \frac{1}{\beta}\right) g_m R_E\right]$$

$$V_\pi = \frac{V_s}{1 + \left(1 + \frac{1}{\beta}\right) g_m R_E}$$

The output voltage is

$$V_o = -I_c R_C = -g_m V_\pi R_C$$

Therefore

$$A_v = \frac{V_o}{V_s} = -\frac{g_m R_C}{1 + \left(1 + \frac{1}{\beta}\right) g_m R_E}$$

Now

$$g_m = \frac{I_C}{V_T} = \frac{1.3 \times 10^{-3}}{26 \times 10^{-3}} = 0.05\ \text{S}$$

$$A_v = -\frac{0.05 \times 2000}{1 + \left(1 + \frac{1}{100}\right)(0.05)(500)} = -\frac{100}{26.25} \approx -3.81$$

$$\boxed{A_v \approx -3.81}$$

Q4.00.3 Descriptive 2000 Ans: $-\sin(\omega t)$ V ● ○ ○

If C_b is short circuited, it behaves as a wire

Since C_b is already considered as a short circuit for midband small-signal analysis, replacing it by an ideal short does not change the AC equivalent circuit

Therefore the voltage gain remains

$$A_v = -g_m R_C$$

Using

$$g_m = \frac{I_C}{V_T} = \frac{1.3 \times 10^{-3}}{26 \times 10^{-3}} = 0.05\ \text{S}$$

$$A_v = -0.05 \times 2000 = -100$$

Hence

$$V_o = A_v V_s$$

$$V_o = -100 \times 0.01 \sin(\omega t)$$

$$V_o = -\sin(\omega t)\ \text{V}$$

$$\boxed{V_o = -\sin(\omega t)\ \text{V}}$$

Q4.00.4 MCQ 2000 2M Ans: D ● ○ ○

Since β is very large, $I_B \approx 0$
Therefore, $I_C \approx I_E$
The Thevenin voltage is

$$V_{TH} = \frac{R_2}{R_1 + R_2} V_{CC} = \frac{5}{10 + 5} \times 15 = 5\ \text{V}$$

Applying KVL

$$V_{TH} = V_{BE} + I_E R_E$$

Hence

$$I_E = \frac{V_{TH} - V_{BE}}{R_E} = \frac{5 - 0.7}{430} = 10\,\text{mA}$$

Since

$$I_C \approx I_E$$

$$I_C = 10\,\text{mA}$$

$$\boxed{(D)\ 10\,\text{mA}}$$

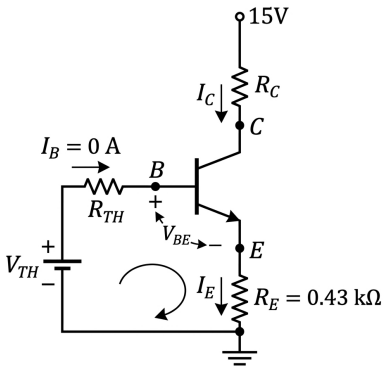

Fig. Solution Q4.00.4

Q4.00.5 MCQ 2000 1M Ans: A ● ○ ○

At high frequencies, the transistor junction capacitances become significant
The base-emitter capacitance (C_π) and base-collector capacitance (C_μ) reduce the current gain with increasing frequency
The current gain decreases approximately as

$$\beta(f) = \frac{\beta_0}{1 + j\left(\frac{f}{f_\beta}\right)}$$

Hence the correct option is

$$\boxed{(A)\ \text{Transistor capacitances}}$$

Q4.99.1 NAT 1999 5M Ans: * ●●○

At midband both capacitors are shorts; the ideal bias source is open for small signals, and the base is at AC ground. The transistor input resistance is

$$r_e = \frac{26\,\text{mV}}{0.5\,\text{mA}} = 52\,\Omega.$$

The collector sees $R_B \parallel R_L = 100\,\text{k}\Omega \parallel 1\,\text{k}\Omega = 990.099$ ohms. Its gain relative to the actual emitter voltage is about +19.04, but the question asks relative to the source voltage. The 50 ohm series input resistor must be included:

$$\frac{v_e}{v_s} = \frac{52}{50+52},$$

$$\frac{v_o}{v_s} = \frac{R_B \parallel R_L}{50+52}.$$

$$\boxed{v_o/v_s \approx +9.7069}$$

The input resistance at the transistor port, excluding the external series resistor, is 52 ohms; viewed by the source including that resistor it is 102 ohms. With input source zero and load excluded, $r_o \to \infty$ gives output resistance 100 kΩ.

Q4.99.2 MCQ 1999 1M Ans: A ●○○

Negative feedback reduces the overall gain of an amplifier
The closed-loop gain is given by

$$A_f = \frac{A}{1 + A\beta}$$

Since $(1 + A\beta) > 1$

$$A_f < A$$

Negative feedback also improves amplifier performance by reducing distortion and noise while increasing bandwidth

$$\boxed{(A)\ \text{Reduces gain}}$$

Q4.99.3 MCQ 1999 1M Ans: A ●○○

The transconductance of the cascode amplifier is,

$$g = \frac{I_o}{V_i}$$

$$g = \frac{I_o}{I_{e2}} \times \frac{I_{e2}}{I_{c1}} \times \frac{I_{c1}}{V_i}$$

For transistor Q_2 in common-base configuration,

$$\frac{I_o}{I_{e2}} = \alpha \approx 1$$

Since the collector of Q_1 is directly connected to the emitter of Q_2, $I_{e2} = I_{c1}$. Also, $\frac{I_{c1}}{V_i} = g_{m1}$. Therefore,

$$g = \alpha \times 1 \times g_{m1} \approx g_{m1}$$

$$\boxed{(A)\ g_{m1}}$$

Q4.99.4 MCQ 1999 2M Ans: C ● ○ ○

For a voltage-series feedback amplifier, the input impedance increases by a factor of $(1 + A\beta)$ and the output impedance decreases by the same factor.

$$R_{if} = R_i(1 + A\beta) \qquad R_{of} = \frac{R_o}{1 + A\beta}$$

Given, $A = 100$, $\beta = 0.99$, $R_i = 1\,\text{k}\Omega$, $R_o = 100\,\Omega$
Therefore

$$1 + A\beta = 1 + 100 \times 0.99 = 100$$

$$R_{if} = 1\,\text{k}\Omega \times 100 = 100\,\text{k}\Omega$$

$$R_{of} = \frac{100}{100} = 1\,\Omega$$

$\boxed{(C)\ 100\,\text{k}\Omega \text{ and } 1\,\Omega}$

Q4.99.5 MCQ 1999 2M Ans: A ● ● ○

The transconductance of the transistor is

$$g_m = \frac{I_C}{V_T} = \frac{1\,\text{mA}}{26\,\text{mV}} = \frac{1}{26}\,\text{A/V}$$

The unity-gain cutoff frequency is

$$f_T = \frac{g_m}{2\pi(C_\pi + C_\mu)}$$

Therefore

$$C_\pi + C_\mu = \frac{g_m}{2\pi f_T} = \frac{1/26}{2\pi(400 \times 10^6)} = 15.3\,\text{pF}$$

Hence

$$C_\pi = 15.3 - 0.3 = 15\,\text{pF}$$

$\boxed{(A)\ 15\,\text{pF}}$

Q4.99.6 MCQ 1999 2M Ans: B ● ○ ○

For a single-pole amplifier, the relationship between rise time and bandwidth is

$$BW = \frac{0.35}{t_r}$$

Given, $t_r = 35\,\text{ns} = 35 \times 10^{-9}\,\text{s}$
Therefore

$$BW = \frac{0.35}{35 \times 10^{-9}} = 10 \times 10^6\,\text{Hz} = 10\,\text{MHz}$$

$\boxed{(B)\ 10\,\text{MHz}}$

Q4.98.1 NAT | 1998 | 5M | Ans: 19.61 | ● ● ○

Set the independent input source to zero. Ignoring the two bias resistors as requested, the base reaches AC ground through $R_s = 1\ \text{k}\Omega$. The resistance looking into the emitter through the transistor is

$$r_{e,\text{look}} = \frac{R_s + r_\pi}{\beta + 1} = 20\,\Omega.$$

This is in parallel with the 1 kΩ emitter resistor:

$$R_o = 1000 \parallel 20 = \frac{20000}{1020}\,\Omega.$$

$$R_o = 19.607843\ldots\,\Omega.$$

$$\boxed{19.61\,\Omega \text{ (two decimals)}}$$

The resistance to two decimal places is 19.61 ohms.

Q4.98.2 MCQ | 1998 | 1M | Ans: B | ● ● ○

The bandwidth of a multistage (cascade) amplifier is

$$\frac{1}{B} = 1.1\sqrt{\frac{1}{B_1^2} + \frac{1}{B_2^2} + \frac{1}{B_3^2}}$$

$$\frac{1}{B} = 1.1\sqrt{\frac{1}{\omega_1^2} + \frac{1}{\omega_2^2} + \frac{1}{\omega_3^2}}$$

Approximating

$$1.1\sqrt{\frac{1}{\omega_1^2} + \frac{1}{\omega_2^2} + \frac{1}{\omega_3^2}} \approx \frac{1}{\omega_1} + \frac{1}{\omega_2} + \frac{1}{\omega_3}$$

$$\boxed{(B)\ \frac{1}{B} = \frac{1}{\omega_1} + \frac{1}{\omega_2} + \frac{1}{\omega_3}}$$

Q4.98.3 MCQ | 1998 | 1M | Ans: D | ● ○ ○

The feedback resistor is connected across the output node. Hence, the output voltage is sampled
At the input, the feedback signal is connected in parallel with the input source. Therefore, the input mixing is shunt mixing
Thus, the given circuit employs voltage-shunt feedback

$$\boxed{(D)\ \text{Voltage shunt}}$$

Q4.98.4 MCQ | 1998 | 1M | Ans: A | ● ○ ○

In a shunt-shunt feedback amplifier:
Shunt mixing at the input decreases the input impedance
Shunt sampling at the output decreases the output impedance

$$\boxed{(A)\ \text{Both input and output impedances decrease}}$$

Q4.98.5 MCQ 1998 1M Ans: D ● ○ ○

The unity-gain cutoff frequency (or gain-bandwidth product) of a BJT is given by

$$f_T = \frac{g_m}{2\pi(C_\pi + C_\mu)}$$

where $g_m = \dfrac{I_C}{V_T}$, C_π is the diffusion capacitance and C_μ is the transition capacitance

$$\boxed{(D)\ \frac{g_m}{2\pi(C_\pi + C_\mu)}}$$

Q4.98.6 MCQ 1998 1M Ans: C ● ○ ○

For a small-amplitude square-wave input, the rise time of the output is related to the upper 3 dB frequency by

$$f_H = \frac{0.35}{t_r}$$

Hence, by measuring the rise time, the upper 3 dB frequency of the amplifier can be estimated

$$\boxed{(C)\ \text{Upper 3-dB frequency}}$$

Q4.97.1 NAT 1997 5M Ans: * ● ● ○

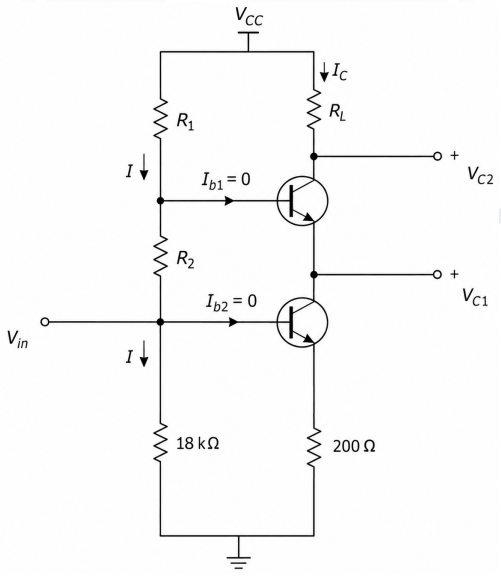

Fig. Solution Q4.97.1

Since β is very high, $I_{B1} = I_{B2} = 0$
Hence the same current I flows through R_1, R_2 and the 18 kΩ resistor
Applying KVL across V_{CC}, R_L and V_{C2}

$$-V_{CC} + I_C R_L + V_{C2} = 0$$

$$-9 + (1\,\text{mA})R_L + 6 = 0$$

$$R_L = 3\,\text{k}\Omega$$

Applying KVL across V_{CC}, R_1, V_{BE} and V_{C1}

$$-V_{CC} + IR_1 + V_{BE} + V_{C1} = 0$$

$$-9 + IR_1 + 0.7 + 3 = 0$$

$$IR_1 = 5.3$$

Applying KVL across V_{CC}, R_1, R_2, V_{BE} and $200\,\Omega$

$$-V_{CC} + IR_1 + IR_2 + V_{BE} + I_E(200) = 0$$

$$-9 + IR_1 + IR_2 + 0.7 + (1\,\text{mA})(200) = 0$$

Substituting $IR_1 = 5.3$

$$-9 + 5.3 + IR_2 + 0.7 + 0.2 = 0$$

$$IR_2 = 2.8$$

Also

$$I = \frac{V_{B1}}{18\,\text{k}\Omega} = \frac{0.7 + 0.2}{18\,\text{k}\Omega} = \frac{0.9}{18\,\text{k}\Omega} = 50\,\mu\text{A}$$

Therefore

$$R_1 = \frac{5.3}{50\,\mu\text{A}} = 106\,\text{k}\Omega$$

$$R_2 = \frac{2.8}{50\,\mu\text{A}} = 56\,\text{k}\Omega$$

$$\boxed{R_L = 3\,\text{k}\Omega,\ R_1 = 106\,\text{k}\Omega,\ R_2 = 56\,\text{k}\Omega}$$

Q4.97.2 MCQ 1997 1M Ans: B ● ○ ○

Without the capacitor, R_E provides emitter degeneration (negative feedback), which reduces the voltage gain and increases the input impedance.
Placing a capacitor across R_E bypasses it for AC signals, thereby removing the negative feedback.
Hence, the voltage gain increases, while the input impedance decreases.

$$\boxed{(B)\ \text{Increase the voltage gain and decrease the input impedance}}$$

Q4.97.3 MCQ 1997 1M Ans: A ● ○ ○

A cascode amplifier consists of a common emitter (CE) stage followed by a common base (CB) stage.
The CE stage provides high current gain, while the CB stage suppresses the Miller effect, resulting in high gain and improved high-frequency response.

$$\boxed{(A)\ \text{A common emitter stage followed by a common base stage}}$$

Q4.97.4 MCQ 1997 2M Ans: C ● ○ ○

In a voltage-series feedback configuration, the feedback is mixed in series at the input. Hence, the input impedance increases.
In a current-shunt feedback configuration, the feedback is mixed in shunt at the input. Hence, the input impedance decreases.
Therefore, the correct matching is $1 \rightarrow (a)$, $2 \rightarrow (b)$

$$\boxed{(C)\ 1\text{-}a,\ 2\text{-}b}$$

Q4.96.1 NAT | 1996 | 5M | **Ans:** 12.85×10^3 | ● ● ○

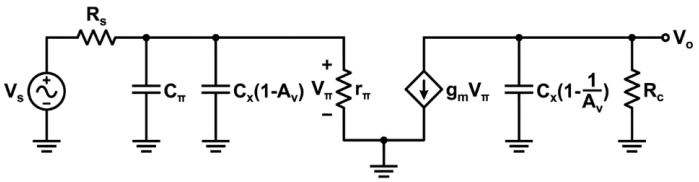

Fig. Solution Q4.96.1

For the intended dominant-pole Miller estimate, combine $C_x = C_C + C_\mu = 25.5$ pF and use the midband base-to-collector gain $-g_m R_C = -50$. Then

$$C_i = C_\pi + C_x(1 + 50) = 1302\,\text{pF},$$

$$R_i = R_s \parallel r_\pi = 9.52381\,\text{k}\Omega,$$

$$f_{Hi} = \frac{1}{2\pi R_i C_i} \approx 12.84\,\text{kHz}.$$

The output Miller capacitance is 26.01 pF, giving a much higher output pole, approximately 612 kHz. The input pole dominates. This is a Miller approximation; the source answer 12.85 kHz is its rounded approximate value, not an exact full-transfer-function cutoff.

Q4.96.2 MCQ | 1996 | 1M | **Ans: B** | ● ○ ○

Both the emitter-base and collector-base junctions are forward biased. Therefore, the transistor is operating in saturation.
Since the collector-base forward bias is greater than the emitter-base forward bias, the carrier flow is predominantly from the collector towards the emitter.
Hence, the transistor operates in reverse saturation mode.

(B) reverse saturation mode

Q4.96.3 MCQ | 1996 | 2M | **Ans: D** | ● ● ○

The overall transconductance of the Darlington pair is,

$$g_m = \frac{I_{c2}}{V_{be}}$$

From the figure,

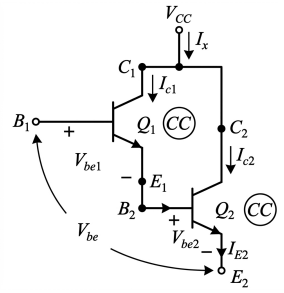

Fig. Solution Q4.96.3

$$V_{be} = V_{be1} + V_{be2}$$

Since

$$I_{c1} = g_{m1}V_{be1}, \qquad I_{c2} = g_{m2}V_{be2},$$
$$V_{be1} = \frac{I_{c1}}{g_{m1}}, \qquad V_{be2} = \frac{I_{c2}}{g_{m2}}$$

Therefore,

$$V_{be} = \frac{I_{c1}}{g_{m1}} + \frac{I_{c2}}{g_{m2}}$$

From the Darlington connection,

$$I_{E1} = I_{B2}, \qquad I_{c1} \approx I_{E1}, \qquad I_{c2} = \beta I_{B2}$$

Hence,

$$I_{c2} \approx \beta I_{c1}$$

Since, $g_m = \frac{I_C}{V_T}$,

$$g_{m2} = \frac{I_{c2}}{V_T} = \frac{\beta I_{c1}}{V_T} = \beta g_{m1}$$

Therefore,

$$\frac{I_{c1}}{g_{m1}} = \frac{I_{c2}/\beta}{g_{m2}/\beta} = \frac{I_{c2}}{g_{m2}}$$

Substituting in the expression for V_{be},

$$V_{be} = \frac{I_{c2}}{g_{m2}} + \frac{I_{c2}}{g_{m2}} = \frac{2I_{c2}}{g_{m2}}$$

Hence,

$$g_m = \frac{I_{c2}}{V_{be}} = \frac{I_{c2}}{2I_{c2}/g_{m2}} = \frac{g_{m2}}{2}$$

$$\boxed{(D)\ 0.5\, g_{m2}}$$

Q4.96.4 MCQ 1996 3M Ans: A ● ○ ○

The characteristics of the given amplifiers are:

- Cascode amplifier is a wideband amplifier due to the reduced Miller effect.
- Differential amplifier provides high common-mode rejection.
- Darlington pair common-collector amplifier has very high input impedance and very high current gain.

Therefore, $A \rightarrow 2$, $B \rightarrow 5$ and $C \rightarrow 4$

$$\boxed{(A)\ A\text{-}2,\ B\text{-}5,\ C\text{-}4}$$

Q4.96.5 MCQ 1996 2M Ans: A ● ○ ○

Given, $f_\beta = 1\,\text{MHz}$ and $\beta_0 = 200$
The unity gain frequency of a BJT is,

$$f_T = \beta_0 f_\beta = (200)(1\,\text{MHz}) = 200\,\text{MHz}$$

The alpha cut-off frequency is,

$$f_\alpha = \frac{f_\beta}{1-\alpha}$$

Since $\alpha = \dfrac{\beta}{1+\beta}$,

$$1 - \alpha = 1 - \frac{\beta}{1+\beta} = \frac{1}{1+\beta}$$

Therefore,

$$f_\alpha = (1+\beta) f_\beta = (1+200)(1\,\text{MHz}) = 201\,\text{MHz}$$

$$\boxed{(A)\ 200\,\text{MHz},\ 201\,\text{MHz}}$$

Q4.96.6 MCQ 1996 2M Ans: D ● ● ○

Since the amplifier has a very large input impedance, the input current flows only through the feedback impedance Z.
Therefore,

$$I_i = \frac{V_i - V_o}{Z}$$

Given the voltage gain of the amplifier,

$$V_o = KV_i$$

Substituting,

$$I_i = \frac{V_i - KV_i}{Z} = \frac{V_i(1-K)}{Z}$$

Hence, the input impedance is

$$Z_{in} = \frac{V_i}{I_i} = \frac{V_i}{V_i(1-K)/Z} = \frac{Z}{1-K}$$

$$\boxed{(D)\ \frac{Z}{1-K}}$$

Q4.95.1 NAT 1995 1M Ans: 3.71 ● ● ○

Given, $\alpha = 0.99$ and $V_{BE} = 0.7\,\text{V}$
Hence,

$$\beta = \frac{\alpha}{1-\alpha} = \frac{0.99}{0.01} = 99$$

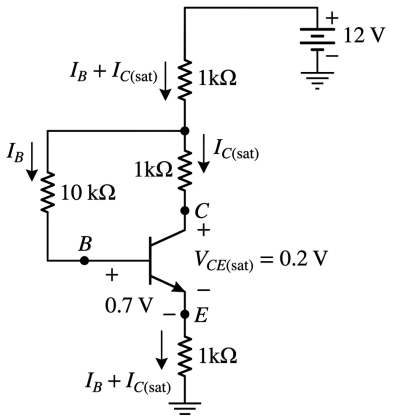

Fig. Solution Q4.95.1

Assume that the transistor is operating in saturation. Thus taking $V_{CE(sat)} = 0.2\,\text{V}$
Applying KVL in the loop containing the base-emitter junction,

$$-12 + 1\,\text{k}(I_B + I_{C(sat)}) + 10\,\text{k}I_B + 0.7 + 1\,\text{k}(I_B + I_{C(sat)}) = 0$$

$$12\,\text{k}I_B + 2\,\text{k}I_{C(sat)} = 11.3 \tag{1}$$

Applying KVL in the loop containing $V_{CE(sat)}$,

$$-12 + 1\,\text{k}(I_B + I_{C(sat)}) + 1\,\text{k}I_{C(sat)} + 0.2 + 1\,\text{k}(I_B + I_{C(sat)}) = 0$$

$$2\,\text{k}I_B + 3\,\text{k}I_{C(sat)} = 11.8 \tag{2}$$

Solving (1) and (2),

$$I_B = 0.321\,\text{mA}, \qquad I_{C(sat)} = 3.71\,\text{mA}$$

The maximum collector current supported by the available base current is

$$\beta I_B = 99 \times 0.321 = 31.779\,\text{mA}$$

Since $\beta I_B > I_{C(sat)}$, the transistor is indeed in saturation.

$$\boxed{I_C = 3.71\,\text{mA}}$$

Q4.95.2 MCQ 1995 1M Ans: B ● ○ ○

In a current-series feedback amplifier, the output current is sampled and the feedback signal is applied in series at the input. Series mixing increases the input impedance, while current sampling increases the output impedance. Hence, both the input and output impedances are high.

$$\boxed{(B)\ \text{Current-series}}$$

Q4.95.3 NAT 1995 1M Ans: 1.59 ● ○ ○

Given, $P = 10\%$, and $f = 50\,\text{Hz}$
The percentage tilt for a single-pole RC-coupled amplifier is,

$$P = \frac{\pi f_L}{f} \times 100\%$$

Therefore,

$$\frac{\pi f_L}{50} \times 100 \leq 10$$

$$f_L \leq \frac{10 \times 50}{\pi \times 100}$$

$$f_L \leq 1.59\,\text{Hz}$$

Hence, the maximum allowable lower cut-off frequency is: 1.59 Hz

$$\boxed{1.59\,\text{Hz}}$$

Q4.95.4 Descriptive | 1995 | 1M | Ans: 1, 10 | ● ○ ○

Given, $A_{OL} = 100$, $f_L = 100\,\text{Hz}$, $f_H = 100\,\text{kHz}$, and $\beta = 0.99$
The desensitivity factor is: $D = 1 + A_{OL}\beta = 1 + 100 \times 0.99 = 100$
The lower cut-off frequency with feedback is: $f_{Lf} = \frac{f_L}{D} = \frac{100}{100} = 1\,\text{Hz}$
The upper cut-off frequency with feedback is: $f_{Hf} = f_H D = 100 \times 10^3 \times 100 = 10^7\,\text{Hz} = 10\,\text{MHz}$

$$\boxed{1\,\text{Hz},\ 10\,\text{MHz}}$$

Q4.94.1 NAT | 1994 | 1M | Ans: 1 | ● ○ ○

Given, $I_C = 1\,\text{mA}$ and $I_B = 25\,\mu\text{A}$
The input resistance of a common-emitter amplifier is:

$$R_i = r_\pi = \frac{\beta}{g_m}$$

Since $\beta = \frac{I_C}{I_B}$ and $g_m = \frac{I_C}{V_T}$,

$$R_i = \frac{I_C/I_B}{I_C/V_T} = \frac{V_T}{I_B}$$

At room temperature, $V_T = 26\,\text{mV}$

$$R_i = \frac{26 \times 10^{-3}}{25 \times 10^{-6}} = 1.04\,\text{k}\Omega \approx 1\,\text{k}\Omega$$

$$\boxed{1\,\text{k}\Omega}$$

Q4.94.2 Descriptive | 1994 | 1M | Ans: Large | ● ○ ○

Harmonic distortion occurs when an amplifier is driven beyond its linear operating region. Increasing the dynamic range allows the amplifier to handle a larger range of input signal amplitudes while maintaining linear operation, thereby reducing harmonic distortion.

$$\boxed{\text{Large}}$$

Q4.93.1 MCQ 1993 2M Ans: D ● ○ ○

The α cut-off frequency is inversely proportional to the base transit time.
Decreasing the base width reduces the base transit time, thereby increasing the α cut-off frequency.

(D) Increase with decrease in the base width

Q4.93.2 MCQ 1993 1M Ans: C ● ○ ○

The given circuit is a common-collector (emitter follower) amplifier.
The mid-band voltage gain of an emitter follower is

$$A_v = \frac{V_o}{V_i} = \frac{I_B(1+h_{fe})R_E}{I_B\left[h_{ie} + (1+h_{fe})R_E\right]}$$

Since $(1+h_{fe})R_E \gg h_{ie}$,

$$A_v = \frac{(1+h_{fe})R_E}{h_{ie} + (1+h_{fe})R_E} \approx 1$$

(C) ≈ 1

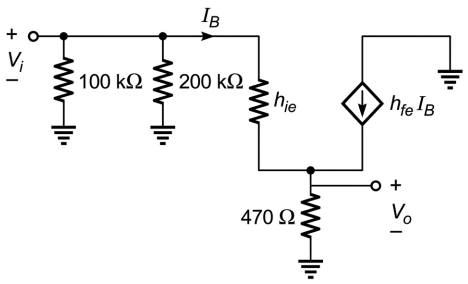

Fig. Solution Q4.93.2

Q4.93.3 MCQ 1993 1M Ans: A ● ○ ○

In an RC-coupled amplifier, the coupling capacitor acts as a high-pass element.
At low frequencies, its reactance is high, which attenuates the signal and limits the low-frequency response. At mid and high frequencies, its reactance is negligible.

(A) Limits the low-frequency response

Q4.93.4 MCQ 1993 1M Ans: C ● ○ ○

The overall bandwidth of an n-stage synchronously tuned amplifier, where each stage has a bandwidth B, is $BW_n = B\sqrt{2^{1/n} - 1}$

(C) $B\sqrt{2^{1/n} - 1}$

Q4.93.5 MSQ 1993 1M Ans: B,D ● ○ ○

Negative feedback does not improve the signal-to-noise ratio at the input since the input signal and input noise are unaffected. However, it improves the signal-to-noise ratio at the output by reducing the effect of internally generated noise. Negative feedback also linearizes the amplifier characteristics and reduces nonlinear distortion.

(B) Improves the signal-to-noise ratio at the output

(D) Reduces distortion

Q4.92.1 NAT 1992 2M Ans: 1 ● ○ ○

Since β is very large, $I_B = 0$

Hence, the voltage across the 1.65 kΩ resistor is obtained using the voltage divider rule.

$$V_{1.65} = 10 \times \frac{1.65}{6.5 + 1.85 + 1.65} = 1.65\ \text{V}$$

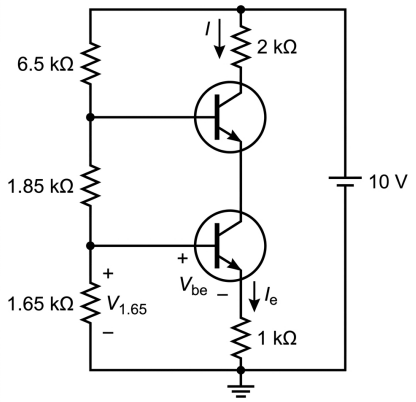

Fig. Solution Q4.92.1

Applying KVL in the input loop,

$$V_{1.65} = V_{BE} + I_E(1\ \text{k}\Omega)$$

$$1.65 = 0.65 + I_E(1\ \text{k}\Omega)$$

$$I_E = 1\ \text{mA}$$

Since β is very large, $I_C \simeq I_E$

$$I = I_C = 1\ \text{mA}$$

1 mA

Q4.91.1 MSQ 1991 2M Ans: A,C ● ○ ○

For a non-inverting amplifier, $A_f = \dfrac{1}{\beta}$

A higher closed-loop gain requires a smaller feedback factor, implying less negative feedback.

Also, for an op-amp, $A_f \times BW$ = constant

Hence, increasing the closed-loop gain decreases the bandwidth.

Therefore, compared to the unity gain amplifier, the amplifier with gain 20 has less negative feedback and less bandwidth.

(A) Less negative feedback

(C) Less bandwidth

Q4.91.2 NAT 1991 2M Ans: 5 ● ○ ○

Since all the transistors are identical and β is very high, the base currents are negligible. Hence, the 5 mA reference current is mirrored to the output branch.

$$I_C = 5\,\text{mA}$$

Applying Ohm's law across the 1 kΩ resistor,

$$V_{DC} = 10 - I_C(1\,\text{k}\Omega) = 10 - (5\,\text{mA})(1\,\text{k}\Omega) = 5\,\text{V}$$

$$\boxed{5\,\text{V}}$$

Q4.90.1 MSQ 1990 2M Ans: A,B ● ○ ○

A common-base (CB) amplifier has low input impedance and a current gain approximately equal to unity.
A common-collector (CC) amplifier has low output impedance and a high current gain.
A common-emitter (CE) amplifier provides high voltage gain and moderate input impedance. Hence, option (C) is incorrect.
The current gain of a CB amplifier is approximately unity, whereas that of a CC amplifier is $(\beta + 1)$. Hence, option (D) is incorrect.

(A) CB amplifier has low input impedance and low current gain

(B) CC amplifier has low output impedance and high current gain

Q4.90.2 MCQ 1990 2M Ans: B ● ● ○

The Thevenin equivalent of the biasing network is

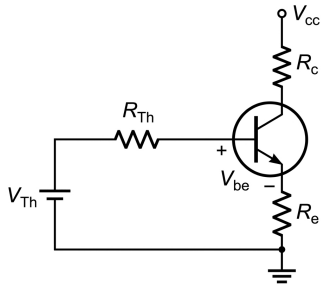

Fig. Solution Q4.90.2

$$V_{Th} = V_{CC}\frac{R_1}{R_1 + R_2}, \qquad R_{Th} = R_1 \parallel R_2 = R_B$$

Applying KVL to the input loop,

$$V_{Th} = I_B R_B + V_{BE} + I_E R_E$$

Since $I_E = I_B + I_C$,

$$V_{Th} = I_B R_B + V_{BE} + (I_B + I_C)R_E$$

Differentiating w.r.t. I_C, keeping β and V_{BE} constant,

$$0 = (R_B + R_E)\frac{\partial I_B}{\partial I_C} + R_E$$

$$\frac{\partial I_B}{\partial I_C} = -\frac{R_E}{R_B + R_E}$$

The stability factor is,

$$S = \frac{1+\beta}{1-\beta\dfrac{\partial I_B}{\partial I_C}}$$

$$S = \frac{1+\beta}{1+\beta\left(\dfrac{R_E}{R_B+R_E}\right)}$$

Since $\beta \gg 1$,

$$S \approx 1 + \frac{R_B}{R_E}$$

For better stability, $S \approx 1$ which requires $\dfrac{R_B}{R_E} \ll 1$ or equivalently,

$$\frac{R_E}{R_B} \gg 1$$

$$\boxed{(B)\ \frac{R_E}{R_B} \gg 1}$$

Q4.89.1 MCQ | 1989 | 2M | Ans: A | ● ○ ○

The voltage-divider bias circuit provides the maximum bias stability.
The voltage divider establishes a nearly constant base voltage, while the emitter resistor provides negative feedback, making the operating point less sensitive to variations in β and temperature. Hence, it offers the maximum bias stability among the given circuits.

$$\boxed{(A)}$$

Q4.89.2 MCQ | 1989 | 2M | Ans: C | ● ● ○

The feedback signal is sampled from the output voltage and is fed back in shunt at the input. Hence, the circuit employs voltage-shunt feedback.
Voltage-shunt feedback decreases both the input impedance and the output impedance.

$$\boxed{(C)\ \text{Voltage-shunt feedback with low input impedance and low output impedance}}$$

Q4.88.1 MCQ | 1988 | 2M | Ans: D | ● ● ○

Since all the capacitors are large, the BJT base is at AC ground.
Hence, the MOSFET operates in common-source configuration and the BJT operates in common-base configuration, forming a cascode stage.
The drain current of the MOSFET is, $i_d = g_{m(\text{MOS})} v_i$
The collector current of the BJT is, $i_c = \alpha i_e \approx \alpha i_d$, where $\alpha = \dfrac{\beta}{\beta+1} = \dfrac{99}{100} = 0.99$
Therefore, the overall transconductance is,

$$g_m = \alpha g_{m(\text{MOS})} = 0.99 \times 2 = 1.98\ \text{mA/V}$$

$$\boxed{(D)\ 1.98\ \text{mA/V}}$$

Q4.88.2 MCQ 1988 2M Ans: D ● ○ ○

For a Darlington pair, the overall current gain is: $h_{FE} = h_{FE1}h_{FE2} + h_{FE1} + h_{FE2}$
Given, $h_{FE1} = h_{FE2} = 100$
Therefore, $h_{FE} = (100)(100) + 100 + 100 = 10200$

$\boxed{(D)\ 10200}$

Q4.88.3 MCQ 1988 2M Ans: C ● ○ ○

The transconductance of a BJT is given by, $g_m = \dfrac{I_C}{V_T}$, where V_T is the thermal voltage, which is approximately constant at a given temperature.
Hence, g_m is directly proportional to the quiescent collector current I_C.
Therefore, increasing I_C increases g_m.

$\boxed{(C)\ g_m \text{ will increase}}$

Q4.88.4 MCQ 1988 2M Ans: D ● ● ○

Since the coupling and bypass capacitors are very large, they act as short circuits for AC analysis.
From the given small-signal model, $h_{re} = 0$. Hence, $I_b = 0$. Therefore, the controlled current source is, $h_{fe}I_b = 0$
The output impedance is the parallel combination of the transistor output resistance and the collector resistance.

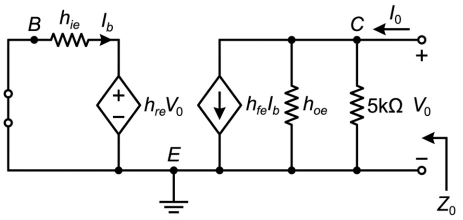

Fig. Solution Q4.88.4

$$r_o = \frac{1}{h_{oe}} = \frac{1}{0.05\ \text{mmho}} = 20\,\text{k}\Omega$$

$$Z_o = r_o \parallel R_C = 20\,\text{k}\Omega \parallel 5\,\text{k}\Omega = \frac{20 \times 5}{20 + 5} = 4\,\text{k}\Omega$$

$\boxed{(D)\ 4\,\text{k}\Omega}$

SOLUTIONS

V

Miscellaneous Topics

Topics

1. CMOS Inverter

Unit Snapshot*

Stream: EC	**Questions:** 1	**MCQ:** 0	**MSQ:** 0	**NAT:** 1
1M: 0	**2M:** 1	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 2M(2019)–2M(2019)

**See preface for how Avg Weightage and Range are calculated.*

PrepFusion

Q5.19.1 NAT | 2019 | 2M | Ans: 0.225 | ● ○ ○

Given,

$$V_{dd} = 3 \text{ V}$$

$$\mu_n C_{ox} = 100\ \mu\text{A/V}^2 \text{ ; } V_{tn} = 0.7 \text{ V for nMOS}$$

$$\mu_p C_{ox} = 40\ \mu\text{A/V}^2 \text{ ; } |V_{tp}| = 0.9 \text{ V for pMOS}$$

From the graph $V_{in} = V_{DD}/2$ and $V_{out} = V_{DD}/2$
The CMOS inverter circuit is given below where,

$$I_{Dp} = I_{Dn} \text{ and } V_{SGp} = V_{GSn} = 1.5$$

and both MOSFETs are in saturation.

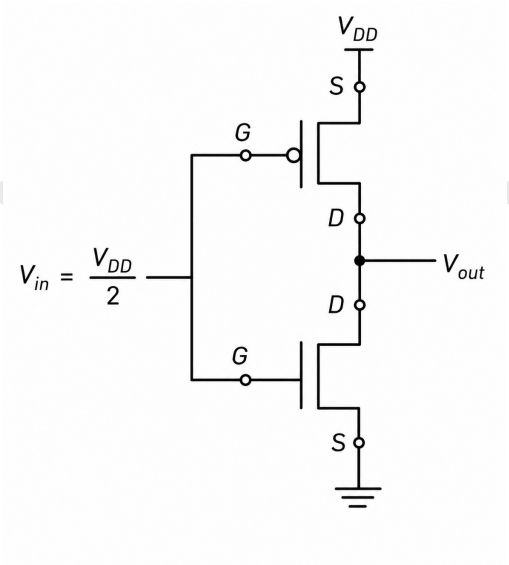

Fig. Solution Q5.19.1

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$$\Rightarrow \frac{1}{2}(\mu_n C_{ox})\left(\frac{W}{L}\right)_n [V_{GSN} - V_{TN}]^2 = \frac{1}{2}(\mu_p C_{ox})\left(\frac{W}{L}\right)_p [V_{SGP} + V_{TP}]^2$$

$$100 \times 10^{-6}\left(\frac{W}{L}\right)_n [1.5 - 0.7]^2 = 40 \times 10^{-6}\left(\frac{W}{L}\right)_p [1.5 - 0.9]^2$$

$$\Rightarrow \frac{\left(\frac{W}{L}\right)_n}{\left(\frac{W}{L}\right)_p} = \frac{40}{100} \times \frac{(0.6)^2}{(0.8)^2} = 0.225$$

Key Points

- At the mid-point (switching threshold) of a CMOS inverter, $V_{in} = V_{out} = \frac{V_{DD}}{2}$.
- Under this condition, both the nMOS and pMOS transistors operate in the saturation region.

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Engineering Mathematics	Linear Algebra, Calculus, Differential Equations, Complex Variables, Probability & Statistics, Numerical Methods, Transform Theory and Vector Calculus — all branches
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